



Choupo

The Open Source Chemical Process Simulator

Tutorials Guide

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How to use this guide

Every Choupo tutorial is a complete, runnable case:

```
source etc/bashrc
runCase tutorials/<category>/<group>/<name>
```

This guide gives each tutorial the CONTEXT the case file cannot: the theory block at the head of each group states the physics and the equations you need (condensed – the Theory Guide carries the full derivations), and each entry tells you **what the case does** and **what to expect** before you run it. The per-tutorial text is assembled from the cases' own headers – the case file is always the source of truth, and reading it IS part of the exercise.

Three reading rules:

1. **Run first, read second, re-run third.** The log at verbosity 3 shows every Newton iteration, every K-value, every announcement – the glass box is the point.
2. **Expect the announcements.** Choupo never silently helps: initial guesses, bounds that bind, model fallbacks, unit overrides – all are printed. If a run is quiet about something, it did not happen.
3. **Golden numbers are commitments.** Cases carrying an **expected** file pin their KPIs; the validation AADs quoted in the entries are locked by the regression suite and re-verified on every change.

Learning paths

The MCFT path (computational transfer phenomena). Start with `flash01_benzene_toluene` (the Rachford–Rice loop, visible), then `flash` non-ideal cases (activity models bend the K-values), `distillation` (stages couple; Wang–Henke vs simultaneous), one `flowsheets` recycle case (tears and convergence), and close with a `gibbs` flame (equilibrium without mechanism). Then the batch side: an `ignition` case for stiffness – run the `_rk4` twin and watch the explicit method die – and a `ctrl` loop for feedback.

The electrolytes path. `pitzer01_nacl` ($\gamma_{\pm}$, φ , a_w of one salt), then `pitzer_gamma_hamer_wu` (the catalogue vs NBS data, zero refit), `pitzer_nacl_sp77_hot` (temperature, to 200°C), speciation (`scaling` cases: SI, pH, Davies-vs-Pitzer forking an engineering answer), the eNRTL series (`enrtl_mixed_nacl_ethanol_esteso` mixed-solvent; `enrtl_multisalt_nacl_kcl` two salts, two formulations, an experimental verdict), and the process payoff: evaporators and crystallisers where all of it sets duties and yields.

The combustion path. `gibbs06_h2_flame_radicals` (the radical pool at equilibrium; the c_p extrapolation trap), `ignition01_h2o2` (GRI kinetics, the stiffness lesson via the `_rk4` twin), `ignition02_h2air_slack` (quantitative: τ_{ign} vs shock-tube data), `nox01_zeldovich_postflame` (thermal NO; kinetics meets Gibbs at 98.9%).

The props-bench path. `estimate_acetone` (CREATE: groups $\rightarrow$ constants, error shown), a `scan` case (SEE), `vle01_consistency` (TEST the data), a `fit` case (Levenberg–Marquardt with identifiability), then the promote-and-use loop into a steady case. The Props Guide is this path’s companion.

Category I: Steady-state flowsheets (choupoSolve)

Solved as one nonlinear system $F(\mathbf{x}) = \mathbf{0}$: units in sequence, recycles as tears, everything announced.

absorption

The theory you need. Gas absorption transfers a solute from gas to liquid across stages; Henry's law ($p_i = H_i x_i$) anchors the equilibrium at dilution. Expect the L/G ratio to control the operating line's slope, and a minimum solvent rate at the pinch with the equilibrium line. *Full derivation: Theory Guide, the absorption section.*

absorber01_NH3_water

tutorials/steady/absorption/absorber01_NH3_water



What it does: NH₃ absorption from air into water (Kremser, 6 stages)

The story: The textbook gas-absorption case: scrubbing ammonia out of an air stream with water, in a counter-current packed/plate column, solved by the Kremser group method (the built-in 'absorber').

Ammonia is very soluble in water (Henry's constant $H \approx 0.99$ bar at 298 K $\rightarrow K \approx 1$ at atmospheric pressure), so a modest water rate and a handful of stages recover most of it — unlike CO₂/water ($H \approx 1640$ bar), where water is a poor solvent.

Feed gas: 100 kmol/h, 10 % NH₃ / 90 % N₂ (N₂ = inert carrier) Solvent: 150 kmol/h pure water ($L/V = 1.5$) Column: 6 theoretical stages, 298.15 K, 1 atm

The model is NON-ISOTHERMAL: ammonia dissolution is strongly exothermic ($\Delta H \approx -34$ kJ/mol), so the heat of absorption heats the column. The energy balance is solved coupled with the mass balance.

Expected: liquid heats up $\approx +17$ K (top 309 K, internal bulge ~ 318 K, bottom 315 K — the classic absorber "temperature bulge") $\text{recovery_NH}_3 \approx 0.675$ (only ~ 68 %, NOT the 97 % an isothermal Kremser would predict — because T rises, H rises, K rises, A falls: the exotherm self-limits the absorption. Real NH₃ / amine absorbers are cooled for this reason.) N₂ stays in the gas ($K_{\text{N}_2} \gg 1 \rightarrow$ no absorption) NH₃ mass balance closes (in = clean-gas NH₃ + rich-liquid NH₃)

Design question ("how many stages for 90 %?") is a DesignSpec on '\$stages', NOT a mode of this unit — the column has the stages it has; the recovery is what comes out.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cleanGas	F	0.0259019084984
stream	gasFeed	F	0.0277777777778
stream	richLiquid	F	0.043542535946
stream	solvent	F	0.0416666666667
kpi	scrubber	A_N2	0.00225907961951
kpi	scrubber	A_NH3	1.53522727273
kpi	scrubber	F_cleanGas	0.0259019084984
kpi	scrubber	F_richLiquid	0.043542535946
kpi	scrubber	K_N2	663.987221631
kpi	scrubber	K_NH3	0.977054034049
kpi	scrubber	L_in	0.0416666666667
kpi	scrubber	L_over_V	1.5
kpi	scrubber	V_in	0.0277777777778
kpi	scrubber	recovery_N2	0

... and 2 more reference values in the case's *expected*.

absorption01_CO2_water

tutorials/steady/absorption/absorption01_CO2_water



What it does: Single-stage CO2 absorption in water (Henry's law)

The story: single-stage CO2 absorption in water, modelled via the v0.28 Henry's-law infrastructure.

Feed: a 5100 kmol/h mixed stream of CO2 / N2 / water already at 298.15 K and 5 bar, with mole fractions matching what you'd obtain by mixing 100 kmol/h of a 10/90 CO2/N2 sour-gas with 5000 kmol/h of liquid water (L/G ~ 50:1 mole). An IsothermalFlash then partitions the species between vapour and liquid.

Why a pre-mixed feed and not Mixer + Flash? The Mixer carries a liquid-enthalpy balance that needs Cp_liq for every component, and CO2 is a permanent gas at 298 K with no liquid-Cp data (Tc = 304 K). Pre-mixing externally keeps the tutorial focused on the Henry's-law K-value selector — the point of the case.

Physics: $H_{\text{CO2}}(298\text{ K}) = 1.64\text{e8 Pa} = 1640\text{ bar}$ (Henry's law) $K_{\text{CO2}} = H / P = 1.64\text{e8} / 5\text{e5} = 328$ $K_{\text{N2}} \gg 1$ (no Henry entry, no Antoine close to 298 K either — falls back to Raoult and Psat extrapolation) $K_{\text{water}} \ll 1$ (most water stays in liquid; $\text{Psat}(298) \sim 3\text{ kPa}$)

Observed result: cleanGas F = 66.0 kmol/h $y_{\text{CO2}} = 0.123$ $y_{\text{N2}} = 0.871$ $y_{\text{water}} = 0.0063$ richLiquid F = 5034 kmol/h $x_{\text{CO2}} = 3.75\text{e-4}$ $x_{\text{N2}} = 6.5\text{e-3}$ $x_{\text{water}} = 0.993$ $V/F = 0.013$ $K_{\text{CO2}} (y/x) = 328$? exactly $H_{\text{CO2}}(298)/P = 1.64\text{e8} / 5\text{e5}$ $K_{\text{water}} = 0.0063$? $\text{Psat}_{\text{water}}(298)/P = 3.17\text{ kPa} / 500\text{ kPa}$ mass balance CO2: 10 in = 8.11 vapour + 1.89 liquid ~ 19 % of feed CO2 absorbed (a single-stage scrubber is far from perfect; an industrial column with 5-10 theoretical stages would recover > 95 %).

Toggling the GUI Units menu to Mass-basis shows the absorption as CO2 mass transfer between the two streams in kg/h, useful for scale-up arithmetic.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cleanGas	F	0.0183200303715
stream	feed	F	1.41666666667
stream	richLiquid	F	1.3983466363
kpi	scrubber	F_alpha	1.3983466363
kpi	scrubber	F_beta	0.0183200303715
kpi	scrubber	F_in	1.41666666667
kpi	scrubber	P	500000
kpi	scrubber	T	298.15
kpi	scrubber	V_over_F	0.0129317861446
kpi	scrubber	phaseSet	0

extract01_ethanol_water_benzene

tutorials/steady/absorption/extract01_ethanol_water_benzene



What it does: Counter-current LL extraction: recover ethanol from water with benzene

The story: Counter-current LL extraction column

Recover ETHANOL (the solute) from a dilute aqueous feed (WATER = the carrier) using BENZENE as the extracting SOLVENT. This is the classic liquid-liquid extraction problem: a multistage counter-current liquid-liquid cascade.

Topology (the ‘extractor’ unit op): the feed (water + ethanol) enters at stage 1; fresh solvent (benzene) enters at stage N; the two liquid phases pass counter-currently — the raffinate (water-rich, ethanol-depleted) flows $1 \rightarrow N$, the extract (benzene-rich, ethanol-enriched) flows $N \rightarrow 1$; EXTRACT leaves stage 1, RAFFINATE leaves stage N.

Each theoretical stage is ONE liquid-liquid equilibrium, solved by the SHARED Gibbs-energy-minimisation LL flash (IsothermalFlash, phaseSet LL) — robust against the $K = 1$ saddle that traps Newton-on-residual methods in LLE (CLAUDE.md sec.9). The cascade is torn by successive substitution (mirroring the distillation bubble-point loop).

Thermo: predictive UNIFAC for both liquid phases (water / ethanol / benzene groups inline), the same model used by the ternary LLE scan tutorial — fully self-contained, no fitted pairs needed.

KPIs to watch: recovery_ethanol (fraction of ethanol leaving in the extract), distribution_ethanol (z_E / z_R), $F_{\text{extract}} / F_{\text{raffinate}}$, massClosure (must be ~ 0).

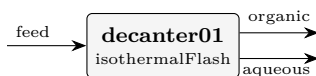
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	extract	F	0.0359293973028
stream	feed	F	0.0277777777778
stream	raffinate	F	0.0251772962492
stream	solvent	F	0.0333333333333
kpi	extractor01	F_extract	0.0359293973028
kpi	extractor01	F_feed	0.0277777777778
kpi	extractor01	F_raffinate	0.0251772962492
kpi	extractor01	F_solvent	0.0333333333333
kpi	extractor01	P	101325
kpi	extractor01	T	298.15
kpi	extractor01	converged	1
kpi	extractor01	distribution_benzene	295.27509389
kpi	extractor01	distribution_ethanol	0.555106921422
kpi	extractor01	distribution_water	0.00706545462823

... and 9 more reference values in the case's *expected*.

extraction01_waterButanol

tutorials/steady/absorption/extraction01_waterButanol



What it does: Water/n-butanol LL flash at 298 K (experimental)

The story: Water + 1-butanol at 25 °C. This system has a well-known liquid-liquid miscibility gap below 125.5 °C (upper critical solution temperature). A 70/30 water/nButanol feed (inside the gap) → splits into a water-rich (aqueous) phase ~98 mol% water and a butanol-rich (organic) phase ~59 mol% nButanol / 41 mol% water.

Demonstrates: ‘phases (...)’ syntax with TWO liquid phases ‘phaseSet LL’ in the IsothermalFlash Michelsen TPD stability test to find a robust initial guess

Water + 1-butanol at 25 °C and 1 atm. The pair has a well-known liquid-liquid miscibility gap below its upper critical solution temperature (125.5 °C), so a feed whose composition lies inside the gap spontaneously splits into two liquid phases. This case feeds a 70/30 water/nButanol mixture (inside the gap) to an isothermal decanter and lets the flash find the two-liquid split.

A single feed splits into two liquids, both at 298.15 K, $v_f = 0$:

The split is found by Michelsen TPD stability testing (to seed a robust initial guess) followed by direct Gibbs-energy minimisation — the LL path the engine uses precisely because a Newton/successive-substitution flash on a symmetric γ -model would collapse to the trivial $K = 1$ saddle.

- phases (...) syntax with TWO liquid phases; - phaseSet LL in the IsothermalFlash; - the Michelsen TPD stability test as the initial-guess generator.

- The LL flash reuses the VL solver’s printout, so the two liquids are shown under Vapor flow V / Liquid flow L headings (and x/y/K labels). Read them as phase- α (organic) and phase- β (aqueous), not as a vapour and a liquid. - The decanter is isothermal and adiabatic in reality (an enthalpy hand-check gives $H_{in} \approx H_{out}$), so the physically correct duty is ≈ 0 . The KPI Q and any utility/cost the run reports for this LL split are a known artifact of the shared flash duty path — do not cost this unit from them.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	aqueous	F	0.0140665296362
stream	feed	F	0.02777777777778
stream	organic	F	0.0137112481416
kpi	decanter01	F_alpha	0.0140665296362
kpi	decanter01	F_beta	0.0137112481416
kpi	decanter01	F_in	0.02777777777778
kpi	decanter01	P	101325
kpi	decanter01	T	298.15
kpi	decanter01	V_over_F	0.493604933098
kpi	decanter01	phaseSet	1
kpi	decanter01	Q_kW	0

stripper01_NH3_water

tutorials/steady/absorption/stripper01_NH3_water



What it does: NH3 stripping from water with air (Kremser, 6 stages, warm)

The story: The inverse of absorber01: instead of scrubbing NH3 OUT of a gas, here we strip NH3 OUT of water with a lean gas (air ~ N2). Counter-current Kremser; the rich liquid enters the top, the stripping gas the bottom.

Run warm (330 K): NH3 is less soluble hot (K rises), so the stripping factor $S = K V/L$ exceeds 1 and the strip is efficient. But desorption is endothermic, so the column cools – which lowers K and self-limits the stripping (the mirror image of the absorber’s exotherm).

Liquid: 100 kmol/h, 5 % NH3 in water, 330 K Gas: 100 kmol/h N2 ($V/L = 1$), 330 K Column: 6 theoretical stages

Expected: $S_{\text{NH3}} > 1 \rightarrow$ most of the NH3 stripped into the gas; stripped liquid nearly NH3-free; rich gas carries the NH3; mass closes 100 %.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	liquidFeed	F	0.0277777777778
stream	richGas	F	0.029140357434
stream	strippedLiquid	F	0.0264151981216
stream	strippingGas	F	0.0277777777778
kpi	stripper	F_richGas	0.029140357434
kpi	stripper	F_strippedLiquid	0.0264151981216
kpi	stripper	K_NH3	3.62862883208
kpi	stripper	K_water	0.170562530402
kpi	stripper	L_in	0.0277777777778
kpi	stripper	S_NH3	3.62862883208
kpi	stripper	S_water	0.170562530402
kpi	stripper	T_bottom	313.524974667
kpi	stripper	T_top	315.973930965
kpi	stripper	V_in	0.0277777777778

... and 6 more reference values in the case’s *expected*.

crystallisation

The theory you need. Crystallisation removes solute when the solution crosses saturation: $m > m_{\text{sat}}(T)$, with $K_{\text{sp}}(T)$ from ΔG_{diss} and its T -dependence from ΔH_{diss} (van ’t Hoff). The heat of crystallisation is ION-DERIVED – $\Delta H_{\text{cryst}} = -(\Delta H_{\text{soln}}^{\infty} + \bar{L}_2(m_{\text{sat}}))$ – never a stored constant (a solid’s formation enthalpy is $\sum \nu_i h_{f,i}^{\text{aq}} - \Delta H_{\text{soln}}$, derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant, $\gamma_{\pm}$ rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

Crystallizer09chatGPT

tutorials/steady/crystallisation/Crystallizer09chatGPT

What it does: Two crystallisers in series separating KCl from potassium bitartrate (KHT, cream of tartar) by manipulating T and adding ethanol. CRYST1 COOLS the hot feed to 5 C $\rightarrow$ KHT (sparingly soluble, steep solubility curve) crystallises while KCl stays dissolved. CRYST2 doses ETHANOL into the KCl-rich mother liquor $\rightarrow$ KCl drops out by drowning-out (eNRTL mixed-solvent). Two salts, two saturation routes (curve vs ion-activity), one declarative property configuration – a stress test of the property system.

The story: ONE operational flowsheet. This case has NO sectors.

The three entries below are unit operations and are resolved under unitOperations/<unitName>/. Units are not subdomains and therefore do not partition 0/ or converged/.

Every physical stream is one named graph edge. The edge name is the stable stream ID and therefore the state filename. Stream role is inferred from topology in the current run domain:

to-only $\rightarrow$ inlet from + to $\rightarrow$ internal from-only $\rightarrow$ outlet

There is no boundary{} and no streams{} state here.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	KCl_cake	F	0.000232421702193
stream	KHT_cake	F	2.54671494088e-05
stream	feed	F	0.02777777777792
stream	freshEthanol	F	0.004166666666667
stream	recoveredEthanol	F	0.02361111111111
stream	stillage	F	0.0307069541328
kpi	cryst1	Q_kW	-93.1162690166
kpi	cryst1	Q_removed	93116.2690166
kpi	cryst1	T_op	278.15
kpi	cryst1	c_sat	0.0034013435
kpi	cryst1	cakeLiquidFlow	2.54671494088e-05
kpi	cryst1	cakeMoisture	0.1
kpi	cryst1	cakeWetness_pct	9.09090909091
kpi	cryst1	crystal_flow	0.00533920523976

...and 37 more reference values in the case's *expected*.

crystalliser01_sugar

tutorials/steady/crystallisation/crystalliser01_sugar

What it does: Cooling crystalliser: sugar solution 80 → 20 C, yield by solubility**The story:** / flowsheetDict (LEAF node)

A hot sugar solution (z_{sucrose} 0.15, ~3.35 kg/kg water, below saturation at 80 C where $c_{\text{sat}} \sim 3.8$) is cooled to 20 C ($c_{\text{sat}} \sim 2.0$ kg/kg). The excess crystallises. Magma = crystals (s[]) + saturated mother liquor.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0261287129062
kpi	cryst	Q_removed	271677.114808
kpi	cryst	T_op	293.15
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.564469958333
kpi	cryst	crystal_mol	0.00164906487154
kpi	cryst	liquorFlow	0.0261287129062
kpi	cryst	solute_dissolved	0.861767541667
kpi	cryst	solute_in	1.4262375
kpi	cryst	yield	0.395775569169

crystalliser02_msmpr

tutorials/steady/crystallisation/crystalliser02_msmpr

What it does: MSMPR sugar crystalliser: population balance by method of moments**The story:** / flowsheetDict (LEAF node)

Continuous MSMPR crystalliser. A hot near-saturated sugar solution is fed to a cooled, well-mixed vessel of working volume V ; the magma (crystals + mother liquor) is withdrawn at the same volumetric rate, so the residence time is $\tau = V / Q$. ‘model MSMPR’ selects the population-balance path (method of moments); the kinetics are referenced from constant/crystallisation.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0266203308749
kpi	cryst	CV	1
kpi	cryst	L_dominant	0.000223952804481
kpi	cryst	L_meanMass	0.000298603739308
kpi	cryst	L_meanNumber	7.46509348269e-05
kpi	cryst	Q_removed	269525.057151
kpi	cryst	T_op	293.15
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.396190602513
kpi	cryst	crystal_mol	0.00115744690287
kpi	cryst	growthRate	1.95272330048e-08
kpi	cryst	liquorFlow	0.0266203308749
kpi	cryst	magmaDensity	302.920529908

... and 10 more reference values in the case's *expected*.

crystalliser03_pbe_size_independent

tutorials/steady/crystallisation/crystalliser03_pbe_size_independent

What it does: Crystalliser FVM-PBE: sanity vs MSMPR (size-independent growth)**The story:** / flowsheetDict (LEAF node)

Same physical setup as crystalliser02_msmpr (hot sugar syrup, 5 m³ vessel at 20 °C, k_g 1e-7, k_b 1e9) – only the solver method changes: ‘model FVM’ discretises the population balance $n(L)$ on N_{bins} evenly-spaced size bins, marching the steady PBE forward in L with first-order upwind. With sizeFactor 0 (the default), $G(L)$ is constant and the FVM result must match the analytical MSMPR exponential.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0266086567804
kpi	cryst	CV	0.999540291345
kpi	cryst	L_dominant	0.000224609375
kpi	cryst	L_meanMass	0.000298633437171
kpi	cryst	L_meanNumber	7.48298114726e-05
kpi	cryst	Q_removed	269576.1605
kpi	cryst	T_op	293.15
kpi	cryst	bins	256
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.400186610054
kpi	cryst	crystal_mol	0.00116912099742
kpi	cryst	growthRate_L0	1.90635340048e-08
kpi	cryst	liquorFlow	0.0266086567804

... and 11 more reference values in the case's *expected*.

crystalliser04_size_dependent_growth

tutorials/steady/crystallisation/crystalliser04_size_dependent_growth

What it does: Crystalliser FVM-PBE with size-dependent growth (sizeFactor > 0)**The story:** / flowsheetDict (LEAF node)

Same hot-sugar feed and 5 m³ vessel at 20 °C as crystalliser03, only ‘sizeFactor’ is non-zero — the size-dependent growth law that breaks the method-of-moments closure but is handled natively by the FVM.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0265894971524
kpi	cryst	CV	1.03562508094
kpi	cryst	L_dominant	0.000232421875
kpi	cryst	L_meanMass	0.000329374475484
kpi	cryst	L_meanNumber	7.45169890057e-05
kpi	cryst	Q_removed	269660.031771
kpi	cryst	T_op	293.15
kpi	cryst	bins	256
kpi	cryst	c_sat	2.026
kpi	cryst	crystal_flow	0.406744893212
kpi	cryst	crystal_mol	0.00118828062534
kpi	cryst	growthRate_L0	1.83025070562e-08
kpi	cryst	liquorFlow	0.0265894971524

... and 11 more reference values in the case's *expected*.

crystalliser05_nacl_pitzer

tutorials/steady/crystallisation/crystalliser05_nacl_pitzer

What it does: NaCl crystalliser: saturation from the PITZER ion activity ($K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

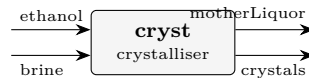
The story: NaCl crystalliser whose SATURATION comes from the Pitzer ion activity ($K_{sp} = (\gamma_{pm} \cdot m_{sat})^2$), via the ‘electrolyte {}’ block – NaCl carries no solubility{} curve, so the electrolyte model is what makes this possible. The feed is a SUPERSATURATED brine (~7.6 mol/kg, as it would leave an evaporator); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess. Self-contained: components live in constant/components/.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	magma	F	0.0271500572444
kpi	cryst	Ksp_activity	38.4846213082
kpi	cryst	Q_removed	2435.55566933
kpi	cryst	T_op	298.15
kpi	cryst	c_sat	0.35905536
kpi	cryst	crystal_flow	0.036683987968
kpi	cryst	crystal_mol	0.000627720533333
kpi	cryst	gamma_pm_sat	1.00970010562
kpi	cryst	liquorFlow	0.0271500572444
kpi	cryst	m_sat	6.144
kpi	cryst	solute_dissolved	0.158116012032
kpi	cryst	solute_in	0.1948
kpi	cryst	yield	0.18831616

crystalliser06_nacl_antisolvent

tutorials/steady/crystallisation/crystalliser06_nacl_antisolvent



What it does: Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises γ_{pm} , and DROPS the saturation molality m_{sat} – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent γ_{pm} .

The story: ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity γ_{pm} (eNRTL mixed-solvent + Born), so the saturation molality m_{sat} COLLAPSES (6.14 $\rightarrow$ ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product $K_{\text{sp}} = (\gamma_{\text{pm}} * m_{\text{sat}})^2$ anchored in pure water, solved with the mixed-solvent γ_{pm} at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine (x_{water} 0.900, x_{NaCl} 0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol $\rightarrow$ salt-free $x_{\text{ethanol}} \sim 0.40$. Self-contained: components live in constant/components/.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

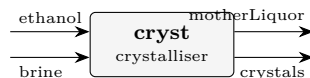
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.02777777777778
stream	crystals	F	0.000329661969914
stream	ethanol	F	0.01666666666667
stream	motherLiquor	F	0.0424240933568
kpi	cryst	Ksp_activity	34.8439846653
kpi	cryst	Q_removed	6559.87377671
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	cakeLiquidFlow	0.000329661969914
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0988038720389
kpi	cryst	crystal_mol	0.00169068911771

... and 8 more reference values in the case's *expected*.

crystalliser07_nacl_antisolvent_cooled

tutorials/steady/crystallisation/crystalliser07_nacl_antisolvent_cooled



What it does: Antisolvent + COOLING crystallisation of NaCl: the same drowning-out brine + ethanol as crystalliser06, but the crystalliser is COOLED to 5 C. The eNRTL is now temperature-dependent ($\tau(T)=\tau_{25}\cdot 298.15/T$, $A_{DH}(T)$ via $\epsilon_{sw}(T)/\rho_w(T)$, Born at T) and $K_{sp}(T)$ shifts by van't Hoff with NaCl's enthalpy of solution – so cooling adds to the antisolvent's effect (yield 61% $\rightarrow$ 70%). Exercises the T-dependent electrolyte path.

The story: brine [cryst @ 5 C] crystals (wet NaCl cake) ethanol motherLiquor (saturated supernatant)

Identical drowning-out feed to crystalliser06 (saturated brine + ethanol, salt-free $x_{ethanol} \sim 0.40$), but the crystalliser is COOLED to 5 C. Two levers now stack: ANTISOLVENT – ethanol lowers the dielectric $\rightarrow$ mixed-solvent $\gamma_{pm}(T)$. COOLING – the electrolyte model is temperature-dependent: $\tau(T) = \tau_{25} \cdot 298.15/T$ (the Δ_g/RT form, like the molecular NRTL), the Debye-Huckel $A_{DH}(T)$ follows $\epsilon_{sw}(T)/\rho_w(T)$, the Born term is at T , and the saturation product $K_{sp}(T)$ shifts by van't Hoff with NaCl's enthalpy of solution (dissolutionEnthalpy in NaCl.dat). Result: $m_{sat} \sim 0.885 \rightarrow 0.681$ mol/kg, yield 61% $\rightarrow$ 70% vs the isothermal 06. (For NaCl alone cooling is weak – solubility is nearly T-flat; the antisolvent carries most of the effect. A salt with a large dH_{sol} , e.g. KNO₃, would gain far more from cooling.)

Self-contained: components live in constant/components/.

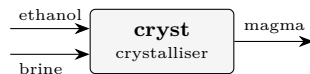
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.02777777777778
stream	crystals	F	0.000380305320939
stream	ethanol	F	0.01666666666667
stream	motherLiquor	F	0.0421245706304
kpi	cryst	Ksp_activity	31.135192385
kpi	cryst	Q_removed	82721.44242
kpi	cryst	T_op	278.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0398065230351
kpi	cryst	cakeLiquidFlow	0.000380305320939
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.113348382739
kpi	cryst	crystal_mol	0.00193956849313

...and 8 more reference values in the case's *expected*.

crystalliser08_nacl_antisolvent_msmpr

tutorials/steady/crystallisation/crystalliser08_nacl_antisolvent_msmpr



What it does: Continuous MSMPR antisolvent crystallisation of NaCl: a saturated brine + ethanol fed to a well-mixed vessel (residence time $\tau = V/Q$). The eNRTL antisolvent supersaturation $S = c/c_{\text{sat}}(\text{eNRTL})$ drives nucleation + growth; the method of moments closes the population balance and returns the CRYSTAL-SIZE DISTRIBUTION $n(L)$ (mean size, CV) – what the equilibrium model cannot give.

The story: CONTINUOUS MSMPR + ANTISOLVENT

brine [cryst MSMPR] magma (crystals WITH a size distribution $n(L)$ ethanol + saturated mother liquor)

Same drowning-out feed as crystalliser06 (saturated brine + ethanol), but the vessel is a CONTINUOUS, well-mixed MSMPR crystalliser of working volume V ; the magma leaves at the same volumetric rate, so the residence time is $\tau = V/Q$.

Where the equilibrium model stops at the YIELD, the population balance goes on to the CRYSTAL-SIZE DISTRIBUTION: the antisolvent supersaturation $S = c / c_{\text{sat}}(\text{eNRTL}, T, x_{\text{ethanol}})$ drives growth $G = k_g (S-1)^g$ and nucleation $B_0 = k_b (S-1)^b$; the method of moments gives $n(L) = n_0 \exp(-L/G\tau)$, the mean sizes ($L_{43} = 4 G\tau$), the CV, and the magma PSD – plotted in the GUI.

The supersaturation here comes from the eNRTL ion-activity K_{sp} (the antisolvent lowers m_{sat}), NOT a solubility curve – the MSMPR now reads the electrolyte saturation, the same helper the equilibrium model uses. Self-contained: components + ion/tau catalogue + kinetics all under constant/.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.0277777777778
stream	ethanol	F	0.0166666666667
stream	magma	F	0.0428576741462
kpi	cryst	CV	1
kpi	cryst	L_dominant	2.02534396647e-05
kpi	cryst	L_meanMass	2.70045862196e-05
kpi	cryst	L_meanNumber	6.75114655489e-06
kpi	cryst	Q_kW	-6.1566687571
kpi	cryst	Q_removed	6156.6687571
kpi	cryst	T_op	298.15
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	crystal_flow	0.092730856228
kpi	cryst	crystal_mol	0.00158677029822
kpi	cryst	growthRate	1.92664299147e-09

... and 12 more reference values in the case's *expected*.

crystalliser09_KHT_KCl_series

tutorials/steady/crystallisation/crystalliser09_KHT_KCl_series



What it does: Two crystallisers in series separating KCl from potassium bitartrate (KHT, cream of tartar) by manipulating T and adding ethanol. CRYST1 COOLS the hot feed to 5 C → KHT (sparingly soluble, steep solubility curve) crystallises while KCl stays dissolved. CRYST2 doses ETHANOL into the KCl-rich mother liquor → KCl drops out by drowning-out (eNRTL mixed-solvent). Two salts, two saturation routes (curve vs ion-activity), one package – a stress test of the property system.

The story: SELECTIVE crystallisation of two salts

feed (hot: KCl + KHT) [cryst1: COOL to 5 C] KHT_cake (cream of tartar)

ethanol [cryst2: ANTISOLVENT] KCl_cake

finalLiquor (EtOH+water+salt)

[recovery: DISTILLATION] recoveredEthanol (recycle-grade)

stillage (water + residual salt)

Two salts with very different solubilities are separated by two DIFFERENT levers, each crystalliser targeting one salt ('solute ...'):

CRYST1 cools the hot feed. Potassium bitartrate (KHT) is sparingly soluble with a STEEP solubility curve, so cooling 50 → 5 C precipitates most of it; KCl, far more soluble and only mildly T-sensitive, stays dissolved. KHT's saturation is its measured solubility CURVE $c_{sat}(T)$ – no ion parameters.

CRYST2 doses ethanol into the KCl-rich mother liquor. KCl drops out by drowning-out: the eNRTL mixed-solvent term raises γ_{pm} as the dielectric falls, collapsing m_{sat} . KCl's saturation is the ion-activity K_{sp} .

The SAME package carries both routes; 'useElec' is decided per solute, so KHT uses its curve while KCl uses the eNRTL. Self-contained.

Basis (per kg water): 55.5 water, 3.0 KCl (below the 5 C saturation ~3.9), and 0.078 KHT (saturated at 50 C).

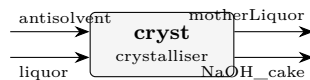
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	KCl_cake	F	0.000232421701759
stream	KHT_cake	F	2.54671494142e-05
stream	feed	F	0.0277777777778
stream	finalLiquor	F	0.0543180652451
stream	freshEthanol	F	0.00416666666667
stream	liquor1	F	0.0277239373137
stream	recoveredEthanol	F	0.0236111111111
stream	stillage	F	0.030706954134
kpi	cryst1	Q_kW	-93.1162690119
kpi	cryst1	Q_removed	93116.2690119
kpi	cryst1	T_op	278.15
kpi	cryst1	c_sat	0.0034013435
kpi	cryst1	cakeLiquidFlow	2.54671494142e-05
kpi	cryst1	cakeMoisture	0.1

... and 39 more reference values in the case's *expected*.

crystalliser10_naoh_antisolvent

tutorials/steady/crystallisation/crystalliser10_naoh_antisolvent



What it does: NaOH antisolvent crystallisation (water + ethanol, eNRTL drowning-out) – the acceptance-case core

The story: H₂O + NaOH + ethanol, every effect that IS calibrated active, every one that is NOT loudly announced: ACTIVE: drowning-out saturation (eNRTL mixed-solvent; Na-OH taus anchored to the Parker-slotted Pitzer surface), BPE-grade water activity, differential-K_{sp} lineage, wet-cake split. ANNOUNCED OMISSIONS (honest): heat of dilution / crystallisation heat use the dH_{cryst} constant (the eNRTL kernel is NOT calorimetrically fitted – per-kernel flag); water-ethanol heat of mixing gated until the H_E refit; the solid is ANHYDROUS NaOH until the NaOH.H₂O hydrate forms entry lands.

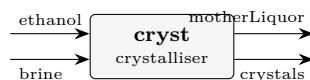
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	NaOH_cake	F	0.000203756483719
stream	antisolvent	F	0.008333333333333
stream	liquor	F	0.01388888888889
stream	motherLiquor	F	0.020384504743
kpi	cryst	K _{sp} _activity	7365.8380048
kpi	cryst	Q_kW	72.7276039114
kpi	cryst	Q_removed	-72727.6039114
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.461538461538
kpi	cryst	c_sat	0.181200024556
kpi	cryst	cakeLiquidFlow	0.000203756483719
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0653535379386

...and 9 more reference values in the case's *expected*.

crystalliser11_kcl_antisolvent_ethanol

tutorials/steady/crystallisation/crystalliser11_kcl_antisolvent_ethanol



What it does: Antisolvent (drowning-out) crystallisation of KCl: a SATURATED aqueous KCl brine (4.76 mol/kg) is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises $\gamma_{\pm}$, and DROPS the saturation molality – so KCl precipitates with NO cooling and NO evaporation. Saturation is the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent $\gamma_{\pm}$ at the feed's salt-free ethanol fraction. KCl-water-ethanol experimental reference: Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

The story: ANTISOLVENT (drowning-out) of KCl

[cryst] crystals (solid KCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual KCl)

The KCl counterpart of crystalliser06 (NaCl). KCl is the second salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)"; this case realises it.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity $\gamma_{\pm}$ (eNRTL mixed-solvent + Born), so the KCl saturation molality m_{sat} COLLAPSES from its aqueous 4.76 mol/kg and KCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product $K_{sp} = (\gamma_{\pm} \cdot m_{sat})^2$ anchored in pure water, solved with the mixed-solvent $\gamma_{\pm}$ at the feed's salt-free ethanol fraction.

Basis: 100 kmol/h saturated aqueous brine (x_{water} 0.9210, x_{KCl} 0.0790; 4.76 mol KCl / kg water) + 60.0 kmol/h ethanol → salt-free $x_{ethanol} \sim 0.39$. Self-contained: components live in constant/components/.

Experimental KCl-water-ethanol reference (the drowning-out anchor): Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

The KCl counterpart of crystalliser06 (NaCl): a saturated aqueous KCl brine is mixed with ethanol, whose low dielectric collapses the KCl solubility, so KCl crystallises with no cooling and no evaporation. KCl is the salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)" — this case realises it.

Ethanol's relative permittivity (24.3) is far below water's (78.5). Dosing it lowers the mixed-solvent dielectric, which raises the mean ionic activity coefficient $\gamma_{\pm}$ (eNRTL mixed-solvent + Born term), so the saturation molality collapses. The saturation is the same ion-activity product $K_{sp} = (\gamma_{\pm} \cdot m_{sat})^2$ anchored in pure water, re-solved with the mixed-solvent $\gamma_{\pm}$ at the feed's salt-free ethanol fraction — the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

100 kmol/h saturated brine (x_{water} 0.921, x_{KCl} 0.079 = 4.76 mol/kg) + 60 kmol/h ethanol → salt-free $x_{ethanol} \approx 0.39$:

The KCl-water-ethanol activity behaviour modelled here is measured in A. Lopes, F. Farelo & M. I. A. Ferra, "Activity Coefficients of Potassium Chloride in Water-Ethanol Mixtures", J. Solution Chem. 28 (1999) 117 — mean activity coefficients of KCl by emf in 5–20 % w/w ethanol, 25–45 °C. That $\gamma_{\pm}$ rise with ethanol content is exactly the driver of the drowning-out captured above.

- The eNRTL τ for K-Cl is the illustrative 1:1-alkali-chloride proxy (= Chen & Evans water-NaCl), so the numbers are qualitative. The Farelo data is the path to a quantitative refit of the KCl-water-ethanol interaction (the paper reports Pitzer $\beta^0/\beta^1/C?$ per solvent composition). - The mass / yield is the reliable output. The crystalliser's heat term carries the shared electrolyte-enthalpy caveat (a dissolved salt's formation datum lives in its ions, so the stream enthalpy falls back to the sensible datum and is announced) — identical to crystalliser06.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.02777777777778
stream	crystals	F	0.000358870836834
stream	ethanol	F	0.01666666666667
stream	motherLiquor	F	0.0426471724268
kpi	cryst	Ksp_activity	16.6554570684
kpi	cryst	Q_kW	-24.7405003096
kpi	cryst	Q_removed	24740.5003096
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.394477317554
kpi	cryst	c_sat	0.0458726802597
kpi	cryst	cakeLiquidFlow	0.000358870836834
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.107234246429

...and 9 more reference values in the case's *expected*.

distillation

The theory you need. A column is N flash stages coupled by counter-current flows. Each stage obeys MESH equations: Material balance, Equilibrium ($y_i = K_i x_i$), Summation ($\sum y = \sum x = 1$), and Heat balance. Choupo ships two solution strategies, selectable per column: Wang–Henke (bubble-point, solving the tridiagonal material balances stage by stage; robust for near-ideal systems, slow near azeotropes) and **simultaneous** (one Newton on ALL the MESH residuals; ~ 5 – 6 iterations even through azeotropes). Expect: reflux and boilup set the split; a minimum reflux below which no stage count reaches the spec; and azeotropes as fixed points no column can cross (x_D pinned at the azeotropic composition). *Full derivation: Theory Guide, the distillation chapter (McCabe–Thiele + MESH).*

column01_benzene_toluene
tutorials/steady/distillation/column01_benzene_toluene



What it does: Benzene/toluene distillation column (Wang-Henke)

The story: Benzene/toluene distillation — classical textbook problem with Raoult, no azeotrope, easy convergence. Used to validate the column code.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	column01	T_top	354.21089412
kpi	column01	T_bottom	382.893530951
kpi	column01	x_D_LK	0.981245045999
kpi	column01	x_B_HK	0.981245183716

column02_simultaneous

tutorials/steady/distillation/column02_simultaneous



What it does: Rigorous distillation by simultaneous MESH Newton (benzene/toluene; validates against column01)

The story: The SAME benzene/toluene column as column01, but solved by the rigorous SIMULTANEOUS-correction (MESH Newton) method instead of the sequential bubble-point (Wang-Henke). Selected by ‘method simultaneous;’.

Pedagogical point: on an ideal system both methods give the SAME answer — this case validates the MESH solver against column01. But the simultaneous Newton converges in ~6 iterations (quadratic) vs ~100 for the bubble-point, and — the reason it exists — it stays stable on azeotropic systems where the sequential method does not.

Expected (identical to column01): $T_{top} \approx 354.2\text{ K}$, $T_{bottom} \approx 382.9\text{ K}$, $x_D(\text{benzene}) \approx 0.981$, $x_B(\text{toluene}) \approx 0.981$.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	R	2
kpi	column01	T_bottom	382.893527916
kpi	column01	T_top	354.210890137
kpi	column01	feedStage	8
kpi	column01	nStages	15
kpi	column01	x_B_HK	0.981245125322
kpi	column01	x_D_LK	0.981245125322

column03_azeotrope_mesh

tutorials/steady/distillation/column03_azeotrope_mesh



What it does: Azeotropic distillation (ethanol/water, NRTL) by the rigorous simultaneous MESH method

The story: Ethanol/water (NRTL) distillation by the rigorous SIMULTANEOUS (MESH Newton) method. This is the payoff of the v0.48 solver: ethanol/water forms an azeotrope at $x_{\text{eth}} \sim 0.894$ (1 atm), and the sequential bubble-point (Wang-Henke) is unstable here – on this very case it does NOT converge (200 outer iters). The MESH Newton converges in ~ 5 iterations to a PHYSICAL distillate ($x_{\text{eth}} \sim 0.80$, below the azeotrope).

Feed: 100 kmol/h, 30/70 ethanol/water, 1 atm. The column feed enthalpy is set by `feedQuality = 1.0` (saturated liquid), NOT by the stream T: the stated $T = 363$ K is ~ 7 K above the 30/70 bubble point (~ 355.5 K at 1 atm), so the stream itself is slightly superheated — the MESH model uses `feedQuality`, so this does not affect the column solution. Column: 16 stages, feed at 8, $R = 2.5$, $D = 25$ kmol/h, method simultaneous.

Expected: converged; $x_D(\text{ethanol}) \sim 0.80$ (< 0.894 azeotrope – physical); $T_{\text{top}} \sim 351.4$ K (the azeotropic temperature); mass closes 100%.

NOTE: asking for a distillate ABOVE the azeotrope ($x_{\text{eth}} > 0.894$) is physically impossible – no method converges, and that is correct. The MESH method does not invent a non-physical "past the azeotrope" answer (which the bubble-point used to do).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0208333333333
stream	distillate	F	0.00694444444444
stream	feed	F	0.0277777777778
kpi	column	B	0.0208333333333
kpi	column	D	0.00694444444444
kpi	column	R	2.5
kpi	column	T_bottom	357.713408957
kpi	column	T_top	351.388958309
kpi	column	feedStage	8
kpi	column	nStages	16
kpi	column	x_{B_HK}	0.868205515093
kpi	column	x_{D_LK}	0.804616545279

column04_multifeed_sidedraw

tutorials/steady/distillation/column04_multifeed_sidedraw



What it does: Simultaneous MESH column: two feeds + a liquid side draw (rigorous MESH)

The story: Multiple feeds as FLOWSHEET STREAMS ('inputs (feed feed2)') + a liquid side draw. operation.feeds maps each input stream to a stage (information follows the streams → the plant-boundary mass + energy balance closes).

The rigorous simultaneous MESH column extended to multiple feeds and side draws — the most-used features of a commercial simulator column, on the glass-box Newton solver.

A benzene / toluene / o-xylene split (light / intermediate / heavy):

- primary feed — 100 kmol/h, benzene/toluene/o-xylene 0.40/0.40/0.20, enters high (stage 5); - second feed — 60 kmol/h, heavier (0.05/0.45/0.50), enters low (stage 11); - liquid side draw — 15 kmol/h pulled from stage 8, between the two feeds.

Mass closes ($100 + 60 = 42 + 15 + 103 = 160$ kmol/h). $Q_{\text{condenser}} = -1436$ kW, $Q_{\text{reboiler}} = +1527$ kW. The side draw does exactly what it should — bleed the intermediate component where it is most enriched on the trays.

The simultaneous MESH is a bubble-point-reduced Newton (CMO flows). The extension generalises the single fixed feed into a per-stage flow staircase: $L[j]$, $V[j]$ are rebuilt by a CMO march that adds each feed's liquid/vapour parts and subtracts each draw; every stage's material balance carries an $F_j z_{\{F,j\}}$ source and $(L_j + U_j)$, $(V_j + W_j)$ outflows. The general balance reduces exactly to the old single-feed equations when there is one feed and no draw — so every existing distillation tutorial is unchanged (verified).

outputs (distillate bottoms <draw1> <draw2> ...) — side-draw streams follow distillate and bottoms, in stage order.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0286111111111
stream	distillate	F	0.0116666666667
stream	feed	F	0.0277777777778
stream	intermediateCut	F	0.0041666666667
kpi	radfracLite	B	0.0286111111111
kpi	radfracLite	D	0.0116666666667
kpi	radfracLite	Q_condenser_kW	-1435.80274013
kpi	radfracLite	Q_reboiler_kW	1516.14164522
kpi	radfracLite	R	3
kpi	radfracLite	T_bottom	395.926996928
kpi	radfracLite	T_top	354.978489481
kpi	radfracLite	feedStage	5
kpi	radfracLite	nFeeds	2
kpi	radfracLite	nSideDraws	1

... and 3 more reference values in the case's expected.

column05_reactive_methylacetate

tutorials/steady/distillation/column05_reactive_methylacetate



What it does: Reactive distillation: methyl acetate synthesis (validation vs P?pken et al. 2001, run S-1)

The story: Both feeds are FLOWSHEET STREAMS ('inputs (feed acidFeed)') – so the plant-boundary mass + energy balance sees them. 'operation.feeds' only maps each stream to a stage (no inline composition – the credo: information follows the streams).

The simultaneous MESH column with an equilibrium reaction on the catalytic stages (rigorous column rung 4), compared to a published experiment.

Mole-conserving, so the CMO internal-flow profile is untouched. Each catalytic stage carries one extra unknown — the molar extent ξ_j — closed by the activity-based equilibrium residual $\sum_i \nu_i \ln(\gamma_i x_i) = \ln K_a(T)$, with γ_i from the liquid activity model and $K_a(T)$ by van't Hoff: $K_a(T) = K_a(298) \cdot \exp(-\Delta h^\circ_r / R \cdot (1/T - 1/298.15))$. Converged by homotopy (solve non-reactive, then switch the reaction on).

> T. P?pken, S. Steinigeweg & J. Gmehling, "Synthesis and Hydrolysis of > Methyl Acetate by Reactive Distillation", Ind. Eng. Chem. Res. 40 (2001) 1566. Run S-1: $N = 25$, reactive stages 11–19, HOAc fed above / > MeOH below the zone, reflux 2.1, $P \approx 1.017$ bar. Measured endpoints, cited, in > constant/experimental/methyl_acetate_RD_profile.csv.

The equilibrium constant and thermodynamics are from the companion paper P?pken, G?tze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601: $K_a(298) \approx 38.7$, $\Delta h^\circ_r \approx -5.67$ kJ/mol $\rightarrow K_a \approx 29$ in the reactive zone ($\sim 65^\circ\text{C}$).

*This is not a tight match, and the reason is physically important: with the correct equilibrium K_a (≈ 29) the model gives the equilibrium ceiling (96 %), and it correctly brackets the measurement from above. The measured 87.6 % is *below* the equilibrium because the real column is kinetically limited — the paper's recommended model is an adsorption-based finite *rate* (per gram of catalyst), not equilibrium. An earlier run with an (incorrect, too low) $K_a = 5.2$ gave 90.7 %, a closer *number* but for the wrong reason; using a wrong constant to fake agreement would be dishonest.*

So the path to a bit-tight match is the kinetic rate model, not the thermo.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	9.13888888889e-06
stream	distillate	F	8.36111111111e-06
stream	feed	F	8.94444444444e-06
kpi	rdColumn	B	9.13888888889e-06
kpi	rdColumn	D	8.36111111111e-06
kpi	rdColumn	Q_condenser_kW	-0.790521316742
kpi	rdColumn	Q_reboiler_kW	0.626813788646
kpi	rdColumn	R	2.1
kpi	rdColumn	T_bottom	363.229492539
kpi	rdColumn	T_top	330.062558898
kpi	rdColumn	conversion	0.960508806863
kpi	rdColumn	feedStage	20
kpi	rdColumn	nFeeds	2
kpi	rdColumn	nSideDraws	0

...and 4 more reference values in the case's expected.

column06_fullMESH

tutorials/steady/distillation/column06_fullMESH



What it does: Full-MESH (Naphtali-Sandholm) distillation – energy-consistent V_j/L_j , seeded from the CMO solution (benzene/toluene; compare column02)

The story: The SAME benzene/toluene column as column02_simultaneous, but solved by the FULL MESH (Naphtali-Sandholm), selected by ‘model fullMESH;’. The full MESH promotes the internal flows V_j , L_j to UNKNOWNs and adds a per-stage total-mass + ENERGY balance, so the flows follow the REAL latent heats instead of the constant-molal-overflow (CMO) assumption behind the ‘simultaneous’ (bubble-point-reduced) method.

Robustness trick: the full MESH is SEEDED from the converged CMO profile — the cheap model bootstraps the rigorous one; ~4 Newton iterations from that seed.

Expected — close to, but NOT equal to, column02 (the gap IS the energy-balance correction the CMO drops): $T_{\text{top}} \sim 354.3$ K, $T_{\text{bottom}} \sim 382.8$ K, $x_D(\text{benzene}) \sim 0.979$ (CMO: 0.981). Benzene/toluene latent heats are similar (30.8 vs 33.2 kJ/mol) so the gap is small here; on a wide-boiling column the full MESH earns its cost.

The same benzene/toluene column as column02_simultaneous, but solved by the full MESH (model fullMESH) instead of the bubble-point/CMO reduction.

The simultaneous method is a bubble-point-reduced MESH: it assumes constant molal overflow (CMO — equimolal latent heats), so the internal vapour/liquid flows V_j , L_j are fixed by a mass march and only the compositions x_j and stage temperatures T_j are solved.

*The full MESH (Naphtali–Sandholm) promotes V_j and L_j to unknowns and adds, per stage, the total-mass balance and the energy balance — so the flows follow the *real* latent heats instead of assuming them equal. The extra per-stage energy equation is the enthalpy residual the CMO never carried:*

Total condenser: $V_0 = 0$, $L_0 = R \cdot D$; reboiler: $L_{\{N-1\}} = B$; their energy balances give $Q_{\text{cond}} / Q_{\text{reb}}$ as results.

The full-MESH is seeded from the converged CMO profile — the cheap model launches the rigorous one. Here it converges in ~4 Newton iterations from that seed (it would be far harder from a cold start).

Benzene ($\Delta h_{\text{vap}} \approx 30.8$ kJ/mol) and toluene (≈ 33.2 kJ/mol) have similar latent heats, so CMO is a good approximation and the two answers nearly coincide — the ~0.002 / ~0.1 K gap IS the energy-balance correction the full-MESH makes and the CMO drops. On a wide-boiling column (a depropaniser) the gap is much larger; that is where the full-MESH earns its cost.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column06	B	0.0138888888889
kpi	column06	D	0.0138888888889
kpi	column06	Q_condenser_kW	-1281.19539111
kpi	column06	Q_reboiler_kW	1299.72437334
kpi	column06	R	2
kpi	column06	T_bottom	382.774927808
kpi	column06	T_top	354.339103085
kpi	column06	feedStage	8
kpi	column06	nFeeds	1
kpi	column06	nSideDraws	0
kpi	column06	nStages	15

... and 2 more reference values in the case's *expected*.

column07_naphtaliSandholm

tutorials/steady/distillation/column07_naphtaliSandholm



What it does: Full-MESH (Naphtali-Sandholm) validation vs AIChE J. 17 (1971) 148, Problem 2: nC4/iC5/nC5

The story: - Choupo -*—————* VALIDATION – Naphtali & Sandholm, AIChE Journal 17(1), 148-153 (1971), DOI 10.1002/aic.690170130, Problem 2 (ideal hydrocarbon distillation). 20 plates; liquid feed ($T = 30^\circ\text{F} = 272.04\text{ K}$, 500 mol each of nC4/iC5/nC5) on plate 10; reflux ratio 1000; 80% recovered in bottoms ($B = 1200$). A liquid sidestream on plate 1 (condenser) takes 30% of the reflux.

EQUIVALENT SPEC (same internal flows, no condenser sidestream): the distillate and the sidestream leave the total condenser with the SAME composition, so they are combined into ONE top product of 300 with reflux $R = L_1/300 = 996.68/300 = 3.3223$. This gives identical $L_1 = 996.68$ and $V_2 = 1296.68$, hence the same column.

Paper result: $B = \text{nC4 } 200.1 / \text{iC5 } 499.9 / \text{nC5 } 500.0$; combined top $\sim 99.97\%$ nC4; plate $T \ 29.7^\circ\text{F}$ (top) $\rightarrow 77.0^\circ\text{F}$ (reboiler) $\sim 272.0 \rightarrow 298.2\text{ K}$.

Reproduces the ideal-hydrocarbon distillation of Naphtali & Sandholm, "Multicomponent Separation Calculations by Linearization", AIChE Journal 17(1), 148–153 (1971) — DOI [10.1002/aic.690170130](https://doi.org/10.1002/aic.690170130) — the paper that introduced the very full-MESH method Choupo's model fullMESH implements.

n-butane / isopentane / n-pentane, 20 plates, liquid feed ($T = 30^\circ\text{F} = 272.04\text{ K}$, 500 mol of each) on plate 10, reflux ratio 1000, 80 % of the feed recovered in the bottoms ($B = 1200$). A liquid sidestream on plate 1 (the condenser) takes 30 % of the reflux.

*Equivalent spec used here. The distillate (≈ 1) and the condenser sidestream (≈ 299) leave the *total* condenser with the same composition, so they are combined into ONE top product of 300 with reflux $R = L_1/300 = 996.68/300 = 3.3223$. This gives identical internal flows ($L_1 = 996.68$, $V_2 = 1296.68$), hence the identical column — without needing a condenser side draw.*

Converges in 3 full-MESH Newton iterations (seeded from the CMO profile).

Products Choupo — top product (300) 99.99 % n-C₄ bottoms n-C₄ / i-C₅ / n-C₅ 200 / 500 / 500 B 1200

*The profile shape — the flat n-butane-rich top, the sharp rise across the feed zone, the pentane-plateau, the reboiler climb — matches stage-by-stage. The $\sim 1\text{--}2\text{ K}$ residual is thermo, not method: Choupo uses fitted Antoine vapour pressures, the paper used the K-value tables in Brown's *Unit Operations* (1950).*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.416666666667
kpi	column07	B	0.333333333333
kpi	column07	D	0.0833333333333
kpi	column07	Q_condenser_kW	-8082.6120755
kpi	column07	Q_reboiler_kW	9374.37429576
kpi	column07	R	3.3223
kpi	column07	T_bottom	296.58946842
kpi	column07	T_top	272.663334648
kpi	column07	feedStage	10
kpi	column07	nFeeds	1
kpi	column07	nSideDraws	0
kpi	column07	nStages	20
kpi	column07	x_B_HK	0.416665691626
kpi	column07	x_D_LK	0.999874145869

column08_radfrac_multidraw

tutorials/steady/distillation/column08_radfrac_multidraw



What it does: rigorous column: multi-feed + multi-side-draw distillation (C3/C4/C5/C6), full-MESH

The story: - Choupo -*—————* Multi-feed, multi-side-draw rigorous column (the most-used rigorous column features at once). A C3/C4/C5/C6 mix split FOUR ways: distillate ~ propane, an upper liquid side draw ~ n-butane, a lower liquid side draw ~ n-pentane, bottoms ~ n-hexane. TWO feeds enter at different heights (a light cut high, a heavy cut low) – so the column sees the realistic case of feeds of different composition. Feeds are flowsheet STREAMS (‘inputs (...)’) so the plant-boundary mass+energy balance sees them (information follows the streams). Validation reference (propose): Hofeling & Seader, AIChE J. 24(6) (1978), DOI 10.1002/aic.690240631 – the Naphtali-Sandholm extension for general multi-feed / multi-draw / interlinked separators.

A C3/C4/C5/C6 depropanizer-style column at 16 bar, split four ways by the rigorous full-MESH, exercising the two most-used rigorous column features at once: TWO feeds (a light cut high, a heavy cut low) and TWO liquid side draws.

Two saturated-liquid feeds (100 kmol/h each, 290 K / 320 K): a light cut on stage 10, a heavy cut on stage 20. 30 stages, $R = 4$. Mass closes exactly ($200 = 45 + 55 + 55 + 45$ kmol/h) and the energy balance closes: the feed-to-product enthalpy gap (1137.7 kW) equals the net duty $Q_{\text{reb}} + Q_{\text{cond}} = 1942.3 - 804.6$ kW. Converges in a few full-MESH iterations from the CMO seed.

This case was first written at 1 atm, and the energy balance would not close — because at 1 atm and 290 K the light hydrocarbons (propane b.p. 231 K, n-butane 272 K) are above their boiling points: the feed streams flash to VAPOUR ($vf = 1.0$), yet the column was told quality 1.0 (saturated liquid). The latent heat of the unrecognised vapour feed was exactly the missing ~1 MW. At 16 bar the feeds are genuinely liquid ($vf = 0$, consistent with quality 1.0) and the balance closes — which is also why a real depropanizer runs at elevated pressure: so the propane distillate condenses against cooling water (here 321.6 K). A feed’s thermal state must be specified consistently with its actual phase.

*Both feeds are real flowsheet streams (inputs (feed1 feed2), mapped to stages by operation.feeds), so the plant-boundary mass + energy balance sees them (*information follows the streams*). Side draws are outputs in stage order: outputs (distillate bottoms drawC4 drawC5).*

Self-consistently validated (mass + energy close, four cuts in volatility order) on the full-MESH solver that IS paper-validated against Naphtali & Sandholm (1971) in column07_naphtaliSandholm. No external multi-feed+multi-draw single-column benchmark with published profiles is readily available in the open literature.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0125
stream	distillate	F	0.0125
stream	drawC4	F	0.0152777777778
stream	drawC5	F	0.0152777777778
stream	feed1	F	0.0277777777778
stream	feed2	F	0.0277777777778
kpi	radfrac	B	0.0125
kpi	radfrac	D	0.0125
kpi	radfrac	Q_condenser_kW	-804.58622364
kpi	radfrac	Q_reboiler_kW	1698.71150765
kpi	radfrac	R	4
kpi	radfrac	T_bottom	454.498489397
kpi	radfrac	T_top	321.647346922
kpi	radfrac	feedStage	10

... and 5 more reference values in the case’s expected.

column09_tray_hydraulics

tutorials/steady/distillation/column09_tray_hydraulics



What it does: Sieve-tray hydraulics: the column that separates, sized so its trays do not flood

The story: column01, asked the question a converged MESH solution never answers: CAN THE TRAYS PASS THE TRAFFIC?

A converged column tells you what separates. It says nothing about whether the vapour rising through the holes will carry the liquid up with it (entrainment flooding), whether the liquid will back up the downcomer until it reaches the tray above (downcomer flooding), or whether it will simply rain through the holes instead of bubbling (weeping). A column that floods separates nothing, and the stage equations never notice.

Add an 'hydraulics {}' block and the column runs a RATING pass on its own converged profile. Two modes, the same equations in opposite directions:

give 'diameter' → RATING. How close to flood is each tray? omit 'diameter' → DESIGN. What diameter keeps every tray at or below 'floodFraction' of its flooding velocity? The widest tray sets the column. That is this case.

Nothing here is guessed. The orifice coefficient C_o is a chart in the literature (Liebson, Kelley & Bullington 1957, against hole-area ratio and plate-thickness/hole-diameter); it is a DECLARED number you own, printed back at you. The weep-point constant $K2$ is a chart too – leave it out and the column tells you the weep check was not performed rather than inventing it.

Read the printed table. The flow parameter F_{LV} climbs below the feed (more liquid, less vapour per unit mass), which lowers Fair's capacity parameter and therefore the flooding velocity – yet the stripping trays are not the tight ones here, because the vapour volumetric flow falls faster than the capacity. The column is sized by whichever tray loses that argument.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column09	B	0.0138888888889
kpi	column09	D	0.0138888888889
kpi	column09	L_rect	0.0277777777778
kpi	column09	L_strip	0.0555555555556
kpi	column09	Q_condenser_kW	-1281.04348002
kpi	column09	Q_reboiler_kW	1279.30480005
kpi	column09	R	2
kpi	column09	T_bottom	382.893530951
kpi	column09	T_top	354.21089412
kpi	column09	V_rect	0.0416666666667
kpi	column09	V_strip	0.0416666666667

... and 11 more reference values in the case's *expected*.

column10_flooding

tutorials/steady/distillation/column10_flooding



What it does: The same column, built too narrow: every tray floods – and it still converges

The story: the SAME column as column09, built 1.10 m wide instead of the 1.322 m that column09 designed for it.

Nothing about the separation changes. The distillate is still 98.12 % benzene, the bottoms still 98.12 % toluene, the MESH equations converge exactly as before, and every number in the stream table is identical. The column is a perfectly good column – on paper.

It floods. The worst tray runs at 115.6 % of its entrainment-flooding velocity: the vapour is fast enough to carry the liquid up to the tray above instead of letting it fall. In a real column the trays would fill, the pressure drop would run away, and the separation printed above would not happen at all.

This is the whole reason hydraulics exists as a separate pass. A converged solution is not a feasible column, and the stage equations have no way of knowing the difference. Widen it (column09), space the trays further apart, or cut the reflux – but do not believe the stream table until a tray can pass the traffic it carries.

Note the downcomer, too: the backup reaches 271 mm against a 275 mm limit. Narrow the column a little further and it would flood from below as well, for an entirely different reason.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column10	B	0.0138888888889
kpi	column10	D	0.0138888888889
kpi	column10	L_rect	0.0277777777778
kpi	column10	L_strip	0.0555555555556
kpi	column10	Q_condenser_kW	-1281.04348002
kpi	column10	Q_reboiler_kW	1279.30480005
kpi	column10	R	2
kpi	column10	T_bottom	382.893530951
kpi	column10	T_top	354.21089412
kpi	column10	V_rect	0.0416666666667
kpi	column10	V_strip	0.0416666666667

... and 11 more reference values in the case's *expected*.

column11_murphree

tutorials/steady/distillation/column11_murphree



What it does: Murphree tray efficiency: 15 REAL trays, not 15 ideal stages – and the purity you actually get

The story: the same 15-stage benzene/toluene column as column09, with ONE line added: murphreeEfficiency 0.70.

Every column before this one is built from EQUILIBRIUM stages: the vapour leaving a tray is in equilibrium with the liquid leaving it. No real tray does that. The vapour spends a finite time in the froth and comes off somewhere between what entered it and what equilibrium would give. Murphree measured the shortfall on the vapour side:

$$y_j = E_MV \cdot y_{j+1}^* + (1 - E_MV) \cdot y_j, y_{j+1}^* = K_j x_j$$

With $E_MV = 1$ the tray is an ideal stage and nothing changes – every existing case still runs byte for byte. With $E_MV < 1$, ‘nStages’ stops counting ideal stages and starts counting REAL TRAYS. Which matters, because the hydraulics pass below sizes those trays: until now we were dimensioning, to the millimetre, a tray we then assumed was perfect.

E_MV is a DECLARED number. It comes from a correlation (O’Connell), a chart, or plant data, and it is yours to own – exactly like the orifice coefficient C_o and the weep constant $K2$ in the hydraulics block. Choupo will not invent it.

Compare against column09 (same column, $E_MV = 1$):

distillate benzene 98.12 % (ideal) → 95.05 % ($E_MV = 0.70$) bottoms toluene 98.12 % (ideal) → 95.05 %

Fourteen real trays do less than fourteen ideal ones. That is the whole content of a tray efficiency, and it is why a column designed on ideal stages and built on real trays comes up short. Sweep E_MV and watch the limit close:

E_MV 0.50 0.70 0.90 0.99 0.999 1.000 x_D 0.9100 0.9505 0.9738 0.9806 0.9812 0.98125 <- the ideal stage

Nothing is switched on at $E_MV = 1$; the effective K collapses to K and the case reproduces column09 to the last bit.

NOTE: the efficiency is folded into an effective K , with the vapour from the tray below lagged one outer pass – the linear solve keeps its tridiagonal shape. At convergence the lag vanishes and the Murphree relation holds exactly. The MESH ‘simultaneous’ method REFUSES the keyword rather than quietly returning ideal stages: there the relation would have to enter the residual and Jacobian.

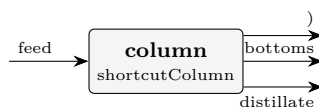
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column11	B	0.0138888888889
kpi	column11	D	0.0138888888889
kpi	column11	L_rect	0.0277777777778
kpi	column11	L_strip	0.0555555555556
kpi	column11	Q_condenser_kW	-1284.29753602
kpi	column11	Q_reboiler_kW	1281.05938025
kpi	column11	R	2
kpi	column11	T_bottom	381.492133108
kpi	column11	T_top	355.136328304
kpi	column11	V_rect	0.0416666666667
kpi	column11	V_strip	0.0416666666667

...and 12 more reference values in the case's *expected*.

shortcut01_benzene_toluene

tutorials/steady/distillation/shortcut01_benzene_toluene



What it does: Shortcut (FUG) design of a benzene/toluene column

The story: Preliminary column design by the Fenske-Underwood-Gilliland (FUG) shortcut method (the built-in 'shortcutColumn'). The same benzene/ toluene separation as the rigorous 'column01', but designed in closed form: you specify the key recoveries and the reflux, and FUG returns the minimum stages, the minimum reflux, the actual stage count and the feed location.

Feed: 100 kmol/h, 50/50 benzene/toluene, saturated liquid, 1 atm. Keys: LK = benzene, HK = toluene ($\alpha \approx 2.4$). Spec: 99 % of the benzene to the distillate, 1 % of the toluene. Reflux: $R = 1.3 \cdot R_{\min}$.

Expected ($\alpha \approx 2.4$): $N_{\min} \approx 10.5$ (Fenske, total reflux) $R_{\min} \approx 1.2$ (Underwood) $N \approx 22$ (Gilliland, at $R = 1.3 R_{\min}$) distillate ~99 % benzene, bottoms ~99 % toluene; mass closes 100 %.

Compare with the rigorous column01 (Wang-Henke): FUG is a fast design estimate; the rigorous column is the verification. FUG does NOT handle azeotropes — it assumes constant relative volatility.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column	B	0.0138888888889
kpi	column	D	0.0138888888889
kpi	column	N_min	10.067785029
kpi	column	N_theoretical	21.5880731317
kpi	column	R	1.68248609289
kpi	column	R_min	1.29422007145
kpi	column	alpha_LK_HK	2.49137883567
kpi	column	feed_stage	10.7940365659
kpi	column	theta	1.42716041594
kpi	column	x_B_HK	0.99
kpi	column	x_D_LK	0.99

drying

solidDryer01_sugar

tutorials/steady/drying/solidDryer01_sugar

What it does: Solid dryer: wet sugar powder dried to sorption equilibrium**The story:** / flowsheetDict (LEAF node)

Wet sugar dried by a REAL hot-air STREAM (no airTemperature/RH parameters, no phantom duty): the hot air BRINGS the heat (its sensible cooling) and CARRIES AWAY the moisture (a humid exhaust). Adiabatic → the outlet T is a RESULT; the plant energy balance closes with real streams.

Wet solid: 10 kmol/h sucrose (3423 kg/h) + 38 kmol/h bound water ($X_{in}=0.20$). Hot air: 800 kmol/h at 420 K, DRY (2 mol% water → RH ~0.4%) – hot dry air, like a real sugar dryer. Its low RH sets a low GAB equilibrium moisture, so the sugar dries to $X_{eq} \sim 0.001$ (~0.1 wt%); the exhaust leaves carrying the evaporated water (838 > 800 kmol/h, the pickup visible in the molar balance).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	drySolid	F	5.80292648834e-05
stream	hotAir	F	0.222222222222
stream	humidExhaust	F	0.232719748513
stream	wetSolid	F	0.0105555555556
kpi	solidDryer	T_out	347.5126791
kpi	solidDryer	X_equilibrium	0.00109945367945
kpi	solidDryer	X_final	0.00109945367945
kpi	solidDryer	X_initial	0.199991235758
kpi	solidDryer	air_in_kmol_h	800
kpi	solidDryer	drySolid_flow	0.95187873054
kpi	solidDryer	moisture_pct_wb	0.109824620861
kpi	solidDryer	water_activity	0.00431158404453
kpi	solidDryer	water_removed	0.189112936126

sprayDryer01_sugar

tutorials/steady/drying/sprayDryer01_sugar



What it does: Spray dryer: 40 wt% sugar solution → powder (rotary atomiser, Ranz-Marshall)

The story: flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	d_droplet	6.04023981348e-05
kpi	dryer	d_droplet_micron	60.4023981348
kpi	dryer	d_particle	3.80926671659e-05

...and 12 more reference values in the case's *expected*.

sprayDryer02_residence_sweep

tutorials/steady/drying/sprayDryer02_residence_sweep



What it does: Spray dryer: chamber-height sweep → the drying curve (residence = L/v ; constant + falling rate → equilibrium)

The story: sprayDryer01's sugar dryer with the drying-chamber HEIGHT swept by system/outerDict (0.01 → 0.3 m, 40 pts): residence time = L / v_{particle} traces the DRYING CURVE – constant-rate then falling-rate, down to the equilibrium moisture.

The base flowsheet is sprayDryer01_sugar's, unchanged: one unit, two feeds, two products.

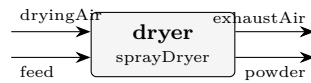
feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; $z_{\text{sucrose}} = 0.03390$

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

sprayDryer03_pressure_nozzle

tutorials/steady/drying/sprayDryer03_pressure_nozzle



What it does: Spray dryer: 40 wt% sugar solution → powder (PRESSURE-SWIRL nozzle at 40 bar, $d_{32}=9575 \cdot dP^{-1/3}$; Ranz-Marshall)

The story: sprayDryer01_sugar – flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; $z_{\text{sucrose}} = 0.03390$

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a PRESSURE-SWIRL nozzle at 40 bar ($d_{32} = 9575 \cdot dP^{-1/3}$ um, the class-average design curve) instead of a wheel.

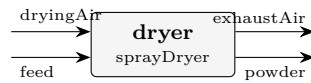
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24532614778
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203070475508
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.2283799688
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

... and 30 more reference values in the case's *expected*.

sprayDryer04_profiles

tutorials/steady/drying/sprayDryer04_profiles



What it does: Spray dryer DISTRIBUTED: axial profiles of moisture, size and temperature along the chamber

The story: sprayDryer01_sugar – flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

...and 30 more reference values in the case's *expected*.

sprayDryer05_whey

tutorials/steady/drying/sprayDryer05_whey



What it does: Spray dryer: whey (prob 5.2) – 500 kg/h, $X_0=5$, rotary co-current, sorption-isotherm residual moisture

The story: sprayDryer01_sugar – flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; $z_{\text{sucrose}} = 0.03390$

dryingAir: ~500 kmol/h of hot AIR (N₂/O₂/Ar) at 180 C – sized to carry the whey water. Whey solids modelled as sucrose (lactose proxy, a disaccharide MW~342).

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.138888888889
stream	exhaustAir	F	0.14529654835
stream	feed	F	0.00649261111111
stream	powder	F	1.73405388823e-05
kpi	dryer	Nu	2.33731434528
kpi	dryer	Pr	0.745230561117
kpi	dryer	Re	0.384509154435
kpi	dryer	Sc	0.602916461126
kpi	dryer	Sh	2.31430911286
kpi	dryer	T_out	382.7774359
kpi	dryer	T_wetbulb	315.824008236
kpi	dryer	X_critical	0.5
kpi	dryer	X_equilibrium	0.0134981947613
kpi	dryer	X_final	0.0134981947613

... and 30 more reference values in the case's *expected*.

sprayDryer06_rea

tutorials/steady/drying/sprayDryer06_rea



What it does: Spray dryer REA (model chen): Reaction Engineering Approach kinetics – smooth rate, no critical-moisture break

The story: sprayDryer01_sugar – flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

... and 30 more reference values in the case's *expected*.

sprayDryer07_design

tutorials/steady/drying/sprayDryer07_design



What it does: Spray dryer DESIGN: give targets (powder size, residence), the sizing pass inverts to the geometry

The story: sprayDryer01_sugar – flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N₂/O₂/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	dryingAir	F	0.0138888888889
stream	exhaustAir	F	0.0148058483113
stream	feed	F	0.000957611111111
stream	powder	F	8.18867205141e-06
kpi	dryer	Nu	2.24583682227
kpi	dryer	Pr	0.746977616919
kpi	dryer	Re	0.203916784363
kpi	dryer	Sc	0.6026290984
kpi	dryer	Sh	2.22885536787
kpi	dryer	T_out	372.025243986
kpi	dryer	T_wetbulb	318.053221267
kpi	dryer	X_critical	0.15
kpi	dryer	X_equilibrium	0.0132756494842
kpi	dryer	X_final	0.0132756494842

...and 31 more reference values in the case's *expected*.

economics

The theory you need. Sizing (stirred tanks, shell-and-tube areas) feeds costing (Guthrie modules); prices are CASE data (constant/economics, dated and cited) – the engine refuses to invent a price. Expect honest refusal when a price is missing, never a silent default.

economics01_esterification_dcf

tutorials/steady/economics/economics01_esterification_dcf



What it does: Esterification plant: CAPEX (Guthrie) → OPEX (COM_d) → discounted-cash-flow appraisal (NPV/IRR/payback)

The story: A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed → reactor (CSTR) → heater → separator (flash) → liqProd (ethyl acetate rich) → vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges (V in $[0.3, 520]$ m³, A in $[10, 1000]$ m²) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing → costing → economics; the economics pass prints the COM_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

A fixed (designed) ethyl-acetate plant taken all the way from equipment sizing to a discounted-cash-flow economic appraisal:

1. *sizing* — StirredTank for the CSTR, ShellTubeHX for the heater, sized from each unit's KPIs.
2. *costing* — Guthrie/Turton purchased + bare-module + total-module cost, CEPCI-updated to the target year, in EUR.
3. *economics* — the new EconomicsPass: it builds the CAPEX ladder (FCI, working capital, TCI), the Turton cost-of-manufacture COM_d, revenue, a straight-line-depreciation cash-flow timeline, then NPV, IRR (bisection bracket + Newton polish), and discounted + simple payback.

Every cost factor and DCF assumption is dict-owned and printed — read the appraisal block and you can re-derive every number by hand.

*Depreciation enters the DCF exactly once (it is **not** in COM_d) — the classic textbook double-count is avoided. Working capital is an outflow at startup and is recovered untaxed at end-of-life.*

Market/time-specific data — product and feed prices, labour rate, FX — lives only in constant/economics, each value dated and primary-cited (an index/bulletin/agency named in a provenance{} block). The EconomicsPass refuses to run if a required price is absent, with a remedy message; set economics.refuseOnMissingPrice 0; in the postDict to switch to defaulted-but-loud. No price ever defaults to a literal in code.

The economics pass also publishes its headline scalars into result.kpis["economics"] (IRR, NPV, paybackYears, FCI, TCI, COM_d), which is what lets a sweep or optimisation read economics as an ordinary response — see economics02_irr_sweep.

economics02_irr_sweep

tutorials/steady/economics/economics02_irr_sweep



What it does: Sweep reactor volume; read economics.IRR/NPV/payback per point (SweepDriver runs the post chain)

The story: A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed $\rightarrow$ reactor (CSTR) $\rightarrow$ heater $\rightarrow$ separator (flash) $\rightarrow$ liqProd (ethyl acetate rich) $\rightarrow$ vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges (V in $[0.3, 520]$ m³, A in $[10, 1000]$ m²) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing $\rightarrow$ costing $\rightarrow$ economics; the economics pass prints the COM_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

The same esterification plant as economics01, but driven by a sensitivity sweep over the reactor volume that reads economic responses at every point:

*The SweepDriver now runs the full post-processing chain (sizing $\rightarrow$ costing $\rightarrow$ economics) on each converged point, so the headline economic scalars the EconomicsPass publishes into result.kpis["economics"] resolve as ordinary unit.kpi responses. A sweep can finally *see cost*:*

A bigger reactor raises conversion (more product, more revenue) but raises CAPEX and feed throughput — the IRR falls monotonically over this range, a trade-off invisible from a single CAPEX number.

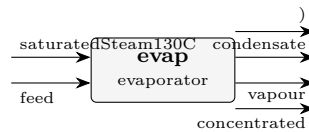
> The range floors at 4 m³ on purpose: below ~3 m³ the lower conversion > drops the mixture bubble point and the fixed-duty heater would cross the > saturation dome (a heater is sensible-only) — the deliberate validity edge.

evaporation

The theory you need. An evaporator boils solvent off a solution. Two pieces of physics beyond a flash: boiling-point elevation, $T_{\text{boil}} = T_{\text{sat}}(P) + \text{BPE}$ with the BPE from the electrolyte activity of water ($\ln a_w = -\varphi \nu m M_w / 1000$); and the heat of dilution/concentration (the relative apparent molar enthalpy $L_\varphi(m, T)$) entering the duty. Expect: the BPE to GROW as the brine concentrates, steam economy below 1 for a single effect, and – for NaCl – the full Silvester–Pitzer T -dependence of $\gamma_\pm$ shaping the answer between 25 and 200°C. *Full derivation: Theory Guide, the electrolyte chapters.*

evaporator01_brine

tutorials/steady/evaporation/evaporator01_brine



What it does: Single-effect evaporator: 35 g/L brine at 1 bar; the fixed steam duty/UA evaporates ~82 % of the feed water ($V/F=0.82$), enriching NaCl ~5.5x (x 0.011 $\rightarrow$ 0.059) – duty-driven, not a target (BPE included)

The story: single-effect brine concentrator (NaCl-water), PHYSICAL-EQUIPMENT model (v0.29+).

We say WHAT THE HARDWARE IS (heat-exchange area, overall HTC, operating pressure on the process side, heating-steam temperature) and the simulator tells us WHAT IT DOES (V/F , duty Q , T_{boil} with BPE, steam demand, economy). There is no "vaporFraction" or "duty" spec to choose — the equipment shape itself IS the spec.

Feed: seawater-equivalent 35 g/L NaCl in water, 100 kmol/h (~ 1.8 kg/s of seawater, the volume of a small medium-capacity desalination unit), 25 °C. At $MW_{\text{NaCl}} = 58.44$ g/mol and $\rho_{\text{water}} \sim 1$ g/cm³:

35 g/L / 58.44 g/mol = 0.599 mol NaCl per litre water = 0.599 mol/kg water (dilute approx) $x_{\text{NaCl_feed}} \approx 0.599 / (0.599 + 55.5) = 0.01067$

Heating side: saturated steam at 130 °C (= 403.15 K, $P \approx 2.7$ bar).

Equipment: heat-exchange area = 20 m² overall $U = 2000$ W/(m²·K) operating pressure = 1 bar (process side)

Expected order-of-magnitude (computed by the solver, not specified): $T_{\text{sat}}(\text{water}, 1 \text{ bar}) \approx 373$ K (~ 100 °C) $\Delta T_{\text{drive}} (\text{steam} - \text{process}) \approx 30$ K $Q = U \cdot A \cdot \Delta T \approx 2000 \cdot 20 \cdot 30 \approx 1200$ kW V evaporated $\approx Q / \Delta H_{\text{vap}} \approx 110$ kmol/h ≈ 1980 kg/h water V/F (mol-basis) ≈ 0.55 Steam consumption $\approx Q / \Delta H_{\text{vap_at_steam}} \approx 2000$ kg/h Economy $\approx V_{\text{evap}} / F_{\text{steam}} \approx 0.99$ $x_{\text{NaCl_outlet}} \approx 0.0237$ ($\sim$ doubled from feed)

Industrial relevance: this is the building block of multi-effect evaporators in seawater desalination (MED, MSF), brine crystallisers, and the sugar industry. Real plants run under vacuum ($T_{\text{boil}} \sim 60$ °C at 0.2 bar) to minimise scaling and use BPE-loaded multi-effect cascades to recover the latent heat — see evaporator02_triple_effect_sugar.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0049888708933
stream	condensate	F	0.0277777777778
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0277777777778
stream	vapour	F	0.0227889068845
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	1.79512624904
kpi	evap	F_conc	0.0049888708933
kpi	evap	F_conc_kmol_h	17.9599352159
kpi	evap	F_conc_mass	0.101856029976
kpi	evap	F_cond	0.500416666667
kpi	evap	F_cond_kg_h	1801.5
kpi	evap	F_in	0.0277777777778
... and 19 more reference values in the case's <i>expected</i> .			

evaporator02_triple_effect_sugar

tutorials/steady/evaporation/evaporator02_triple_effect_sugar



What it does: Triple-effect cocurrent evaporator: 22500 kg/h sugar solution 5%→25% (MCFT Ex. 10.4)

The story: MCFT Ex. 10.4: triple-effect cocurrent evaporator concentrating a 22 500 kg/h sucrose solution from 5 wt% sucrose toward ~ 25 wt%, using 200 kPa saturated steam to drive Effect 1 and a 15 kPa vacuum on Effect 3.

Cascade (cocurrent: liquid 1→2→3, generated vapour drives the next effect's heating side):

$F=22\,500\text{ kg/h}$ $L1\ L2\ L3$ $x=0.05 \rightarrow [1] \rightarrow [2] \rightarrow [3] \rightarrow \hat{} \hat{} \hat{} || \text{Steam } V1\ V2\ V3 \rightarrow \text{condenser}$
 200 kPa (Effect 1's (Effect 2's $T_{\text{sat}} \sim 393\text{K}$ vapour drives vapour drives Effect 2) Effect 3) $P1 = 80\text{ kPa}$ $P2 = 45\text{ kPa}$ $P3 = 15\text{ kPa}$

PHYSICAL-EQUIPMENT model (v0.29+). Each effect is its own physical evaporator with an 'area' and a 'U'; the simulator computes V/F , Q , T_{boil} and steam demand from the energy balance. Effect 1 consumes utility steam; the vapour $V1$ that Effect 1 produces is declared as the heating-side input to Effect 2, and similarly $V2 \rightarrow$ Effect 3. The flowsheet runs SEQUENTIALLY left-to-right: no outer iteration is needed because the inputs of each effect are produced by the preceding one (or by the utility stream).

Equipment we picked (the "spec"): $A_1 = A_2 = A_3 = 100\text{ m}^2$ $U_1 = 2319\text{ W}/(\text{m}^2\cdot\text{K})$ (Effect 1, clean steam) $U_2 = 2194\text{ W}/(\text{m}^2\cdot\text{K})$ (Effect 2) $U_3 = 1296\text{ W}/(\text{m}^2\cdot\text{K})$ (Effect 3, lowest pressure $\rightarrow$ lowest k)

The textbook target $L_3 \sim 4500\text{ kg/h}$ at $x = 0.25$ corresponds to evaporating $\sim 18\,000\text{ kg/h}$ total water; that target is NOT enforced here — the equipment dimensions ARE the spec, the outlet concentration is the RESULT. A later tutorial wraps this case in a DesignSpec block that manipulates A_i (or, more commonly, U_i at fixed $A_i = A$) to drive L_3 to the design target.

1. Cocurrent multi-effect literally just chains single-effect evaporators. Choupo's flowsheet engine handles the cascading naturally — the V_i stream of Effect i appears as the heating-side input of Effect $i+1$, and the V_i mass flow is computed by Effect i 's energy balance.
2. Economy: in single-effect (evaporator01_brine) the economy was $\sim 0.83\text{ kg evap} / \text{kg steam}$. Running this cascade gives total $V_{\text{evap}} / \text{utility steam} \sim 2.0$ ($= 17\,972 / 8\,917$) — more than a doubling, simply by chaining the heat through three effects. Industrial design targets ~ 2.5 with optimised area allocation and pressure cascade.
3. BPE matters more in Effect 3 (most concentrated solute, lowest U); the student SEES T_{boil} deviate further from $T_{\text{sat}}(\text{water}, P_3)$ in the per-effect run header.

Honest simplification for v0.29:

The Evaporator unit op reads only T from its heating-steam input (and ignores the F). This means the F of V_1 that effect 2 actually CONSUMES on its heating chest is computed from effect 2's own energy balance and is NOT constrained to equal the F of V_1 that effect 1 PRODUCED. In a real plant those two flows must close (all of V_1 condenses in effect 2's chest, with no other path). Here the v0.29 model treats the three energy balances independently — there is a residual mismatch in the V_i mass that the cascade quietly absorbs.

Closing the inter-effect mass balance turns the cascade into a coupled system (one constraint per link) that wants an outer DesignSpec wrap: manipulate (T_{boil_i}) or (A_i) to enforce " V_i produced == V_i consumed at effect $i+1$'s chest". That is the v0.30 multi-effect tutorial.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1	F	0.24340099744
stream	L2	F	0.150822370581
stream	L3	F	0.0533879416888
stream	V1	F	0.0870984081566
stream	V2	F	0.0925786268589
stream	V3	F	0.0974344288922
stream	cond1	F	0.1375
stream	cond2	F	0.0870984081566
stream	cond3	F	0.0925786268589
stream	feed	F	0.330499405597
stream	utilitySteam_200kPa	F	0.1375
kpi	effect1	A	100
kpi	effect1	A_m2	100
kpi	effect1	BPE	0.107002096415
<i>...and 81 more reference values in the case's expected.</i>			

evaporator03_co_current

tutorials/steady/evaporation/evaporator03_co_current



What it does: Co-current forward-feed triple-effect evaporator, user-defined components (MCFT 10.38, case a)

The story: triple-effect evaporator, forward-feed, user-defined components, MCFT Ex. 10.38 (a).

Problem (textbook gives): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) P_3 such that effect 3 boils at 325 K $\Rightarrow P_3 = 13.5$ kPa U_1 3100 W/(m²·K) U_2 2500 W/(m²·K) U_3 1700 W/(m²·K) Spec productConc = 40 wt% solids Equal areas $A_1 = A_2 = A_3 = A$

Question (in system/outerDict): find A, P_1 , P_2 and the live- steam consumption.

Wiring (liquid + vapour BOTH flow 1 $\rightarrow$ 2 $\rightarrow$ 3):

feed (294 K, 10 %) productConc (40 %) | ^ v | +-----+ L1to2 +-----+ L2to3 +-----+ | effect1
|----->| effect2 |----->| effect3 | | P_1 | | P_2 | | P_3 | +---+---+ +---+---+ +---+---+ ^ ^ ^
liveSteam V1 (from 1) V2 (from 2) (170 kPa) chest of 2 chest of 3

$V_3 \rightarrow$ condenser (outside the dict) cond1, cond2, cond3 $\rightarrow$ condensate terminals

Notes for the student: Thermo properties (MW, Cp, Antoine, ...) of 'agua' and 'solidos' live in constant/-components/<name>.dat — these are USER-DEFINED components, NOT from the standard catalogue. Open them to see the textbook's Cp = 4.18 kJ/(K·kg) assumption baked in, and to confirm BPE is set to zero (no K_b on 'agua'). Forward-feed with COLD feed (294 K) penalises effect 1: most of its duty goes into sensible heating, not evaporation. Case (b) of the textbook pre-heats the feed to 355 K — swap T_{feed} below to compare economies.

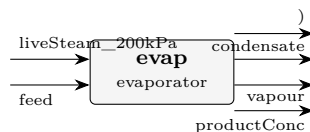
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1to2	F	0.0469536109199
stream	L2to3	F	0.0295647943065
stream	V1	F	0.0161887935578
stream	V2	F	0.0173888166134
stream	V3	F	0.018462595509
stream	cond1	F	0.0262363636123
stream	cond2	F	0.0161887935578
stream	cond3	F	0.0173888166134
stream	feed	F	0.0631424044778
stream	liveSteam	F	0.0262363636123
stream	productConc	F	0.0111021987975
kpi	effect1	A	16.5383245281
kpi	effect1	A_{m2}	16.5383245281
kpi	effect1	BPE	0

... and 81 more reference values in the case's *expected*.

evaporator04_orange_juice_simple

tutorials/steady/evaporation/evaporator04_orange_juice_simple



What it does: Single-effect evaporator concentrating orange juice 10% to 50% solids (MCFT Ex. 10.2 a-c, without BPE)

The story: Single-effect evaporator concentrating orange juice.

Problem (textbook gives): Feed 25 000 kg/h orange juice, 20 °C (293 K), 10 wt% solids Target product Conc = 50 wt% solids P_process 14 kPa (vacuum; keeps T_boil low for thermolabile juice) Heating saturated steam at 2 bar (200 kPa) U 2840 W/(m²·K) Cp_solution 4.18 kJ/(kg·K) (built into ‘solidos’ Cp_mol = 752.4 J/(mol·K)) BPE negligible for parts (a)-(c). Part (d) uses the Capriste-Lozano correlation — skipped here; a separate tutorial picks it up later.

Wiring (the simplest credo case):

```

feed (25000 kg/h, 10 % at 293 K) productConc (5000 kg/h, 50 %) | ^ v | +-----+
+-----+ | evaporator (single effect) | | P = 14 kPa area = $A U = 2840 W/m^2/K | +-----+ +-----+ +-----+ +-----+ +-----+
+-----+ +-----+ +-----+ +-----+ +-----+ | v v liveSteam (200 kPa) vapour condensate (vapor de aquecimento) (20000 kg/h)
(= F_steam) -> condenser

```

What the textbook asks: a) Heat-transfer area $\rightarrow$ KPI effect1.A_needed b) Heating-steam flow $\rightarrow$ KPI effect1.F steam kg h c) Capacity (V evaporated) + economy $\rightarrow$ KPI effect1.F vap, .economy

How the case answers in Choupo

The DesignSpec in 'system/outerDict' manipulates \$F_steam (the only free variable here, with \$A merely "for sizing") until the product outlet hits 5000 kg/h. Then the evaporator reports A_needed, the economy, all duties.

Expected results after convergence (textbook ballpark): productConc : 5000 kg/h @ 50 wt% solids (exact)
Capacity V_evap : 20000 kg/h water (mass balance) Q_duty : ~ 14.1 MW F_steam : ~ 23 000 kg/h
saturated steam at 2 bar A_needed : ~ 73 m² Economy : ~ 0.87 (single-effect; cold feed eats sensible)

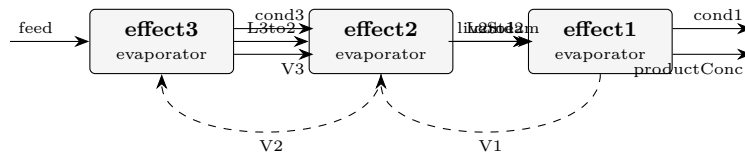
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	0.360686102898
stream	feed	F	0.350791135988
stream	liveSteam_200kPa	F	0.360686102898
stream	productConc	F	0.0424059629635
stream	vapour	F	0.308385173024
kpi	evap	A	75.5133330137
kpi	evap	A_m2	75.5133330137
kpi	evap	BPE	0
kpi	evap	F_conc	0.0424059629635
kpi	evap	F_conc_kmol_h	152.661466669
kpi	evap	F_conc_mass	1.38888555242
kpi	evap	F_cond	6.49776014371
kpi	evap	F_cond_kg_h	23391.9365174
kpi	evap	F_in	0.350791135988

...and 19 more reference values in the case's expected.

evaporator05_counter_current

tutorials/steady/evaporation/evaporator05_counter_current



What it does: Counter-current backward-feed triple-effect evaporator (MCFT 10.38 variant)

The story: COUNTER-CURRENT (BACKWARD-FEED) triple-effect evaporator, Mode-2, credo-pure.

Problem (textbook gives, MCFT 10.38 numerics, counter-current variant): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) entering EFFECT 1 (hot end) P_3 (cold end) such that T_3 = 325 K → 13.5 kPa U_1 3100 W/(m²·K) U_2 2500 W/(m²·K) U_3 1700 W/(m²·K) Spec productConc = 40 wt% solids (exits from effect 1) Equal areas A_1 = A_2 = A_3 = A (single \$A by reference)

Wiring (LIQUID flows 3 → 2 → 1 ; VAPOUR flows 1 → 2 → 3 ; OPPOSITE directions => counter-current):

```
feed (294 K, 10 %) productConc (40 %, exits effect 1) | ^ v | +-----+ L3to2 +-----+ L2to1 +-----+ |
effect3 |<-----| effect2 |<-----| effect1 | | COLD | | MID | | HOT | +---+---+ +---+---+ +---+---+
^^^ | | V2 (from 2) V1 (from 1) liveSteam chest of 3 chest of 2 (170 kPa)
```

v V3 → condenser cond1, cond2, cond3 → condensate terminals

Both V_1 and V_2 flow OPPOSITE to the liquid: effect 1 produces V_1 which feeds effect 2's chest; effect 2 produces V_2 which feeds effect 3's chest. In the directed graph this creates a CYCLE:

```
effect3 -L3to2-> effect2 -L2to1-> effect1 ^ | | V2 V1 | | v +- effect2 <- effect1 (V1 chest) +
```

So we DECLARE V_1 and V_2 as TEAR STREAMS. The flowsheet executive runs effect3 → effect2 → effect1 once per pass, with the initial V_1 / V_2 guesses from the streams block; at end of pass the newly-produced V_1, V_2 are compared with the old. Wegstein (or plain direct substitution) iterates until they match.

In Mode 2 each effect computes T_boil and P_op locally from the chest T and the hardware (A, U) — so the only thing that the Wegstein loop actually iterates is V_i.F (the produced flows). T and P_op stabilise IMMEDIATELY (single pass). Convergence in ~3-5 Wegstein passes.

Design unknowns (in outerDict): \$A common heat-exchange area \$F_liveSteam live-steam flow

Constraints: productConc.F_mass = 0.3125 kg/s (40 wt%) effect3.P = 13.5 kPa (textbook terminal vacuum)

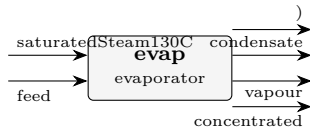
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L2to1	F	0.0319496046345
stream	L3to2	F	0.0496904316429
stream	V1	F	0.020847035226
stream	V2	F	0.0177408270084
stream	V3	F	0.0134519728349
stream	cond1	F	0.0227907734916
stream	cond2	F	0.0208470355984
stream	cond3	F	0.0177408244446
stream	feed	F	0.0631424044778
stream	liveSteam	F	0.0227907734916
stream	productConc	F	0.0111025694085
kpi	effect1	A	17.1719452001
kpi	effect1	A_m2	17.1719452001
kpi	effect1	BPE	0

... and 81 more reference values in the case's *expected*.

evaporator06_nacl_pitzer

tutorials/steady/evaporation/evaporator06_nacl_pitzer



What it does: Single-effect NaCl evaporator with the boiling-point elevation from the PITZER electrolyte model (a_w), not the ideal K_b*m. Concentrated brine → BPE several degrees, the real value.

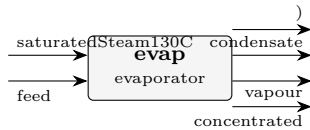
The story: / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the PITZER water activity (a_w), not the ideal K_b*m: concentrated brine is a REAL electrolyte, and the BPE follows from the osmotic coefficient. The thermo arrives via the aqueousNaCl_pitzer property package (see constant/propertyDict) – swap to evaporator07’s eNRTL to compare.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0200845615144
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00769321626341
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	5.05207301886
kpi	evap	F_conc	0.0200845615144
kpi	evap	F_conc_kmol_h	72.3044214517
kpi	evap	F_conc_mass	0.417969209015
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
... and 19 more reference values in the case’s expected.			

evaporator07_nacl_enrtl

tutorials/steady/evaporation/evaporator07_nacl_enrtl



What it does: Single-effect NaCl evaporator with the boiling-point elevation from the eNRTL water activity (electrolyte-NRTL) – the same slot as model pitzer, one word changed. eNRTL is the model for salt in a non-ideal molecular solvent.

The story: / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the eNRTL water activity (a_w), not the ideal K_b*m – the SAME flowsheet as evaporator06, with only the electrolyte model swapped (pitzer → eNRTL): run both and compare the BPE the two G^E surfaces predict.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0200845615144
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00769321626341
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	3.86220786107
kpi	evap	F_conc	0.0200845615144
kpi	evap	F_conc_kmol_h	72.3044214517
kpi	evap	F_conc_mass	0.417969209015
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
...and 19 more reference values in the case's expected.			

evaporator08_naoh_dilution_heat

tutorials/steady/evaporation/evaporator08_naoh_dilution_heat



What it does: Single-effect NaOH evaporator: the heat of dilution IN the duty (fitted L_phi, Parker-validated)

The story: the heat of dilution, IN the duty

Single-effect caustic (NaOH) evaporator – the McCabe enthalpy-concentration classic. NaOH is the FIRST calorimetrically fitted salt (Silvester-Pitzer dbeta/dT vs Parker NSRDS-NBS 2, rel-AAD 4.97%), so this evaporator's duty CLIMBS THE MEASURED L_phi CURVE: concentrating caustic absorbs extra heat beyond sensible+latent, and the run prints it as its own named line:

heat of dilution = X kW (L_phi: m a → b mol/kg, calorimetric fit)

Contrast evaporator06 (whose CASE-LOCAL pair overlay omits the T-slots, so ITS NaCl runs calorimetricFit false – the standards Na-Cl record is fitted since the Parker refit): same machinery, but there the dilution heat is OMITTED and the run says so loudly. The feed (m ~ 2.0) concentrates to m ~ 4-5 mol/kg – inside the Parker fit window (<= 6), so no extrapolation flag; push the steam rate up and watch the [electrolyte] EXTRAPOLATED advisory fire.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0118369724866
stream	condensate	F	0.0208333333333
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0208333333333
stream	vapour	F	0.0159408052912
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	6.44765466966
kpi	evap	F_conc	0.0118369724866
kpi	evap	F_conc_kmol_h	42.6131009516
kpi	evap	F_conc_mass	0.234614448234
kpi	evap	F_cond	0.3753125
kpi	evap	F_cond_kg_h	1351.125
kpi	evap	F_in	0.0277777777778

... and 20 more reference values in the case's *expected*.

flash

The theory you need. An isothermal flash splits a feed of composition z_i into vapour y_i and liquid x_i at fixed (T, P) . Phase equilibrium is $y_i = K_i x_i$ with $K_i = \gamma_i P_i^{\text{sat}}/P$ (γ - φ formulation), and the vapour fraction β solves the Rachford–Rice equation

$$\sum_i \frac{z_i (K_i - 1)}{1 + \beta (K_i - 1)} = 0,$$

monotone in β – Newton converges in a handful of iterations, all of them visible at verbosity 3. Expect: a bubble point when the equation's root hits $\beta = 0$, a dew point at $\beta = 1$, and non-ideal systems ($\gamma_i \neq 1$: NRTL/Wilson/UNIQUAC/UNIFAC) bending K_i with composition. Liquid–liquid and three-phase cases replace successive-substitution with direct Gibbs-energy minimisation (the symmetric γ -model trap: Newton collapses onto the trivial $K = 1$ root). *Full derivation: Theory Guide, the VLE and flash chapters.*

adiabaticFlash01_benzene_toluene

tutorials/steady/flash/adiabaticFlash01_benzene_toluene



What it does: Adiabatic flash, benzene/toluene 40/60, 380K/3bar $\rightarrow$ 1atm

The story: Adiabatic flash: sub-cooled 40/60 benzene/toluene liquid at 380 K, 3 bar expanded to 1 atm. ΔT and V/F come from the energy balance.

The outlet pressure is a plain editable value – tweak it in the GUI's Properties box (transient tinkering) and re-run, or edit it here.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	liquid	F	0.0262953073608
stream	vapor	F	0.00148247041701

bubbleT01_ethanol_water
tutorials/steady/flash/bubbleT01_ethanol_water



What it does: Bubble-T of 30/70 ethanol/water at 1 atm (NRTL)

The story: Bubble-T of a 30/70 ethanol/water liquid at 1 atm (NRTL gamma, non-ideal). A saturation calc, not a flash: the unit solves $\sum(K_i x_i) = 1$ for T at the given P (no product streams). Expected: T_bubble ~ 354.4 K with y_ethanol ~ 0.58 – ethanol enriched in the incipient vapour well beyond its 0.30 in the liquid.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
kpi	bubble01	P	101325
kpi	bubble01	T_bubble	354.424819099
kpi	bubble01	converged	1
kpi	bubble01	y_ethanol	0.581915403358
kpi	bubble01	y_water	0.418084596646

bubbleT02_uniquac_ethanol_water

tutorials/steady/flash/bubbleT02_uniquac_ethanol_water



What it does: Bubble-T of 30/70 ethanol/water at 1 atm (UNIQUAC) – validates the UNIQUAC model against the NRTL/Wilson twins

The story: Bubble-T of a 30/70 ethanol/water liquid at 1 atm (UNIQUAC).

Same conditions as bubbleT01 (NRTL): a strongly non-ideal binary whose bubble temperature and vapour composition are set by the activity coefficients. The UNIQUAC result should land close to the NRTL/Wilson twins ($T_{\text{bubble}} \sim 354$ K, the vapour enriched in ethanol towards the azeotrope, $y_{\text{ethanol}} \sim 0.58$).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
kpi	bubble02	P	101325
kpi	bubble02	T_bubble	353.469206954
kpi	bubble02	converged	1
kpi	bubble02	y_ethanol	0.582711434226
kpi	bubble02	y_water	0.417288565785

dewT01_ethanol_water

tutorials/steady/flash/dewT01_ethanol_water



What it does: Dew-T of 30/70 ethanol/water vapour at 1 atm (NRTL)

The story: Dew-T of a 30/70 ethanol/water vapour at 1 atm (NRTL gamma, non-ideal). The mirror of bubbleT01: the unit solves $\sum(y_i / K_i) = 1$ for T at the given P (a saturation calc, no product streams); the incipient liquid is water-enriched. Dew-T sits ABOVE the bubble-T of the same overall composition – the two bound the two-phase window.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778

flash01_benzene_toluene

tutorials/steady/flash/flash01_benzene_toluene



What it does: Isothermal flash benzene/toluene at 370 K, 1 bar (Raoult)

The story: THE first tutorial: isothermal flash of a 40/60 benzene/toluene feed (100 kmol/h) at 370 K, 1 bar, ideal Raoult K-values ($K_i = P_{\text{sat}_i}/P$). 370 K sits inside the two-phase window (bubble ~368 K, dew ~374 K), so the feed splits into a benzene-enriched vapour and a toluene-enriched liquid – the Rachford-Rice solve is printed iteration by iteration at verbosity 3.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	flash01	V_over_F	0.30398357314
kpi	flash01	T	370
stream	vapor	F	0.00844398814279
stream	liquid	F	0.019333789635

flash02_ethanol_water

tutorials/steady/flash/flash02_ethanol_water



What it does: Isothermal flash 50/50 ethanol/water at 355 K (NRTL)

The story: isothermal flash, ethanol/water, NRTL

NOTE (golden re-recorded): the operation pressure formerly read a BARE 'P 1.01325;' (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to $O(1e5)$, so the flash silently reported a fully superheated vapour ($V/F = 1$, $\text{phaseSet} = 0$) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the 'bar' unit restores the physically-correct partial vaporisation; the 'expected' golden was re-recorded accordingly (was the wrong all-vapour result).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	liquid	F	0.00562323865754
stream	vapor	F	0.0221545391202
kpi	flash02	F_alpha	0.00562323865754
kpi	flash02	F_beta	0.0221545391202
kpi	flash02	F_in	0.0277777777778
kpi	flash02	P	101325
kpi	flash02	T	355
kpi	flash02	V_over_F	0.797563408329
kpi	flash02	phaseSet	0

flash03_wilson_ethanol_water

tutorials/steady/flash/flash03_wilson_ethanol_water



What it does: Isothermal flash 50/50 ethanol/water at 355 K (Wilson activity model)

The story: same conditions as flash02, but Wilson activity model

NOTE (golden re-recorded): the operation pressure formerly read a BARE 'P 1.01325;' (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to O(1e5), so the flash silently reported a fully superheated vapour (V/F = 1, phaseSet = 0) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the 'bar' unit restores the physically-correct partial vaporisation; the 'expected' golden was re-recorded accordingly (was the wrong all-vapour result).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0277777777778
stream	liquid	F	0.0063767199082
stream	vapor	F	0.0214010578696
kpi	flash03	F_alpha	0.0063767199082
kpi	flash03	F_beta	0.0214010578696
kpi	flash03	F_in	0.0277777777778
kpi	flash03	P	101325
kpi	flash03	T	355
kpi	flash03	V_over_F	0.770438083305
kpi	flash03	phaseSet	0

flash04_molarFlows_input

tutorials/steady/flash/flash04_molarFlows_input



What it does: Isothermal flash, feed declared via molarFlows (benzene 40 + toluene 60 kmol/h)

The story: same benzene/toluene flash as flash01, but the feed is declared via per-species molar flows.

Instead of writing $F = 100 \text{ kmol/h} + \text{composition } \{ \text{benzene } 0.40; \text{toluene } 0.60; \}$, this case writes the absolute molar flow of each species:

molarFlows { benzene 40 kmol/h; toluene 60 kmol/h; }

The parser sums to $F_{\text{total}} = 100 \text{ kmol/h}$ and derives $x_{\text{benzene}} = 0.4$, $x_{\text{toluene}} = 0.6$. Physically equivalent to flash01 – and the result table is identical to printer precision ($V/F = 0.304$, $L = 69.6 \text{ kmol/h}$, $V = 30.4 \text{ kmol/h}$).

Demonstrates the (d) form of stream declaration; see userGuide "Composition basis" for the five accepted forms.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.019333789635
stream	vapor	F	0.00844398814279
kpi	flash04	F_alpha	0.019333789635
kpi	flash04	F_beta	0.00844398814279
kpi	flash04	F_in	0.02777777777778
kpi	flash04	P	100000
kpi	flash04	T	370
kpi	flash04	V_over_F	0.30398357314
kpi	flash04	phaseSet	0

flash05_ternary_massFlows

tutorials/steady/flash/flash05_ternary_massFlows



What it does: Ternary flash benzene/toluene/n-hexane, feed in mass flows (5000 kg/h)

The story: ternary benzene/toluene/n-hexane flash, feed declared via per-species MASS flows.

This is the (e) form of stream input from the v0.27 grammar (see userGuide "Composition basis"): instead of writing F total plus a composition, the student writes the absolute mass flow of each species and the parser sums + converts to molar:

$$m_i \text{ [kg/s]} \rightarrow F_mol,i \text{ [kmol/s]} = m_i / MW_i \quad F_total \text{ [kmol/s]} = \text{Sigma } F_mol,i \quad x_i = F_mol,i / F_total$$

Feed composition by mass (5000 kg/h total): benzene 1500 kg/h (30 wt%) MW = 78.11 kg/kmol toluene 2500 kg/h (50 wt%) MW = 92.14 kg/kmol n-hexane 1000 kg/h (20 wt%) MW = 86.18 kg/kmol

Conversion to molar: $F_benzene = 1500 / 78.11 = 19.203$ kmol/h $F_toluene = 2500 / 92.14 = 27.131$ kmol/h $F_nHexane = 1000 / 86.18 = 11.604$ kmol/h $F_total = 57.938$ kmol/h $x_benzene = 0.3315$; $x_toluene = 0.4682$; $x_nHexane = 0.2003$

At T = 365 K, P = 1 bar the n-hexane (Tb = 342 K) is well above its boiling point and goes preferentially to the vapour; toluene (Tb = 384 K) is sub-cooled and stays preferentially in the liquid; benzene (Tb = 353 K) partitions. Result:

molar basis F = 57.94 kmol/h V/F = 0.494 L = 29.34 kmol/h V = 28.60 kmol/h

mass basis F = 5000.0 kg/h V/F_{mass} = 0.487 L = 2567.3 kg/h V = 2432.7 kg/h

Mass fractions tell the partition story directly:

feed: w_benzene 0.300 w_toluene 0.500 w_nHexane 0.200 liquid: w_benzene 0.244 w_toluene 0.624 w_nHexane 0.133 vapor: w_benzene 0.360 w_toluene 0.369 w_nHexane 0.271

Mass balance closes per species — e.g. n-hexane: feed: $5000 * 0.200 = 1000$ kg/h liquid: $2567 * 0.133 = 341$ kg/h vapor: $2433 * 0.271 = 659$ kg/h $\rightarrow 341 + 659 = 1000$.

Toggle the units menu in the GUI to Mass-basis to see the F and composition flip back to kg/h + w_i, matching the input statement symmetrically.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0160941453232
stream	liquid	F	0.00814913402453
stream	vapor	F	0.00794501129864
kpi	flash05	F_alpha	0.00814913402453
kpi	flash05	F_beta	0.00794501129864
kpi	flash05	F_in	0.0160941453232
kpi	flash05	P	100000
kpi	flash05	T	365
kpi	flash05	V_over_F	0.493658478851
kpi	flash05	phaseSet	0

flash06_sweep_T

tutorials/steady/flash/flash06_sweep_T



What it does: Flash benzene/toluene at 1 bar: T swept 360→385 K across the bubble→dew window (V/F S-curve + duty)

The story: the SAME 40/60 benzene/toluene isothermal flash as flash01 (100 kmol/h, 1 bar, Raoult); the T value below is only the BASE point – system/outerDict sweeps it 360 → 385 K across the bubble→dew window (~368..374 K) and reads the V/F S-curve + duty as responses.

flash07_gridsweep_TP`tutorials/steady/flash/flash07_gridsweep_TP`

What it does: 2-D grid sweep: flash V/F over $T \times P$ (the flash envelope)

The story: one isothermal flash, benzene/toluene 40/60.

The base point is 370 K, 1 bar. The gridSweep in system/outerDict overrides BOTH operation.T and operation.P across a grid, so V/F traces the flash envelope ($V/F=0$ bubble line, $V/F=1$ dew line) over the $T \times P$ plane.

flash08_co2_water_package

tutorials/steady/flash/flash08_co2_water_package

What it does: Carbonated water flash whose thermo is an INLINE property-package MANIFEST: constant/propertyDict IS the full record (components, methods per phase, solvent+solutes, the CO2-water Henry pair declared+verified). CO2 dissolves by the unsymmetric Henry convention; water stays Raoult. Read the run log: the assembly announces itself.

The story: the property-package convention, minimal.

Soda water at 298 K: 1 bar flash of a CO2-rich feed. The THERMO is not a pile of flat keys – the case's constant/propertyDict IS the full INLINE manifest (name flash08_co2Water) and the ThermoPackage-Builder assembles everything from it: liquid = solution.henryDilute (water on the Raoult rung, CO2 on the infinite-dilution Henry rung – ONE Gibbs surface, two conventions), vapour = ideal gas, the CO2-water Henry pair declared and VERIFIED at assembly (delete the pair path from the package and the run REFUSES, naming the file). Expect $K_{\text{CO2}} = H(298)/P \sim 1.6\text{e3}$ (Sander): a huge K sets the vapour's PURITY, not the split – the vapour is nearly pure CO2 ($y_{\text{CO2}} \sim 0.97$) but its AMOUNT is tiny ($V/F \sim 4\text{e-4}$), so only ~40% of the fed CO2 leaves in it; the liquid keeps $x_{\text{CO2}} \sim y_{\text{CO2}}/K \sim 6\text{e-4}$.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	flatWater	F	0.000277662593734
stream	gasOff	F	1.15184043824e-07
stream	sodaWater	F	0.000277777777778
kpi	node	F_alpha	0.000277662593734
kpi	node	F_beta	1.15184043824e-07
kpi	node	F_in	0.000277777777778
kpi	node	P	101325
kpi	node	Q	0
kpi	node	Q_kW	0
kpi	node	T	298.15
kpi	node	V_over_F	0.000414662557767
kpi	node	phaseSet	0

flash09_n2ch4_stryjek

tutorials/steady/flash/flash09_n2ch4_stryjek

What it does: phi-phi VLE VALIDATION vs Stryjek, Chappelaar & Kobayashi (1974), J. Chem. Eng. Data 19:334 (DOI 10.1021/je60063a023), Table III, -190 F = 149.82 K. The case carries its thermo as an INLINE manifest: liquid+vapour both eos.SRK (one cubic, two roots: $K = \phi_L/\phi_V$) with the N2-CH4 interaction constant $k_{ij} = 0.0289$ (a small measured correction to the mixing rule, from the Knapp/DECHEMA data compilation). Measured at 299 psia: $x_{N2}=0.1503$, $y_{N2}=0.4579$ – compare the Flash Result table in the log with these measured values (agreement ~2%).

The story: the phi-phi world pinned to a MEASURED isotherm

Nitrogen-methane at 149.82 K (-190.00 F), 20.616 bar (299 psia) – a point of Stryjek, Chappelaar & Kobayashi (1974), J. Chem. Eng. Data 19:334, Table III (DOI 10.1021/je60063a023). MEASURED there: $x_{N2} = 0.1503$ $y_{N2} = 0.4579$ $K_{N2} = 3.047$ $K_{CH4} = 0.638$ The feed $z_{N2} = 0.30$ sits inside the envelope, so the flash must split into exactly those phases if the thermodynamics is right. The K-values come from $K = \phi_L/\phi_V$ – the SAME SRK cubic evaluated at its liquid and vapour roots, with $k_{ij}(N2,CH4) = 0.0289$ – a small MEASURED correction to the cubic mixing rule, from the Knapp/DECHEMA VLE data compilation (announced in this log). Neighbouring measured points bracket the model: $K_{N2} = 3.047$ at 299 psia and 2.241 at 396.5 psia – the engine interpolates between them consistently (try P 22 bar in this dict: $K_{N2} \sim 2.9$). This is the quantitative anchor of the phi-phi world, the same standard as Wiebe-1934 for the Henry world.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	liq	F	0.000145628387466
stream	vap	F	0.000132149390312
kpi	node	F_alpha	0.000145628387466
kpi	node	F_beta	0.000132149390312
kpi	node	F_in	0.000277777777778
kpi	node	P	2061600
kpi	node	Q	0
kpi	node	Q_kW	0
kpi	node	T	149.82
kpi	node	V_over_F	0.475737805121
kpi	node	phaseSet	0

flash_polymer_devolatilisation

tutorials/steady/flash/flash_polymer_devolatilisation



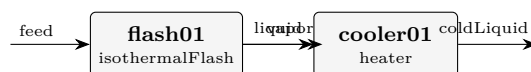
What it does: Devolatilise a polystyrene melt: flash off residual toluene; the polymer (fixed-MW pseudo-component, role nonvolatile) stays liquid

The story: polymer melt + residual solvent → flash (devolatiliser).

The flash boils off the residual toluene (vapour) and leaves the polymer in the liquid. polystyrene is 'role nonvolatile', so its $K = 0$: it stays 100 % in the liquid by construction (no P_{sat} needed for a nonvolatile lump).

flash_pseudoComponent_petCut

tutorials/steady/flash/flash_pseudoComponent_petCut



What it does: Flash of a petroleum pseudo-component (Riazi-Daubert lump) + toluene, with a heater closing the energy balance

The story: feed → isothermal flash → heater on the liquid.

The flash splits the feed into vapour + liquid (the pseudo-component participates in VLE through its critical{} + Ambrose-Walton Psat). The heater then adds sensible heat to the liquid; energyBalance reports the implied duty, and it CLOSES because the lump carries gasIdeal + liquidPure Cp. No reaction → no formation datum needed (matching the lump's omitted gibbsFormation).

leaf01_flash

tutorials/steady/flash/leaf01_flash

What it does: Leaf-node form: isothermal flash benzene/toluene (fractal base case)**The story:** / flowsheetDict (LEAF NODE – fractal base case)

A leaf carries ‘type’ + ‘operation’ (the unit) + ‘boundary’ (its interface) + ‘streams’ (the default feeds, so it runs isolated). No ‘units (...)’ list. Compare flash01_benzene_toluene, which writes the SAME flash as a one-element units list – identical physics, two roupas.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.019333789635
stream	vapor	F	0.00844398814279
kpi	flash	F_alpha	0.019333789635
kpi	flash	F_beta	0.00844398814279
kpi	flash	F_in	0.02777777777778
kpi	flash	P	100000
kpi	flash	T	370
kpi	flash	V_over_F	0.30398357314
kpi	flash	phaseSet	0

vllc01_waterButanol

tutorials/steady/flash/vllc01_waterButanol



What it does: Liquid-liquid flash of water + 1-butanol at 320 K, 1.013 bar – the real miscibility gap from the cited Winkelman et al. (2009) UNIQUAC LLE fit: water-rich ~98.3 mol% water, butanol-rich ~46 mol% butanol (their Figs 1-2). See README.

The story: liquid-liquid flash of water + 1-butanol with the cited Winkelman et al. (2009) UNIQUAC LLE parameters.

Water + 1-butanol is THE textbook partially-miscible binary. At 320 K and 1.013 bar an equimolar mixture is well below the bubble point (water boils at 373 K, 1-butanol near 390 K), so the equilibrium is GENUINELY two LIQUIDS – no vapour. A simple ‘phaseSet LL’ flash (Michelsen TPD pre-check, then direct Gibbs-energy minimisation, multi-start Nelder-Mead) finds the split.

THE PARAMETERS (cited, NOT fabricated – see constant/propertyDict and README.md): the UNIQUAC interaction parameters are the LLE-specific fit of

J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, Fluid Phase Equilibria 284 (2009) 71-79, Table 2 (p.73); structural r,q from their Table 1 (p.72).

with the full temperature dependence $A_{ij}(T) = a_{ij} + b_{ij}T + c_{ij}T^2$ (their Eq. 10), $\tau_{ij} = \exp(-A_{ij}/T)$ (their Eq. 7), valid 273-363 K. The same numbers, fully provenanced, also live in data/standards/binaryPairs/UNIQUAC/nButanol-water.dat.

a REAL liquid-liquid miscibility gap, reproduced from a cited primary source – the water-rich phase is ~98.3 mol% water (~1.7 mol% butanol) and the butanol-rich phase is ~54 mol% water (~46 mol% butanol), matching Winkelman’s Figs 1-2; the IsothermalFlash LL branch: a Michelsen TPD stability test that flags the feed as unstable, then a direct Gibbs-minimisation that resolves the two coexisting compositions; honest phase labelling – the two output streams are named by their actual composition (waterRich / butanolRich), with alphaRich/betaRich hints steering which liquid lands in which slot.

HISTORICAL NOTE: an earlier version of this case carried only a DECHEMA VLE-fit NRTL set, which (being a VLE fit) has NO miscibility gap, so the flash found a single liquid – it was kept as an honest "VLE-fit-misses-LLE" lesson pending a cited LLE set. Winkelman et al. (2009) IS that cited set, so the case now shows the genuine split. (The VLE-blind NRTL lesson lives on in the NRTL catalogue file’s provenance.)

*Water + 1-butanol (n-butanol) is *the* textbook partially-miscible binary. Below the bubble point it splits into two coexisting liquids: a water-rich phase carrying only a couple of mol % butanol, and a butanol-rich phase holding roughly half water. So a flash of a feed *inside the miscibility gap* should give two liquids — and here it does.*

At 320 K, 1.013 bar (well below the bubble point, so no vapour) a feed of 80 mol % water / 20 mol % butanol splits into:

*with split fraction $\beta \approx 0.41$ into the butanol-rich phase. The solver reports the Michelsen TPD instability ($tm_{min} \approx -0.45$, *unstable = YES*) and then a direct Gibbs-energy minimisation that lands the two compositions.*

An independent re-implementation of the UNIQUAC γ (fresh code, isoactivity Newton) reproduces the engine’s split to 4 decimals across 298–350 K, confirming the parameters are entered correctly. The agreement with the published figures is essentially exact — the UNIQUAC fit is doing its job.

constant/propertyDict carries the LLE-specific UNIQUAC for water/1-butanol, cited per value:

- Primary source: J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, *"Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"*, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013. - Binary interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation $A_{ij}(T) = a_{ij} + b_{ij}T + c_{ij}T^2$ (their Eq. 10) and $\tau_{ij} = \exp(-A_{ij}/T)$ (their Eq. 7): - $A_{1,3}$ (butanol→water): $a =$

155.31, $b = 1.0822$, $10^? \cdot c = -43.711 - A_{3,1}$ (water→butanol): $a = -579.36$, $b = 2.7517$, $10^? \cdot c = -6.7700$ - Structural r , q — their Table 1 (p.72) (value Magnussen et al.): 1-butanol $r = 3.9243$, $q = 3.668$; water $r = 0.9200$, $q = 1.400$.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	butanolRich	F	0.0114431411907
stream	feed	F	0.0277777777778
stream	waterRich	F	0.016334636587
kpi	decanter01	F_alpha	0.016334636587
kpi	decanter01	F_beta	0.0114431411907
kpi	decanter01	F_in	0.0277777777778
kpi	decanter01	P	101300
kpi	decanter01	T	320
kpi	decanter01	V_over_F	0.411953082867
kpi	decanter01	phaseSet	1

vlle03_audit_artificial

tutorials/steady/flash/vlle03_audit_artificial



What it does: Algorithm-audit synthetic VLLE: 3 components with identical P_{sat} and exaggerated NRTL. Should converge to 3-phase by construction.

The story: algorithm-correctness audit for the VLLE Gibbs-min branch.

Three synthetic components A, B, C with identical Antoine ($P_{\text{sat}} \approx 0.5$ bar at 350 K). NRTL parameters chosen to GUARANTEE three-phase coexistence at $T = 350$ K, $P = 1$ bar:

A-B: $\tau_{AB} = \tau_{BA} = 5$ (very strong, near-immiscible) A-C: $\tau_{AC} = \tau_{CA} = 5$ (very strong, near-immiscible) B-C: $\tau_{BC} = \tau_{CB} = 0$ (ideal, fully miscible)

Expected behaviour by symmetry and construction: one liquid phase rich in A (the "isolated" component); another liquid phase rich in B+C (they pair together); a vapour phase carrying some of each (P_{sat} is equal so all are equally volatile, $\ln(P/P_{\text{sat}}) \approx +0.301$ at 1 bar).

Equimolar feed (1/3, 1/3, 1/3). This case is the litmus test for the VLLE solver: if it cannot find 3-phase here, the bug is in the algorithm, not in any real-system NRTL parameter set.

What to expect (golden-locked):

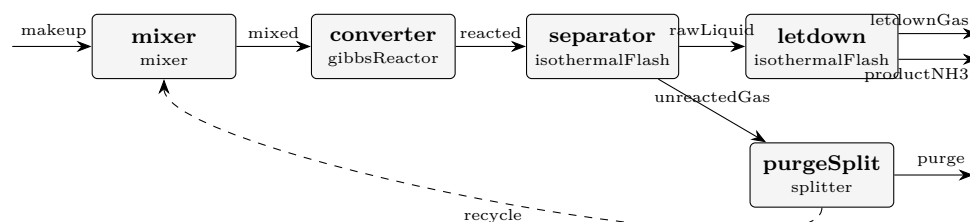
kind	unit/stream	key	expected
stream	aqueous	F	0.011307571384
stream	feed	F	0.0277777777778
stream	organic	F	0.0113091924076
stream	vapor	F	0.00516101398624
kpi	decanter01	F_alpha	0.0164702063938
kpi	decanter01	F_beta	0.011307571384
kpi	decanter01	F_in	0.0277777777778
kpi	decanter01	F_vapor	0.00516101398624
kpi	decanter01	P	100000
kpi	decanter01	T	350
kpi	decanter01	V_over_F	0.407072569823
kpi	decanter01	beta_vapor	0.185796503505
kpi	decanter01	phaseSet	2

flowsheets

The theory you need. Recycle turns a flowsheet into a fixed-point problem: guess the tear stream, march the units, compare. Successive substitution converges linearly (and diverges past unit gain); Wegstein accelerates it; Newton-on-tears solves the tear residuals with a Jacobian. Choupo announces the tear choice, every iteration's residual, and any bound that binds – no silent crutch. Expect: purge streams controlling inert build-up (close the loop with no purge and watch the accumulation), and convergence behaviour that CHANGES with the tear location. *Full derivation: Theory Guide, the flowsheet-convergence chapter.*

ammonia01_synthesis_loop

tutorials/steady/flowsheets/ammonia01_synthesis_loop



What it does: Complete ammonia synthesis loop (Haber-Bosch) with recycle and purge: makeup 1:3 N₂:H₂ (+ Ar inert) → mix with recycle → SRK Gibbs converter at 700 K / 200 bar (real-gas fugacity) → chill-flash at 250 K to condense the NH₃ product → split the unreacted gas into a small Ar-bleeding PURGE and the recycle. The recycle tear + purge are the whole point: high overall N₂/H₂ utilisation despite a modest per-pass conversion, with the inert Ar held at steady state by the purge.

The story: the classic Haber-Bosch loop, recycle + purge

makeup (N₂:H₂ = 1:3 + trace Ar) [mixer] recycle [converter] gibbsReactor (SRK) N₂ + 3H₂ ↔ 2NH₃, 700 K, 200 bar (fugacity-corrected equilibrium ~30% NH₃) [separator] isothermalFlash 250 K, 200 bar (chill: NH₃ condenses out) rawLiquid unreactedGas [purgeSplit] splitter [letdown] flash 20 bar purge recycle (bleeds Ar) (~95% back to the mixer) productNH₃ letdownGas (97.8% NH₃) (H₂-rich)

THE LOOP. Per-pass conversion is equilibrium-limited (~30% NH₃ at the reactor outlet), but the RECYCLE lifts overall N₂/H₂ utilisation high – the reason the loop exists. The reactor equilibrium AND the separator both use SRK FUGACITY (real gas at 200 bar): switch the package to idealGas and the conversion falls. Mass closes to 100% on N, H and Ar.

THE PURGE. Ar enters with the makeup and would accumulate without end; the purge bleeds it, holding Ar at steady state (Ar_{in} = Ar_{out}). Delete the purge and the recycle fills with Ar until the loop chokes – try it.

THE LET-DOWN. At 200 bar the SRK flash leaves a lot of N₂/H₂ DISSOLVED in the liquid (a real high-pressure effect, over-stated here because there is no N₂/H₂-NH₃ binary interaction parameter – SRK over-predicts light-gas solubility). So the raw separator liquid is only ~80% NH₃. A let-down drum (flash to 20 bar) boils the dissolved gas off → 97.8% NH₃ product.

HONEST SIMPLIFICATIONS. (1) The loop is isobaric at 200 bar; the makeup compression and the feed/effluent heat integration are folded into the unit duties (reported), not drawn as separate machines. (2) The let-down gas (letdownGas, H₂-rich) is RECOMPRESSED AND RECYCLED in a real plant; here it exits as a secondary stream to keep ONE clean recycle tear – so the H₂ loss reads higher than industrial. Recovering it (a recompressor + a second tear) is the efficiency refinement. The topology keeps the loop's ESSENCE: reaction, separation, recycle, purge, product finishing.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	letdownGas	F	1.18571362082e-06
stream	makeup	F	0.000277777777778
stream	mixed	F	0.0005681436407
stream	productNH3	F	0.000131387462567
stream	purge	F	1.52824159741e-05
stream	rawLiquid	F	0.000132573176187
stream	reacted	F	0.000438221495669
stream	recycle	F	0.000290365903507
stream	unreactedGas	F	0.000305648319481
kpi	converter	F_in_kmol_h	2.04531710652
kpi	converter	F_out_kmol_h	1.57759738441
kpi	converter	N_in_mol_s	0.5681436407
kpi	converter	N_out_mol_s	0.438221495669
kpi	converter	P	20000000

...and 33 more reference values in the case's *expected*.

ammonia02_full_plant

tutorials/steady/flowsheets/ammonia02_full_plant

What it does: Full-scale ammonia plant (~59 t/h NH₃, 1400 t/day): explicit syngas + recycle COMPRESSORS, feed-effluent HEAT EXCHANGER, ADIABATIC converter (the reaction heats itself – exothermic), water cooler + refrigerated separator, product let-down, and the recycle + purge. SRK fugacity throughout; dissolved H₂/N₂ by measured Henry constants (Wiebe 1934, Larson-Black 1925 via IUPAC SDS 5/6).

The story: the complete Haber-Bosch plant, ~50 t/h NH₃

makeup 25 bar [makeupComp 25→200] recycle [recycleComp] [mixer] mixed (cool) cold side [FEHE] hot effluent (heat recovery) preheatedFeed (~600 K) [converter] gibbsReactor mode ADIABATIC N₂+3H₂→2NH₃ exothermic → self-heats to ~720 K hotEffluent hot side of FEHE → cooledEffluent [waterCooler] (cooling water) ~310 K [separator] flash 250 K/200 bar (refrig.) rawLiquid unreactedGas [letdown] [purgeSplit] productNH₃ gas purge recycle (tear)

THE REACTOR HEAT. N₂+3H₂→2NH₃ is EXOTHERMIC – there is NO external reactor heating. The converter runs ADIABATIC: the exotherm raises the gas T (feed ~705 K in, ~848 K out), and the feed-effluent exchanger (FEHE) recovers that heat to preheat the incoming feed to catalyst light-off. Self-sustaining once lit. The compressors set the 200-bar synthesis pressure; the water cooler (cooling-water utility) + refrigerated flash condense the NH₃. Every machine is explicit and its duty/work is reported.

CAPACITY ~59 t/h NH₃ (~1400 t/day) at 98.9% product purity – a world-scale single train.

DISSOLVED GASES BY HENRY'S LAW. H₂ and N₂ are supercritical at the separator (no physical Psat), so their dissolved amounts follow MEASURED Henry constants (pairs H₂-NH₃ from Wiebe 1934, N₂-NH₃ from Larson & Black 1925 as evaluated in IUPAC SDS Vol 5/6), with the SRK vapour fugacity at 200 bar: $K = H(T)/(\phi_V P)$. The raw liquid carries ~0.5% H₂ + 0.2% N₂ – physical, vs the ~13% a Psat extrapolated above T_c used to fake. KNOWN RESIDUAL: Ar has no measured Ar-NH₃ Henry pair yet, so its dissolved ~1% is a Raoult-extrapolation artefact (~7x over by analogy with N₂) – the product's main impurity is overstated.

THE SIMULATION STRATEGY (how you build a plant like this). 1. PICK THE THERMO BY THE PHYSICS, per phase ("the two worlds"): the 200-bar GAS side needs real-gas fugacity → a cubic EoS (SRK) declared once in the package; the reactor folds phi into its equilibrium automatically. The DISSOLVED gases are supercritical solutes in a condensable solvent → the unsymmetric convention (Henry / Krichevsky-Kasarnovsky), NOT Raoult: pair files H₂-NH₃ + N₂-NH₃ fitted to measured solubility, 'solvent NH₃'; in the package, 'role solute;' overlays on H₂/N₂ in this case's constant/. 2. TEAR THE LOOP WHERE INFORMATION FLOWS BACKWARD: one MASS tear (recycle, guessed then Newton-converged) and one ENERGY tear (hotEffluent – the FEHE preheats the feed with the reactor's own product heat, so the converter's outlet is needed upstream of the converter). Both declared in tearStreams; the solver announces every iteration. 3. LET THE EXOTHERM DO THE HEATING: the converter is ADIABATIC (mode adiabatic; T is a guess, not a spec) – the reaction self-heats 705 → 848 K and the FEHE recycles that heat to light off the feed. No phantom reactor heater exists anywhere. 4. MAKE EVERY MACHINE EXPLICIT so it can be SIZED AND COSTED: compressors by shaft power, exchangers by area, the converter + drums by vessel design bases (space velocity / residence time) in system/postDict → Guthrie costs → the DCF economics in constant/economics. Every unit operation carries its own sizing – the doctrine. 5. VALIDATE AGAINST MEASUREMENT AT EVERY JOINT: the reactor equilibrium against the ideal-vs-SRK shift (gibbs10), the dissolved gas against Wiebe 1934 / Larson-Black 1925 (x_H₂ at 400 atm: 0.0287 computed vs 0.0282 measured), the balances against 100% element closure.

HONEST SIMPLIFICATIONS. (1) The converter is a SINGLE adiabatic bed, so it runs hotter (~575 C) than a real multi-bed converter with interbed quench cooling (which stays near the 400-500 C equilibrium sweet spot for more NH₃ per pass). (2) The makeup compressor is one stage (real: multi-stage with intercooling), so its discharge is hot. (3) The let-down gas and the loop pressure drops are as in ammonia01. (4) 'water' is in the package ONLY as the cooling-water utility fluid (it is not a reaction species).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	chilledEffluent	F	10.5918567473
stream	compMakeup	F	2.2222222222
stream	compRecycle	F	9.3296744801
stream	cooledEffluent	F	10.5918567473
stream	cw	F	138.888888889
stream	cwOut	F	138.888888889
stream	hotEffluent	F	10.5918563175
stream	letdownGas	F	0.00792840799315
stream	makeup	F	2.2222222222
stream	mixed	F	11.5518967023
stream	preheatedFeed	F	11.5518967023
stream	productNH3	F	0.965707281277
stream	purge	F	0.28854663174
stream	rawLiquid	F	0.97363568927

*... and 108 more reference values in the case's **expected**.*

composite01_two_flashes

tutorials/steady/flowsheets/composite01_two_flashes

What it does: Composite node: two flashes in series (fractal step 2)**The story:** / flowsheetDict (COMPOSITE node)

children : the child unit ops (folders flash1/, flash2/ – leaf nodes). connections : the wiring. ‘from’/‘to’ endpoints are either a bare name (this node’s boundary inlet/outlet) or ‘<child>/<port>’. The engine flattens to namespaced streams (flash1.liquid → flash2.feed).

Two flashes in series, each a GENUINE vapour/liquid split. flash1 (370 K) splits the feed; its liquid (benzene-lean) feeds flash2 (373 K), which is still below that liquid’s dew point, so flash2 produces a real vapour AND a real liquid ‘bottoms’ – not a total vaporiser.

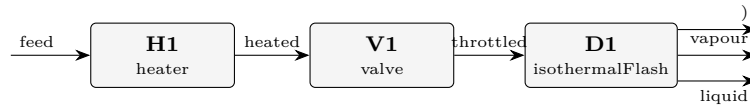
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.00370154472079
stream	feed	F	0.0277777777778
stream	vapor1	F	0.0212163063101
stream	vapor2	F	0.00285992674691
kpi	flash1	F_alpha	0.0065614714677
kpi	flash1	F_beta	0.0212163063101
kpi	flash1	F_in	0.0277777777778
kpi	flash1	P	100000
kpi	flash1	T	370
kpi	flash1	V_over_F	0.763787027163
kpi	flash1	phaseSet	0
kpi	flash2	F_alpha	0.00370154472079
kpi	flash2	F_beta	0.00285992674691
kpi	flash2	F_in	0.0065614714677

... and 4 more reference values in the case’s *expected*.

credo01_valve_heater_drum

tutorials/steady/flowsheets/credo01_valve_heater_drum



What it does: Info-follows-streams credo: heater (Q) + valve (P) set the state; the flash drum (no knobs) just separates

The story: "Information follows the streams" — every change of state is a **VISIBLE** box on the stream, and the separator has no operating knobs. Read the chain top-to-bottom, the way the plant is built:

feed (10 bar liquid) → [H1 heater : Q] the heat box. Its only knob is the duty; T_{out} | is a result. Stays liquid at 10 bar (well below | the bubble point). → [V1 valve : P] the pressure box. Its only knob is the | downstream pressure; the let-down to 1 bar is | isenthalpic, so the hot liquid flashes and T and | vf come out as RESULTS. → [D1 flash : {}] the separator. NO knobs — ‘operation {}’ is | empty. It inherits T and P from the stream V1 | hands it and just splits the phases. → liquid + vapour

Compare with flash01_benzene_toluene, where the same separation is asked as ‘isothermalFlash; operation { T 370; P 1 bar; }’. Those two numbers on the flash were, physically, a hidden heater (T) and a hidden valve (P). Here they are visible boxes, and the drum is honestly knob-less.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	H1	T _{out}	413.08978161
kpi	V1	T _{out}	369.30464285
kpi	V1	vf	0.20555173
kpi	V1	dP	900000
stream	liquid	F	0.0220680075
stream	vapour	F	0.0057097703

plant01_two_sectors`tutorials/steady/flowsheets/plant01_two_sectors`

What it does: Recursive plant: factory of two composite sectors (fractal step 3)

The story: the RECURSIVE node (fractal step 3).

A FACTORY whose children are SECTORS that are themselves composite (each a flowsheet of one flash). Recursion: plant $\rightarrow$ sector $\rightarrow$ unit. Tests nested namespacing (secA.fa, secB.fb) and multi-level cabling (feed $\rightarrow$ secA $\rightarrow$ secA/liquid $\rightarrow$ secB). The whole plant runs from one command.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0
stream	feed	F	0.02777777777778
stream	vaporA	F	0.0212163063101
stream	vaporB	F	0.0065614714677

plant02_basename_twins

tutorials/steady/flowsheets/plant02_basename_twins

What it does: basename twins: a top-level ‘feed’ and a DISTINCT sector-internal ‘A.feed’ coexist

The story: the ADVERSARIAL alias case of design forum #87-P0.

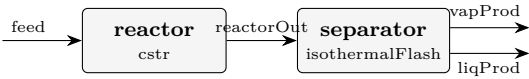
A top-level stream ‘feed’ (the domain inlet) and a sector-INTERNAL edge also named ‘feed’ (A.feed – a completely different stream between A’s two heaters) coexist. Identity is the PATH, never the basename: the manifest must carry BOTH (0/feed and 0/A/feed). The old basename filter silently dropped the top-level one from the manifest, so a legitimate case reported ORPHAN on its own inlet.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001000000000002
stream	hot	F	0.001000000000002
stream	out	F	0.001000000000002
kpi	h	F	0.001000000000002
kpi	h	P	100000
kpi	h	Q	2000
kpi	h	Q_kW	2
kpi	h	Q_per_mol	1999.99999997
kpi	h	T_in	298.15
kpi	h	T_out	366.662038108
kpi	h	dT	68.5120381076

process01_reactor_flash

tutorials/steady/flowsheets/process01_reactor_flash



What it does: Two-unit flowsheet: CSTR → isothermal flash

The story: Two-unit flowsheet: feed → CSTR → flash → (liquid, vapour)

feed reactorOut reactor separator (CSTR) (T 355, P 1 bar) liqProd vapProd

Reaction kinetics are in constant/reactions Solver defaults are in system/solverDict

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	liqProd	F	0.000108248328241
stream	reactorOut	F	0.000277777777778
stream	vapProd	F	0.000169529449537
kpi	reactor	Da_kTau	1.10929132154
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	X_limiting	0.52590711876
kpi	reactor	k	0.00357499942014
kpi	reactor	tau_s	310.291329081
kpi	reactor	xi_mol_per_s	0.0730426553834
kpi	separator	F_alpha	0.000108248328241

...and 6 more reference values in the case's *expected*.

process02_reactor_hx_flash

tutorials/steady/flowsheets/process02_reactor_hx_flash



What it does: Three-unit flowsheet: CSTR → heater → flash

The story: Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~362 K) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator]
vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	hotStream	F	0.000277777777778
stream	liqProd	F	0.000186109936751
stream	reactorOut	F	0.000277777777778
stream	vapProd	F	9.16678410264e-05
kpi	heater	F	0.000277777777778
kpi	heater	P	101325
kpi	heater	Q	400
kpi	heater	Q_kW	0.4
kpi	heater	Q_per_mol	1440
kpi	heater	T_in	350
kpi	heater	T_out	361.946251976
kpi	heater	dT	11.9462519765
kpi	reactor	Da_kTau	1.10929132154

...and 15 more reference values in the case's *expected*.

process02_with_design

tutorials/steady/flowsheets/process02_with_design



What it does: process02 with full reports: balances + sizing + Guthrie costing

The story: Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~362 K) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator]
vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

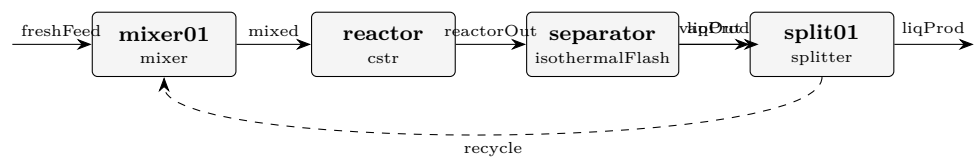
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	hotStream	F	0.0002777777777778
stream	liqProd	F	0.000186109936751
stream	reactorOut	F	0.0002777777777778
stream	vapProd	F	9.16678410264e-05
kpi	heater	F	0.0002777777777778
kpi	heater	P	101325
kpi	heater	Q	400
kpi	heater	Q_kW	0.4
kpi	heater	Q_per_mol	1440
kpi	heater	T_in	350
kpi	heater	T_out	361.946251976
kpi	heater	dT	11.9462519765
kpi	reactor	Da_kTau	1.10929132154

...and 15 more reference values in the case's expected.

process03_recycle

tutorials/steady/flowsheets/process03_recycle



What it does: Recycle loop: the tear stream is solved by Newton (5 iters, |r|2~5e-6)

The story: the classic Reactor-Separator-Recycle architecture

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge

(the splitter is fed by liqProd)

Tear stream: ‘recycle’ — declared explicitly; the Flowsheet runs NEWTON on the tear variables (F, z..., T) until the post-pass value matches the input guess to a tolerance. (‘recycleSolver Wegstein;‘ selects the fixed-point accelerator instead — try both and compare the iteration counts.)

The CSTR converts ~25% per pass (small volume); without recycle most ethanol would simply leave in liqProd. With ~80% recycle, the overall ethanol conversion approaches 100% at steady state.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	freshFeed	F	0.0002777777777778
stream	liqOut	F	0.000364610626883
stream	liqProd	F	0.000145844250753
stream	mixed	F	0.000496545446603
stream	reactorOut	F	0.000496545446603
stream	recycle	F	0.00021876637613
stream	vapProd	F	0.00013193481972
kpi	reactor	Da_kTau	3.53861103626
kpi	reactor	F_in_kmol_h	1.78756360777
kpi	reactor	F_out_kmol_h	1.78756360777
kpi	reactor	T	354.303008007
kpi	reactor	V_R	0.02
kpi	reactor	X_limiting	0.779668274718
kpi	reactor	k	0.00478796954448

...and 9 more reference values in the case's *expected*.

process04_research_workflow

tutorials/steady/flowsheets/process04_research_workflow



What it does: Case-local binary pair file (research workflow pattern)

The story: isothermal flash of ethanol/water at 355 K, 1 atm

Identical to flash02_ethanol_water, but this case demonstrates the RESEARCH WORKFLOW: parameters loaded from case-local binaryPairs file rather than declared inline. See README.md for the full discussion.

The research workflow for managing thermodynamic parameters: instead of declaring NRTL pairs (...) inline in the thermoPackage, the case declares only the model and lets the simulator load the parameters from a file inside the case directory.

This is the pattern used when you have your own fitted parameters that you do not want to mix into the standards database. The case remains self-contained and reproducible: the parameters travel with the case.

When the simulator builds the NRTL model and discovers there are no inline pairs in thermoPackage, it falls back to file lookup:

1. Case-local – <case>/constant/binaryPairs/NRTL/ethanol-water.dat ? used here 2. Standards – \$CHOUP0_HOME/data/standards/binaryPairs/NRTL/ethanol-water.dat

In this case the local file exists, so it wins. If you delete the local file, the run still succeeds because the standards file is the same DECHEMA set.

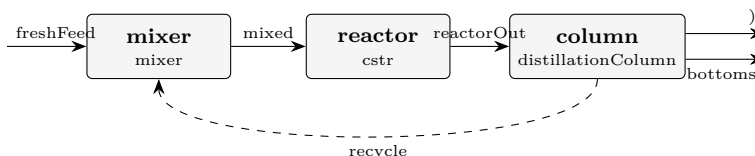
The result is numerically identical to flash02_ethanol_water (same NRTL parameters; just a different way of supplying them).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.00562323865754
stream	vapor	F	0.0221545391202
kpi	flash01	F_alpha	0.00562323865754
kpi	flash01	F_beta	0.0221545391202
kpi	flash01	F_in	0.02777777777778
kpi	flash01	P	101325
kpi	flash01	T	355
kpi	flash01	V_over_F	0.797563408329
kpi	flash01	phaseSet	0

process05_isomerization_recycle

tutorials/steady/flowsheets/process05_isomerization_recycle



What it does: Isomerization neoPentane→nPentane: mixer + CSTR + distillation with recycle

The story: Reactor + distillation with recycle of the unconverted reactant — the canonical flowsheet for exercising the reports{} layer (stream table, mass balance with closure, energy balance) on a non-trivial topology with a tear stream.

freshFeed (neoPentane) [mixer] recycle (neoPentane) [reactor] neoPentane → nPentane (CSTR) [column] (distillation, $\alpha \sim 2.5$) distillate bottoms (neoPentane, (nPentane, recycled) (recycle) the PRODUCT

Chemistry: neoPentane → nPentane is an isomerization — 1:1 in moles AND mass (identical MW 72.15). So at steady state the fresh feed (1 kmol/h neoPentane) equals the product (1 kmol/h nPentane in the bottoms), and the overall MASS BALANCE CLOSES TO 100 %. The recycle carries per-pass unconverted neoPentane back to the reactor, so the OVERALL conversion is ~100 % (product is 99 % nPentane) even though the per-pass conversion is only ~80 %.

Tuning notes (honest): The loop runs at 5 bar so every neoPentane/nPentane stream is a genuine LIQUID at its temperature. neoPentane's Antoine here boils at ~275 K at 1 bar ($P_{\text{sat}}(320 \text{ K}) \sim 4.3 \text{ bar}$), so a 1 bar loop would flash the feed + recycle to VAPOUR — the phase is a CONSEQUENCE of (T,P,z) (Duhem), not a silent liquid default. At 5 bar the bubble point clears ~326 K, so the warm reactor (~327 K) and the recycled distillate stay liquid, as the flowsheet intends. The reaction uses a LOW activation energy on purpose — a high-Ea rate would make the per-pass conversion collapse against the recycle temperature and destabilise the loop. Low Ea decouples conversion from the recycle temperature. 'distillateRate' is FIXED here (the column's spec). It cannot perfectly track the per-pass unconverted neoPentane, so the recycle carries some nPentane (product) along — exactly why real recycle loops need a purge. Closing the recycle PURITY would be a DesignSpec manipulating distillateRate; left out here to keep the case a single pass for reviewing the reports.

This is the case to open in the GUI and to read the reports/ CSVs from: streamTable.csv, massBalance.csv (+ _byUnit), energyBalance_byUnit.csv.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.000277777777778
stream	freshFeed	F	0.000277777777778
stream	mixed	F	0.000363888888889
stream	reactorOut	F	0.000363888888889
stream	recycle	F	8.61111111111e-05
kpi	column	B	0.000277777777778
kpi	column	D	8.61111111111e-05
kpi	column	L_rect	0.000172222222222
kpi	column	L_strip	0.000536111111111
kpi	column	R	2
kpi	column	T_bottom	364.62708689
kpi	column	T_top	350.366212454
kpi	column	V_rect	0.000258333333333
kpi	column	V_strip	0.000258333333333

... and 13 more reference values in the case's expected.

proxy01_gas_loop

tutorials/steady/flowsheets/proxy01_gas_loop

□



What it does: Proxy step 1: gas WGS loop with one recycle (mixer + Gibbs reactor + splitter)

The story: The GUI/credo bench in one case: TWO recycles + a shaft-work energy wire (W) + a utility STREAM (cooling water). Toggle the stream classes in the canvas: energy (the W wire), recycle (the two tears), utility (the cooling-water edge).

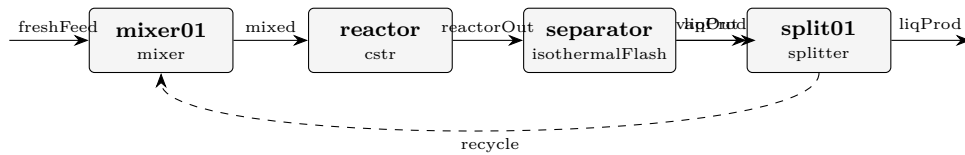
feed M1 mixer recycle1 (hot) recycle2 (cold, dry) R1 gibbsReactor (WGS) S1 splitter recycle1 (30 % hot gas back) toFlash CW (cooling water, UTILITY) HX1 warmCW (HX1: sensible cooling of the gas) F1 flash { } (no knobs: inherits T,P; condenses the water out) productWater (liquid) product dryGas (vapour) S2 splitter recycle2 (30 % dry gas back) productGas T1 turbine (power recovery) G1 electricLoad (W wire) expandedProduct

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	CW	F	0.0166666666667
stream	cooledGas	F	0.000375486726088
stream	dryGas	F	0.000325696494167
stream	expandedProduct	F	0.000227987545917
stream	feed	F	0.000277777777778
stream	mixed	F	0.000536409608696
stream	productGas	F	0.000227987545917
stream	productWater	F	4.97902319203e-05
stream	reacted	F	0.000536409608697
stream	recycle1	F	0.000160922882609
stream	recycle2	F	9.77089482502e-05
stream	toFlash	F	0.000375486726088
stream	warmCW	F	0.0166666666667
kpi	F1	F_alpha	4.97902319203e-05
... and 64 more reference values in the case's <i>expected</i> .			

recycle_autoinit_tear

tutorials/steady/flowsheets/recycle_autoinit_tear



What it does: Recycle with NO authored tear guess – the solver auto-seeds the tear from the feed aggregate (announced), then converges

The story: same Reactor-Separator-Recycle as process03_recycle; this case's LESSON is the TEAR-SEED CONTRACT of the stream-state architecture:

the tear 'recycle' has an AUTHORED state file, 0/recycle – the seed is a first-class case file, owned by the author; completeness is FATAL before the solve: delete 0/recycle and choupoSolve REFUSES (MISSING, zero iterations run); 'bin/choupo-init0' refuses to INVENT a recycle seed (propagation is only honest on an acyclic graph) – it names the missing file and leaves the decision to you.

Compare the seed in 0/recycle with process03_recycle's: a recycle seed is a NUMERICAL aid the student owns, honest and visible ("no silent crutch", CLAUDE.md). (Historical note, marked legacy: on the retired 'streams {}' reader path a missing tear guess was auto-seeded from the feed aggregate, announced on the log; the 0/ contract replaced that with the authored, refuse-if-missing seed above.)

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge

(the splitter is fed by liqProd)

Tear stream: 'recycle' — declared explicitly; the Flowsheet runs Wegstein on (F, z..., T) until the post-pass value matches the input guess to a tolerance.

The CSTR converts ~25% per pass (small volume); without recycle most ethanol would simply leave in liqProd. With ~80% recycle, the overall ethanol conversion approaches 100% at steady state.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	freshFeed	F	0.0002777777777778
stream	liqOut	F	0.000364608639339
stream	liqProd	F	0.000145843455736
stream	mixed	F	0.000496542956892
stream	reactorOut	F	0.000496542956892
stream	recycle	F	0.000218765183603
stream	vapProd	F	0.000131934317553
kpi	mixer01	F_out	0.000496542956892
kpi	mixer01	P_out	100000
kpi	mixer01	T_out	354.302988011
kpi	mixer01	n_in	2
kpi	mixer01	vf	0
kpi	reactor	Da_kTau	3.53861380168
kpi	reactor	F_in_kmol_h	1.78755464481

... and 16 more reference values in the case's *expected*.

gas-solid

bagFilter01_dust

tutorials/steady/gas-solid/bagFilter01_dust



What it does: Bag filter (baghouse) on a dusty gas: >99% capture + cake pressure drop

The story: The high-efficiency complement to cyclone01 (same dusty gas: N2 carrying 50 kg/h of silica, PSD 1..50 μm). Where the cyclone captured ~82 % by SIZE (fines escape), the fabric filter's dust cake captures ~99.9 % across all sizes — the price is a much larger pressure drop, dominated by the cake ($dP = \mu V (K1 + K2 W)$).

Face velocity $V = Q_{\text{gas}} / \text{filterArea}$ is the air-to-cloth ratio (~1.5 m/min here); with a cake load $W = 0.5 \text{ kg/m}^2$ the pressure drop is ~2.7 kPa. The classic train is cyclone (cheap pre-cleaner) THEN bag filter (polish) — this case is the polish stage on the same dust.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, PSD 1..50 μm Filter: 40 m^2 cloth

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	baghouse	airToCloth_m_min	1.53857549963
kpi	baghouse	arealDustLoad	0.5
kpi	baghouse	dP_filter	2745.33935569
kpi	baghouse	efficiency	0.999334225794
kpi	baghouse	faceVelocity	0.0256429249938
kpi	baghouse	filterArea	40
kpi	baghouse	mu_gas	1.78433840676e-05
kpi	baghouse	penetration	0.000665774205525
kpi	baghouse	rho_gas	1.13794383208
kpi	baghouse	solidsCaptured_mass	0.0138796420249
kpi	baghouse	solidsEscaped_mass	9.24686396563e-06

... and 1 more reference values in the case's *expected*.

cyclone01_dust_removal

tutorials/steady/gas-solid/cyclone01_dust_removal



What it does: Cyclone dust removal: silica from N2 (Lapple, $D_c = 0.5$ m)

The story: A dusty gas (N2 carrying 50 kg/h of silica) passes through a cyclone. The cyclone separates BY SIZE: coarse particles spin to the wall and fall out (capturedSolids), fine particles follow the gas (cleanGas). This is the whole point of the PSD — a cyclone is a size classifier, not a simple splitter.

Lapple cut diameter $d_{50} \sim 2.7$ μm for this geometry/flow. The grade efficiency $\eta(d) = 1/(1+(d_{50}/d)^2)$: $\sim 12\%$ at 1 μm , $\sim 93\%$ at 10 μm . Overall collection $\sim 82\%$ of the dust mass; the escaping fines are the sub- d_{50} tail (why a cyclone is a pre-cleaner, followed by a bag filter).

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, $\rho_p = 2650$ kg/m³, PSD 1..50 μm Cyclone: body diameter 0.5 m, 5 effective turns

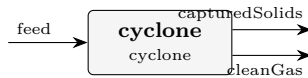
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	2.71094994999e-06
kpi	cyclone	d50_micron	2.71094994999
kpi	cyclone	efficiency	0.820454419117
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
kpi	cyclone	rho_particle	2650
kpi	cyclone	solidsCaptured_mass	0.0113952002655
kpi	cyclone	solidsEscaped_mass	0.00249368862337

...and 1 more reference values in the case's *expected*.

cyclone02_leith_licht

tutorials/steady/gas-solid/cyclone02_leith_licht



What it does: Cyclone dust removal – Leith-Licht back-mixing model

The story: The same dusty gas as cyclone01, but the grade efficiency uses the Leith-Licht back-mixing model instead of Lapple. Accounting for turbulent re-entrainment in the vortex, Leith-Licht predicts a **SMALLER** cut diameter (finer collection) for the same geometry/flow.

Leith-Licht cut diameter $d_{50} \sim 1.11 \mu\text{m}$ here. The grade efficiency is $\sim 48\%$ at $1 \mu\text{m}$, $\sim 94\%$ at $10 \mu\text{m}$. Overall collection $\sim 88\%$ of the dust mass — higher than Lapple's 82% because more of the sub-micron tail is captured.

Gas: 150 kmol/h N_2 , 300 K , 1 atm Dust: 50 kg/h silica , $\rho_p = 2650 \text{ kg/m}^3$, PSD $1..50 \mu\text{m}$ Cyclone: body diameter 0.5 m , 5 effective turns

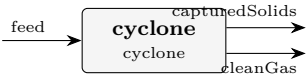
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	1.10984825815e-06
kpi	cyclone	d50_micron	1.10984825815
kpi	cyclone	efficiency	0.880048347849
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
kpi	cyclone	rho_particle	2650
kpi	cyclone	solidsCaptured_mass	0.0122228937201
kpi	cyclone	solidsEscaped_mass	0.00166599516876

...and 1 more reference values in the case's *expected*.

cyclone03_iozia_leith

tutorials/steady/gas-solid/cyclone03_iozia_leith



What it does: Cyclone dust removal – Iozia-Leith data-fitted logistic (beta)

The story: The same dusty gas as cyclone01, now through the REAL Iozia-Leith (1989) FLOW-PATTERN model: the cut diameter d50 is NOT the Lapple residence-time surrogate but falls out of the gas flow pattern – the critical particle sits at the core radius where the tangential velocity peaks, and $d50 = \sqrt{9 \mu Q / (\pi z_c \rho_p V_{t_max}^2)}$ (their eq 4), with V_{t_max} , the core diameter and its length regressed from the cyclone’s OWN geometry (eqs 5-7). The grade curve stays logistic, $\eta(d) = 1/(1+(d50/d)^\beta)$, $\beta = 3.5$. (See the forum note: the 1989 eq-5 prefactor 6.1 is a decimal typo for 0.61 – 6.1 gives a supersonic V_{t_max} , against the paper’s own Figure 4; we use the Figure-4-consistent 0.61.)

Flow-pattern d50 ~ 1.0 μm – FINER than Lapple’s 2.71 μm : a Stairmand-shaped cyclone spins harder than the residence-time picture assumes, so it collects smaller particles. Overall collection RISES accordingly. Geometry defaults to the Stairmand high-efficiency ratios (add an operation.geometry {} block to override a/b/De/S/h/H/B).

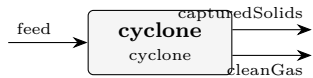
Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, $\rho_p = 2650 \text{ kg/m}^3$, PSD 1..50 μm Cyclone: body diameter 0.5 m, 5 effective turns, $\beta = 3.5$

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	Q_gas	1.02571699975
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	9.86636018962e-07
kpi	cyclone	d50_micron	0.986636018962
kpi	cyclone	dP_cyclone	4903.83536027
kpi	cyclone	efficiency	0.96703441953
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	loading	0.0118992372589
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
... and 5 more reference values in the case’s expected.			

cyclone04_high_loading

tutorials/steady/gas-solid/cyclone04_high_loading



What it does: Cyclone high dust loading – Muschelknautz mass-loading effect

The story: The same cyclone geometry as cyclone01, but with a HEAVY dust load — 2000 kg/h of silica (inlet loading ~ 0.476 kg dust / kg gas, ~40x the ~0.012 kg/kg of cyclone01). At high loading the Muschelknautz model captures the mass-loading effect: above a critical loading the dust forms a wall strand that sweeps out particles independently of the single-particle cut, so the collection efficiency RISES sharply. Muschelknautz overall collection ~ 98.4 % of the dust mass — far above the ~82 % a dilute Lapple cyclone gives, because the high solids loading itself enhances separation. Grade efficiency ~ 90 % even at 1 um. Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 2000 kg/h silica, ?_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns, cLimit = 0.05

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	cyclone	bodyDiameter	0.5
kpi	cyclone	d50	1e-09
kpi	cyclone	d50_micron	0.001
kpi	cyclone	dP_cyclone	4903.83536027
kpi	cyclone	efficiency	0.983932479817
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	32.8229439921
kpi	cyclone	loading	0.475969490356
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5
kpi	cyclone	rho_gas	1.13794383208
... and 4 more reference values in the case's <i>expected</i> .			

cyclone05_dirgo_leith_validation

tutorials/steady/gas-solid/cyclone05_dirgo_leith_validation



What it does: VALIDATION vs Dirgo & Leith (1985): a Stairmand high-efficiency cyclone ($D=0.305$ m, 0.139 m³/s, ~ 15 m/s inlet, mineral-oil aerosol) run through the flow-pattern Iozia-Leith model. Swap the model slot to see the $\sim 3\times$ d50 spread across correlations on ONE real cyclone – why you validate against data.

The story: VALIDATION against Dirgo & Leith (1985), Aerosol Sci. Technol. 4:401 – the measured fractional-efficiency baseline for a Stairmand high-efficiency cyclone. Their pilot-plant cyclone: body diameter $D = 0.305$ m, design air flow 0.139 m³/s (300 cfm) $\rightarrow$ inlet velocity ~ 15 m/s; test aerosol a nebulised mineral oil ($\rho_p = 860$ kg/m³), particles ≥ 1 μ m.

This case reproduces THOSE conditions and runs the flow-pattern Iozia-Leith (1989) model. Re-run it with ‘model Lapple;’ / ‘LeithLicht;’ / ‘Barth;’ / ‘Muschelknautz;’ and compare each model’s d50 and grade curve against the paper’s measured fractional-efficiency curves (their Figures 5-6). THE LESSON: on this ONE real cyclone the models disagree by $\sim 3\times$ – Iozia-Leith 2.2 μ m, Leith-Licht 2.6 μ m (the fine, high-efficiency end) Lapple / Muschelknautz 6.2 μ m, Barth 6.5 μ m (the coarse end) – which is EXACTLY why you validate against data rather than trust one correlation. Dirgo & Leith’s own finding: the theories BRACKET the measured curve and no single one is universally best (they consistently underpredicted efficiency at the highest inlet velocity, 25 m/s). The geometry defaults to the Stairmand high-efficiency ratios (a/D 0.5 , b/D 0.2 , D_e/D 0.5 , S/D 0.5 , h/D 1.5 , H/D 4 , B/D 0.375) – exactly Dirgo & Leith’s cyclone – so the Iozia-Leith flow pattern (V_{t_max} , core) is evaluated on the real test geometry; $V_i = Q/(ab) \sim 15$ m/s reproduces their design point.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.005638888888889
stream	feed	F	0.005638888888889
kpi	cyclone	Q_gas	0.138813700633
kpi	cyclone	bodyDiameter	0.305
kpi	cyclone	d50	2.24296780859e-06
kpi	cyclone	d50_micron	2.24296780859
kpi	cyclone	dP_cyclone	648.673413872
kpi	cyclone	efficiency	0.702163610002
kpi	cyclone	eulerNumber	8
kpi	cyclone	inletVelocity	11.9377544216
kpi	cyclone	loading	0.00879253984647
kpi	cyclone	mu_gas	1.78433840676e-05
kpi	cyclone	numberOfTurns	5

...and 5 more reference values in the case’s *expected*.

gasSolidSplitter01_ideal

tutorials/steady/gas-solid/gasSolidSplitter01_ideal



What it does: Ideal gas-solid splitter: specified 90% solids recovery (no efficiency physics)

The story: The same dusty gas as cyclone01 / bagFilter01 (N₂ + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	capturedSolids	F	0
stream	cleanGas	F	0.0416666666667
stream	feed	F	0.0416666666667
kpi	separator	gasCarryover	0
kpi	separator	gasToCleanGas	0.0416666666667
kpi	separator	gasToSolids	0
kpi	separator	solidsCaptured_mass	0.0125
kpi	separator	solidsEscaped_mass	0.00138888888889
kpi	separator	solidsIn_mass	0.0138888888889
kpi	separator	solidsRecovery	0.9

gibbs

The theory you need. Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A} \mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as λ_E , the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~100–250 K when they are not (dissociation is endothermic). Above ~1500 K the catalogue's polynomial c_p fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

Equilibrium maps. Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes T and P in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The **temperatureApproach** ΔT evaluates the *reaction* equilibrium at $T + \Delta T$ while enthalpy and the energy balance stay at the physical T – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on **nh3_equilibrium_map**, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium NH₃ mole fraction crosses 30 %; report T to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in T at fixed P . *Full derivation:* Theory Guide, the Gibbs-equilibrium chapter, *Equilibrium maps*.

gibbs01_water_gas_shift

tutorials/steady/gibbs/gibbs01_water_gas_shift



What it does: Gibbs reactor: water-gas shift $\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$ at 800 K

The story: Water-gas shift reactor at 800 K, 1 bar. $\text{CO} + \text{water} \rightleftharpoons \text{CO}_2 + \text{H}_2$ Equimolar feed. The Gibbs-minimisation reactor gives CO conversion ~ 67 % (CO 0.50 $\rightarrow$ 0.164 mole fraction at equilibrium; $\text{CO}_2 = \text{H}_2 = 0.336$), consistent with the independent $K(800 \text{ K})$ calculation in Smith, Van Ness, Abbott (Table 13.4).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	products	F	0.000277777777778
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	1
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_out_mol_s	0.277777777778
kpi	wgs	P	100000
kpi	wgs	T	800
kpi	wgs	converged	1
kpi	wgs	lambda_C	5549.05961227
kpi	wgs	lambda_H	-60095.3190791
kpi	wgs	lambda_O	-294758.913033
kpi	wgs	y_CO	0.16409231685
kpi	wgs	y_CO2	0.33590768315

... and 2 more reference values in the case's *expected*.

gibbs02_steam_reforming

tutorials/steady/gibbs/gibbs02_steam_reforming



What it does: Gibbs reactor: steam reforming of methane at 1000 K

The story: Industrial methane steam reforming feed: methane plus excess steam (steam-to-carbon ratio = 3, i.e. 1 mol CH₄ + 3 mol H₂O per pass). Expected at 1000 K, 1 bar: near-complete CH₄ conversion, hydrogen yield ~ 2.7 mol H₂ per mol CH₄ (reforming + WGS combined).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0011111111111111
stream	syngas	F	0.0016586091548
kpi	reformer	F_in_kmol_h	4
kpi	reformer	F_out_kmol_h	5.97099295728
kpi	reformer	N_in_mol_s	1.111111111111
kpi	reformer	N_out_mol_s	1.6586091548
kpi	reformer	P	100000
kpi	reformer	T	1000
kpi	reformer	converged	1
kpi	reformer	lambda_C	-33527.5812363
kpi	reformer	lambda_H	-75161.3018601
kpi	reformer	lambda_O	-309185.611918
kpi	reformer	y_CH4	0.0024289965616
kpi	reformer	y_CO	0.0979819433622
... and 3 more reference values in the case's expected.			

gibbs03_methane_combustion

tutorials/steady/gibbs/gibbs03_methane_combustion



What it does: Gibbs reactor: methane combustion in stoichiometric air at 1800 K

The story: Stoichiometric combustion of methane in air: $\text{CH}_4 + 2 (\text{O}_2 + 3.76 \text{ N}_2) \rightarrow \text{CO}_2 + 2 \text{ H}_2\text{O} + 7.52 \text{ N}_2$ (ideally) At 1800 K some CO and H2 appear from partial dissociation.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.00292222222222
stream	flueGas	F	0.00292404303636
kpi	burner	F_in_kmol_h	10.52
kpi	burner	F_out_kmol_h	10.5265549309
kpi	burner	N_in_mol_s	2.92222222222
kpi	burner	N_out_mol_s	2.92404303636
kpi	burner	P	100000
kpi	burner	T	1800
kpi	burner	converged	1
kpi	burner	lambda_C	-357484.903199
kpi	burner	lambda_H	-201140.892694
kpi	burner	lambda_N	-201371.845813
kpi	burner	lambda_O	-268835.220885
kpi	burner	y_CH4	7.51542148893e-20

...and 6 more reference values in the case's expected.

gibbs04_wgs_temperature_sweep`tutorials/steady/gibbs/gibbs04_wgs_temperature_sweep`

What it does: Water-gas shift equilibrium swept over $T = 600..1200$ K

The story: / flowsheetDict Water-gas shift equilibrium swept over $T = 600..1200$ K (outerDict sweep on the Gibbs reactor): CO conversion falls as T rises – exothermic equilibrium – reproducing the textbook $K(T)$ curve point by point.

gibbs05_adiabatic_flame
tutorials/steady/gibbs/gibbs05_adiabatic_flame



What it does: Adiabatic flame of lean CH4 in air ($\varphi = 0.5$)

The story: Lean methane combustion at $\varphi = 0.5$ (100 % excess air): 1 mol CH4 + 4 mol O2 + 15.04 mol N2 Feed at ambient 298.15 K, 1 bar. Reactor finds its own adiabatic flame temperature.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.005566666666667
stream	flueGas	F	0.00556667001791
kpi	burner	F_in_kmol_h	20.04
kpi	burner	F_out_kmol_h	20.0400120645
kpi	burner	N_in_mol_s	5.56666666667
kpi	burner	N_out_mol_s	5.56667001791
kpi	burner	P	100000
kpi	burner	T	1491.58454904
kpi	burner	converged	1
kpi	burner	lambda_C	-432004.832024
kpi	burner	lambda_H	-204860.137156
kpi	burner	lambda_N	-162961.775795
kpi	burner	lambda_O	-186456.945423
kpi	burner	outerIterations	3
...and 7 more reference values in the case's expected.			

gibbs06_h2_flame_radicals

tutorials/steady/gibbs/gibbs06_h2_flame_radicals



What it does: Stoichiometric H₂/air adiabatic flame – the radical pool at equilibrium

The story: THE RADICAL DATABASE AT WORK: stoichiometric H₂/air adiabatic flame, equilibrium by Gibbs minimisation over the FULL radical pool (H, O, OH, HO₂, H₂O₂ – GRI-Mech 3.0 / Burcat NASA-7 thermo) plus equilibrium thermal NO.

1 H₂ + 0.5 O₂ + 1.88 N₂ at 298.15 K, 1 atm, adiabatic.

THE ANSWER (golden-locked): T_{ad} = 2385.8 K – the literature gives 2380-2390 K (Glassman, Combustion, 4th ed.; Law, Combustion Physics). Equilibrium pool: OH 0.55 %, H 0.18 %, O 0.06 %, NO 0.27 % – all inside the published equilibrium ranges. NO here is the Zeldovich CEILING (what kinetics approaches with residence time, never exceeds).

TWO LESSONS THIS CASE TEACHES: 1. THE RADICALS' WORK: endothermic dissociation (H₂O → OH + H, O₂ → 2 O) soaks up enthalpy near 2400 K. Without the radical species the same reactor answers ~2640 K – the first-year textbook overshoot. 2. THE EXTRAPOLATION TRAP (why constant/components/ overrides exist here): the catalogue's poly3 Cp fits are 250-1500 K PROCESS fits; extrapolated to 2641 K, N₂'s cubic turns NEGATIVE (-33 J/mol/K) and water's reads 45 instead of ~56. The case-local overrides switch the four stable species to the NASA-7 two-range form (GRI-Mech 3.0, valid 200-3500 K) – the same source as the radical entries. Run it once with the overrides removed to SEE the 250 K error a silent extrapolation buys.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	burnt	F	0.000808562123724
stream	feed	F	0.000938888888889
kpi	flame	F_in_kmol_h	3.38
kpi	flame	F_out_kmol_h	2.91082364541
kpi	flame	N_in_mol_s	0.938888888889
kpi	flame	N_out_mol_s	0.808562123724
kpi	flame	P	101325
kpi	flame	T	2385.83679448
kpi	flame	converged	1
kpi	flame	lambda_H	-240335.177161
kpi	flame	lambda_N	-277509.500244
kpi	flame	lambda_O	-344071.838893
kpi	flame	outerIterations	3
kpi	flame	y_H	0.00181958387904

...and 9 more reference values in the case's *expected*.

gibbs06_thermal_nox
tutorials/steady/gibbs/gibbs06_thermal_nox



What it does: Methane combustion at 1800 K with thermal NOx formation

The story: / flowsheetDict Methane combustion at 1800 K with thermal NOx: the Gibbs reactor finds the full equilibrium including NO/NO2 – the EQUILIBRIUM ceiling on NOx, to contrast with the KINETIC path (nox01_zeldovich_postflame, where the rate, not equilibrium, controls).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.00292222222222
stream	flueGas	F	0.00292416183331
kpi	burner	F_in_kmol_h	10.52
kpi	burner	F_out_kmol_h	10.5269825999
kpi	burner	N_in_mol_s	2.92222222222
kpi	burner	N_out_mol_s	2.92416183331
kpi	burner	P	100000
kpi	burner	T	1800
kpi	burner	converged	1
kpi	burner	lambda_C	-355589.713611
kpi	burner	lambda_H	-200666.346513
kpi	burner	lambda_N	-201373.229482
kpi	burner	lambda_O	-269787.239957
kpi	burner	y_CH4	9.68349314225e-20
...and 7 more reference values in the case's expected.			

gibbs07_wgs_cooled
tutorials/steady/gibbs/gibbs07_wgs_cooled



What it does: Cooled WGS Gibbs reactor (350 K): water condenses, shifting the equilibrium (multi-phase)

The story: multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <=> CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The ‘elementPotential’ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated ($y_{\text{H}_2\text{O,P}} > P_{\text{sat}}(350\text{ K}) \sim 0.42\text{ bar}$), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas’ (the dry-ish product gas) + ‘condensate’ (liquid water).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.76244802832e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240153299998
kpi	wgs	F_condensate_kmol_h	0.135448129019
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864551879994
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0376244802832
kpi	wgs	N_out_mol_s	0.277777780282
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	50085.9480936
kpi	wgs	lambda_H	-24736.1727481

...and 8 more reference values in the case’s *expected*.

gibbs08_wgs_cooled_reactiveflash

tutorials/steady/gibbs/gibbs08_wgs_cooled_reactiveflash



What it does: Cooled WGS via reactiveFlash (B1) – cross-checks the elementPotential V+L result

The story: the gibbs07_wgs_cooled problem solved by ‘model reactiveFlash;’ (route B1: an equilibrium-reaction + isothermal-flash inner loop) instead of the default elementPotential (A). Cross-checks the two-phase V+L answer by an independent method.

The physics is gibbs07’s: multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <=> CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The ‘elementPotential’ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated ($y_{\text{H}_2\text{O.P}} > P_{\text{sat}}(350\text{ K}) \sim 0.42\text{ bar}$), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas’ (the dry-ish product gas) + ‘condensate’ (liquid water).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.76244802827e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240153299999
kpi	wgs	F_condensate_kmol_h	0.135448129018
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864551879995
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0376244802827
kpi	wgs	N_out_mol_s	0.277777780281
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	50085.9480935
kpi	wgs	lambda_H	-24736.1727482

...and 8 more reference values in the case’s *expected*.

gibbs09_wgs_cooled_directmin

tutorials/steady/gibbs/gibbs09_wgs_cooled_directmin



What it does: Cooled WGS via directMin (B2, Nelder-Mead direct G-min) – cross-checks A/B1 to ~0.3%

The story: the gibbs07_wgs_cooled problem solved by ‘model directMin;’ (route B2: direct Nelder-Mead minimisation of the total Gibbs energy G) – the third independent route, cross-checking A (elementPotential, gibbs07) and B1 (reactiveFlash, gibbs08) to ~0.3%.

The physics is gibbs07’s: multi-phase (gas + liquid) Gibbs reactor.

$\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$, excess steam (1 CO + 3 H₂O), 350 K, 1 bar.

The ‘elementPotential’ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated ($y_{\text{H}_2\text{O.P}} > \text{Psat}(350\text{ K}) \sim 0.42\text{ bar}$), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas’ (the dry-ish product gas) + ‘condensate’ (liquid water).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	3.77165683291e-05
stream	feed	F	0.000277777777778
stream	gas	F	0.000240061209449
kpi	wgs	F_condensate_kmol_h	0.135779645985
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	0.864220354015
kpi	wgs	N_in_mol_s	0.277777777778
kpi	wgs	N_liq_mol_s	0.0377165683291
kpi	wgs	N_out_mol_s	0.277777777778
kpi	wgs	P	100000
kpi	wgs	T	350
kpi	wgs	converged	1
kpi	wgs	lambda_C	0
kpi	wgs	lambda_H	0
... and 11 more reference values in the case’s <i>expected</i> .			

gibbs10_ammonia_fugacity

tutorials/steady/gibbs/gibbs10_ammonia_fugacity



What it does: Ammonia synthesis equilibrium at 200 bar with the REAL-gas fugacity correction: $\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$ at 700 K, 200 bar over an SRK equation of state. At this pressure the ideal-gas equilibrium (30.0 % NH_3) is wrong – the Gibbs reactor now folds the SRK fugacity coefficients into the equilibrium ($K_{\text{phi}} \sim 0.75$, favouring the fewer-moles NH_3 side), giving 32.4 % NH_3 , toward the measured ~35-40 %. Switch the equationOfState to idealGas and watch NH_3 drop back to 30.0 % – the fugacity correction is the whole point of a high-pressure reactor.

The story: the high-pressure equilibrium done RIGHT

$\text{N}_2 + 3 \text{H}_2 \rightleftharpoons 2 \text{NH}_3$ at 700 K and 200 bar, a stoichiometric feed. The Gibbs reactor minimises the REAL-gas Gibbs energy: it pulls the fugacity coefficients from the package EoS (SRK here) and adds $\ln(\phi_i)$ to each species' chemical potential. At 200 bar this MATTERS – the fugacity ratio $K_{\text{phi}} = \phi_{\text{NH}_3}^2 / (\phi_{\text{N}_2} \phi_{\text{H}_2}^3) \sim 0.75$ shifts the equilibrium toward NH_3 (the fewer-moles side): 30.0 % (ideal) $\rightarrow$ 32.4 % (SRK), toward the measured ~35-40 %. Set the package EoS to idealGas and NH_3 falls back to 30.0 %. (A cornerstone of the classic ammonia synthesis loop: the reactor is only quantitative at pressure once fugacity is in.)

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	products	F	0.000209823048593
kpi	ammonia	F_in_kmol_h	1
kpi	ammonia	F_out_kmol_h	0.755362974933
kpi	ammonia	N_in_mol_s	0.277777777778
kpi	ammonia	N_out_mol_s	0.209823048593
kpi	ammonia	P	20000000
kpi	ammonia	Q_kW	-3.68175302075
kpi	ammonia	T	700
kpi	ammonia	converged	1
kpi	ammonia	lambda_H	-34945.3954982
kpi	ammonia	lambda_N	-59416.3950887
kpi	ammonia	y_H2	0.507099864715
kpi	ammonia	y_N2	0.169033288238

... and 1 more reference values in the case's *expected*.

heat

The theory you need. Heat exchange couples streams without mixing them: $Q = U A \Delta T_{\text{lm}}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

Exchanger DESIGN vs rating. A shell-and-tube poses two inverse questions: RATE it (geometry given $\rightarrow$ find U , area, ΔP , duty – model geometry;) or DESIGN it (duty given $\rightarrow$ size the geometry – model design;). U is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` $\rightarrow$ `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Heat-exchanger design.*

condenser01_film_nusselt

tutorials/steady/heat/condenser01_film_nusselt



What it does: Horizontal-tube steam condenser: Nusselt film HTC → duty (geometry mode)

The story: A horizontal-tube steam condenser whose duty is NOT specified – it EMERGES from the physics of the condensing film, exactly as a real condenser's does.

Saturated steam at 1 bar enters a bundle of horizontal tubes cooled by cooling water. The 'phaseChanger' runs in 'model geometry;' (the seventh resolution): a Nusselt (1916) LAMINAR FILM coefficient h_{cond} is computed on the tube, the wall conducts, and the cooling-water film carries the heat away. Because the condensing-film coefficient itself depends on the driving $dT_{\text{film}} = T_{\text{sat}} - T_{\text{wall}}$, the unit solves a 1-D energy balance at the wall ($q_{\text{cond}} == q_{\text{cool}}$) – the wall temperature is the unknown. Then the duty is $Q = h_{\text{cond}} * A * dT_{\text{film}}$ and the outlet is the dome-flash at that duty.

steam (saturated vapour, 1 bar, $vf = 1$) → condenser (model geometry: NusseltFilm + cooling-water film) → condensate (two-phase, $vf \sim 0.55$) + Q (RESULT)

PARTIAL CONDENSATION: with this tube area the duty $Q \sim -507$ kW removes only part of the latent heat, so the outlet is NOT fully condensed — it leaves at the dome as a wet two-phase stream, $vf \sim 0.55$ (about 45 % of the vapour condensed) at T_{sat} . Fully condensing it would need more area; here the area is fixed and the outlet quality is the result.

Method (glass-box): h_{cond} : Nusselt film, horizontal tube, coefficient $0.729 h = 0.729 [k_l^3 \rho_l (\rho_l - \rho_v) g h_{\text{fg}}] / (\mu_l D dT_{\text{film}})^{1/4}$ $h_{\text{fg}} = h_{\text{fg}} + 0.68 cp_l dT_{\text{film}}$ (Rohsenow subcooling corr.) (Incropera & DeWitt, Fund. of Heat & Mass Transfer, Ex. 10.4) wall : carbon-steel cylindrical resistance $r_o \ln(r_o/r_i)/k$ coolant: a cooling-water film coefficient h (Dittus-Boelter order) the three resistances + which CONTROLS are printed; the duty is cross-checked against $U * A * dT$.

All film/saturation properties are IAPWS-IF97 water. The case is SELF-CONTAINED (its own constant/-components/water.dat). ASSUMPTION stated plainly: the coolant is treated as ISOTHERMAL at its inlet T (its ~ 10 K heat-up is neglected, so $dT = T_{\text{sat}} - T_{\text{in}}$ slightly OVERSTATES the true LMTD – the computed area errs a few % small). The utility allocation still reports the real cooling-water flow and duty.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	condensate	F	0.0277777777778
stream	steam	F	0.0277777777778
kpi	condenser	F	0.0277777777778
kpi	condenser	P	100000
kpi	condenser	Q	-506700.51677
kpi	condenser	Q_UAdT_kW	506.617311982
kpi	condenser	Q_kW	-506.70051677
kpi	condenser	R_coolant	0.00014880952381
kpi	condenser	R_film	0.000349003587291
kpi	condenser	R_wall	4.35883467862e-05
kpi	condenser	Re_film	15.1166609728
kpi	condenser	T_in	372.76
kpi	condenser	T_out	372.755918611
kpi	condenser	T_wall	325.855275434

... and 8 more reference values in the case's *expected*.

heatExchanger01_water_water

tutorials/steady/heat/heatExchanger01_water_water



What it does: Two-stream heat exchanger (eps-NTU): hot/cold water recovery

The story: Two-stream heat exchanger by the effectiveness-NTU method (the built-in ‘heatExchanger‘). A hot water stream is cooled by a cold water stream in a counter-current unit — the canonical heat-recovery problem.

Hardware (the credo): $\text{area} + U$. The outlet temperatures and the duty are RESULTS. "I need $T_{\text{hot_out}} = X$ " would be a DesignSpec on \$area.

Hot : 100 kmol/h water at 360 K Cold : 120 kmol/h water at 300 K Hardware: $A = 10 \text{ m}^2$, $U = 500 \text{ W}/(\text{m}^2\text{K})$, counter-current

Expected (eps-NTU, $C_{p_water} \sim 75 \text{ J/mol/K}$): $C_{\text{hot}} = 100/3.6 * 75 \sim 2083 \text{ W/K}$ ($= C_{\text{min}}$) $C_{\text{cold}} = 120/3.6 * 75 \sim 2500 \text{ W/K}$ $NTU = U.A/C_{\text{min}} \sim 2.4$, $C_r \sim 0.83$, effectiveness ~ 0.75 $Q \sim 93 \text{ kW}$ $T_{\text{hot_out}} \sim 315 \text{ K}$ (cooled $\sim 45 \text{ K}$) $T_{\text{cold_out}} \sim 337 \text{ K}$ (heated $\sim 37 \text{ K}$) $LMTD \sim 22 \text{ K}$, and $U.A.LMTD \sim Q$ (cross-check) energy balance: hot stream loses what the cold stream gains.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cold	F	0.03333333333333
stream	coldOut	F	0.03333333333333
stream	hot	F	0.02777777777778
stream	hotOut	F	0.02777777777778
kpi	hx	C_cold	2516.66666667
kpi	hx	C_hot	2097.22222222
kpi	hx	C_r	0.833333333333
kpi	hx	LMTD	18.7584806629
kpi	hx	NTU	2.38410596026
kpi	hx	Q	93792.4033146
kpi	hx	Q_kW	93.7924033146
kpi	hx	T_cold_in	300
kpi	hx	T_cold_out	337.268504628
kpi	hx	T_hot_in	360
... and 5 more reference values in the case's expected.			

heatExchanger02_geometry_U

tutorials/steady/heat/heatExchanger02_geometry_U



What it does: Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m².K), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

The story: A shell-and-tube heat exchanger whose overall coefficient U and area are COMPUTED from the bundle geometry plus per-side convective correlations (the 'model geometry;' path), instead of being specified. This is the glass-box answer to "where does U come from?".

Hot water (shell side) is cooled by cold water (tube side), clean service.

Method: tube-side h_i : Gnielinski $Nu = (f/8)(Re-1000)Pr/[1+12.7 \sqrt{f/8}(Pr^{2/3}-1)]$ (Incropera & DeWitt, Fund. of Heat and Mass Transfer, 6th ed.) shell-side h_o : Kern method $Nu = 0.36 Re_s^{0.55} Pr^{1/3}$ (Kern, Process Heat Transfer, 1950) on the bundle equivalent diameter D_e and crossflow mass flux G_s . U on the OUTSIDE area: $1/U = 1/h_o + r_o \ln(r_o/r_i)/k_{wall} + r_o/(r_i h_i)$ area = $\pi d_o L n_{tubes}$ PRESSURE DROP (Kern, both sides): tube $f = 0.079 Re^{-0.25}$ (Blasius), $dP = (4 f L n_p/d_i + 4 n_p)(\rho u^2/2)$ – friction + turnaround losses; shell $f = \exp(0.576 - 0.19 \ln Re_s)$, $dP = f G_s^2 D_s (N_b+1) / (2 \rho D_e)$ over the N_b baffles. (Kern, Process Heat Transfer, 1950.)

Properties are IF97 water (ρ , μ , λ) at each stream temperature.

Expected order of magnitude (clean water-water): $U \sim 2000-3000$ W/(m².K), the SHELL side controlling, both sides turbulent ($Re \sim 2-4 \times 10^4$); tube-side $dP \sim 9$ kPa, shell-side $dP \sim 35$ kPa (shell > tube: baffled crossflow is more dissipative – the duty-vs-pumping trade-off you design around, NOT just the area). The exact numbers are pinned by –record – they are RESULTS.

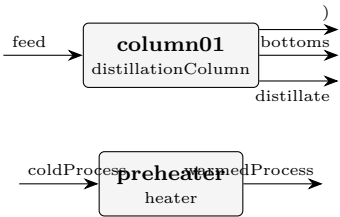
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	15.5339177262
kpi	hx	NTU	1.26058061182
kpi	hx	Nu_shell	171.621364606
kpi	hx	Nu_tube	287.732215939
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	2065604.42087

... and 22 more reference values in the case's *expected*.

heatlink01_condenser_to_heater

tutorials/steady/heat/heatlink01_condenser_to_heater



What it does: Forward heat-link: column condenser preheats a downstream stream

The story: feed column01 distillate bottoms (condenser heat) preheater warmedProcess coldProcess

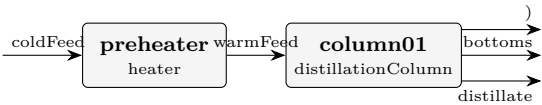
The preheater is heated by the column’s rejected condenser duty (~1.28 MW), warming a 1500 kmol/h benzene stream from 320 K by ~23 K. column01 is listed FIRST so it solves before the preheater reads its condenser KPI (forward heat-link).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	coldProcess	F	0.416666666667
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
stream	warmedProcess	F	0.416666666667
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0277777777778
kpi	column01	L_strip	0.0555555555556
kpi	column01	Q_condenser_kW	-1281.04348002
kpi	column01	Q_reboiler_kW	1279.30480005
kpi	column01	R	2
kpi	column01	T_bottom	382.893530951
kpi	column01	T_top	354.21089412
...and 14 more reference values in the case's expected.			

heatlink02_condenser_feed_preheat

tutorials/steady/heat/heatlink02_condenser_feed_preheat



What it does: Feedback heat-link (energy tear): column condenser preheats its OWN feed

The story: FEEDBACK heat-link == an ENERGY TEAR.

coldFeed preheater column01 distillate bottoms (condenser heat)

The preheater (UPSTREAM of the column) is heated by the column’s CONDENSER duty (DOWN-STREAM) → circular. The Flowsheet detects the feedback, turns the recycle outer loop on, and converges the duty by successive substitution (Wegstein). Sized so the ~1 MW condenser preheats the large 1000 kmol/h feed by a modest ~25 K (stays liquid).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.263888888889
stream	coldFeed	F	0.277777777778
stream	distillate	F	0.0138888888889
stream	warmFeed	F	0.277777777778
kpi	column01	B	0.263888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0208333333333
kpi	column01	L_strip	0.298611111111
kpi	column01	Q_condenser_kW	-1067.88365467
kpi	column01	Q_reboiler_kW	1846.42629755
kpi	column01	R	1.5
kpi	column01	T_bottom	365.99791143
kpi	column01	T_top	354.558892551
kpi	column01	V_rect	0.0347222222222
...and 13 more reference values in the case’s expected.			

hx01_heater

tutorials/steady/heat/hx01_heater



What it does: Heater: 50/50 ethanol/water from 300 K to ~338 K (sensible)

The story: Stand-alone heater — credo-pure (v0.34+).

Hardware spec: $Q = 100$ kW (the absolute thermal power delivered by the heater – think of it as the electrical rating).

Streams IN $\rightarrow$ OP (Q only) $\rightarrow$ Streams OUT

Feed is 100 kmol/h of 50/50 ethanol/water at 300 K. The simulator solves the enthalpy balance $H_{\text{out}} = H_{\text{in}} + Q/F$ to compute T_{out} as a RESULT. With $Q = 100$ kW the outlet temperature lands near 338 K ($C_{p,\text{mix}} \sim 94$ J/(mol*K) at this composition); the exact value falls out of the Newton-1D in T.

NOTE (duty reduced 160 $\rightarrow$ 100 kW). At 160 kW the outlet reached ~360 K, ABOVE the 50/50 ethanol/water bubble point at 1 atm – the old single-phase heater silently reported a physically-impossible "superheated liquid". The new saturation guard refuses that (a heater is sensible-only); a duty that boils belongs to a 'phaseChanger' / 'boiler' (see steady/heat/phasechange01_partial_condenser and steady/power/rankine02_water). 100 kW keeps this the clean sensible-heat lesson it was meant to be.

If you instead need a specific T_{out} (say "I want exactly 338 K"), wrap the heater in a DesignSpec outerDict that manipulates \$Q with a target on 'heater01.T_out'. The unit op itself never specifies its outlet state – that is the credo we adopted in v0.33.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	hotStream	F	0.02777777777778
kpi	heater01	F	0.02777777777778
kpi	heater01	P	101325
kpi	heater01	Q	100000
kpi	heater01	Q_kW	100
kpi	heater01	Q_per_mol	3600
kpi	heater01	T_in	300
kpi	heater01	T_out	338.338658147
kpi	heater01	dT	38.338658147

hxWorkflow1_design_from_duty

tutorials/steady/heat/hxWorkflow1_design_from_duty



What it does: PART 1 of 2 – DESIGN: the DUTY TARGET is given (cool the hot shell 330→315 K) and the exchanger GEOMETRY is SOLVED for (model design;). Glass-box Kern sizing: fix the tube choices, then find the tube count (and the shell diameter it implies, Kern/Sinnott bundle) so $U \cdot A = Q_{\text{req}}/\text{LMTD}$; U/dP then rated with Gnielinski (tube) + Kern (shell). The detailed-design (sizing) step, glass-box. Part 2 (hxWorkflow2_rate_designed) rates the bundle this case produces.

The story: PART 1 of 2: DESIGN from the duty

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 1. DESIGN mode ('model design;') – detailed sizing, glass-box. Instead of GIVING the tube count and shell diameter (that is 'model geometry;'), the RATING problem: geometry → U , here we give the DUTY TARGET (cool the hot shell stream from 330 K to 315 K) and the engine SOLVES for the geometry: fix the tube choices (OD/ID/length/pitch/passes/material), then find the number of tubes – and the shell diameter it implies (Kern/Sinnott bundle correlation, the chosen pattern + passes) so the exchanger DELIVERS THE DUTY – the design loop sizes N by the eps-NTU duty (incl. the 1-2 penalty for a 2-pass U-tube), not merely $U \cdot A = Q/\text{LMTD}$, so the outlet actually reaches the target. It then rates that designed bundle exactly like a given one (same Gnielinski tube + Kern shell correlations) and reports U , area, and the pressure drop.

DO I HAVE TO SPECIFY EVERYTHING? No – and this is the whole point. A HX poses two OPPOSITE questions, and this is the DESIGN one:

RATING ('model geometry;'): you GIVE the full hardware (tube count, shell diameter, ...) and ask "what outlet T do I get?". The temperature is a RESULT. DESIGN ('model design;'), THIS case): you GIVE the outlet T you NEED and ask "what hardware delivers it?". The GEOMETRY is the result.

So HOW does the outlet reach the temperature you need? You do NOT tune the geometry by hand until it works. You DECLARE the target and the engine sizes the hardware to hit it. Look at the two blocks below:

```

'design { spec { stream hotShell; T 315 K; } }' <- WHAT YOU WANT: the required outlet temperature.
THIS is where the final temperature is set. 'geometry { tubeID ...; passes 2; tubePattern ...; wallMaterial ...; }' <- the FIXED CHOICES a designer commits to (a standard tube, the number of passes, the layout, the metal). Notice what is MISSING: there is NO 'nTubes' and NO 'shellID'. Those are exactly what the engine SOLVES.

```

So the student specifies the REQUIREMENT (the outlet T) + the design CHOICES (tube size, passes, pattern, material) – NOT the answer. The engine returns the tube count, the shell diameter, U , the pressure drop, and confirms the outlet lands on the target. You said "cool this to 315 K with 1/2-inch steel tubes in a 2-pass U-tube"; it answers "then you need 105 tubes in a 0.31 m shell – here is U , the dP , and yes it leaves at 315 K".

This is the workflow that matches industrial practice: establish the DUTY, THEN design the exchanger from it – you do NOT invent the geometry up front. (Standard practice splits this into a shortcut that gets the duty + a detailed-design step that gets the geometry; Choupo's design mode folds both into one honest step – it computes the duty from your target and ANNOUNCES it, then sizes.) PART 2 (hxWorkflow2_rate_designed) then RATES the bundle this case produced – a detailed-rating step – to verify it and probe off-design (turn the coolant down and watch the outlet drift off the target; the geometry is now FIXED, only the performance moves).

Water-water, clean service, 2-pass U-tube (the general arrangement). Expect: ~100 tubes, a ~0.3 m shell; rated $U \sim 4000 \text{ W}/(\text{m}^2 \cdot \text{K})$ (2 tube passes → high tube velocity), and the delivered hot outlet 315 K (the target, reproduced by the 1-2 eps-NTU on the designed geometry). Numbers are RESULTS pinned by –record. Single-phase; the 1-shell/2-tube-pass eps-NTU carries the built-in LMTD F-correction ($F < 1$), so a 2-pass needs more area than pure counter-current.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	16.7868881533
kpi	hx	NTU	1.0053658663
kpi	hx	Nu_shell	195.331265319
kpi	hx	Nu_tube	416.603846461
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	1888506.69951

...and 22 more reference values in the case's *expected*.

hxWorkflow2_rate_designed

tutorials/steady/heat/hxWorkflow2_rate_designed



What it does: Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band ($\sim 2000\text{--}3000\text{ W/(m}^2\text{K)}$), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

The story: PART 2 of 2: RATE the designed exchanger

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 2.

Part 1 (heatExchanger03_design_sizing) gave the engine a DUTY TARGET (cool the hot shell $330 \rightarrow 315\text{ K}$) and it SIZED the bundle: 105 tubes, shell ID 0.306 m (Kern/Sinnott, 2-pass U-tube). That is the detailed-design step.

Part 2 – THIS case – is the detailed-rating / checking step: take the DESIGNED geometry as a FIXED given ('model geometry;') and RATE it – confirm it delivers the target outlet, and read the honest pressure drop. This is the verification a real project always does after sizing: the design loop used a bundle correlation for the shell diameter and rounded the tube count UP, so the rated exchanger is slightly OVER-surfaced – the outlet comes out a touch below 315 K (good: margin), and that is what you check here.

OFF-DESIGN (try it): drop the cold-tube flow F to $\sim 6000\text{ kmol/h}$ and re-run – less coolant $\rightarrow$ the hot stream leaves WARMER than 315 K, and the shell dP falls. That is the rating question ("will my exchanger cope when the plant turns down?") the design step cannot answer. The geometry is now FIXED; only the performance moves.

Numbers are RESULTS pinned by –record. Same clean water-water service, same Gnielinski (tube) + Kern (shell) correlations, single-phase, 2-pass U-tube (1-2 eps-NTU).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	coldOut	F	2.22027777778
stream	coldTube	F	2.22027777778
stream	hotOut	F	1.66527777778
stream	hotShell	F	1.66527777778
kpi	hx	C_cold	167630.972222
kpi	hx	C_hot	125728.472222
kpi	hx	C_r	0.750031277368
kpi	hx	LMTD	16.7876807682
kpi	hx	NTU	1.00522927337
kpi	hx	Nu_shell	195.283521062
kpi	hx	Nu_tube	416.603846461
kpi	hx	Pr_shell	3.16405821902
kpi	hx	Pr_tube	5.87094963164
kpi	hx	Q	1888394.47754

... and 22 more reference values in the case's *expected*.

mheatx01_two_hot_one_cold

tutorials/steady/heat/mheatx01_two_hot_one_cold



What it does: Multi-stream heat exchanger (MHeatX): 2 hot + 1 cold, balance closes

The story: Multi-stream heat exchanger (the commercial detailed-rating tools MHeatX analogue): TWO hot water streams are cooled by ONE cold water stream inside a single adiabatic shell — the canonical multi-stream heat-recovery problem.

THE SPEC (glass-box, "information follows the streams"): every inlet carries its OWN outlet target temperature. The unit does NOT rate hardware; it VERIFIES the first law:

$Q_k = F_k (h_{k,out} - h_{k,in})$ per stream $Q_{hot} = \sum Q_k$ over the streams that cool (< 0 , released)
 $Q_{cold} = \sum Q_k$ over the streams that heat (> 0 , absorbed) imbalance = $\sum Q_k$ over ALL streams (external duty)

For a truly adiabatic shell the imbalance is ~ 0 ; a non-zero value is the external duty the topology would have to supply, REPORTED honestly (never a silently-invented duty). The unit also builds the hot/cold composite curves and reports the INTERNAL minimum-approach temperature (the in-unit pinch).

Streams (water, liquid): hot1 : 100 kmol/h, 380 $\rightarrow$ 330 K (gives up ~ 105 kW) hot2 : 60 kmol/h, 360 $\rightarrow$ 330 K (gives up ~ 38 kW) cold : 340 kmol/h, 300 $\rightarrow$ 320 K (takes up ~ 143 kW)

The cold flow is sized so the bundle is adiabatic: cold absorbs what the two hot streams release. Both hot outlets (330 K) stay above the cold outlet (320 K), so the composite curves never cross — $dT_{min} > 0$.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cold	F	0.094444444444444
stream	coldOut	F	0.094444444444444
stream	hot1	F	0.027777777777778
stream	hot1Out	F	0.027777777777778
stream	hot2	F	0.016666666666667
stream	hot2Out	F	0.016666666666667
kpi	mhx	Q_cold	142611.111111
kpi	mhx	Q_cold_kW	142.611111111
kpi	mhx	Q_hot	-142611.111111
kpi	mhx	Q_hot1_kW	-104.861111111
kpi	mhx	Q_hot2_kW	-37.75
kpi	mhx	Q_hot_kW	-142.611111111
kpi	mhx	balanceCloses	1
kpi	mhx	dT_min	30

...and 4 more reference values in the case's *expected*.

phasechange01_partial_condenser

tutorials/steady/heat/phasechange01_partial_condenser



What it does: Partial condenser: benzene/toluene vapour cooled to quality 0.4

The story: A superheated benzene/toluene vapour enters a condenser that is asked to land its outlet at a vapour QUALITY of 0.4 (40 % of the moles stay vapour, 60 % condense). The duty Q is the RESULT – the unit solves it as a built-in self-target (no outerDict, no variables block) and ANNOUNCES that it did so. Because the mixture is multicomponent, the condensed liquid is toluene-RICH and the remaining vapour benzene-RICH – the partial- condensation lesson: a partial condenser is also a SEPARATION.

feedVapour (superheated, 1 bar, 390 K, $vf = 1$) → condenser (outletQuality 0.4) → wetOut (two-phase, $0 < vf < 1$)

Same outlet vocabulary as rankine02_water's IF97 boiler/condenser, on the GENERIC (Antoine + Raoult) backend.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	condenser	vf_out	0.399999998746
kpi	condenser	T_out	367.494289759
kpi	condenser	Q_kW	-630.508868406
kpi	condenser	Tsat	352.826063711
stream	wetOut	F	0.0277777777778
stream	wetOut	vf	0.399999998746

reboiler_water_copper

tutorials/steady/heat/reboiler_water_copper



What it does: Steam-heated copper-surface pool reboiler: Rohsenow nucleate flux → duty, Zuber CHF ceiling (geometry mode)

The story: A steam-heated copper-surface POOL REBOILER whose duty is NOT specified – it EMERGES from the physics of NUCLEATE BOILING, exactly as a real reboiler’s does.

Saturated-liquid water at 1 bar sits in a pool boiling on a bundle of horizontal copper tubes heated INSIDE by condensing steam. The ‘phaseChanger’ runs in ‘model geometry;’ with a ‘boiling { ... }’ block (the boiling resolution): a Rohsenow (1952) NUCLEATE-boiling flux q'' is computed on the tube surface, the copper wall conducts, and the heating-medium film delivers the heat. Because the nucleate flux itself depends on the wall superheat $dT_{\text{excess}} = T_{\text{surface}} - T_{\text{sat}}$ (and on its CUBE), the unit solves a 1-D energy balance at the surface ($q_{\text{boil}} == q_{\text{heat}}$) – the surface temperature is the unknown. Then the duty is $Q = q'' * A$ and the outlet is the dome-flash at that duty.

water (saturated liquid, 1 bar, $vf = 0$) → reboiler (model geometry: Rohsenow nucleate flux + steam film) → wet vapour (partially boiled) + Q (RESULT)

HONESTY-AS-STRUCTURE (the binding constraint): the nucleate flux is +/-100% uncertain – C_{sf} is surface-FINISH lab data (polish, oxidation), NOT a fluid property. There is NO default C_{sf} and NO catalogue in v1: it is STUDENT-SUPPLIED here (water-copper 0.013, polished) WITH a mandatory citation. the result carries the $[q/2, 2q]$ scatter band ($C_{\text{sf}} \pm 25\%$, cubed). the RELIABLE figure is the Zuber (1959) CHF ceiling, printed FIRST. A design above CHF is HARD-REFUSED (burnout) – a pool boiler cannot operate above the critical heat flux.

Method (glass-box): q_{boil} : Rohsenow nucleate flux, water-copper, C_{sf} 0.013, s 1.0 $q'' = \mu_l h_{\text{fg}} \sqrt{g(\rho_l - \rho_v)/\sigma} [cp_l dT_{\text{excess}} / (C_{\text{sf}} h_{\text{fg}} Pr_l^s)]^3$ (Rohsenow 1952; Incropera & DeWitt, Ch.10 Table 10.1) q_{CHF} : Zuber 1959, $q''_{\text{max}} = 0.149 h_{\text{fg}} \rho_v^{0.5} [\sigma g (\rho_l - \rho_v)]^{0.25}$ (~1.26 MW/m² for water at 1 atm) wall : copper cylindrical resistance $r_o \ln(r_o/r_i)/k$ medium : a condensing-steam film coefficient h (high) the CHF margin SAFE/BURNOUT + the cube sensitivity are printed.

All film/saturation properties are IAPWS-IF97 water (incl. σ via R1-76). The case is SELF-CONTAINED (its own constant/components/water.dat).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	wetVapour	F	0.02777777777778
kpi	reboiler	F	0.02777777777778
kpi	reboiler	P	100000
kpi	reboiler	Q	111804.644945
kpi	reboiler	Q_kW	111.804644945
kpi	reboiler	T_in	372
kpi	reboiler	T_out	372.755918611
kpi	reboiler	T_surface	381.390508143
kpi	reboiler	Tsat	372.755918611
kpi	reboiler	area	1.25663706144
kpi	reboiler	chf_margin	0.0709724846287
kpi	reboiler	dT_excess	8.6345895313
kpi	reboiler	q_CHF	1253960.82217

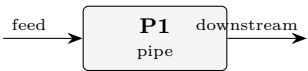
...and 4 more reference values in the case’s *expected*.

hydraulics

The theory you need. Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves' intersection.

pipe01_water_line

tutorials/steady/hydraulics/pipe01_water_line



What it does: Pipe pressure drop: 2000 kmol/h water in a 0.1 m x 50 m commercial-steel line (Churchill friction)

The story: A single PIPE element. Water enters at 5 bar and flows through 50 m of 0.1 m commercial-steel pipe with two elbows, a globe valve, and a 2 m rise. The pressure DROP is a COMPUTED RESULT of the geometry and the flow — the pipe has NO pressure knob (information follows the streams: a pipe only dissipates, a pump would raise P, a valve would set a let-down P).

feed (5 bar liquid, 2000 kmol/h) → [P1 pipe] D, L, roughness, dz, fittings → ΔP (result) → downstream (P reduced by the computed ΔP, T held)

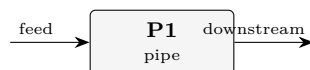
Friction factor: Churchill (default) — one explicit correlation across laminar / transition / turbulent. Re ~ 1.5e5 here, so the flow is turbulent and all three ΔP terms (friction, fittings, elevation) matter.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	downstream	F	0.555555555556
stream	feed	F	0.555555555556
kpi	P1	F	0.555555555556
kpi	P1	P_in	500000
kpi	P1	P_out	462942.299189
kpi	P1	dP_elevation	17171.7847864
kpi	P1	dP_fittings	10942.8641074
kpi	P1	dP_friction	8943.05191699
kpi	P1	deltaP	37057.7008108
kpi	P1	density	875.517367621
kpi	P1	frictionFactor	0.0192870918591
kpi	P1	head_loss_m	2.31611521712
kpi	P1	regime	2
kpi	P1	reynolds	149291.31749
...and 2 more reference values in the case's <i>expected</i> .			

pipe02_airwater_twophase

tutorials/steady/hydraulics/pipe02_airwater_twophase



What it does: Gas-liquid two-phase pressure drop: air (N₂) + water in a horizontal 0.05 m x 10 m line, Lockhart-Martinelli. The two-phase path activates automatically because the feed flashes two-phase ($0 < V/F < 1$).

The story: GAS-LIQUID two-phase pressure drop

feed (air N₂ + water, two-phase at 2 bar / 25 °C) → [P1 pipe] D, L, roughness, horizontal → ΔP (result) → downstream (P reduced by the two-phase ΔP)

The two-phase path activates AUTOMATICALLY: the feed flashes two-phase at (T, P) (N₂ is supercritical → all vapour; water → mostly liquid), so $0 < V/F < 1$ and the pipe routes to the gas-liquid models instead of the single-phase liquid path. Model: Lockhart-Martinelli (Chisholm multiplier + Butterworth holdup). Try 'model homogeneous;' in twoPhase to compare.

The Lockhart-Martinelli correlation was developed and VALIDATED on isothermal air-water (and similar gas-liquid) pipe data – Lockhart & Martinelli, Chem. Eng. Prog. 45 (1949) 39 – so air-water is exactly the case it is trusted on.

Basis: N₂ 3 kmol/h (the fast light gas) + water 200 kmol/h (the slow heavy liquid) → a liquid-dominated two-phase flow ($j_G \sim$ a few m/s, $j_L \sim 0.5$ m/s).

Air (N₂) + water flowing through a horizontal 0.05 m × 10 m commercial-steel line. The pipe's two-phase path activates automatically: the feed flashes two-phase at 2 bar / 25 °C (N₂ is supercritical → all vapour; water → mostly liquid), so $0 < V/F < 1$ and the pipe routes to the gas-liquid models instead of the single-phase liquid path.

The two teaching points: 1. The multiplier $\varphi_L^2 = 8.6$: two-phase friction is far above either phase alone — the classic Lockhart-Martinelli amplification, visible in the printout via X and the Chisholm C. 2. Slip → holdup: the homogeneous (no-slip) model would put the liquid fraction at $j_L/(j_L+j_G) \approx 0.12$; L-M gives 0.36 because the liquid moves slower than the gas and accumulates. Switch twoPhase { model homogeneous; } to see the difference for yourself.

The Lockhart-Martinelli correlation (Chem. Eng. Prog. 45 (1949) 39) was developed and validated on isothermal air-water (and similar gas-liquid) pipe data — so air-water is exactly the system it is trusted on. The model structure here (X, φ_L^2 , the tt/vt/tv/vv C-selection, the Butterworth 1975 void fraction) reproduces that framework.

Switch twoPhase { model ...; } and compare — the spread IS a lesson (two-phase ΔP correlations routinely disagree by ~50 %):

Layer 2 (Friedel, BeggsBrill) needs surface tension — supplied here by transport { surfaceTension { model BrockBird; } } (from water's Tc/Pc/Tb). Beggs-Brill also classifies the flow pattern (segregated / intermittent / distributed) and corrects holdup for pipe inclination (here horizontal, so $\psi = 1$).

*- Adiabatic: the inlet vapour fraction is held along the pipe (no flashing / condensation in the line). Flashing flow is a future increment. - Liquid density: $\rho_L = 877$ kg/m³ comes from the Rackett branch, ~12 % low for water (a known anomaly, also in pipe01_water_line). It biases the *absolute* ΔP ; the correlation *structure* is unaffected. - Liquid viscosity via operation { viscosity ...; } (water's value): the flash leaves a trace of dissolved N₂ in the liquid and the Vogel model has no N₂ entry.*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	downstream	F	0.0563888888889
stream	feed	F	0.0563888888889
kpi	P1	F	0.0563888888889
kpi	P1	P_in	200000
kpi	P1	P_out	193263.21239
kpi	P1	Re_Gs	27493.3230716
kpi	P1	Re_Ls	28761.0631705
kpi	P1	X_martinelli	2.67141881475
kpi	P1	dP_elevation	0
kpi	P1	dP_friction	6736.78760984
kpi	P1	deltaP	6736.78760984
kpi	P1	holdup_liquid	0.36001183622
kpi	P1	jG	4.29950133573
kpi	P1	jL	0.583736217791

...and 6 more reference values in the case's *expected*.

membranes

The theory you need. Solution-diffusion (Wijmans & Baker 1995): inside a dense membrane the pressure is uniform, so permeation is concentration-driven. Linearised: $J_w = A_w(\Delta P - \Delta\pi)$ and $J_s = B_s(c_m - c_p)$ – water driven by NET pressure, solute blind to pressure; (A_w, B_s) are MATERIAL constants from `data/standards/membranes/`. Concentration polarisation (film model) makes the wall concentration $c_m = c_b e^{J_w/k}$ exceed the bulk – the flux–polarisation–osmotic feedback is THE nonlinearity of RO/NF, solved per module segment. Expect: rejection improving with pressure ($c_p = B_s c_m / (J_w + B_s)$), polarisation eating driving force at high flux, and charged-membrane (Donnan) effects on multivalent ions that plain solution-diffusion cannot see – that is what the DSPM-DE sibling model is for. *Full derivation: Theory Guide, the membrane chapter.*

membrane01_RO_NaCl_seawater
tutorials/steady/membranes/membrane01_RO_NaCl_seawater



What it does: Spiral-wound RO of 35 g/L NaCl feed at 55 bar (canonical seawater desalination)

The story: the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With $c_{\text{NaCl}} = 35/58.44 = 0.599$ kmol/m³ and c_{water} in seawater $\approx 1025 \text{ kg/m}^3 \div 18 \text{ kg/kmol} = 56.94 \text{ kmol/m}^3$, the mole fractions are $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$ and $z_{\text{water}} = 0.9896$. Feed molar flow $F = 575 \text{ kmol/h}$ ($\approx 10 \text{ m}^3/\text{h}$ of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m² active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~ 30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~ 22 bar). Permeate pressure 1 bar.

Mass transfer: $k_{\text{film}} = 5\text{e-}5 \text{ m/s}$ (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives $\sim 5 \times 10^{??} \text{ m/s}$; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

Expected (with the SW30HR parameters): water recovery $\approx 18 \%$ (single pass) observed NaCl rej. $\approx 99.5 \%$ (typical for SW30HR) average water flux $\approx 20\text{--}25 \text{ LMH}$ feed $\Delta P = 2 \text{ bar}$ (set linearly via $dP_{\text{feed_total}}$)

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	ROunit	R_obs_NaCl	0.997859093278
kpi	ROunit	water_recovery	0.157762204919
kpi	ROunit	J_w_avg	1.32725173967e-05

membrane02_NF_sugar

tutorials/steady/membranes/membrane02_NF_sugar



What it does: NF270 partial rejection of glucose at 10 bar (canonical NF: 10 g/L sugar feed)

The story: loose-NF separation of glucose from water, the canonical NF test that pedagogically distinguishes NF from RO:

RO tight membrane, high pressure, near-100 % rejection NF (here) loose membrane, low pressure, PARTIAL rejection (~50 %)

Feed: 10 g/L glucose (a typical pharma / food-processing feed). $c_{\text{glucose}} = 10 / 180.16 = 0.0555 \text{ kmol/m}^3$
 With pure-water $c_{\text{water}} = 1000/18 = 55.56 \text{ kmol/m}^3$, $z_{\text{glucose}} = 0.0555 / (0.0555 + 55.56) = 9.97\text{e-}4$,
 $z_{\text{water}} = 0.999003$. Feed $F = 200 \text{ kmol/h}$ ($\approx 3.6 \text{ m}^3/\text{h}$ of dilute solution).

Module: NF270-style spiral-wound element, 35 m^2 area, 1 m length. Feed pressure 10 bar (an order of magnitude less than the SW30HR seawater case — the price of loose rejection is that membrane permeability A_w is $10\times$ higher, so the same operating flux is achieved with $10\times$ less driving pressure). Permeate at atmospheric. Same $k_{\text{film}} = 5\text{e-}5 \text{ m/s}$.

Expected (with the NF270 parameters): water recovery $\approx 20\text{-}25 \%$ observed glucose rej. $\approx 50\text{-}65 \%$ (? the SW30HR's 99 %+ !) average water flux $\approx 30\text{-}40 \text{ LMH}$

The partial rejection is the defining NF feature: the membrane is not tight enough to fully reject glucose (MW 180, NF270 nominal MWCO 200 Da), so a fraction of the sugar leaks through with the water — and that leakage is what makes NF useful for fractionation rather than just total rejection.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.05555555555556
stream	permeate	F	0.0149607433177
stream	retentate	F	0.0405951326871
kpi	NFunit	J_w_avg	5.40656892958e-05
kpi	NFunit	R_obs_glucose	0.662821422531
kpi	NFunit	dP_feed	100000
kpi	NFunit	water_recovery	0.267701129953

membrane03_concentration_polarization

tutorials/steady/membranes/membrane03_concentration_polarization



What it does: Concentration polarization study: sweep k_{film} over a SW30HR seawater RO case

The story: Same SW30HR seawater RO case as membrane01_RO_NaCl_seawater, but here we use the SweepDriver (see outerDict) to vary the mass-transfer coefficient k_{film} over a decade and plot how observed rejection and average flux change.

The physics: $c_m / c_b = \exp (J_w / k_{\text{film}})$ (film model)

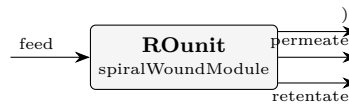
When k_{film} is large (good cross-flow mixing, e.g. high-velocity channel + good spacer), the boundary layer is thin and the concentration at the membrane wall is close to the bulk: $c_m \approx c_b$. $\Delta\pi$ then reflects the bulk concentration, fluxes are high, and observed rejection is close to the intrinsic value.

When k_{film} is small (poor mixing, low velocity, dead zones), the solute concentrates at the wall: $c_m \gg c_b$. $\Delta\pi$ rises sharply, J_w collapses, and a larger fraction of the wall solute leaks through, dropping observed rejection.

This sweep illustrates why membrane-module designers obsess over spacer geometry and feed velocity — they are not adjusting the membrane itself, they are adjusting k_{film} .

membrane04_spacer_schock_miquel

tutorials/steady/membranes/membrane04_spacer_schock_miquel



What it does: Seawater RO with k_{film} computed from spacer-channel hydraulics (Schock-Miquel Sherwood); k_{film} falls ~ 6.6 to $\sim 5.6 \times 10^{-5}$ m/s along the channel

The story: the same seawater RO as membrane01, but k_{film} is COMPUTED from the spacer-channel hydraulics (Schock-Miquel Sherwood correlation) instead of being a fixed number. The ‘massTransfer’ block makes k_{film} a selectable model: $Sh = 0.065 Re^{0.875} Sc^{0.25}$ with Re from the local crossflow velocity $u = Q_b(z)/A_{\text{channel}}$ — so k_{film} FALLS along the channel as the retentate flow drops (the profile column ‘ k_{film} ’ shows this). At the inlet $u \sim 0.26$ m/s gives $k_{\text{film}} \sim 7 \times 10^{-5}$ m/s, a touch above membrane01’s fixed 5×10^{-5} (better mixing $\rightarrow$ less polarisation $\rightarrow$ slightly higher recovery/rejection). Set ‘model constant; value ...;’ to recover the membrane01 behaviour.

membrane01_RO_NaCl_seawater – the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With $c_{\text{NaCl}} = 35/58.44 = 0.599$ kmol/m³ and c_{water} in seawater ≈ 1025 kg/m³ $\div 18$ kg/kmol = 56.94 kmol/m³, the mole fractions are $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$ and $z_{\text{water}} = 0.9896$. Feed molar flow $F = 575$ kmol/h (≈ 10 m³/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m² active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~ 30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~ 22 bar). Permeate pressure 1 bar.

Mass transfer: $k_{\text{film}} = 5 \times 10^{-5}$ m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives $\sim 5 \times 10^{??}$ m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

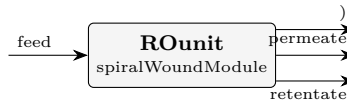
Expected (with the SW30HR parameters): water recovery ≈ 18 % (single pass) observed NaCl rej. ≈ 99.5 % (typical for SW30HR) average water flux ≈ 20 –25 LMH feed $\Delta P = 2$ bar (set linearly via $dP_{\text{feed_total}}$)

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0286373914761
stream	retentate	F	0.131086212764
kpi	ROunit	J_w_avg	1.47407124897e-05
kpi	ROunit	R_obs_NaCl	0.99810213206
kpi	ROunit	dP_feed	26065.9064084
kpi	ROunit	water_recovery	0.175213731874

membrane05_train

tutorials/steady/membranes/membrane05_train



What it does: Multi-element membrane train: 3 SW30HR elements in series (seawater RO)

The story: a MULTI-ELEMENT TRAIN: 3 SW30HR spiral-wound elements in series, seawater RO (the same feed + per-element geometry as membrane04, which is the single element). ‘elements 3;’ chains them: the retentate of element 1 is the feed of element 2, and so on; the permeate is collected over the whole train. Both the spacer k_{film} and the spacer pressure drop (Schock-Miquel) are computed from the local hydraulics, so they fall element by element as the flow drops.

The pedagogical point is the per-element DEGRADATION reported in the KPIs (recovery_elemK , c_ret_elemK): element 1 sees fresh feed (high P , low c) and recovers the most; the last element sees concentrated, lower-pressure brine (P dropped by friction + the inter-element connector loss) and recovers far less. The train recovery is the cumulative sum — higher than a single element, but with diminishing returns down the train.

Feed: 35 g/L NaCl seawater, $z_{\text{NaCl}} = 0.01040$, $F = 575 \text{ kmol/h}$ ($\sim 10 \text{ m}^3/\text{h}$), 55 bar, 298.15 K — the same feed as membrane01/04. Each of the 3 elements is $35 \text{ m}^2 / 1 \text{ m}$, SW30HR, permeate at 1 bar, with a 0.2 bar inter-element connector loss.

What this run produces (golden master): train water recovery 35.3 % (3 elements, cumulative) observed NaCl rejection 99.68 % train-average water flux 34.85 LMH ($9.68\text{e-}6 \text{ m/s}$) total feed-side $\Delta P \sim 0.99 \text{ bar}$ (Schock-Miquel friction + 2 connector losses)

The per-element DEGRADATION (the teaching point) reads off the KPIs: recovery_elem1 17.2 % c_ret_elem1 0.697 (fresh feed) recovery_elem2 28.4 % c_ret_elem2 0.805 (cumulative) recovery_elem3 35.3 % c_ret_elem3 0.891 (cumulative) The retentate concentration climbs element to element while each successive element adds less recovery: a single SW30HR element at this feed recovers $\sim 17 \%$, the second adds $\sim 11 \%$, the third only $\sim 7 \%$ — the diminishing return of stacking elements in series at fixed feed.

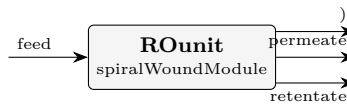
NOTE (known issue): this train case currently has an open mass-balance discrepancy under investigation; it is run-only in the regression and the KPIs above are the present golden values, not a validated material balance.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0590874153104
stream	retentate	F	0.100637449271
kpi	ROunit	$J_{\text{w_avg}}$	$1.01384186683\text{e-}05$
kpi	ROunit	$R_{\text{obs_NaCl}}$	0.99694467035
kpi	ROunit	c_ret_elem1	0.68349184216
kpi	ROunit	c_ret_elem2	0.793343136295
kpi	ROunit	c_ret_elem3	0.881961644632
kpi	ROunit	dP_feed	101143.55107
kpi	ROunit	elements	3
kpi	ROunit	recovery_elem1	0.175213731874
kpi	ROunit	recovery_elem2	0.289843097471
kpi	ROunit	recovery_elem3	0.361527335552
kpi	ROunit	water_recovery	0.361527335552

membrane06_pitzer

tutorials/steady/membranes/membrane06_pitzer



What it does: Seawater RO with the Pitzer osmotic model (vs membrane01 van't Hoff)

The story: the canonical seawater RO (same as membrane01) but with the PITZER osmotic-pressure model instead of van't Hoff. van't Hoff ($\phi = 1$) over-estimates the osmotic pressure of a real brine: for NaCl the osmotic coefficient is $\phi \sim 0.92$ at seawater concentration (and dips further before rising at high molality). The Pitzer ion-interaction model captures $\phi(I)$, so the osmotic back-pressure is $\sim 7\text{--}8\%$ LOWER $\rightarrow$ a slightly HIGHER water flux and recovery than membrane01 (15.55 %). The 'osmotic' block makes it a selectable model with the NaCl coefficients (calibratable); remove the block to recover van't Hoff.

membrane01_RO_NaCl_seawater – the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With $c_{\text{NaCl}} = 35/58.44 = 0.599$ kmol/m³ and c_{water} in seawater ≈ 1025 kg/m³ $\div 18$ kg/kmol = 56.94 kmol/m³, the mole fractions are $z_{\text{NaCl}} = 0.599 / (0.599 + 56.94) = 0.01040$ and $z_{\text{water}} = 0.9896$. Feed molar flow $F = 575$ kmol/h (≈ 10 m³/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m² active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~ 30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~ 22 bar). Permeate pressure 1 bar.

Mass transfer: $k_{\text{film}} = 5\text{e-}5$ m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives $\sim 5 \times 10^{??}$ m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

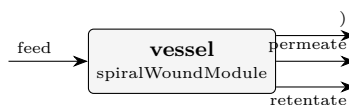
Expected (with the SW30HR parameters): water recovery $\approx 18\%$ (single pass) observed NaCl rej. $\approx 99.5\%$ (typical for SW30HR) average water flux $\approx 20\text{--}25$ LMH feed $\Delta P = 2$ bar (set linearly via $dP_{\text{feed_total}}$)

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.159722222222
stream	permeate	F	0.0284588599363
stream	retentate	F	0.13126472432
kpi	ROunit	J_w_avg	1.46488544181e-05
kpi	ROunit	R_obs_NaCl	0.997986472373
kpi	ROunit	dP_feed	200000
kpi	ROunit	water_recovery	0.174121871794

membrane07_scaling_si

tutorials/steady/membranes/membrane07_scaling_si



What it does: Brackish RO, 6 x 8-inch modules: per-module scaling audit (SI at wall vs bulk)

The story: WHERE along the vessel does scaling start?

Brackish RO on the SAME representative groundwater analysis as tutorials/props/electrolyte/scaling_ro_brackish (~1500 mg/L TDS):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026
Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003

Here the analysis is a PROCESS STREAM: the ions are flowsheet components (case-local constant/components/, MW from ions.dat) and the mg/L totals become mole fractions against water ($x_i = m_i / 55.55$; $m_i = (\text{mg/L})/\text{MW}$ at $\rho \sim 1 \text{ kg/L}$):

Ca 3.7731e-5 Mg 1.9998e-5 Na 2.8425e-4 K 5.5253e-6 Cl 2.5897e-4 SO4 4.6852e-5 HCO3 5.3989e-5 water = balance

HARDWARE BY NOMINAL DIAMETER (no 'area'): one pressure vessel with 6 x 8-inch standard 40-inch elements – 'moduleDiameter 8 in; nModules 6;'. The per-module area (37.2 m², 400 ft² class) comes from the engine's nominal table and the derivation is announced: "6 x 37.2 m² (8in/40in standard element) = 223.2 m²".

THE SCALING AUDIT ('scaling {}'): at each module exit the local BULK concentrate AND the membrane-WALL composition (c_m from the module's own film-model polarisation) are speciated (Davies) and the calcite / gypsum saturation indices recorded. The WALL crosses SI=0 modules before the bulk does – concentration polarisation, normally an invisible flux penalty, is seen here as THE scaling driver.

pH 7.0 GIVEN, not solved: the raw well water is already calcite- supersaturated (scaling_ro_brackish solves the raw feed to pH 7.72, SI_calcite = +0.24 – it would scale in the PIPE, let alone the vessel). Real brackish plants dose acid for exactly that reason; here the feed is dosed and CONTROLLED at pH 7.0, and the audit asks the honest follow-up: with the dosing in place, where along the vessel does the concentrate STILL reach calcite saturation? ('pH solve;' is also accepted – it re-solves electroneutrality per module, the undosed scenario.)

Expected story (KPIs locked by the golden master; recovery 55 %): calcite WALL crosses SI=0 at module 3 of 6 ($r_{cum} = 0.31$); calcite BULK is still undersaturated there (-0.27) and only crosses in module 6 (+0.03) – the wall leads the bulk by three full modules: polarisation IS the scaling driver; gypsum stays undersaturated everywhere (max wall SI = -0.73).

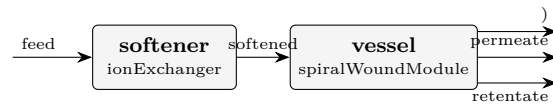
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.107212342596
stream	retentate	F	0.0872324772828
kpi	vessel	J_w_avg	8.653382114e-06
kpi	vessel	R_obs_Ca	0.999474633761
kpi	vessel	R_obs_Cl	0.994770268575
kpi	vessel	R_obs_HCO3	0.996072731843
kpi	vessel	R_obs_K	0.993471097929
kpi	vessel	R_obs_Mg	0.999605925236
kpi	vessel	R_obs_Na	0.994770268575
kpi	vessel	R_obs_SO4	0.999737250092
kpi	vessel	SI0_module_calcite	3
kpi	vessel	SI0_module_gypsum	-1
kpi	vessel	SI0_recovery_calcite	0.305177946028

...and 18 more reference values in the case's *expected*.

membrane08_softened_scaling

tutorials/steady/membranes/membrane08_softened_scaling



What it does: Softener + brackish RO: ion-exchange removes the scaling driver upstream of the vessel

The story: SOFTEN the feed, and the scaling is GONE.

THE CONTRAST (read membrane07_scaling_si first – the BEFORE):

membrane07 the SAME brackish groundwater (~1500 mg/L TDS, Ca 84 / Mg 27 mg/L) goes straight into a 6 x 8-inch brackish-RO vessel. As the concentrate builds along the train the membrane-WALL composition crosses calcite SI = 0 at MODULE 3 of 6 (r_cum = 0.31) – concentration polarisation is THE scaling driver; the vessel would foul.

membrane08 here a cation-exchange SOFTENER ('ionExchanger', SAC resin in the Na form, Gaines-Thomas) sits UPSTREAM of the SAME vessel. Ca and Mg drop ~99% onto the resin (Na rises eq-for-eq – the salt penalty), so the water entering the vessel is essentially hardness-free. With Ca gone the calcite IAP collapses and the per-module WALL SI_calcite stays NEGATIVE through ALL 6 modules – the scaling driver is removed at source.

ZERO change to the membrane. The spiralWoundModule is byte-for-byte the membrane07 vessel (module-Diameter 8 in; nModules 6; the same scaling{} audit); it simply receives a BETTER feed. Removing the hardness, not chasing the supersaturation in the vessel, is the cure – the engineering lesson.

HONESTY (printed, non-suppressible): the softener returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass, NOT a bed in service (breakthrough / regeneration are TRANSIENT and not modelled). See the banner on every run.

Data: self-contained case-local constant/electrolyte/ snapshot (USGS PHREEQC, public domain: speciation.dat + minerals.dat + ions.dat + exchange.dat) + the SAC_Na resin nameplate + the SW30HR membrane – the same snapshots as membrane07 / softener_brackish.

Expected story (golden master): softener removes ~99.99% hardness (Ca/Mg leakage ~ 1e-2 mg/L, Na added ~ 6.7 meq/L at the given pH 7.0 closure – the eq-for-eq salt penalty for the full hardness load); the vessel's maxSI_calcite_wall is NEGATIVE (-3.68: no SI = 0 crossing in any module, SI0_module_calcite = -1), vs +0.27 / crossing at module 3 in membrane07.

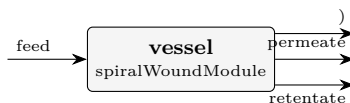
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.1054509275
stream	retentate	F	0.0890050837601
stream	softened	F	0.194455667987
kpi	softener	CEC_utilised_pct	100
kpi	softener	Ca_leakage_mgL	0.0098339472306
kpi	softener	I	0.0243975946959
kpi	softener	Mg_leakage_mgL	0.00549119043267
kpi	softener	Na_added_meqL	6.71876194262
kpi	softener	Na_added_mgL	154.464337061
kpi	softener	bedVolume_m3	1.5
kpi	softener	dH_exchange_Jkgw	21.9905304514
kpi	softener	hardness_in_mgL_CaCO3	320.96282036
kpi	softener	hardness_out_mgL_CaCO3	0.0471722805225

... and 34 more reference values in the case's expected.

membrane09_index_vs_rigorous

tutorials/steady/membranes/membrane09_index_vs_rigorous



What it does: Seawater RO at high recovery: industry calcite indices (LSI / Stiff-Davis / Ryznar) vs the rigorous activity SI at the membrane WALL

The story: the bag-of-indices industry shortcut is NOT safe at the membrane wall in a high-recovery concentrate.

THE LESSON (activity vs concentration – the whole point of the case):

The industry calcite-scaling indices read every plant data-sheet ships – LSI Langelier saturation index (Langelier 1936) Stiff-Davis brine-corrected LSI (Stiff & Davis 1952) RSI Ryznar stability index (Ryznar 1944) are built from ION CONCENTRATIONS plus an EMPIRICAL ionic-strength correction (a published pK-shift / chart-fit K). Choupo's rigorous SI_calcite instead forms real ION ACTIVITIES $a_i = \gamma_i m_i$ from the Davies model:

$LSI = pH - \log K_{cc}(T) - p[Ca^{2+}] - p[HCO_3^-]$ (CONCENTRATIONS) $SI = pH - \log K_{cc}(T) - p\{a_{Ca^{2+}}\} - p\{a_{HCO_3^-}\}$ (ACTIVITIES)

They share the SAME thermodynamic anchor $\log K_{cc}(T)$ (the HCO_3 -basis calcite $\log K$, $= pK_2 - pK_{sp}$) – so the ONLY difference is concentration vs activity. At LOW ionic strength $\gamma \rightarrow 1$ and $LSI == SI$ (validation); as the activity coefficients fall away from 1 the two DIVERGE.

WHERE IT BITES – THE WALL. Concentration polarisation piles solute against the membrane wall: the LOCAL ionic strength at the wall runs far above the bulk. There the empirical index over-predicts calcite saturation (it cannot see the gamma reduction), while the rigorous activity SI does not. Reading the index off the bulk analysis – as the industry shortcut does – therefore MISLEADS at exactly the place scaling actually starts.

THE CASE: standard surface seawater ($S = 35$, the major-ion subset), one pressure vessel of 7 x 8-inch SW elements, pushed to high recovery. Acid is dosed and CONTROLLED at pH 7.0 (real SWRO practice). At each module exit the BULK concentrate AND the membrane-WALL composition are speciated (Davies) and BOTH the rigorous SI_calcite AND the empirical LSI / Stiff-Davis / Ryznar are recorded, bulk + wall.

THE STORY (locked by the golden master): - This is already a SEAWATER concentrate (bulk $I \sim 0.68$ mol/kg at module 1, far above the $I \rightarrow 0$ regime where $LSI == SI$), so the empirical index OVER-PREDICTS calcite saturation at EVERY module, including the first: at module 1 $LSI(wall) = +0.98$ vs rigorous $SI_calcite(wall) = +0.20$ (gap ~ 0.78); $LSI(bulk) = +0.56$ vs $SI(bulk) = -0.28$ (gap ~ 0.84). At this ionic strength there is no point where index and activity-SI agree to hundredths – to SEE that limit you must drop to a brackish / low-I feed (the index family is only validated there). - the gap widens further into the vessel: at the deepest module $LSI(wall)$ over-predicts the rigorous $SI_calcite(wall)$ by ~ 0.78 – the gap ($LSI - SI$) printed in the audit footer is the headline number. Stiff-Davis pulls PART of the way back toward the rigorous SI (its empirical brine K), but not all the way – it is still a chart fit, not the real gammas. - Ryznar ($RSI = 2$ pHs - pH) reads the same saturation pHs as LSI on the practical 0..14 stability scale.

Engineering take-away (Choupo's transparency differentiator): the concentration-based index family is convenient but UNSAFE for high-recovery membrane scaling – the rigorous activity SI, visible term-by-term, is the honest call.

Sibling cases: membrane07_scaling_si (bulk vs wall SI, brackish); tutorials/props/electrolyte/pitzer_vs_davies_ro_concent (Davies vs Pitzer activity fork). Index kernel + primary sources: src/thermo/electrolyte/ScalingIndices.{H,cpp}.

PRIMARY SOURCES: Langelier, W.F. (1936) J. AWWA 28(10) 1500-1521. Stiff, H.A. & Davis, L.E. (1952) Trans. AIME 195, 213-216. Ryznar, J.W. (1944) J. AWWA 36(4) 472-486.

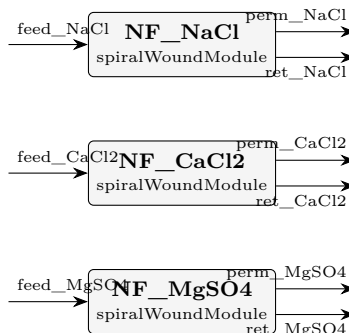
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.25
stream	permeate	F	0.0675548243983
stream	retentate	F	0.182446484545
kpi	vessel	J_w_avg	4.67400706783e-06
kpi	vessel	LSI_calcite_bulk_end	0.76535387412
kpi	vessel	LSI_calcite_wall_end	0.985330125244
kpi	vessel	RSI_calcite_wall_end	5.02933974951
kpi	vessel	R_obs_Ca	0.999255215369
kpi	vessel	R_obs_Cl	0.992602201838
kpi	vessel	R_obs_HCO3	0.994441275819
kpi	vessel	R_obs_K	0.990769979906
kpi	vessel	R_obs_Mg	0.999441306528
kpi	vessel	R_obs_Na	0.992602201838
kpi	vessel	R_obs_SO4	0.99962746766

...and 25 more reference values in the case's *expected*.

membrane10_dspmde_divalent

tutorials/steady/membranes/membrane10_dspmde_divalent



What it does: DSPM-DE NF: divalent (MgSO_4) > mixed (CaCl_2) > monovalent (NaCl) rejection at 10 bar

The story: the charge-aware NF signature, glass-box.

THREE parallel NF270 elements, one per salt, all on the SAME membrane (NF270_dspmde, with the DSPM-DE charged-pore tier) at the SAME 10 bar / 5 mM feed. The point is the REJECTION ORDERING that the Donnan-Steric Pore Model with Dielectric Exclusion produces from first principles:

$R_{\text{obs}}(\text{MgSO}_4) > R_{\text{obs}}(\text{CaCl}_2) > R_{\text{obs}}(\text{NaCl})$ (2:2 divalent) (2:1 mixed) (1:1 monovalent)

WHY (Bowen-Welfoot 2002): the membrane carries a NEGATIVE fixed charge ($X_d = -40 \text{ mol/m}^3$). The CO-ION (the anion, here Cl^- / SO_4^{2-}) is excluded by Donnan; a DIVALENT co-ion (SO_4^{2-}) is excluded far more strongly than a monovalent one (z^2 in the Donnan + Born partition). So MgSO_4 – whose co-ion is divalent – is rejected most; NaCl – whose co-ion is the monovalent Cl^- – least. This is the classic NF fingerprint that plain solution-diffusion (one B_s per salt) cannot predict mechanistically.

All data is case-local (self-contained): the salts carry an electrolyte{} ion decomposition, the ions carry the radius/D0 transport tier, and the membrane carries the poreModel{} block.

‘transport DSPM-DE;’ selects the charged-pore law; drop it (or set ‘transport solutionDiffusion;’) to compare against the baseline.

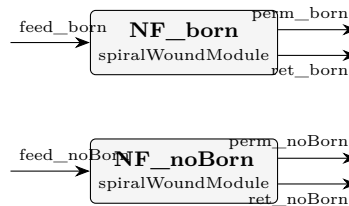
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed_CaCl2	F	0.05555555555556
stream	feed_MgSO4	F	0.05555555555556
stream	feed_NaCl	F	0.05555555555556
stream	perm_CaCl2	F	0.0191775929422
stream	perm_MgSO4	F	0.020040469745
stream	perm_NaCl	F	0.0208297354228
stream	ret_CaCl2	F	0.0363780274704
stream	ret_MgSO4	F	0.0355151987788
stream	ret_NaCl	F	0.034725847724
kpi	NF_CaCl2	J_w_avg	6.91063443929e-05
kpi	NF_CaCl2	R_obs_CaCl2	0.704588151782
kpi	NF_CaCl2	R_obs_MgSO4	1
kpi	NF_CaCl2	R_obs_NaCl	1
kpi	NF_CaCl2	dP_feed	100000

... and 13 more reference values in the case's expected.

membrane11_dspmde_born_toggle

tutorials/steady/membranes/membrane11_dspmde_born_toggle



What it does: DSPM-DE NF: Born dielectric-exclusion toggle (same NaCl feed, poreDielectric on vs off)

The story: SEE the dielectric-exclusion (Born) term.

The SAME NaCl feed (50 mM, 10 bar) through TWO NF270 elements that differ in ONE thing only:

NF_born : membrane NF270_dspmde – poreDielectric 35 (Born ON) NF_noBorn : membrane NF270_dspmde_noBorn – poreDielectric absent (Born OFF)

Everything else – pore radius, porosity/thickness, fixed charge X_d , feed, pressure, area, k_{film} – is IDENTICAL. The ONLY physics that changes is the Born dielectric-exclusion term

$$dW_i = (z_i^2 e^2 / 8 \pi \epsilon_0 r_i) (1/\epsilon_p - 1/\epsilon_b)$$

(Bowen & Welfoot 2002, Eq. 16). With $\epsilon_p < \epsilon_b$ an ion pays an extra solvation-energy penalty to enter the low-dielectric pore, so it is excluded MORE strongly. The student reads the Born contribution straight off the two $R_{\text{obs}}(\text{NaCl})$ KPIs: $R_{\text{obs}}(\text{NF_born}) > R_{\text{obs}}(\text{NF_noBorn})$.

This is the pedagogical core of DSPM-"DE": the DE (dielectric exclusion) is what distinguishes the 2002 Bowen-Welfoot model from the 1997 steric-Donnan DSPM. Toggling poreDielectric is a single, visible switch.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed_born	F	0.05555555555556
stream	feed_noBorn	F	0.05555555555556
stream	perm_born	F	0.0103299082671
stream	perm_noBorn	F	0.0112254914132
stream	ret_born	F	0.0452257284146
stream	ret_noBorn	F	0.0443301389741
kpi	NF_born	J_w_avg	3.7229087272e-05
kpi	NF_born	R_obs_NaCl	0.861027895843
kpi	NF_born	dP_feed	100000
kpi	NF_born	water_recovery	0.185615581216
kpi	NF_noBorn	J_w_avg	4.0473218085e-05
kpi	NF_noBorn	R_obs_NaCl	0.65954044001
kpi	NF_noBorn	dP_feed	100000
kpi	NF_noBorn	water_recovery	0.201790063872

optimisation

The theory you need. The outer driver wraps the whole simulation in $f(\mathbf{x})$: sweeps map it, Nelder–Mead descends it, parameter estimation fits it to data (Levenberg–Marquardt, with identifiability diagnostics – χ_{red}^2 , correlation, conditioning). Expect every function evaluation to be a full, honest flowsheet pass.

designSpec01_triple_equal_areas

tutorials/steady/optimisation/designSpec01_triple_equal_areas



What it does: DesignSpec: equal-area triple-effect sugar evaporator (manipulate \$A so L3 = 4500 kg/h)

The story: Same physics as evaporator02_triple_effect_sugar (MCFT Ex. 10.4 — three cocurrent effects on a 22 500 kg/h, 5 wt% sucrose feed). The twist: instead of HARD-CODING each effect’s area at 100 m², we declare A as a CASE-LEVEL VARIABLE and use the outer DesignSpec driver to find the equal-areas value that delivers L_3 ~ 4500 kg/h (25 wt% sucrose), the textbook design target.

This tutorial showcases two new pieces of v0.30 machinery:

- \$references in the dict (‘variables { ... }’ block + ‘\$A’ syntax inside any scalar). The same A is shared by the three effects without copy-paste — which makes it trivially a single degree of freedom for the outer solver.
- The ‘designSpec’ outer driver: a Newton-ND wrap on Flowsheet::solve, manipulating the case’s \$variables to satisfy named targets. Here it is a 1-D Newton in \$A on the residual L3.F - 0.001249 kmol/s. Convergence in 4-5 evaluations.

F=22 500 kg/h L1 L2 L3 (target ~ 4500 kg/h) x=0.05 →[1] → [2] → [3] → ^ ^ ^ | V1 V2 V3
→ condenser Steam (drives 2) (drives 3) 200 kPa P1 = 80 P2 = 45 P3 = 15 kPa

Constraints we leave fixed: - P_i pressures (textbook cascade) - U_i overall HTC’s per effect (textbook values) - Feed and steam conditions - Equal areas: enforced trivially by the single \$A variable

Read the run log to see the Newton converge in <10 evaluations. At the optimum, the case ends with the converged final state on stdout, exactly like a hand-tuned evaporator02 run — but the area was DERIVED, not specified.

Initial guess A_0 = 100 m² (the educated guess from evaporator02); the solver should home in on ~ 110-120 m² to hit the target.

Target tolerance: ±5 kg/h on L_3, encoded via tol = 1.4e-6 kmol/s (= 5 kg/h ? 3600 s/h ? 18 kg/kmol).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	L1	F	0.242853688633
stream	L2	F	0.150128936321
stream	L3	F	0.0529527028239
stream	V1	F	0.087645716964
stream	V2	F	0.0927247523111
stream	V3	F	0.0971762334975
stream	cond1	F	0.138858068999
stream	cond2	F	0.087645716964
stream	cond3	F	0.0927247523111
stream	feed	F	0.330499405597
stream	utilitySteam_200kPa	F	0.138858068999
kpi	effect1	A	108.007004701
kpi	effect1	A_m2	108.007004701
kpi	effect1	BPE	0.107244152342
... and 81 more reference values in the case’s expected.			

fitNRTL01_ethanol_water

tutorials/steady/optimisation/fitNRTL01_ethanol_water

fit_NRTL_ethanol_water
 fitParameters

What it does: Fit NRTL ethanol/water pair against 1 atm bubble-T data (Levenberg-Marquardt)

The story: fitNRTL01 – Levenberg-Marquardt regression of the ethanol-water NRTL pair against 1 atm bubble-T data, on the CANONICAL fit engine (fitParameters, choupProps). This case previously ran the condemned FitBinaryPair outer driver in choupSolve, which abused a dummy flowsheet and steered the feed composition through the legacy streams{} block per data row – the design forum (#53) ordered it moved here, WITH a golden: its χ^2 once degraded 100x with nothing going red, because a fit without numerical verification does not belong in a green corpus.

Data: transcribed from constant/experiments/ethanol-water-101kPa (the raw- data home; DECHEMA / Gmehling-Onken Vol. I Part 1, pedagogical subset). Starting point: the catalogue NRTL values in constant/propertyDict.

The parameter-estimation workflow: regress the four NRTL binary interaction parameters of the ethanol-water pair (a_{ij} , b_{ij} , a_{ji} , b_{ji} ; α held fixed) against experimental bubble-temperature data at 1.01325 bar, using hand-rolled Levenberg-Marquardt inside the fitParameters engine (successor of the retired fitBinaryPair driver).

** See an outer Marquardt loop (3-up/5-down trust region on λ) wrapped around an inner Newton-1D solving $\sum K x = 1$ for each (x_i, T) pair. * Watch λ oscillate as the algorithm switches between Gauss-Newton (λ small) and gradient-descent (λ large) regimes. * Watch some LM steps get rejected — χ^2 gets worse, the step is reversed, λ is multiplied by 5, retry. The whole "trust region" idea visible in one column of the log.*

Note how minimal the forward case is: one stream, one bubbleT unit. The outer driver is what does the work — iterating over the dataset internally and looping LM over the four parameters.

Four keywords (plus optional maxIter, tolerance, fdStep, lambda0). The driver:

1. locates the pair in thermoPackage.activityModel.pairs by name, 2. reads the dataset (a two-column flat list of x_i T_i), 3. finds the bubbleT unit's input stream from the flowsheetDict, 4. for each LM step: - 8 forward evals to build a central-difference Jacobian over 4 params $\times$ N data points, - solve the 4×4 normal equations with Marquardt damping, - try the step; accept if χ^2 decreased, otherwise raise λ and retry.

After the run header you get (the iteration trace below is illustrative – it shows the SHAPE of an LM run, accept/reject and λ adaptation; the actual run converges to $\chi^2 = 1.0426$, RMS = 0.3079 K with fitted $a_{ij} = 9.328$, $b_{ij} = -3609.18$):

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	fit_NRTL_ethanol_water	chi2	1.04279
diag	fit_NRTL_ethanol_water	chi2_reduced	0.14897
diag	fit_NRTL_ethanol_water	cond_JtJ	3.55369e+11
diag	fit_NRTL_ethanol_water	converged	1
diag	fit_NRTL_ethanol_water	corr_computed	1
diag	fit_NRTL_ethanol_water	dof	7
diag	fit_NRTL_ethanol_water	identifiable	0
diag	fit_NRTL_ethanol_water	invertible	1
diag	fit_NRTL_ethanol_water	iter	24
diag	fit_NRTL_ethanol_water	max_abs_corr	0.999967
diag	fit_NRTL_ethanol_water	max_abs_resid	0.625729
diag	fit_NRTL_ethanol_water	n_data	11
diag	fit_NRTL_ethanol_water	n_params	4
diag	fit_NRTL_ethanol_water	params_at_bound	0

... and 4 more reference values in the case's expected.

optim01_column_reflux

tutorials/steady/optimisation/optim01_column_reflux



What it does: Optimise reflux ratio + feedStage of benzene/toluene column to minimise reboiler vapour V_strip

The story: Benzene/toluene distillation, used by the optimisation driver. Same column as sensitivity01_column_reflux; we add looser tolerances on the outer loop so the simplex can probe odd feedStages without blowing up.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0416666666667
kpi	column01	L_strip	0.0694444444444
kpi	column01	R	3
kpi	column01	T_bottom	383.299688974
kpi	column01	T_top	353.770171691
kpi	column01	V_rect	0.0555555555556
kpi	column01	V_strip	0.0555555555556
kpi	column01	feedStage	9
kpi	column01	nStages	15
...and 2 more reference values in the case's expected.			

optim02_process_cost

tutorials/steady/optimisation/optim02_process_cost



What it does: Minimise total module Guthrie cost of reactor + heater by varying reactor V_R and heater Q

The story: Same reactor → heater → separator flowsheet as process02_with_design. The optimisation driver varies two design variables:

reactor V_R in $[0.002, 0.010]$ m³ heater Q in $[0, 500]$ W (credo-pure: hardware, not outlet T)

In v0.34 the heater is credo-pure: it specifies Q (the absolute thermal power, hardware), not T_{out} (which would be a DesignSpec disguised as a unit-op parameter). The Nelder-Mead simplex therefore varies the heater's Q directly; the heater's T_{out} becomes a KPI the student can plot, not a control variable.

Initial guesses are placeholders – overwritten by Nelder-Mead.

A small reactor → heater → separator process. The OptimizationDriver (Nelder-Mead) tunes two design variables to minimise the total Guthrie module cost of the two costed units (reactor + heater):

The heater is credo-pure: it specifies Q (the thermal power, hardware), not an outlet T . T_{out} is a reported KPI, not a control variable.

The optimum is not at the lower corner. Nelder-Mead converges (8 moves, simplexInit 0.15) to:

Increasing Q monotonically lowers the objective: the simplex walks the cost from $\sim 2.0e6$ EUR at $Q \approx 170$ W down to $\sim 9.35e5$ EUR at $Q = 500$ W.

The dominant cost is the heat exchanger (~ 916 k of the ~ 935 k total). Its area follows $A = Q / (U \cdot \text{LMTD}) = Q / 6000$ m², so $Q = 500$ W gives only $A \approx 0.083$ m². The Turton shell-and-tube correlation used by the Guthrie model is valid for $A \in [10, 1000]$ m²; our exchanger is two orders of magnitude below S_{min} . In that extrapolated tail the correlation

*turns *upward* as A shrinks (the quadratic term dominates for $\log_{10} A < 0$), so a larger area is cheaper. The optimiser correctly exploits this: pushing Q to its upper bound grows A toward the correlation's valid range and slides down the cost curve. The reactor volume settles at an interior 0.00807 m³ — large enough for adequate conversion ($X \approx 0.64$), small enough that its vessel cost (~ 19 k) stays a rounding error next to the exchanger.*

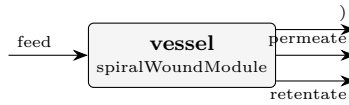
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	hotStream	F	0.0002777777777778
stream	liqProd	F	7.65621619403e-05
stream	reactorOut	F	0.0002777777777778
stream	vapProd	F	0.000201215615838
kpi	heater	F	0.0002777777777778
kpi	heater	P	101325
kpi	heater	Q	500
kpi	heater	Q_kW	0.5
kpi	heater	Q_per_mol	1800
kpi	heater	T_in	350
kpi	heater	T_out	364.85579492
kpi	heater	dT	14.8557949202
kpi	reactor	Da_kTau	1.79046552367

... and 15 more reference values in the case's expected.

optim04_membrane_recovery_scaling

tutorials/steady/optimisation/optim04_membrane_recovery_scaling



What it does: Brackish RO: maximise water recovery up to the scaling limit (SQP, scaling constraints active at the optimum)

The story: HOW HARD can we push recovery before the membrane scales?

The flagship CONSTRAINED-OPTIMISATION case. Same brackish-RO vessel and scaling physics as tutorials/steady/membranes/membrane07_scaling_si (6 x 8-inch elements, the ~1500 mg/L groundwater analysis, the bulk-vs-wall saturation-index audit), but now we ASK the engine the design question the audit only diagnosed:

maximise water_recovery varying feed pressure and the (continuous) module count subject to NO mineral scaling at the membrane wall: maxSI_gypsum_wall <= 0 maxSI_calcite_wall <= 0 (rigorous Davies saturation index – smoother than the Stiff-Davis/Langelier chart fit)

Recovery and wall saturation pull AGAINST each other: every extra % recovery concentrates the retentate and drives the WALL composition (bulk x polarisation) further into supersaturation. The optimum therefore sits ON the scaling limit – at least one of the two constraints is ACTIVE, with a strictly positive Lagrange multiplier (the shadow price: how much more recovery one unit of scaling head-room would buy). That is exactly what the SQP trace prints.

GOLDEN-CHECKED regression: the SQP now converges to a CERTIFIED KKT optimum ($\|grad L\|_{inf} \sim 3e-8$, all feasibility 0, complementarity $\sim 1e-8$, calcite-wall constraint ACTIVE with $\lambda \sim 1.14 > 0$). Because this is a certified stationary point – not a guessed answer – its KPIs are pinned in ‘expected’, which ALSO guards against a regression to the singular-QP failure a readiness review caught earlier (a "QP subproblem failed: KKT system singular – LICQ failure" at iteration 1 that left the design infeasible). The KPIs still inherit the polarisation + Davies model assumptions; the golden locks the answer THIS model gives at the certified optimum, not an independent ground truth. (If the engine ever fails to converge again, the optimum block prints an explicit "NO KKT certificate ... NOT a certified optimum" block – never the misleading all-zeros certificate it used to forge on the failure path.)

Two design variables, written through \$references so the SQP can drive them and the trace shows their canonical-SI values.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.194444444444
stream	permeate	F	0.0800476772619
stream	retentate	F	0.11439691471
kpi	vessel	J_w_avg	6.05900819134e-06
kpi	vessel	LSI_calcite_bulk_end	0.376636925312
kpi	vessel	LSI_calcite_wall_end	0.568802059189
kpi	vessel	RSI_calcite_wall_end	5.86239588162
kpi	vessel	R_obs_Ca	0.999415347979
kpi	vessel	R_obs_Cl	0.994183822598
kpi	vessel	R_obs_HCO3	0.995631569104
kpi	vessel	R_obs_K	0.99274024463
kpi	vessel	R_obs_Mg	0.999561447433
kpi	vessel	R_obs_Na	0.994183822598
kpi	vessel	R_obs_SO4	0.999707589243

... and 25 more reference values in the case's *expected*.

optim05_reactor_npv

tutorials/steady/optimisation/optim05_reactor_npv



What it does: Esterification CSTR sized & operated for MAX NPV by hand-rolled SQP, subject to a reactor-size cap and a capital-budget cap (active-constraint shadow prices)

The story: CONSTRAINED ECONOMIC optimisation: size & operate an esterification CSTR plant for MAXIMUM NPV.

Same esterification plant as the economics01 flagship –

feed → reactor (CSTR) → heater → separator (flash) → vapProd (ethyl-acetate-rich | overhead, the prime | product @ 1.55 EUR/kg) → liqProd (water/acid-rich bottoms, low-value byproduct @ 0.30 EUR/kg)

– but instead of a FIXED design we ask the engine the design question:

maximise economics.NPV varying V_R (reactor volume, m^3) T_{feed} (feed/reactor T, K) subject to $\text{reactor.V}_R \leq V_{\text{max}}$ (a physical SIZE cap) $\text{economics.FCI} \leq \text{FCI}_{\text{max}}$ (a CAPITAL-BUDGET cap)

THE ECONOMIC TRADE-OFF (why this is a genuine optimum, not a corner)

The heater runs at a FIXED partial duty, so the downstream flash SPLITS the reactor effluent into a light, ethyl-acetate-rich VAPOUR overhead (sold high) and a heavy, water/acid-rich LIQUID bottoms (sold cheap). Esterification conserves total mass; what conversion buys you is the RESHUFFLING of that mass toward the valuable overhead. Both knobs raise conversion – and the CSTR is ISOTHERMAL at the feed temperature (CSTR.cpp: $T = \text{feed.T}$), so:

V_R (reactor volume) – bigger V_R raises the residence time $\tau = V_R/Q$, so conversion $X = k\tau/(1 + k\tau)$ climbs, more acetic acid + ethanol turn into ethyl acetate, and the priced overhead grows and enriches (REVENUE up). But a bigger vessel costs more steel: the StirredTank sizing → Guthrie costing drives FCI (CAPEX) UP. Revenue rises with DIMINISHING returns as X saturates toward 1, while CAPEX keeps climbing – so an NPV maximum exists.

T_{feed} (feed/reactor temperature) – higher T raises the Arrhenius rate constant $k = A \cdot \exp(-E_a/RT)$, so X climbs too. T_{feed} is the cheap conversion ACCELERATOR; its bounds are kept inside the window where the flash splits cleanly (below the temperature that boils the bottoms dry).

We deliberately set the size cap TIGHTER than where the NPV-greedy design would otherwise stop, so at the optimum the reactor-size constraint BINDS.

HOW TO READ THE SHADOW PRICES (the whole point of the tutorial)

The SQP prints, at the certified optimum, a Lagrange multiplier (λ) for each inequality. A constraint with $\lambda > 0$ is ACTIVE (binding) and its multiplier is the SHADOW PRICE – the marginal NPV you would gain by relaxing that cap by one unit:

reactor. V_R cap ACTIVE → $\lambda = d(\text{NPV})/d(V_{\text{max}})$ [EUR of NPV per extra m^3 of reactor head-room]. economics.FCI cap → $\lambda = d(\text{NPV})/d(\text{FCI}_{\text{max}})$ [EUR of NPV per extra EUR of capital you may spend].

A constraint with $\lambda \sim 0$ is SLACK (not binding) – relaxing it buys nothing. Here the FCI cap is deliberately generous: it stays SLACK so the student sees BOTH an active and an inactive constraint side by side. The trace marks the active set and the KKT certificate proves the point is a true constrained optimum ($\| \text{grad } L \| \sim 0$, complementarity ~ 0).

The two design variables are case-level \$references so the SQP can drive them through ‘variables.<name>’ and the trace shows their canonical-SI values.

Take the *economics01* ethyl-acetate plant and stop **fixing** the design — let the engine choose it. The hand-rolled SQP sizes and operates the reactor to maximise the net present value, subject to inequality caps, and prints the shadow prices (Lagrange multipliers) of the binding caps.

The objective *economics.NPV* is a KPI published by the *EconomicsPass* in the *system/postDict* chain (*sizing* → *costing* → *economics*). The optimiser runs that whole post chain on every evaluation, so the *economics* scalars (*economics.NPV*, *economics.FCI*, ...) are reachable as ordinary objective/constraint paths — exactly like a sweep reads them.

The heater runs at a fixed partial duty, so the flash **splits** the reactor effluent into a light ethyl-acetate-rich vapour overhead (the prime product at 1.55 €/kg) and a heavy water/acid-rich liquid bottoms (a low-value byproduct at 0.30 €/kg). Esterification conserves total mass; what conversion buys you is the reshuffling of that mass toward the valuable overhead.

* *V_R* — a bigger reactor raises the residence time $\tau = V_R/Q$, so the isothermal-CSTR conversion $X = k\tau/(1+k\tau)$ climbs and the priced overhead grows. But a bigger vessel costs more steel — the *StirredTank sizing* → *Guthrie costing* drives *FCI* (CAPEX) up. Revenue rises with diminishing returns (*X* saturates toward 1) while CAPEX keeps climbing → an NPV maximum exists. * *T_{feed}* — a higher feed temperature raises the Arrhenius rate $k = A \cdot \exp(-E_a/RT)$, the cheap conversion **accelerator**. Its bounds stay inside the window where the flash splits cleanly (below the ~353 K edge that boils the bottoms dry).

We deliberately set the size cap tighter than where the NPV-greedy design would otherwise stop, so the reactor-size constraint binds at the optimum.

The optimiser drives the money-losing initial design ($V_R = 4 \text{ m}^3$, $T_{\text{feed}} = 350 \text{ K} \rightarrow \text{NPV} \approx -164 \text{ M€}$) to a profitable, certified optimum:

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.138888888889
stream	hotStream	F	0.138888888889
stream	liqProd	F	0.0017296965168
stream	reactorOut	F	0.138888888889
stream	vapProd	F	0.137159192372
kpi	economics	COM_d	223170524.398
kpi	economics	FCI	522154.982187
kpi	economics	NPV	477589097.525
kpi	economics	TCI	600478.229515
kpi	economics	WC	78323.2473281
kpi	economics	paybackYears	0.00848797903819
kpi	economics	revenue	321660167.134
kpi	heater	F	0.138888888889
kpi	heater	P	101325

... and 25 more reference values in the case's *expected*.

optim06_integer_stages

tutorials/steady/optimisation/optim06_integer_stages



What it does: The classic N-vs-R trade solved in ONE run: nStages declared integer → exhaustive enumeration, one full SQP per stage count, ALL values on the table

The story: sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0313732166828
kpi	column01	L_strip	0.0591509944605
kpi	column01	Q_condenser_kW	-1391.66557074
kpi	column01	Q_reboiler_kW	1389.90800504
kpi	column01	R	2.25887160116
kpi	column01	T_bottom	382.835800828
kpi	column01	T_top	354.273383646
kpi	column01	V_rect	0.0452621055716
kpi	column01	V_strip	0.0452621055716

...and 7 more reference values in the case's expected.

pareto01_purity_energy

tutorials/steady/optimisation/pareto01_purity_energy



What it does: The purity-energy PARETO FRONT by epsilon-constraint: minimise V_strip subject to x_D_LK >= eps, eps swept 0.96..0.994 – each front point one auditable SQP run

The story: sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.052352377807
kpi	column01	L_strip	0.0801301555848
kpi	column01	Q_condenser_kW	-2035.41756863
kpi	column01	Q_reboiler_kW	2033.86732819
kpi	column01	R	3.7693712021
kpi	column01	T_bottom	383.489767144
kpi	column01	T_top	353.562992445
kpi	column01	V_rect	0.0662412666959
kpi	column01	V_strip	0.0662412666959

...and 7 more reference values in the case's expected.

sensitivity01_column_reflux`tutorials/steady/optimisation/sensitivity01_column_reflux`

What it does: Sweep reflux ratio $R = 1.5..5.0$ (10 pts) across the column01 case; responses: x_D purity + V_{strip} (reboiler-duty proxy)

The story: the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies `units[0].operation.refluxRatio` over $R = 1.5 .. 5.0$ (10 points) and reads T_{top} , T_{bottom} , x_{D_LK} , x_{B_HK} and V_{strip} (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R .

sensitivity02_feasibility_map

tutorials/steady/optimisation/sensitivity02_feasibility_map



What it does: Feasibility MAP: the reflux sweep with declared constraints – every row marked (value, bound, residual, satisfied), nothing filtered; the feasible window between the purity spec and the boilup budget read off the CSV

The story: sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0138888888889
stream	distillate	F	0.0138888888889
stream	feed	F	0.0277777777778
kpi	column01	B	0.0138888888889
kpi	column01	D	0.0138888888889
kpi	column01	L_rect	0.0424382716049
kpi	column01	L_strip	0.0702160493827
kpi	column01	Q_condenser_kW	-1730.95769425
kpi	column01	Q_reboiler_kW	1729.37389301
kpi	column01	R	3.05555555556
kpi	column01	T_bottom	383.383107764
kpi	column01	T_top	353.679332066
kpi	column01	V_rect	0.0563271604938
kpi	column01	V_strip	0.0563271604938
... and 7 more reference values in the case's <i>expected</i> .			

power

The theory you need. Gas (Brayton) and steam (Rankine) cycles, alone and combined. Compressor and turbine hardware is specified by SHAFT WORK and isentropic efficiency – outlet pressure is a RESULT, not an input (the anti-DesignSpec credo). Expect: pressure ratio setting Brayton efficiency, the combined cycle’s gain coming from bottoming the gas turbine’s exhaust exergy, and every energy flow visible in the ledger. *Full derivation: Theory Guide, the power-cycle chapter.*

brayton01_gas_turbine

tutorials/steady/power/brayton01_gas_turbine



What it does: Closed Brayton (gas-turbine) cycle: compress air → combust (gas-aware heater) → expand to feed P

The story: a closed-loop Brayton (gas-turbine) cycle, the showcase for the gas-aware heater (v0.65).

feed (air, 1 bar, 298 K) → compressor (+10 kW, eta 0.80) : 1 → ~9.9 bar, 298 → 633 K → combustor (heater, +20 kW) : add heat at constant P; the air is gas (vf=1) so the heater integrates the ideal-gas H_{ig} (N₂/O₂ carry no liquid Cp) → 633 → ~1246 K → turbine (-W_{turb}, eta 0.85) : expand back to 1.013 bar

The turbine shaft power is NOT guessed: a DesignSpec (system/outerDict) manipulates \$W_{turb} until the turbine discharges at the feed pressure, closing the cycle. Net shaft work and thermal efficiency are computed \$variables (v0.64), evaluated after the solve from the unit KPIs.

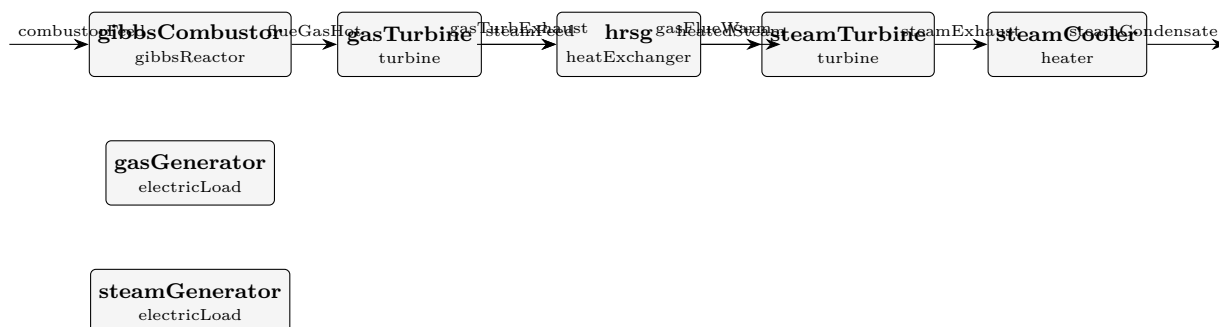
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	compressed	F	0.001
stream	exhaust	F	0.001
stream	feed	F	0.001
stream	hot	F	0.001
kpi	combustor	F	0.001
kpi	combustor	P	991200.306811
kpi	combustor	Q	20000
kpi	combustor	Q_kW	20
kpi	combustor	Q_per_mol	20000
kpi	combustor	T_in	633.374172563
kpi	combustor	T_out	1246.40394079
kpi	combustor	dT	613.029768231
kpi	compressor	F	0.001
kpi	compressor	P_in	100000

...and 27 more reference values in the case's *expected*.

combined01_brayton_rankine

tutorials/steady/power/combined01_brayton_rankine



What it does: Combined-cycle: Brayton (CH₄ + air, Gibbs combustor) + HRSG + Rankine + 2 generators

The story: - Choupo -*—————* combined01_brayton_rankine – combined-cycle power plant

Two thermodynamic cycles linked by a HRSG (heat-recovery steam generator). Top cycle: gas turbine on combustion products of CH₄ in air, made with a Gibbs equilibrium reactor. Bottom cycle: simplified Rankine on steam (gas-only, see rankine01_water). HRSG transfers heat from the gas-turbine exhaust to the Rankine working fluid. Both turbines drive their own ElectricLoad generators.

Pre-mixed combustion air (N₂/O₂/CH₄, 8 bar, 600 K) → gibbsCombustor (T 1300 K, P 8 bar) → flueGasHot (8 bar, 1300 K) → gasTurbine (-600 kW, eta 0.85) → gasTurbExhaust (~4.0 bar, ~1131 K) → HRSG hot side — counter — HRSG cold side (steam) → gasFlueWarm (~4.0 bar, ~1051 K) ^

v heatedSteam (5 bar, ~1131 K) v steamTurbine (-50 kW, eta 0.85) v steamExhaust (~2.6 bar, ~1011 K) → steamCooler → condensate

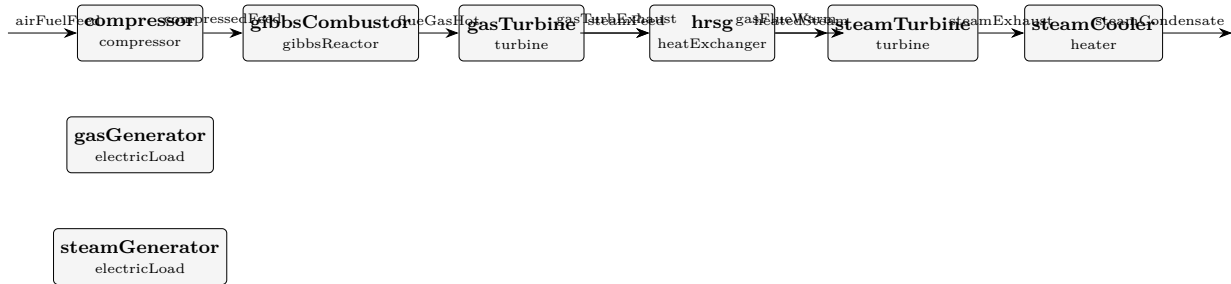
(T/P annotations above are this run's results; gas-only Rankine has no phase change, so the steam side heats to nearly the gas-exhaust T.)

Shaft routing (the v0.0.44 electricLoad unit): gasTurbine.shaft –wire→ gasGenerator (electricLoad, eta 0.97) steamTurbine.shaft –wire→ steamGenerator (electricLoad, eta 0.97)

Simplifications (deliberate, documented for v0.0.44): Pre-mixed combustion feed: a real Brayton has a separate compressor + air/fuel mixer. The current Choupo Mixer is liquid-only; gas-aware Mixer + Compressor closing this loop ship in v0.0.45. Rankine bottom cycle is gas-only (no liquid leg, no phase change). A multi-phase Boiler unit op is also on the v0.0.45 roadmap.

combined02_brayton_rankine_shaft

tutorials/steady/power/combined02_brayton_rankine_shaft



What it does: Combined cycle with SHAFT-SPLIT: gas turbine drives the compressor + net to the generator

The story: - Choupo -*—————* combined02_brayton_rankine_shaft – combined cycle, FULL Brayton with a SHAFT-SPLIT node.

Upgrades combined01: a real COMPRESSOR now sits on the gas-turbine shaft. The gas turbine’s gross work drives the compressor AND the net goes to the generator – a SHAFT NODE that SUMS work inputs:

$\text{gasGenerator.W} = \text{gasTurbine.shaft (+gross)} + \text{compressor.shaft (-load)}$

(Both rotating units publish a ‘shaft’ port with expression “-W_shaft”; the turbine’s $W_{\text{shaft}} < 0 \rightarrow +$, the compressor’s $W_{\text{shaft}} > 0 \rightarrow -$, so the generator nets to the delivered power.)

airFuelFeed (1 bar) $\rightarrow$ compressor ($\rightarrow \sim 8$ bar) $\rightarrow$ gibbsCombustor (1300 K) $\rightarrow$ gasTurbine $\rightarrow$ exhaust $\rightarrow$ HRSG hot side $\rightarrow$ flue | \{\} | (shaft) (shaft) (counter) | \{\} | compressor gasGenerator(net) steamFeed $\rightarrow$ heatedSteam $\rightarrow$ steamTurbine $\rightarrow$ steamGenerator $\rightarrow$ steamCooler $\rightarrow$ condensate

Bottom Rankine stays gas-only (phase-change boiler is the next step).

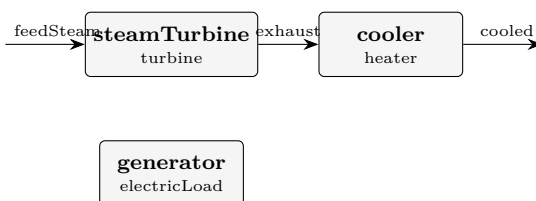
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	airFuelFeed	F	0.1025
stream	compressedFeed	F	0.1025
stream	flueGasHot	F	0.1025
stream	gasFlueWarm	F	0.1025
stream	gasTurbExhaust	F	0.1025
stream	heatedSteam	F	0.01
stream	steamCondensate	F	0.01
stream	steamExhaust	F	0.01
stream	steamFeed	F	0.01
kpi	compressor	F	0.1025
kpi	compressor	P_in	100000
kpi	compressor	P_out	828817.033759
kpi	compressor	T_in	300
kpi	compressor	T_out	592.090873473

... and 89 more reference values in the case’s *expected*.

rankine01_water

tutorials/steady/power/rankine01_water



What it does: Simplified gas-phase Rankine: superheated steam → turbine (-10 kW) → cooler (gas-only, no phase change)

The story: - Choupo -*—————* rankine01_water – simplified Rankine (gas-phase only)

A pedagogical Rankine cycle that demonstrates: a turbine extracting work from superheated steam, an ElectricLoad consuming the turbine's shaft via an energy wire, a thermal efficiency derived from the unit KPIs.

Simplification (documented, deliberate): the cycle runs entirely in the gas phase — there is no liquid leg and no phase change. A real Rankine has a pump + boiler that crosses the saturation line; that needs a multi-phase Boiler unit op which is on the v0.0.45 roadmap.

For the closed Rankine demonstration here:

feedSteam (superheated, 5 bar, 800 K, 1 mol/s) → steamTurbine (W_{shaft} -10 kW, eta 0.85) → exhaust (vapour, ~0.53 bar, ~531 K) → cooler (heater, Q -5 kW) → cooled (vapour, ~388 K)

Shaft routing (the new electricLoad unit, v0.0.44): steamTurbine.shaft —wire—> generator (electricLoad, eta 0.97)

What the turbine produces (this run): W_{generated} = 10 kW (the shaft delivers this; W_{isen} would be 11.765 kW at eta = 1) P_{out} = 0.528 bar (a RESULT — the gas expands here) T_{out} = 530.9 K (258 degC) T_{out_isen} = 480.9 K (ideal isentropic discharge) expansion ratio = 9.47

W_{shaft} = -10 kW is the realistic specification for 1 mol/s of steam at this 5 → ~0.5 bar expansion (an early draft over-specified -300 kW, which would have demanded a physically absurd expansion).

The cooler duty is deliberately bounded: the turbine exhaust at 0.528 bar has T_{sat} = 355.5 K, so the sensible-only 'heater' must stop ABOVE the saturation dome. Q = -5 kW removes the desuperheat down to ~388 K (~33 K of margin above T_{sat}) and the cooled stream stays vapour (vf = 1) – a legitimate gas-phase desuperheater. Push the duty past ~6 kW and the outlet would cross the dome; a sensible-only heater correctly refuses (it asks for a phaseChanger/condenser). Fully closing the cycle (condensing back to a liquid leg + pump) needs the multi-phase Boiler/Condenser unit op on the v0.0.45 roadmap; this gas-phase demonstration stops at the desuperheater by design.

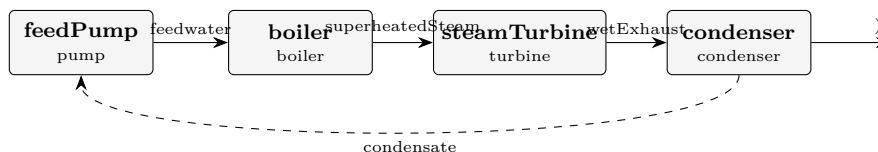
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	cooled	F	0.001
stream	exhaust	F	0.001
stream	feedSteam	F	0.001
kpi	cooler	F	0.001
kpi	cooler	P	52827.3976295
kpi	cooler	Q	-5000
kpi	cooler	Q_kW	-5
kpi	cooler	Q_per_mol	-5000
kpi	cooler	T_in	530.878445241
kpi	cooler	T_out	388.087132345
kpi	cooler	dT	-142.791312896
kpi	generator	W_electric	9700
kpi	generator	W_electric_kW	9.7
kpi	generator	W_losses	300

...and 17 more reference values in the case's *expected*.

rankine02_water

tutorials/steady/power/rankine02_water



What it does: Closed IF97 Rankine cycle: pump → boiler → turbine → condenser → recycle

The story: a CLOSED Rankine steam cycle on IAPWS-IF97

recycle (condensate) [feedPump] [boiler] [steamTurbine] [condenser] raise P cross the extract work cross the dome to 80 bar dome up, (W_shaft, eta), DOWN, land on (liquid) superheat exhaust ~0.1 bar saturated liquid

A real Rankine: the working fluid crosses the saturation dome TWICE – up in the boiler (subcooled liquid → superheated vapour) and down in the condenser (wet vapour → saturated liquid). Pure water on IF97 supplies the steam-table h/s/v surface, so:

the boiler is a dome-crossing phaseChanger driven to superheatedVapour (its duty Q_boiler is the result), the turbine expands on the IF97 entropy surface (H_real/S_real routed), the condenser is a dome-crossing phaseChanger driven to saturatedLiquid (its duty Q_condenser is the result), the pump returns the condensate to boiler pressure (closed-form v*dP).

The loop is CLOSED: the condensate is the tear stream, recycled to the pump. Wegstein/Newton drives the post-pass condensate to match the guess.

First-law check (KPIs): Q_boiler + Q_condenser + W_pump + W_turbine ~ 0.

Contrast: rankine01_water runs the SAME architecture on the ideal gas, in the gas phase only (no liquid leg, no phase change) – the pedagogical baseline. rankine02 is the steam-table truth.

What to expect (golden-locked):

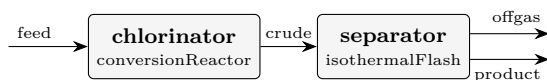
kind	unit/stream	key	expected
kpi	boiler	T_out	773.15
kpi	boiler	vf_out	1.0
kpi	boiler	Tsat	568.159121229
kpi	boiler	Q_kW	47.8559413747
kpi	condenser	vf_out	0.0
kpi	condenser	Tsat	446.515894355
kpi	condenser	Q_kW	-39.017165456
kpi	steamTurbine	P_out	858590.397228
kpi	steamTurbine	W_gen_kW	9.0
stream	superheatedSteam	vf	1.0
stream	condensate	vf	0.0

reactors

The theory you need. Batch reactors integrate $dn_i/dt = \nu_i r V$ from a charged initial state – no in/out flows; time replaces residence time. Adiabatic runs couple the energy balance $\sum_i n_i c_{p,i} dT/dt = -\Delta H_{rxn} r V$, so composition and temperature evolve together. Stiffness appears the moment fast and slow reactions coexist (radical chains: μs radical lifetimes vs ms induction) – the explicit RK4 integrator then needs Δt at the FASTEST scale or blows up, while Rosenbrock23 (L-stable, linearly implicit) steps at the scale of the SOLUTION. The stiffness ratio is printed; watch it. *Full derivation: Theory Guide, the batch/ODE chapter.*

chlorophenol01_chlorination

tutorials/steady/reactors/chlorophenol01_chlorination



What it does: Chlorophenol: classic phenol chlorination ($\text{PhOH} + \text{Cl}_2 \rightarrow 4\text{-CP} + \text{HCl}$) + HCl flash

The story: the CLASSIC chlorophenol process.

Direct chlorination of phenol with chlorine gas – the industrial route to the chlorophenols (here the mono, 4-chlorophenol):

chlorinator : $\text{C}_6\text{H}_5\text{OH} + \text{Cl}_2 \rightarrow \text{C}_6\text{H}_4\text{ClOH} + \text{HCl}$ (exothermic, ~ 130 kJ/mol) separator : a flash at 80 C, 1 atm splits the volatile HCl (+ excess Cl_2) off the liquid phenol / chlorophenol.

Feed: 1.1 kmol/h phenol + 1.0 kmol/h Cl_2 (slight phenol excess $\rightarrow$ Cl_2 limiting). phenol + 4-chlorophenol are JOBACK ESTIMATES (glass-box, unvalidated).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	crude	F	0.0005833333333333
stream	feed	F	0.0005833333333333
stream	offgas	F	0.000274077911624
stream	product	F	0.00030925542171
kpi	chlorinator	F_in_kmol_h	2.1
kpi	chlorinator	F_out_kmol_h	2.1
kpi	chlorinator	T	353
kpi	chlorinator	conversion	0.9
kpi	chlorinator	extent_kmol_h	0.9
kpi	separator	F_alpha	0.00030925542171
kpi	separator	F_beta	0.000274077911624
kpi	separator	F_in	0.0005833333333333
kpi	separator	P	101325
kpi	separator	T	353

... and 2 more reference values in the case's *expected*.

conversion01_hda

tutorials/steady/reactors/conversion01_hda



What it does: Conversion reactor: HDA toluene+H2 $\rightarrow$ benzene+CH4 at 75%/pass (gas-basis, no kinetics, no Vliq)

The story: A single gas-phase conversion reactor: toluene + H2 $\rightarrow$ benzene + CH4, 75 % of the limiting reactant (toluene) per pass. H2 is fed in excess (3:1), as in the real HDA.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.111111111111
stream	product	F	0.111111111111
kpi	reactor	F_in_kmol_h	400
kpi	reactor	F_out_kmol_h	400
kpi	reactor	Q_kW	-1007.67392146
kpi	reactor	T	900
kpi	reactor	conversion	0.75
kpi	reactor	dHrxn_kJ_per_mol	-48.3683482303
kpi	reactor	extent_kmol_h	75

conversion02_parallel_selectivity

tutorials/steady/reactors/conversion02_parallel_selectivity



What it does: Conversion reactor, PARALLEL network $A \rightarrow B$ / $A \rightarrow C$: selectivity as a SPEC (75 %)

The story: a stoichiometric (conversion) reactor with a PARALLEL network, $A \rightarrow B$ / $A \rightarrow C$.

Multi-reaction grammar 'reactions (a_to_b a_to_c);'. Each reaction's extent is a SPEC, not a result: 'conversions ({ reaction ...; conversion ...; } ...)' gives the fraction of the FEED A that each leg consumes. 60 % of A goes the wanted way and 20 % the side way, so:

total A conversion = 80 % selectivity $B/(B+C) = 0.36 / 0.48 = 75$ %

This is the honest reactor when you have neither rate data (cstr/pfr) nor an equilibrium (equilibriumReactor) – just a catalyst whose split you measured. A single-reaction conversionReactor cannot express a selectivity at all.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	product	F	0.0001888888888889
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.68
kpi	reactor	T	350
kpi	reactor	extent_a_to_b_kmol_h	0.12
kpi	reactor	extent_a_to_c_kmol_h	0.04
kpi	reactor	nReactions	2

cstr01_first_order
tutorials/steady/reactors/cstr01_first_order



What it does: CSTR with pseudo-first-order Arrhenius esterification ($\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$), ~53% conversion of ethanol

The story: CSTR single-reaction tutorial. The reaction kinetics live in constant/reactions; here the unit references it by name.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	X_limiting	0.52590711876
kpi	reactor	T	350
kpi	reactor	k	0.00357499942014

cstr02_reversible_wgs

tutorials/steady/reactors/cstr02_reversible_wgs



What it does: Reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) in a CSTR

The story: water-gas-shift reaction in a CSTR with reverse rate from thermodynamic K_{eq} .

Feed: equimolar $\text{CO} + \text{H}_2\text{O}$ at 600 K, 10 bar. The forward Arrhenius kinetics drives a high conversion in the early residence times; the reverse term (auto-derived from $g_{\text{pure_ig}}$ of the four species at $T = 600$ K) caps the conversion at the Le Chatelier equilibrium.

At $T = 600$ K: $dG_{\text{rxn}} = g_{\text{CO}_2} + g_{\text{H}_2} - g_{\text{CO}} - g_{\text{H}_2\text{O}} \approx -16.3 \text{ kJ/mol}$ $K_{\text{eq}} = \exp(16300 / (R \cdot 600)) \approx 26.5$ For equimolar $\text{CO} + \text{H}_2\text{O}$, $K_{\text{eq}} = X^2 / (1-X)^2 \rightarrow X_{\text{eq}} = 0.838$

V_R is picked large enough ($1\text{e-}3 \text{ m}^3$ for a 1 kmol/h feed) that the kinetics are fast relative to residence; we see the conversion approach (but not exceed) $X_{\text{eq}} = 0.838$ — exactly the thermodynamic ceiling.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	feed	vf	1
stream	products	F	0.0002777777777778
stream	products	vf	1
kpi	wgs	Da_kTau	145.857299066
kpi	wgs	F_in_kmol_h	1
kpi	wgs	F_out_kmol_h	1
kpi	wgs	T	600
kpi	wgs	V_R	0.001
kpi	wgs	X_limiting	0.841431955275
kpi	wgs	k	1.08521882096
kpi	wgs	tau_s	134.403584096
kpi	wgs	xi_mol_per_s	0.116865549344

cstr03_series_selectivity

tutorials/steady/reactors/cstr03_series_selectivity



What it does: CSTR with a SERIES network $A \rightarrow B \rightarrow C$: multi-reaction, the selectivity problem

The story: a CSTR with a SERIES reaction network, $A \rightarrow B \rightarrow C$, solved as ONE coupled system.

The multi-reaction grammar ‘reactions (a_to_b b_to_c);’ names both steps from constant/reactions; the reactor solves the R coupled design equations $xi_j = r_j(xi) V_R$ together by the multivariate Newton. A single-reaction CSTR cannot express this at all: the intermediate B is BOTH a product and a reactant, so its yield peaks in residence time and SELECTIVITY becomes the design question.

Feed: pure A at 350 K. Try sweeping V_R (an outerDict sweep) and watch the B yield rise, peak, and fall as the over-reaction $B \rightarrow C$ takes over.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	product	F	0.000199218293117
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.717185855222
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	k_a_to_b	0.00214522777035
kpi	reactor	k_b_to_c	0.000536306942589
kpi	reactor	nReactions	2
kpi	reactor	newtonResidual	6.93889390391e-18
kpi	reactor	tau_s	225
kpi	reactor	xi_a_to_b_mol_per_s	0.0392797423303
kpi	reactor	xi_b_to_c_mol_per_s	0.0126884461066

cstr04_adiabatic

tutorials/steady/reactors/cstr04_adiabatic



What it does: ADIABATIC CSTR: T_{out} solved WITH the extents; $Q=0$ by construction (elements datum)

The story: an ADIABATIC continuous stirred-tank reactor.

The outlet temperature is an UNKNOWN, solved WITH the reaction extents:

$x_{i,j} = r_j(x_i, T) V_R$ (R design equations) $H_{\text{out}}(T, x_i) = H_{\text{in}}$ (one enthalpy balance)

There is NO dH_{rxn} source term. With the enthalpy on the ELEMENTS datum, the heat of reaction already lives inside H: as reactants become products the formation enthalpies differ and T rises on its own. $Q_{\text{kW}} = 0$ by construction.

A non-isothermal CSTR can have up to THREE steady states (multiplicity). The Newton lands on ONE of them, selected by ' T_{guess} ' – sweep it to find the ignited and extinguished branches. That honesty is the point: the reactor reports which root it found, it does not pretend there is only one.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	$F_{\text{in_kmol_h}}$	1
kpi	reactor	$F_{\text{out_kmol_h}}$	1
kpi	reactor	Q_{kW}	4.70379291073e-12
kpi	reactor	T	350
kpi	reactor	T_{guess}	350
kpi	reactor	T_{out}	454.152948902
kpi	reactor	V_R	0.005
kpi	reactor	dT_{rise}	104.152948902
kpi	reactor	$k_{\text{esterification_etac}}$	0.88934531923
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	3.5527136788e-15
kpi	reactor	τ_{s}	310.291329081

... and 1 more reference values in the case's *expected*.

cstr05_multiplicity

tutorials/steady/reactors/cstr05_multiplicity



What it does: CSTR multiplicity: three steady states – the reactor prints them all, T_guess picks the branch

The story: an ADIABATIC CSTR with THREE steady states.

Eliminate the extents (at a fixed T the design equations are monotone) and what remains is ONE equation in the temperature:

$$\phi(T) = H_{\text{out}}(T) - H_{\text{in}} - Q_{\text{ext}}(T) = 0$$

the textbook "heat generated equals heat removed". The generation curve $G(T)$ is a sigmoid (the rate saturates once the reactant is gone); the removal line $R(T) = F c_p (T - T_{\text{in}})$ is straight (adiabatic: the only sink is the sensible heat that leaves with the product). A straight line can cut a sigmoid THREE times: the extinguished state, an unstable middle state, and the ignited state.

The feed enters COLD (300 K), where the rate is negligible. Nothing forces this reactor to ignite – and nothing forces it to stay cold either.

Choupo scans, finds them all, prints them, and reports the one nearest 'T_guess'. Change T_guess and the reactor moves to another branch – the same equipment, the same feed, a different answer. That is not a numerical defect; it is the physics, and it is why real reactors need a start-up procedure.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	8.68283223099e-12
kpi	reactor	T	300
kpi	reactor	T_guess	380
kpi	reactor	T_out	405.903598573
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	105.903598573
kpi	reactor	k_esterification_etac	0.406108484365
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	8.32667268469e-16
kpi	reactor	steadyStates	3

... and 2 more reference values in the case's *expected*.

ctr06_jacketed

tutorials/steady/reactors/ctr06_jacketed



What it does: JACKETED CSTR: the coolant carries the reaction heat away – $Q = UA (T_c - T_{out})$

The story: the same esterification as ctr04, now with a COOLING JACKET instead of an adiabatic wall.

The energy balance gains one term, and only one:

$$H_{out}(T) = H_{in} + UA (T_{coolant} - T_{out})$$

There is still NO dH_{rxn} source: on the elements datum the heat of reaction is already inside H . What the jacket does is remove some of it, so the outlet settles BELOW the adiabatic 431.96 K of ctr04.

Check it by hand from the printed KPIs: Q_{kW} should equal $UA (T_c - T_{out})$, and it is NEGATIVE – heat leaves the reactor. Cool harder (raise UA , lower $T_{coolant}$) and the outlet slides down the generation curve; cool too hard and the reactor extinguishes. Nothing here is switched on by a flag; it is the same one equation in T , with one more term.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-1.85832125175
kpi	reactor	T	350
kpi	reactor	T_guess	400
kpi	reactor	T_out	394.33285007
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	44.3328500695
kpi	reactor	k_esterification_etac	0.0534254733127
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	1.38777878078e-16
kpi	reactor	steadyStates	1

... and 2 more reference values in the case's *expected*.

cstr07_lhhw_methylAcetate

tutorials/steady/reactors/cstr07_lhhw_methylAcetate



What it does: LHHW on activities: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

The story: an isothermal CSTR packed with Amberlyst 15.

The rate constants are reported per GRAM of dry catalyst, so the reactor needs to know how much catalyst a cubic metre of it holds: 'catalystLoading'. It is a declared, visible number, not a fudge – change the packing and the reactor changes with it.

Run it and read the KPI 'k_esterification_meoac' against the conversion: the forward rate constant says one thing, the adsorption denominator says another, and the outlet is where they settle.

This reactor is KINETICALLY limited, which is the point – if it sat at equilibrium the denominator would be invisible. Sweep V_R and watch it climb to the activity-based equilibrium and stop:

V_R = 0.005 m3 X = 49.02 % <- this case V_R = 0.020 m3 X = 63.74 % V_R = 0.100 m3 X = 72.92 %
V_R = 1.000 m3 X = 76.18 % <- equilibrium

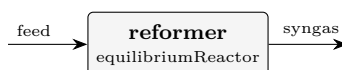
The plateau is not asserted anywhere in the dict. It is where $k_1 a'_{\text{HOAc}} a'_{\text{MeOH}} = k_{-1} a'_{\text{MeOAc}} a'_{\text{H}_2\text{O}}$, i.e. exactly the $K_a = 25.50$ that the authors' own Eq 17 recovers from the same four rate constants and four adsorption constants at 333.15 K. Two independent equations of the paper agreeing is a cheap and honest check that the parameters were entered right.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	reactorOut	F	0.0002777777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-2.09513270004
kpi	reactor	T	333.15
kpi	reactor	T_out	333.15
kpi	reactor	V_R	0.005
kpi	reactor	k_esterification_meoac	0.00280769603355
kpi	reactor	nReactions	1
kpi	reactor	newtonResidual	1.2490009027e-16
kpi	reactor	tau_s	366.524129505
kpi	reactor	xi_esterification_meoac_mol_per_s	0.0680930737737

equil01_reforming

tutorials/steady/reactors/equil01_reforming



What it does: Equilibrium reactor (REquil): SMR + WGS to SIMULTANEOUS equilibrium at 1100 K

The story: the EQUILIBRIUM reactor (REquil) with MULTIPLE simultaneous reactions.

Steam-methane reforming: two independent reactions are driven to SIMULTANEOUS chemical equilibrium at 1100 K, 1 bar – no kinetics, no residence time. The two extents are a RESULT of each reaction's $K_p(T)$ (from the gibbsFormation data), solved together by the multivariate Newton:

smr : $\text{CH}_4 + \text{H}_2\text{O} \leftrightarrow \text{CO} + 3 \text{H}_2$ (strongly endothermic) wgs : $\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 + \text{H}_2$ (mildly exothermic)

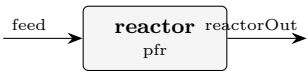
The stoichiometry lives in constant/reactions and is referenced BY NAME – the same ‘reactions (r1 r2)’; grammar the batch / dynamic reactors take.

Feed: 1 kmol/h CH_4 + 3 kmol/h steam (steam-to-carbon = 3). At 1100 K, 1 bar the reforming is near-complete (Le Chatelier favours the mole-increasing SMR at low pressure); the shift then sets the CO/CO₂ split. This is the classic syngas problem – the reactor a ‘conversionReactor’ cannot do (it needs the extents GIVEN) and a ‘cstr’/‘pfr’ cannot do without rate data.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001111111111111
stream	syngas	F	0.00166593958576
kpi	reformer	F_in_kmol_h	4
kpi	reformer	F_out_kmol_h	5.99738250873
kpi	reformer	Kp_smr	312.740523794
kpi	reformer	Kp_wgs	0.978425617548
kpi	reformer	Q_kW	59.6424648629
kpi	reformer	T	1100
kpi	reformer	conversion_smr	0.998691254363
kpi	reformer	conversion_wgs	0.109768673641
kpi	reformer	extent_smr_kmol_h	0.998691254363
kpi	reformer	extent_wgs_kmol_h	0.329306020923
kpi	reformer	nReactions	2
kpi	reformer	newtonResidual	3.2596148003e-13

pfr01_first_order
tutorials/steady/reactors/pfr01_first_order



What it does: PFR with the same kinetics as cstr01 (RK4 integration)

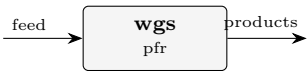
The story: PFR with the same kinetics and conditions as cstr01. For a first-order reaction the PFR gives $X = 1 - \exp(-k\tau)$. With $k\tau = 1.10$: $X_{\text{PFR}} = 1 - \exp(-1.10) = 0.667$ (vs CSTR $X = 0.524$)

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	reactorOut	F	0.000277777777778

pfr02_reversible_wgs

tutorials/steady/reactors/pfr02_reversible_wgs



What it does: Reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) in a PFR; approaches the same X_{eq} as the CSTR

The story: the reversible water-gas shift in a PLUG-FLOW reactor, the PFR sibling of cstr02_reversible_wgs.

Same reaction, same feed (equimolar $\text{CO} + \text{H}_2\text{O}$ at 600 K, 10 bar), same reverse term auto-derived from g_pure_ig. The pedagogical point: a reversible reaction relaxes toward the THERMODYNAMIC equilibrium conversion regardless of reactor type — with enough volume the PFR approaches the very same ceiling the CSTR hit,

$K_{\text{eq}}(600\text{ K}) \sim 26.5 \rightarrow X_{\text{eq}} \sim 0.84$ (equimolar feed),

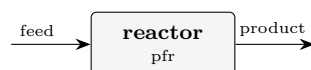
not the $X = 1$ a forward-only PFR would chase. V_{R} is taken large enough that the axial profile flattens onto X_{eq} well before the exit.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	feed	vf	1
stream	products	F	0.000277777777778
stream	products	vf	1

pfr03_series_selectivity

tutorials/steady/reactors/pfr03_series_selectivity



What it does: PFR with the SAME series network $A \rightarrow B \rightarrow C$: no back-mixing $\rightarrow$ MORE of the intermediate B

The story: the SAME series network $A \rightarrow B \rightarrow C$ as cstr03, now in a PFR. Compare them.

Multi-reaction grammar 'reactions (a_to_b b_to_c);'; the PFR marches $dF_i/dV = \sum_j \nu_{ij} r_j(F)$ by RK4 – an IVP, no Newton. At the SAME residence time ($\tau = 225$ s):

species CSTR PFR compA (feed) 40.8 % 14.1 % compB (want) 52.8 % 79.2 % <– the point compC (over) 6.4 % 6.8 %

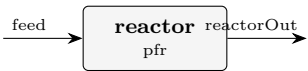
The PFR converts far more A AND keeps far more of the intermediate B: with no back-mixing, the freshly made B is not held in a vessel at low A concentration where the over-reaction $B \rightarrow C$ dominates. For positive-order kinetics the PFR beats the CSTR – the classic result a single-reaction reactor cannot show.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	product	F	0.000176612457804
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	0.635804848095
kpi	reactor	T	350
kpi	reactor	V_R	0.005
kpi	reactor	k_a_to_b	0.00214522777035
kpi	reactor	k_b_to_c	0.000536306942589
kpi	reactor	nReactions	2
kpi	reactor	nSteps	200
kpi	reactor	tau_s	225

pfr04_adiabatic

tutorials/steady/reactors/pfr04_adiabatic



What it does: ADIABATIC PFR: the exothermic esterification heats its own stream – T(V) is a RESULT

The story: an ADIABATIC plug-flow reactor. The temperature is a RESULT, marched with the species by the same RK4:

$$dF_i/dV = \sum_j \nu_{ij} r_j(F,T) \quad dT/dV = \sum_j (-\Delta H_j(T)) r_j / \sum_i F_i c_{p,i}(T)$$

The exothermic esterification heats its own stream, which accelerates the Arrhenius rate, which releases more heat – the positive feedback every reactor engineer must see. T(V) is in the axial profile: plot it.

‘thermalMode’ takes ‘isothermal’ (default – T imposed, duty a result), ‘adiabatic’ (here), or ‘heatExchange’ (a jacket: U, areaPerVolume, T_coolant). A non-isothermal reactor needs the heat of reaction, so every reacting species must carry ‘gibbsFormation’ – a fictitious toy species cannot be used.

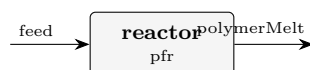
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.0002777777777778
stream	reactorOut	F	0.0002777777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	Q_kW	-1.42108547152e-14
kpi	reactor	T	350
kpi	reactor	T_max	454.513818186
kpi	reactor	T_min	350
kpi	reactor	T_out	454.513818186
kpi	reactor	V_R	0.005
kpi	reactor	dT_rise	104.513818186
kpi	reactor	hotSpot_dT	104.513818186
kpi	reactor	k_esterification_etac	0.00357499942014
kpi	reactor	nReactions	1

... and 2 more reference values in the case's *expected*.

pfr_polyesterification

tutorials/steady/reactors/pfr_polyesterification



What it does: PFR step-growth polyesterification — Carothers/Flory chain statistics

The story: Step-growth polyesterification in a PFR. A molten AB-type hydroxy-acid monomer self-condenses to polycaprolactone, releasing water. As the monomer conversion p rises along the reactor, the Carothers relation $X_n = 1/(1-p)$ makes the average chain length climb steeply, and the polydispersity $PDI = 1+p$ approaches 2 (the most-probable distribution).

Try a ‘sweep’ over operation. V_R and watch X_n , M_n and PDI climb and the Flory-Schulz distribution broaden – the whole story in one glass-box run.

A molten AB-type hydroxy-acid monomer (6-hydroxyhexanoic acid) self- condenses to polycaprolactone in a plug-flow reactor, releasing one water per ester bond:

As the monomer conversion p climbs along the reactor, the Carothers relation makes the average chain length rise steeply and the polydispersity approach 2 — the most-probable (Flory–Schulz) distribution.

and emits the Flory–Schulz distribution as a profile (chainLength vs n_x , w_x) so the GUI plots it: $n(x)=(1-p)p^{x-1}$ decays monotonically, $w(x)=x(1-p)^2 p^{x-1}$ peaks near $x \approx X_n$.

Run a sweep over operation. V_R (reactor volume $\rightarrow$ residence time $\rightarrow p$) and watch PDI climb toward 2, X_n and M_n shoot up, and the weight distribution broaden. The whole Carothers/Flory story in one glass-box run.

Lives inside the reaction. Absent ? the reactor behaves as an ordinary PFR. p is taken from the reactor’s own conversion of limitingReactant (single source of truth). The distribution + $PDI = 1+p$ are valid because a PFR has a uniform residence time; on a CSTR the residence-time distribution broadens both, so the distribution is gated to PFR/batch only.

- The chain-statistics math is exact and standard: P.J. Flory, **Principles of Polymer Chemistry**, Cornell Univ. Press (1953), Ch. VIII–IX (the most-probable distribution and $PDI = 1+p$); G. Odian, **Principles of Polymerization**, 4th ed., Ch. 2; H.S. Fogler, **Elements of Chemical Reaction Engineering**, 5th ed., ?9. - The monomer/repeat component properties are ILLUSTRATIVE (clearly labelled in each .dat) — plausible round numbers for a teaching case, not curated literature values. The Arrhenius rate is illustrative, tuned so $Da = k\tau \approx 4$ gives $p \approx 0.98$. Curate these before any quantitative use.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	polymerMelt	F	0.000550395491925
kpi	reactor	Da_kTau	3.98587252785
kpi	reactor	F_out_kmol_h	1.98142377093
kpi	reactor	Mn_kg_kmol	6144.41174045
kpi	reactor	Mw_kg_kmol	12174.6834809
kpi	reactor	PDI	1.98142377093
kpi	reactor	T	450
kpi	reactor	V_R	0.2
kpi	reactor	X_limiting	0.981423770929
kpi	reactor	Xn	53.8322388335
kpi	reactor	Xw	106.664477667
kpi	reactor	p_conversion	0.981423770929
kpi	reactor	tau_s	5829.95951417

ryield01_pyrolysis

tutorials/steady/reactors/ryield01_pyrolysis



What it does: Yield reactor (empirical mass-yield model): toluene hydro-dealkylation as a LUMPED mass-yield slate – no stoichiometry, no kinetics, overall mass closes

The story: A single YIELD reactor (empirical mass-yield model). The HDA chemistry (toluene + H₂ → benzene + CH₄) is treated here NOT with a balanced reaction but as an empirical MASS-YIELD slate: per unit mass of TOTAL feed the reactor makes benzene/CH₄/H₂ in fixed mass fractions. This is the right tool when only the product distribution is known (lumped / empirical chemistry). No stoichiometry, no kinetics – the OVERALL MASS BALANCE always closes because the yields are on a normalised (Sum = 1) basis.

A single yieldReactor (the Choupo analogue of an empirical-yield reactor): the outlet is fixed by an empirical mass-yield slate rather than by stoichiometry or kinetics. This is the right tool when only the product distribution is known (lumped / empirical chemistry) — the overall mass balance always closes because the yields are given on a normalised (sum = 1) mass basis.

fed as 25 mol% toluene / 75 mol% H₂ at 900 K, 35 bar. (The folder name ryield01_pyrolysis is a legacy label; the slate below is HDA, not thermal cracking — a pyrolysis example would use a coke/olefin product distribution.)

The yields are illustrative, chosen to sum to 1.0 kg/kg of total feed and to demonstrate the mechanics of a yield reactor, not measured for a specific HDA unit:

- Mass in = mass out (9818.9 kg/h in and out) — the headline property of a yield reactor. - Molar flow grows (400 → ~1503 kmol/h) because the slate makes lighter species, while mass is conserved. - Product composition ~0.91 H₂ / 0.052 benzene / 0.041 CH₄ (mole fractions), isothermal at 900 K with an endothermic duty Q ~ 3760 kW.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.111111111111
stream	product	F	0.417464865223
kpi	reactor	F_in_kmol_h	400
kpi	reactor	F_out_kmol_h	1502.8735148
kpi	reactor	M_in_kg_h	9818.9
kpi	reactor	M_out_kg_h	9818.9
kpi	reactor	Q_kW	3759.14055046
kpi	reactor	T	900

rotating

The theory you need. Compressors, turbines, pumps: specified by shaft work W_s and isentropic efficiency η_s ; the outlet state follows from $h_{\text{out}} = h_{\text{in}} + W_s/\dot{n}$ with the isentropic reference path. Outlet pressure is a RESULT (hardware, not a spec).

compressor01_air

tutorials/steady/rotating/compressor01_air



What it does: Adiabatic air compressor: 10 kW shaft, eta 0.75 — P_out and T_out are RESULTS

The story: The simplest credo-pure rotating-equipment case. An adiabatic compressor takes ambient air (1 bar, 25 degC) and a fixed shaft power of 10 kW with isentropic efficiency 0.75.

Hardware (the only things you specify): W_shaft = 10 kW motor rating delivered to the gas eta = 0.75 isentropic efficiency (machine characteristic)

RESULTS the simulator computes: P_out the discharge pressure the machine reaches T_out the discharge temperature (real, with losses) ratio P_out / P_in T_out_isen isentropic discharge T (diagnostic) dS_gen entropy generated by the irreversibility

Hand check (ideal gas, Cp ~ 29.2 J/(mol*K), gamma ~ 1.40): F = 3.6 kmol/h = 1 mol/s w_real = 10000 W / 1 mol/s = 10000 J/mol w_isen = 0.75 * 10000 = 7500 J/mol T_out_isen = 298.15 + 7500/29.2 ~ 555 K ratio = (T_isen/T_in)^(gamma/(gamma-1)) = (555/298)^3.5 ~ 8.8 T_out = 298.15 + 10000/29.2 ~ 640 K

"I need exactly 8 bar at discharge" is a DESIGN question — it goes in a DesignSpec that manipulates \$W_shaft with target compressor.P_out.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	out	F	0.001
kpi	compressor01	F	0.001
kpi	compressor01	P_in	100000
kpi	compressor01	P_out	890211.642995
kpi	compressor01	T_in	298.15
kpi	compressor01	T_out	633.374206884
kpi	compressor01	T_out_isen	551.37943898
kpi	compressor01	W_isen	7500
kpi	compressor01	W_isen_kW	7.5
kpi	compressor01	W_shaft	10000
kpi	compressor01	W_shaft_kW	10
kpi	compressor01	dS_gen	4.22615345389
kpi	compressor01	dT	335.224206884

... and 3 more reference values in the case's expected.

compressor02_designspec_pout

tutorials/steady/rotating/compressor02_designspec_pout



What it does: Air compressor sized to hit $P_{out} = 8$ bar exactly, via DesignSpec on W_{shaft}

The story: Same air compressor as compressor01, but now answering the DESIGN question: "what shaft power do I need to discharge at exactly 8 bar?"

The credo forbids specifying P_{out} on the unit op (it's a result). Instead the outer DesignSpec manipulates the hardware variable W_{shaft} until the computed $compressor.P_{out}$ hits the 8-bar target. This is the rotating-equipment analogue of sizing an evaporator's area to hit a product concentration.

Converges to $W_{shaft} \sim 9.36$ kW (slightly below compressor01's 10 kW, which over-shot to 8.90 bar). At the solution $T_{out} \sim 612$ K (339 degC) and $W_{isen} \sim 7.02$ kW ($\eta * W_{shaft}$, $\eta = 0.75$).

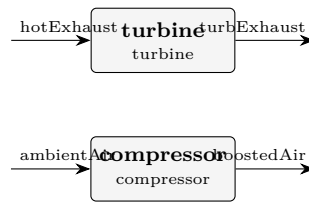
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	out	F	0.001
kpi	compressor	F	0.001
kpi	compressor	P_{in}	100000
kpi	compressor	P_{out}	800004.793643
kpi	compressor	T_{in}	298.15
kpi	compressor	T_{out}	612.41049696
kpi	compressor	T_{out_isen}	535.384868081
kpi	compressor	W_{isen}	7017.30522226
kpi	compressor	W_{isen_kW}	7.01730522226
kpi	compressor	W_{shaft}	9356.40696301
kpi	compressor	W_{shaft_kW}	9.35640696301
kpi	compressor	dS_{gen}	4.08116534481
kpi	compressor	dT	314.26049696

... and 3 more reference values in the case's *expected*.

compressorTurbine01_turbo_booster

tutorials/steady/rotating/compressorTurbine01_turbo_booster

**What it does:** Turbo-booster: turbine shaft drives a compressor (energy wire)**The story:** turbo-booster: turbine drives compressor through an ENERGY WIRE (v0.80)

Two physically independent gas streams, one shared shaft:

hotExhaust → turbine —shaft—+

ambientAir → compressor <—+

Hardware specs (the only knobs): turbine $W_{\text{shaft}} = -5$ kW (NEGATIVE: work extracted) $\eta = 0.80$
 compressor W_{shaft} is OMITTED (the wire provides it) $\eta = 0.75$

The wire injects $+W_{\text{turb}}$ into the compressor before its solve(), so the compressor's W_{shaft} is whatever the turbine produces — here, 5 kW (sign flipped: $-W_{\text{shaft}}$ on the producer's expression → the consumer reads $+W_{\text{shaft}}$, the usual sign convention).

Hand check (ideal gas, $C_p \sim 29$ J/(mol*K), $\gamma \sim 1.40$): Turbine: 1 mol/s, 800 K, 5 bar → extract 5 kW (= 5000 J/mol) $w_{\text{isen}} = w_{\text{real}} / \eta = 5000 / 0.80 = 6250$ J/mol $T_{\text{isen}} = 800 - 6250/29 \sim 585$ K → ratio = $(585/800)^{3.5} \sim 0.35$ $P_{\text{out}} \sim 5 * 0.35 \sim 1.74$ bar $T_{\text{out}} (\text{real}) = 800 - 5000/29 \sim 627$ K (warmer than T_{isen} : the losses stay as heat in the gas)

Compressor: 0.5 mol/s, 298 K, 1 bar, receiving 5 kW $w_{\text{real}} = 5000 / 0.5 = 10000$ J/mol $w_{\text{isen}} = 0.75 * 10000 = 7500$ J/mol $T_{\text{isen}} = 298 + 7500/29 \sim 557$ K → ratio = $(557/298)^{3.5} \sim 8.7$ $P_{\text{out}} \sim 8.7$ bar $T_{\text{out}} (\text{real}) = 298 + 10000/29 \sim 643$ K

The wire makes the energy flow visible at the case level — a single edge in the flowsheet that the GUI can render dashed.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	ambientAir	F	0.0005
stream	boostedAir	F	0.0005
stream	hotExhaust	F	0.001
stream	turbExhaust	F	0.001
kpi	compressor	F	0.0005
kpi	compressor	P_in	100000
kpi	compressor	P_out	890211.642995
kpi	compressor	T_in	298.15
kpi	compressor	T_out	633.374206884
kpi	compressor	T_out_isen	551.37943898
kpi	compressor	W_isen	3750
kpi	compressor	W_isen_kW	3.75
kpi	compressor	W_shaft	5000
kpi	compressor	W_shaft_kW	5

... and 19 more reference values in the case's *expected*.

pump01_water

tutorials/steady/rotating/pump01_water



What it does: Water pump: 2 kW shaft, eta 0.65 — incompressible, P_out / T_out are RESULTS

The story: An incompressible-liquid pump. Liquid water enters at 1 bar, 298 K; the pump shaft delivers 2 kW with efficiency 0.65.

Hardware (the only OP parameters): W_shaft = 2 kW motor power delivered to the liquid eta = 0.65 pump efficiency

Because the liquid is incompressible ($v \sim \text{const}$), the discharge is closed-form — no EoS, no Newton: $w_{\text{real}} = W_{\text{shaft}} / F_{\text{mol_s}}$ $w_{\text{isen}} = \eta * w_{\text{real}}$ (the useful $v*dP$ part) $dP = w_{\text{isen}} / v_{\text{molar}}$ $P_{\text{out}} = P_{\text{in}} + dP$ (RESULT) $dT = (1 - \eta) * w_{\text{real}} / C_{p_liq}$ (dissipation)

Hand check (water: $v \sim 1.8\text{e-}5 \text{ m}^3/\text{mol}$, $C_{p_liq} \sim 75 \text{ J}/(\text{mol}\cdot\text{K})$): $F = 36 \text{ kmol/h} = 10 \text{ mol/s}$ $w_{\text{real}} = 2000 \text{ W} / 10 \text{ mol/s} = 200 \text{ J/mol}$ $w_{\text{isen}} = 0.65 * 200 = 130 \text{ J/mol}$ $dP = 130 / 1.8\text{e-}5 \sim 7.2\text{e}6 \text{ Pa} = 72 \text{ bar}...$ (so $P_{\text{out}} \sim 73 \text{ bar}$) $dT = 0.35 * 200 / 75 \sim 0.9 \text{ K}$ (tiny — pumps barely heat)

"Pump to exactly 50 bar" is a DesignSpec on \$W_shaft targeting pump.P_out (converges in one step — the relation is linear).

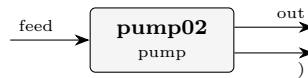
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.01
stream	out	F	0.01
kpi	pump01	F	0.01
kpi	pump01	P_in	100000
kpi	pump01	P_out	7294244.60432
kpi	pump01	T_in	298.15
kpi	pump01	T_out	299.077152318
kpi	pump01	W_hydraulic	1300
kpi	pump01	W_shaft	2000
kpi	pump01	W_shaft_kW	2
kpi	pump01	dP	7194244.60432
kpi	pump01	dT	0.927152317881
kpi	pump01	eta	0.65
kpi	pump01	head_m	735.848502288

...and 1 more reference values in the case's expected.

pump02_pressure_spec

tutorials/steady/rotating/pump02_pressure_spec



What it does: Water pump specified by DISCHARGE PRESSURE (P_out 30 bar) — the shaft power W_shaft is the RESULT (closed-form inverse of pump01)

The story: The SAME incompressible-liquid pump as pump01, specified the OTHER way: by the discharge pressure it must reach. In a real plant you rarely know a pump's shaft power up front — you know the HEADER it feeds (here 30 bar). So you state P_out and the shaft power is the RESULT.

This is NOT a DesignSpec. For an incompressible liquid the pump relation is invertible in CLOSED FORM,

$$\Delta P = \eta \cdot W_{\text{shaft}} / Q_{\text{vol}} \Leftrightarrow W_{\text{shaft}} = Q_{\text{vol}} \cdot \Delta P / \eta ,$$

with $Q_{\text{vol}} = F \cdot v$ set by the feed. Give P_out, get W_shaft — one shot, no Newton, no outer loop. (pump01 gives W_shaft and gets P_out; the two cases are exact inverses of one another.)

Specify EXACTLY ONE of { W_shaft | P_out | dP }. A pressure let-down (P_out < P_in) is not a pump — use a 'valve'.

Hand check (water: $v \sim 1.8\text{e-}5 \text{ m}^3/\text{mol}$): $F = 36 \text{ kmol/h} = 10 \text{ mol/s}$ $Q_{\text{vol}} = 10 * 1.8\text{e-}5 = 1.8\text{e-}4 \text{ m}^3/\text{s}$
 $dP = 30 - 1 \text{ bar} = 2.9\text{e}6 \text{ Pa}$ $W_{\text{shaft}} = Q_{\text{vol}} * dP / \eta = 1.8\text{e-}4 * 2.9\text{e}6 / 0.65 \sim 803 \text{ W} \sim 0.80 \text{ kW}$

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.01
stream	out	F	0.01
kpi	pump02	F	0.01
kpi	pump02	P_in	100000
kpi	pump02	P_out	3000000
kpi	pump02	T_in	298.15
kpi	pump02	T_out	298.523735099
kpi	pump02	W_hydraulic	524.03
kpi	pump02	W_shaft	806.2
kpi	pump02	W_shaft_kW	0.8062
kpi	pump02	dP	2900000
kpi	pump02	dT	0.373735099338
kpi	pump02	eta	0.65
kpi	pump02	head_m	296.620531272

... and 1 more reference values in the case's *expected*.

turbine01_air

tutorials/steady/rotating/turbine01_air



What it does: Adiabatic gas turbine: extract 10 kW from hot air, eta 0.85 — P_{out} / T_{out} are RESULTS

The story: The mirror of compressor01. Hot compressed air (10 bar, 900 K) drives an adiabatic turbine; the shaft extracts 10 kW with isentropic efficiency 0.85.

Hardware (the only things you specify): W_{shaft} = -10 kW shaft power extracted (NEGATIVE = out) eta = 0.85 isentropic efficiency

Sizing note: at 1 mol/s and 900 K inlet, the enthalpy available down to ambient bounds the extractable work to ~15 kW. The 10 kW asked for here expands the gas to ~1.32 bar (ratio ~7.6); asking for much more (e.g. 30 kW) would drive the isentropic discharge below 0 K — physically impossible, and the solver says so.

RESULTS the simulator computes (this run): P_{out} ~1.32 bar the discharge pressure the gas expands to T_{out} ~583 K the (real) exhaust temperature T_{out_isen} ~524 K the ideal (isentropic) discharge T ratio ~7.59 P_{in} / P_{out} (expansion ratio) W_{generated} +10 kW what the shaft delivers to the grid

Per the credo, P_{out} is a RESULT. If the turbine must exhaust at a fixed back-pressure (a condenser, the atmosphere), wrap the case in a DesignSpec that manipulates \$W_{shaft} until turbine.P_{out} hits it.

What to expect (golden-locked):

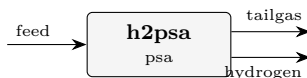
kind	unit/stream	key	expected
stream	exhaust	F	0.001
stream	hotAir	F	0.001
kpi	turbine01	F	0.001
kpi	turbine01	P _{in}	1000000
kpi	turbine01	P _{out}	131729.126973
kpi	turbine01	T _{in}	900
kpi	turbine01	T _{out}	582.935206477
kpi	turbine01	T _{out_isen}	524.463853734
kpi	turbine01	W _{gen_kW}	10
kpi	turbine01	W _{generated}	10000
kpi	turbine01	W _{shaft}	-10000
kpi	turbine01	dS _{gen}	3.19034018436
kpi	turbine01	dT	-317.064793523
kpi	turbine01	dT _{isen}	-375.536146266

... and 2 more reference values in the case's *expected*.

separation

psa01_h2_psa

tutorials/steady/separation/psa01_h2_psa



What it does: H2-PSA: 20-bar syngas (H₂/CO₂/CH₄/CO/N₂) over zeolite 5A → high-purity H₂ raffinate + CO₂-rich tail gas (equilibrium cyclic-steady-state shortcut)

The story: the canonical first PSA case: hydrogen recovery from steam-methane-reformer / shifted syngas.

Feed: a typical PSA feed gas (mol fractions) H₂ 0.70 (the LIGHT product → raffinate) CO₂ 0.18 (the strongest adsorbate → tail gas) CH₄ 0.05 CO 0.04 N₂ 0.03 F = 1000 kmol/h, 20 bar, 313 K.

Adsorbent: zeolite 5A (Ca-LTA), the classic H₂-PSA sieve. CO₂ » CH₄ > CO ~ N₂ » H₂ on the competitive (extended) Langmuir isotherm, so the heavy impurities load onto the bed at P_{high} (20 bar) and are blown down with the tail at P_{low} (1.2 bar), while H₂ barely adsorbs and sweeps through as the high-pressure raffinate product.

Operation: P_{high} 20 bar / P_{low} 1.2 bar swing; isothermal 313 K bed; purgeRatio 0.15 (a Skarstrom-style countercurrent light-product sweep that cleans the regeneration at the cost of product); eta 0.80 (the equilibrium-utilisation / LUB derating — the equilibrium ceiling is eta = 1); bedCapacity 2.0 kg adsorbent per mol of feed throughput (the lumped throughput knob; cycle time / bed volume are deliberately collapsed into it by the equilibrium shortcut).

Expected (this model, golden-checked): H₂ recovery ~ 74 % (single train, with the 0.15 purge sweep) H₂ purity ~ 98-99 % in the raffinate tail gas CO₂-enriched ~ 18 % → ~ 38 %

The run header prints, out loud, what the equilibrium shortcut does NOT model (breakthrough / cycle time, thermal swing, blowdown gas loss, hydraulics, isotherm-validity window) — "all models are wrong, some useful".

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.277777777778
stream	hydrogen	F	0.145276064457
stream	tailgas	F	0.13250171332
kpi	h2psa	Dq_CH4	0.0260273101613
kpi	h2psa	Dq_CO	0.0367785976251
kpi	h2psa	Dq_CO2	1.74280135323
kpi	h2psa	Dq_H2	0.0186248522203
kpi	h2psa	Dq_N2	0.00987823175641
kpi	h2psa	eta_used	0.8
kpi	h2psa	pressureRatio	16.6666666667
kpi	h2psa	productivity	0.0715324198173
kpi	h2psa	purgeRatio	0.15
kpi	h2psa	purity_H2	0.984779152498
kpi	h2psa	recovery_H2	0.735762032407

... and 5 more reference values in the case's *expected*.

solids

The theory you need. Solids carry mass and enthalpy without VLE: crystallisers, dryers, cyclones, gas-solid splitters. A dryer's air is a REAL stream (two inlets, adiabatic mixing, outlet humidity from the psychrometrics); a cyclone's cut size comes from the selected model (Lapple... Muschelknautz). Expect solid streams flagged separately (F_{solid}) and energy balances that close across phase change.

pneumaticConveyor01_sand

tutorials/steady/solids/pneumaticConveyor01_sand

What it does: Dilute-phase pneumatic conveying of sand in nitrogen: the glass-box dP breakdown + the saltation-margin safety check. Particle velocity from a mechanistic FORCE BALANCE (drag = wall friction + gravity); solids friction from Yang (1974, AIChE J 20:605); Rizk (1973) saltation. Loading ~2 kg/kg (in the Yang voidage range); solids friction DOMINATES the dP – conveying solids IS a pressure-drop problem. $u_g/u_{salt} = 1.72$ (safe margin above plugging).

The story: / flowsheetDict (single unit)

Operation block is HARDWARE only (the credo): pipe diameter, length, rise. Every velocity, the suspended density and the pressure drop are RESULTS.

A single pneumaticConveyor unit: silica sand (~200 μm) carried by nitrogen up a 100 mm line, 50 m long, with a 10 m vertical rise.

*The total conveying-line pressure drop as a sum of physical contributions (glass-box) — Rhodes, *Introduction to Particle Technology*, Ch. 8:*

- superficial gas velocity $u_g = 21.9$ m/s, - particle velocity $u_p = 21.1$ m/s (Hinkle slip), - single-particle terminal velocity $u_t = 1.03$ m/s, - saltation velocity $u_{salt} = 3.6$ m/s (Rizk) — u_g is well above it, so transport is stable. The conveyor warns aloud (verbosity ≥ 1) if u_g ever falls below u_{salt} (risk of solids settling and blocking the line) — a first-class, announced design check, never a silent clip.

Solids loading = 0.20 kg solid / kg gas (dilute phase), suspended-solids density $\rho_s = 0.47$ kg/m³.

- particle velocity: Hinkle $u_p = u_g(1 - 0.68 d_p^{0.92} \rho_p^{0.5} \rho_g^{-0.2} D^{-0.54})$, with an honest $u_g - u_t$ fallback if the correlation leaves the dilute regime; - gas friction: Fanning (Blasius turbulent / 16/Re laminar); - solids friction: Hinkle $f_p = (3/8)(\rho_g/\rho_p) C_D (D/d_p)((u_g - u_p)/u_p)^2$, C_D from Schiller–Naumann at the slip Reynolds number; - saltation: Rizk (1973).

Operation block is hardware only (the credo): pipe diameter D , length L , vertical rise dz . Every velocity, the suspended density and the pressure drop are results. Particle density comes from the solid component's solid { ρ_p ; } (silica = 2650 kg/m³); the size from the inlet PSD (Sauter mean for the drag). The carrier-gas viscosity needs a transport { model Chung; } block in the thermoPackage (for the friction Reynolds numbers).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.0138888888889
stream	conveyed	F	0.0138888888889
kpi	node	P_out	234394.335968
kpi	node	accelerationLength	16.8207521299
kpi	node	deltaP	65605.6640315
kpi	node	deltaP_bends	0
kpi	node	deltaP_gasAccel	361.754574941
kpi	node	deltaP_gasFriction	1492.10736304
kpi	node	deltaP_solidsAccel	1383.86777275
kpi	node	deltaP_solidsFriction	61751.3873786
kpi	node	deltaP_static	616.546942159
kpi	node	nBends	0
kpi	node	particleWallFriction	0.4
kpi	node	rho_gas	3.39180174393

... and 10 more reference values in the case's *expected*.

pneumaticConveyor02_bends

tutorials/steady/solids/pneumaticConveyor02_bends

What it does: A SHORT (8 m) conveying line routed through 6 bends – the lesson: A FITTING ADDS PRESSURE DROP WITHOUT ADDING LENGTH. With long-radius sweeps the 6 bends cost ~10% of the total dP; swap them for blind tees (type blindTee;) and their share jumps to ~25% – the erosion-vs-pressure trade made visible. (On this short line the straight-pipe solids friction still carries most of the dP; the bends’ share GROWS as lines get shorter and bendier.) Same gas/solids as pneumaticConveyor01 (which runs 30 m / 6 m rise, bend-free) – compare the PER-METRE dP, not the totals.

The story: a fitting adds pressure drop without adding length

The same sand-in-N₂ service as pneumaticConveyor01 (loading ~2), on a SHORTER line: 100 mm, 8 m, 2 m rise, routed through 6 long-radius bends. (Case 01 runs 30 m / 6 m rise, bend-free – compare dP PER METRE, not totals.) Each bend decelerates the solids to ~0.75 u_p ; the gas re-accelerates them downstream – the same momentum work as the line entry, once per bend: $dP_{bend} = m\dot{S} (u_p - u_{p,bend}) / A$. Run both cases and compare the dP breakdown: the bends add ~10% (long-radius) to ~25% (blind tees) of the dP while adding NO length. Swap a bend ‘type longRadius;’ for ‘type blindTee;’ and watch its re-acceleration cost jump (ratio 0.75 → 0.25) – the blind tee trades pressure for near-zero erosion. Solids friction: Yang (1974); particle velocity: force balance; Rizk saltation.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.01388888888889
stream	conveyed	F	0.01388888888889
kpi	node	P_out	279210.165228
kpi	node	accelerationLength	15.5453372639
kpi	node	deltaP	20789.8347721
kpi	node	deltaP_bends	2070.63576081
kpi	node	deltaP_gasAccel	361.754574941
kpi	node	deltaP_gasFriction	397.89529681
kpi	node	deltaP_solidsAccel	1380.42384054
kpi	node	deltaP_solidsFriction	16373.2628912
kpi	node	deltaP_static	205.862407804
kpi	node	nBends	6
kpi	node	particleWallFriction	0.4
kpi	node	rho_gas	3.39180174393
... and 10 more reference values in the case’s <i>expected</i> .			

thermo

The theory you need. Consistency tests interrogate DATA, not models: the van Ness point test checks measured (x, y, T, P) sets against the Gibbs–Duhem constraint (residuals in $\ln(\gamma_1/\gamma_2)$); Herington’s area test is its integral cousin. A dataset that fails is not “bad” – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

air01_heater_expand

tutorials/steady/thermo/air01_heater_expand



What it does: Predefined mixture: heat a stream fed with ‘components (air)’, which expands to N2/O2/Ar at load time

The story: showcase for TRACK A predefined-mixture expansion.

The thermoPackage lists ‘components (air)’. At load time the loader splices ‘air’ into N2 + O2 + Ar (see constant/propertyDict). A single heater then adds 5 kW to a 1 mol/s air stream at 1 bar, 298.15 K. The air is gas (vf = 1), so the heater integrates the ideal-gas mixture H_ig over the three real components → a correct MULTICOMPONENT result.

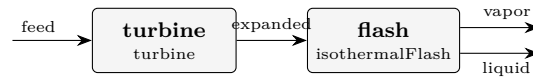
Z-SEEDING NOTE (glass-box honesty): The expansion captures the member mole fractions (N2 0.7808 / O2 0.2095 / Ar 0.0093) as a per-component seed on the ThermoPackage (‘mixtureSeedVector()’). The feed below STATES the same fractions explicitly so the case is self-evident and the result is unambiguous – a stream that declares its own composition always wins over the seed. The seed exists so a future feed that OMITs its composition can be filled from the mixture defaults; that wiring lives in the stream-construction path (Flowsheet / Composition), not in this case.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.001
stream	hot	F	0.001
kpi	airHeater	F	0.001
kpi	airHeater	P	100000
kpi	airHeater	Q	5000
kpi	airHeater	Q_kW	5
kpi	airHeater	Q_per_mol	5000
kpi	airHeater	T_in	298.15
kpi	airHeater	T_out	468.511847495
kpi	airHeater	dT	170.361847495

perUnitThermo01_sr_k_nrtl

tutorials/steady/thermo/perUnitThermo01_sr_k_nrtl



What it does: Per-unit thermo: turbine SRK (real gas) → flash NRTL (non-ideal liquid)

The story: Demonstrates PER-UNIT thermo models: the global package (Raoult) is the default, but each unit can override it with a ‘thermo { ... }’ block that replaces the model it cares about. Components stay global.

turbine → equationOfState SRK (real-gas expansion, high P) flash → activityModel NRTL (non-ideal ethanol/water liquid)

Pedagogical caveat: the stream that crosses from the turbine (SRK) into the flash (NRTL + idealGas) is re-interpreted by a different model – a small thermodynamic inconsistency at the boundary. Real practice keeps one consistent method per section; mixing is an advanced, deliberate act.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	expanded	F	0.0138888888889
stream	feed	F	0.0138888888889
stream	liquid	F	0.00854830793289
stream	vapor	F	0.005340580956
kpi	flash	F_alpha	0.00854830793289
kpi	flash	F_beta	0.005340580956
kpi	flash	F_in	0.0138888888889
kpi	flash	P	100000
kpi	flash	T	353
kpi	flash	V_over_F	0.384521828832
kpi	flash	phaseSet	0
kpi	turbine	F	0.0138888888889
kpi	turbine	P_in	600000
kpi	turbine	P_out	553956.972451

... and 11 more reference values in the case's *expected*.

thermoTest

model1_lumped_evaporator

tutorials/steady/thermoTest/model1_lumped_evaporator



What it does: thermoTest model 1 – NaCl as a LUMPED molecular component: the boiling-point elevation is the IDEAL $K_b \cdot m$ (van’t Hoff), BPE ~ 2.11 K. Compare model2 (same evaporator, Pitzer $a_w \rightarrow$ BPE ~ 5.07 K): the REPRESENTATION axis, lumped vs dissociated, on one operation.

The story: thermoTest model1_lumped_evaporator / flowsheetDict NaCl treated as a LUMPED molecular component: the boiling-point elevation is the IDEAL van’t Hoff $K_b \cdot m$ (~2.11 K). Model 1 of the thermoTest ladder – the deliberately-too-simple baseline the pitzer (model 2) and eNRTL rungs correct. Same evaporator, climbing thermo.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0200845615144
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00769321626341
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	2.11135909531
kpi	evap	F_conc	0.0200845615144
kpi	evap	F_conc_kmol_h	72.3044214517
kpi	evap	F_conc_mass	0.417969209015
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778
...and 19 more reference values in the case's <i>expected</i> .			

model2_pitzer_evaporator

tutorials/steady/thermoTest/model2_pitzer_evaporator



What it does: thermoTest model 2 – NaCl as a PITZER electrolyte (Na+ + Cl-, aqueous-inf-dilution reference): the boiling-point elevation comes from the real water activity a_w , BPE ~ 5.07 K vs the ideal 2.11 K of model1.

The story: BPE NOTE (2026-07-02, a story in three acts): (1) frozen 25 C surface → BPE 5.07 K; (2) LINEAR calorimetric T-slots (Parker fit) → 5.58 K – the 25 C derivative extrapolated straight to the boil; (3) the FULL Silvester & Pitzer 1977 T-functions (eqs 30-32, Table IV verbatim in Na-Cl.dat) + $A_{\phi}(T)$ curated against their Table II → BPE 5.06 K. $\beta_0(T)$ has a MAXIMUM near 150 C (their Fig 4): the linear slope overshoots where the real curve has already bent over. The lesson: a first derivative is not a function – validity windows are data.

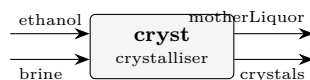
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrated	F	0.0203403273092
stream	condensate	F	0.0125
stream	feed	F	0.0277777777778
stream	saturatedSteam130C	F	0.0125
stream	vapour	F	0.00743745046862
kpi	evap	A	20
kpi	evap	A_m2	20
kpi	evap	BPE	5.0621680882
kpi	evap	F_conc	0.0203403273092
kpi	evap	F_conc_kmol_h	73.225178313
kpi	evap	F_conc_mass	0.422576829808
kpi	evap	F_cond	0.2251875
kpi	evap	F_cond_kg_h	810.675
kpi	evap	F_in	0.0277777777778

... and 19 more reference values in the case's *expected*.

model3_enrtl_antisolvent_crystalliser

tutorials/steady/thermoTest/model3_enrtl_antisolvent_crystalliser



What it does: Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises γ_{pm} , and DROPS the saturation molality m_{sat} – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product K_{sp} anchored in pure water, solved with the mixed-solvent γ_{pm} .

The story: thermoTest/model3_enrtl_antisolvent_crystalliser – step 3 of the thermoTest arc (same {NaCl, CaSO₄, ethanol, water} system, five property models): here the package selects eNRTL MIXED-SOLVENT, the only model in the arc that can reach the ethanol-lowered dielectric. The flowsheet is crystalliser06_nacl_antisolvent's, unchanged:

ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity γ_{pm} (eNRTL mixed-solvent + Born), so the saturation molality m_{sat} COLLAPSES (6.14 → ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product $K_{\text{sp}} = (\gamma_{\text{pm}} * m_{\text{sat}})^2$ anchored in pure water, solved with the mixed-solvent γ_{pm} at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine (x_{water} 0.900, x_{NaCl} 0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol → salt-free $x_{\text{ethanol}} \sim 0.40$. Self-contained: components live in constant/components/.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	brine	F	0.02777777777778
stream	crystals	F	0.000329661969914
stream	ethanol	F	0.01666666666667
stream	motherLiquor	F	0.0424240933568
kpi	cryst	Ksp_activity	34.8439846653
kpi	cryst	Q_kW	-6.55987377671
kpi	cryst	Q_removed	6559.87377671
kpi	cryst	T_op	298.15
kpi	cryst	antisolvent_xfrac	0.399920015997
kpi	cryst	c_sat	0.0517444790334
kpi	cryst	cakeLiquidFlow	0.000329661969914
kpi	cryst	cakeMoisture	0.1
kpi	cryst	cakeWetness_pct	9.09090909091
kpi	cryst	crystal_flow	0.0988038720389

... and 9 more reference values in the case's *expected*.

model5_nrtl_flash

tutorials/steady/thermoTest/model5_nrtl_flash



What it does: thermoTest model 5 – water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model. This is the antisolvent (ethanol) and the solvent (water) in their own role – the same substances that dissolve/precipitate NaCl in models 2-4, here separated by distillation.

The story: thermoTest model5_nrtl_flash / flowsheetDict Water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model, selected through the ethanolWater_NRTL property package. Model 5 of the thermoTest ladder – the molecular gamma-phi anchor beside the electrolyte rungs.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.02777777777778
stream	liquid	F	0.0170966158658
stream	vapor	F	0.010681161912
kpi	flash	F_alpha	0.0170966158658
kpi	flash	F_beta	0.010681161912
kpi	flash	F_in	0.02777777777778
kpi	flash	P	100000
kpi	flash	Q	0
kpi	flash	Q_kW	0
kpi	flash	T	353
kpi	flash	V_over_F	0.384521828832
kpi	flash	phaseSet	0

userops

The theory you need. User-defined operations: the extension seam. A custom property op or unit compiled into the binary via the explicit factory – the tutorial shows the registration pattern (`registerBuiltins`) end to end.

userOp01_yield_reactor

tutorials/steady/userops/userOp01_yield_reactor



What it does: User-defined yieldReactor (case-local code/): neoPentane → nPentane, X=0.8

The story: A single unit op — but a USER-DEFINED one ('yieldReactor'), written in this case's code/ folder, not in the Choupo source tree. Build + run it with:

bin/buildCase tutorials/steady/userOp01_yield_reactor

(NOT runCase: the type 'yieldReactor' does not exist in the stock binary; buildCase compiles the case's code/ against libchoupo.a.)

The op converts a SPECIFIED fraction of the reactant to the product (no kinetics) — a "yield reactor", which the built-in CSTR/PFR are not. Here: 80 % of neoPentane isomerised to nPentane. Because the two are isomers (same MW), the mass balance closes to 100 %.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	feed	F	0.000277777777778
stream	out	F	0.000277777777778
kpi	reactor	F_in_kmol_h	1
kpi	reactor	F_out_kmol_h	1
kpi	reactor	T	320
kpi	reactor	X	0.8
kpi	reactor	converted_kmol_h	0.8

userOp02_component_splitter

tutorials/steady/userops/userOp02_component_splitter



What it does: User-defined componentSplitter (case-local code/): shortcut benzene/toluene split

The story: A USER-DEFINED unit op ('componentSplitter'), written in this case's code/ folder. Build + run with:

runCase tutorials/steady/userOp02_component_splitter

(runCase sees the code/ folder and compiles it first.)

The op is a SHORTCUT separator: a per-component recovery to the overhead. Here a benzene/toluene feed is split so 95 % of the benzene and only 8 % of the toluene leave overhead — an enriched light cut, the kind of stand-in you place before sizing a real column. Two outlet streams (overhead, bottoms); mass closes to 100 % (every component is conserved across the split).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	bottoms	F	0.0134722222222
stream	feed	F	0.0277777777778
stream	overhead	F	0.0143055555556
kpi	splitter	F_bottoms_kmol_h	48.5
kpi	splitter	F_overhead_kmol_h	51.5
kpi	splitter	overhead_fraction	0.515

utilities

The theory you need. Utility streams (steam levels, cooling water, refrigerants, electricity) carry duty into and out of the plant at declared temperature levels and prices; allocation is by T -level feasibility, announced.

utility01_dowtherm_preheat

tutorials/steady/utilities/utility01_dowtherm_preheat



What it does: Dowtherm A loop preheats a hydrocarbon feed (eps-NTU)

The story: - Choupo -*—————* utility01_dowtherm_preheat

A hydrocarbon feed (50/50 n-hexane / n-octane) needs to be heated to ~530 K before downstream conversion. HP steam can deliver at most ~525 K (T_{sat} at 40 barg) – not enough; the next escalation up the temperature ladder is a hot-oil loop, the textbook choice in the 250-400 C range being Dowtherm A (delivered here at 623 K).

NOTE: the 25 m² / $U=600$ exchanger is deliberately generous – $NTU=13.8$, $C_r=0.06$ give effectiveness ~ 1.00 , so the cold outlet is PINNED to the 623 K oil inlet, OVERSHOOTING the 530 K target by ~93 K. The case shows a pinned-effectiveness exchanger; shrink the area to land near 530 K.

Hardware (the credo): area + U + flow direction. The duty Q , the outlet temperatures, and the Dowtherm flowrate fall out of the effectiveness-NTU solution.

Process : 40 kmol/h at 320 K, 5 bar (mixed C6 + C8 hydrocarbon) Utility : Dowtherm A loop, delivery 623 K (350 C), ‘utility dowthermA;’ pulls the rest of the conditions from the catalogue Hardware: $A = 25 \text{ m}^2$, $U = 600 \text{ W}/(\text{m}^2\cdot\text{K})$, counter-current

The ‘utilities’ report aggregates the hot-side mass flow and emits reports/utilities/consumption.csv with MW and EUR/h delivered. The inspection question for the student: how much Dowtherm A do we actually consume? At what monetary cost per hour?

utility02_hitec_csp_heater

tutorials/steady/utilities/utility02_hitec_csp_heater



What it does: HITEC molten salt loop sensibly heats pressurised (40 bar) boiler feedwater to ~244 C, just below T_{sat} , for a downstream steam generator (eps-NTU, no phase change), ~208 kW duty

The story: - Choupo -*—————* utility02_hitec_csp_heater

When the duty needs a hot side above Dowtherm A's ceiling (~370 C), the next escalation up the ladder is a molten salt loop – HITEC is the textbook ternary ($\text{NaNO}_3\text{-NaNO}_2\text{-KNO}_3$) used for solar-thermal storage and high-T process heat down to its 142 C melting point.

Process here: pressurised water at 40 bar pre-heated towards its T_{sat} (~250 C) for a downstream HP steam generation step. Hot side: HITEC at 773 K (500 C) delivery, 'utility hitecSalt;' pulls the loop conditions from data/standards/utilities/hitecSalt.dat.

Hardware: $A = 0.85 \text{ m}^2$, $U = 800 \text{ W}/(\text{m}^2\cdot\text{K})$, counter-current. The Q and T_{out} fall out of eps-NTU. The area is deliberately sized so the cold outlet lands at ~517 K (244 C) – a sensible pre-heat that stops JUST below $T_{\text{sat}}(40 \text{ bar}) \approx 523.5 \text{ K}$ (250 C): the water leaves as a hot liquid ($vf = 0$), no phase change, exactly as a feedwater pre-heater ahead of the steam generator should. Oversize the area and the cold outlet would try to cross the saturation dome – a separate boiler's job, not this sensible-only eps-NTU unit's.

Inspection: how much HITEC do we circulate to deliver the duty? At what hourly cost? Compare to the equivalent steam-generated route via fired heater – the salt loop trades higher Capex for cleaner T-control and the absence of decomposition byproducts in the process side.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	boilerFeed	F	0.0222222222222
stream	boilerOut	F	0.0222222222222
stream	boilerOut	T	517.400926398
stream	saltReturn	F	0.0833333333333
stream	saltSupply	F	0.0833333333333
kpi	boilerHX	C_cold	1677.77777778
kpi	boilerHX	C_hot	12795.4797468
kpi	boilerHX	C_r	0.131122694184
kpi	boilerHX	LMTD	306.566828203
kpi	boilerHX	NTU	0.405298013245
kpi	boilerHX	Q	208465.443178
kpi	boilerHX	Q_kW	208.465443178
kpi	boilerHX	T_cold_in	393.15
kpi	boilerHX	T_cold_out	517.400926398

... and 6 more reference values in the case's *expected*.

utility03_glycol_brine_chiller

tutorials/steady/utilities/utility03_glycol_brine_chiller



What it does: PG-30 brine sensibly cools an ethanol vapour stream from 330 K to ~273 K (eps-NTU, no phase change), ~20.8 kW duty

The story: - Choupo -*—————* utility03_glycol_brine_chiller

When the process T sits below the freezing margin of plain chilled water (~5 C), the next escalation DOWN the temperature ladder is a secondary brine loop – propylene-glycol-30 protects sub-zero coils in food, pharma and freeze-cryst applications.

Process: a stream of ethanol vapour from the top of a fermentation column, to be sensible-cooled towards 285 K (~12 C). Utility: PG-30 brine, ‘utility refrigerationPG;’ pulls the loop conditions from data/standards/utilities/refrigerationPG.dat (delivery 0 C, return 5 C).

Hardware: $A = 8 \text{ m}^2$, $U = 350 \text{ W}/(\text{m}^2\cdot\text{K})$ (lower U on the vapour side is realistic for a film condenser-style geometry), counter-current.

Inspection: how much brine flow is required? At 6 EUR/GJ the refrigeration cost is sharply higher per GJ than cooling water – is it justified by the process T? Compare with what cooling water alone could deliver.

Category II: The property bench (choupoProps)

Questions asked of the thermodynamics BEFORE a simulation uses it – CREATE, SEE, TEST, FIT, PROMOTE.

adsorption

The theory you need. The isotherm bench: evaluating a curated adsorption-equilibrium record (Henry-limit, saturation, pin and unit-invariance gates run aloud) and REGRESSING one from data – the multi-temperature van't Hoff Langmuir fit with identifiability verdicts: single- T data refuses ΔH_{ads} , Henry-regime data yields only the product $q_{\text{max}}b$, and a non-converged fit refuses its verdict by name. *Full derivation: Theory Guide, the adsorption chapter.*

isotherm01_eval_13x_co2

tutorials/props/adsorption/isotherm01_eval_13x_co2

eval_13x_co2
 isothermEval

What it does: Adsorption isotherm bench: CO2/zeolite13X q(T,p) grid + Henry/saturation/pin/unit-invariance gates

The story: One isothermEval operation: the curated CO2/zeolite13X langmuir record on a 3-T x 4-p grid (K, Pa – grid values are RAW canonical SI), with the unit-invariance twin (gate d) declared case-locally in constant/. Temperatures match the Jaschik 2020 dataset the FIT tutorial (isotherm02_fit_jaschik) regresses – eval here, fit there.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	eval_13x_co2	T0	2.931e+02
diag	eval_13x_co2	T1	3.131e+02
diag	eval_13x_co2	T2	3.331e+02
diag	eval_13x_co2	b_T0	1.227e+01
diag	eval_13x_co2	b_T1	3.772e+00
diag	eval_13x_co2	b_T2	1.337e+00
diag	eval_13x_co2	henry_T0	6.133e-04
diag	eval_13x_co2	henry_T1	1.886e-04
diag	eval_13x_co2	henry_T2	6.683e-05
diag	eval_13x_co2	qsat	5.000e+00
diag	eval_13x_co2	tRef	2.981e+02
diag	eval_13x_co2	gate_henry	1.000e+00
diag	eval_13x_co2	gate_henry_maxdev	1.227e-07
diag	eval_13x_co2	gate_saturation	1.000e+00

*...and 7 more reference values in the case's **expected**.*

isotherm02_fit_jaschik

tutorials/props/adsorption/isotherm02_fit_jaschik

fit_jaschik
fitParametersrecovery_multiT
fitParametersgate_singleT
fitParametersgate_henry
fitParameters

What it does: Isotherm regression: CO₂/13X Jaschik 2020 (CC BY) multi-T Langmuir fit + recovery/single-T/Henry identifiability gates

The story: Four fitParameters(kind isotherm) operations: the REAL Jaschik fit plus the three identifiability gates on zero-noise synthetics generated from a KNOWN surface (q_{max} 4.6 mol/kg, b_{ref} 3.0e-4 1/Pa @ 298.15 K, dH_{ads} -38000 J/mol) – so every gate has an exact truth to recover or to honestly refuse. Inline synthetic data is canonical SI (K, Pa, mol/kg); the vendored Jaschik dataset declares its own units (bar, mol/kg) and is converted at the data boundary.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	fit_jaschik	converged	1.00000e+00
diag	fit_jaschik	identifiable	1.00000e+00
diag	fit_jaschik	n_data	1.32000e+02
diag	fit_jaschik	n_params	3.00000e+00
diag	fit_jaschik	dof	1.29000e+02
diag	fit_jaschik	single_T	0.00000e+00
diag	fit_jaschik	henry_regime	0.00000e+00
diag	fit_jaschik	endothermic_suspicious	0.00000e+00
diag	fit_jaschik	fit.q_max.value	4.31448e+00
diag	fit_jaschik	fit.q_max.stderr	5.17246e-02
diag	fit_jaschik	fit.b_ref.value	4.06758e-04
diag	fit_jaschik	fit.b_ref.stderr	2.90989e-05
diag	fit_jaschik	fit.dH_ads.value	-3.73835e+04
diag	fit_jaschik	fit.dH_ads.stderr	1.90453e+03

...and 32 more reference values in the case's *expected*.

isotherm03_fit_refusals
tutorials/props/adsorption/isotherm03_fit_refusals

gate_henry_nonconverged fitParameters	gate_nonconverged fitParameters	gate_langfull_nonconverged fitParameters
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What it does: Fit refusal gates: non-converged fits refuse their verdicts by name

The story: Two ADVERSARIAL fitParameters(kind isotherm) operations – see the controlDict header. Synthetic data generated EXACTLY from the known surface (q_max 4.6 mol/kg, b_298 3.0e-4 1/Pa, dH_ads -38000 J/mol, tRef 298.15 K) – zero noise, so the ONLY failure mode under test is the strangled optimiser, never the data.

compare

The theory you need. Method comparison runs the SAME property through different models side-by-side – UNIFAC vs NRTL, Joback vs measured, ideal vs EOS. The gap between curves IS the modelling uncertainty; picking a method is a decision the case author owns, made with eyes open. *Full derivation: Theory Guide, the Props Guide.*

compare_activity_etoh_water

tutorials/props/compare/compare_activity_etoh_water



What it does: Compare 3 activity models (ideal / NRTL / Wilson) for γ_{ethanol} in equimolar mixture with water vs T

The story: COMPARE 3 activity-coefficient models for γ_{ethanol} in an equimolar ethanol/water mixture, T = 280 K to 360 K, P = 1 atm.

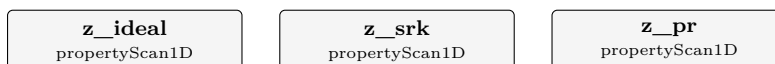
ideal : $\gamma = 1$ always (Raoult's law assumption). NRTL : strong non-ideality, $\gamma_{\text{ethanol}} > 1$ (azeotrope behaviour). Wilson : similar to NRTL for this binary, slightly different curve.

Pedagogical point: a student studying ethanol/water VLE will see that "ideal" is a bad approximation here (γ_{ethanol} grows to ~2-3 at low T), and that NRTL/Wilson give similar non-ideal corrections. The choice between the two is then decided by which fits the project's experimental data better (workflow Mode B, see tutorials/props/README.md).

Each op overrides the global activityModel block via 'thermo {}'. The three CSVs (gamma_ideal.csv, gamma_nrtl.csv, gamma_wilson.csv) are auto-overlaid by the GUI.

compare_eos_co2

tutorials/props/compare/compare_eos_co2



What it does: Compare 3 EoS (idealGas / SRK / PR) for Z of CO2 near critical ($T=310$ K, P from 1 to 100 bar)

The story: COMPARE 3 equations of state for the same property at the same conditions. This is what choupProps is **for**: helping a student or researcher CHOOSE the model that best describes the system at hand — not just look at pretty Z curves of one EoS at a time.

System: pure CO2 at $T = 310$ K (just above its critical temperature $T_c = 304.13$ K), pressure swept from 1 bar to 100 bar. Near the critical pressure ($P_c = 7.38$ MPa) the compressibility plunges as the fluid turns liquid-like; the three curves read (at $P = 73.7$ bar, the swept point nearest P_c):

idealGas: $Z = 1$ always (textbook reference; ignores attraction). SRK : $Z = 0.538$ at P_c . PR : $Z = 0.509$ at P_c — systematically lower than SRK (more attraction), and essentially on the experimental value. NIST : the **experimental** Z near the critical point is ~ 0.49 .

The lesson: at and just above T_c , BOTH cubics track CO2 well — PR sits almost exactly on the ~ 0.49 datum and SRK is only ~ 0.05 high. This is a regime where cubic EoS are at their best, not their worst. Both Z curves keep falling past P_c to a shallow minimum (SRK ~ 0.30 , PR ~ 0.275 near 98 bar) before rising again; the SRK-vs-PR gap is the diagnostic a student should read off the overlay.

Each operation overrides the global thermoPackage with a different EoS via the 'thermo {}' per-op block. The three CSVs are independent, so the GUI overlays them on the same Plotly canvas (same X axis "P", same Y "Z"). The result is one figure with three curves, ready for the project report.

compare_kinetics_order

tutorials/props/compare/compare_kinetics_order

rate_order1
kinetics1D**rate_order2**
kinetics1D

What it does: Rate-law order from DATA: 1st- vs 2nd-order fits overlaid on measured $c(t)$ — the data picks the order

The story: The "see, then decide" of CHEMICAL KINETICS: fit a 1st- and a 2nd-order rate law to the SAME measured $c(t)$ and overlay both on the data. No 'k' is given, so each op REGRESSES its own k from the points (provenance = fitted). The data here are second-order – the 1st-order curve misses the tail, the 2nd lands on the points. The student picks the order from the evidence.

compare_psat_ambrose_walton

tutorials/props/compare/compare_psat_ambrose_walton

psat_aw_vs_antoine
propertyScan1D

What it does: Psat: Ambrose-Walton corresponding-states PREDICTION vs fitted Antoine DATA, for benzene (non-polar, good) and ethanol (associating, degrades) — see the error, then decide

The story: Ambrose-Walton corresponding states (Ambrose & Walton 1989; Poling et al. 5th ed., Eq. 7-2.7) predicts vapour pressure from T_c , P_c and omega ALONE – exactly the constants a Joback + Lee-Kesler estimate produces. So an estimated component becomes flashable with NO measured Psat data. But it is an ESTIMATE, and the glass-box rule is: SEE the error before you trust it.

This sweeps $T = 270 \rightarrow 360$ K and tabulates four curves: Psat_benzene – fitted Antoine (catalogue DATA) } non-polar pair Psat_benzeneAW – Ambrose-Walton PREDICTION } Psat_ethanol – fitted Antoine (catalogue DATA) } associating pair Psat_ethanolAW – Ambrose-Walton PREDICTION }

Open ‘psat_aw_vs_antoine.csv’ (GUI plot, or gnuplot). What you SEE: Near the normal boiling point (~350-360 K) BOTH agree to ~1 % – corresponding states is anchored there and is excellent. Towards the cold end (270 K) the curves separate, and MORE for the associating ethanol than for non-polar benzene: benzene AW vs Antoine: +3.3 % average, +8.9 % at 270 K ethanol AW vs Antoine: +5.2 % average, -16 % at 270 K So corresponding states is a solid FREE first predictor (it makes an estimated component flashable with no data), best near T_b and degrading at low reduced temperature – the degradation worse for hydrogen-bonding species. The lesson: prediction gets you started; for an associating species, or far below T_b , fit Antoine to data (choupoProps vaporPressureFit) before any design number.

compare_transport_water

tutorials/props/compare/compare_transport_water

mu_iapws
propertyScan1Dmu_generic
propertyScan1Dk_iapws
propertyScan1Dk_generic
propertyScan1D

What it does: Compare water transport: IAPWS (IF97 R12-08/R15-11) vs generic correlations (Andrade / SatoRiedel), 280-370 K

The story: COMPARE water transport properties computed TWO WAYS on the SAME water, over the saturated-liquid window $T = 280\text{--}370\text{ K}$:

(a) IAPWS reference – water flagged ‘pureFluids { water { method IF97; } }’ so the transport accessors route to the matching IAPWS releases: viscosity $\rightarrow$ R12-08 (industrial) thermal conductivity $\rightarrow$ R15-11 (background term) evaluated at the saturation state (the legacy accessors carry no P, so $p = p_{\text{sat}}(T)$ – saturated liquid). These are THE international standard reference values (verified to $1\text{e-}8$ by IAPWSTransport::verify(), which fires on every run).

(b) generic correlations – the same property KEY, but with a generic transport block and NO pureFluids flag, so the accessors fall to: viscosity $\rightarrow$ Andrade $\ln(\mu) = A + B/T$ thermal conductivity $\rightarrow$ SatoRiedel fitted from the per-component .dat (T_c/T_b + the Andrade pair).

Because BOTH ops emit the same property key (‘viscosity_liquid’, ‘thermal_conductivity_liquid’) the two curves overlay directly – the student SEES where the cheap generic correlation deviates from the IAPWS reference.

THE LESSON (the actual computed gaps): VISCOSITY – Andrade tracks IAPWS within a few percent over the window ($\sim 7\%$ at 280 K, $\sim 2\text{--}4\%$ mid-range, $\sim 3\%$ at 370 K). The 2-parameter $\ln(\mu)=A+B/T$ cannot follow the full curvature, but a few % is fine for a Reynolds number; it is NOT good enough for a precision steam duty. CONDUCTIVITY – SatoRiedel is a corresponding-states estimate and it FAILS BADLY for water: it underpredicts IAPWS by $\sim 50\text{--}60\%$ across the window. Water is the textbook SatoRiedel failure (strong hydrogen bonding, anomalous k); this is exactly why "SEE before you commit" matters – a blind generic pick would have halved the heat-transfer coefficient. For water conductivity use IAPWS R15-11, full stop. Generic correlations are CHEAP and GENERAL (any compound with the handful of .dat parameters); IAPWS is EXACT but water-only. For a water-dominated process use the IF97 + IAPWS route; for a screening sweep over many fluids the generic correlations are the right tool. SEE the gap, then decide – do not pick blindly.

The GUI renders the CSVs via the existing compare machinery (overlay of the *_iapws / *_generic curves vs T) – no new Explorer kind.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	mu_iapws	n_failures	0
diag	mu_iapws	n_points	46
diag	mu_generic	n_failures	0
diag	mu_generic	n_points	46
diag	k_iapws	n_failures	0
diag	k_iapws	n_points	46
diag	k_generic	n_failures	0
diag	k_generic	n_points	46

compare_visc_liquid_ammonia

tutorials/props/compare/compare_visc_liquid_ammonia

mu_andrade
propertyScan1D

What it does: Liquid-viscosity Andrade for ammonia, 235-300 K (cross-check μ @239.85K $\sim$ 0.255 mPa.s)

The story: Liquid-viscosity Andrade correlation for pure ammonia (NH₃), 235-300 K. Coefficients live in data/standards/components/NH3.dat (PROVISIONAL — a 2-point Andrade fit; see that file's provenance block).

Validation anchor: $\mu_{\text{NH}_3(\text{liquid})}$ @ 239.85 K (normal b.p., -33.3 degC) is approx 0.255 mPa.s. This 2-point PROVISIONAL Andrade fit gives $\sim$ 0.277 mPa.s at 240 K – about 9% ABOVE the anchor, i.e. it does NOT yet pass through the cross-check point. That gap is the honest state of a provisional fit (see the provenance block in NH3.dat); refitting against more low-T data would close it. Note this case scans a SINGLE method (Andrade) – it is a single-correlation curve, not a multi-method comparison. Ammonia is liquid only under pressure in this T window; viscosity_liquid is a pure-T correlation, so the state P is immaterial to the value.

compare_visc_liquid_water

tutorials/props/compare/compare_visc_liquid_water

mu_andrade
propertyScan1D**mu_vogel**
propertyScan1D

What it does: Compare 2 liquid viscosity correlations (Andrade vs Vogel) for water, 280-360 K

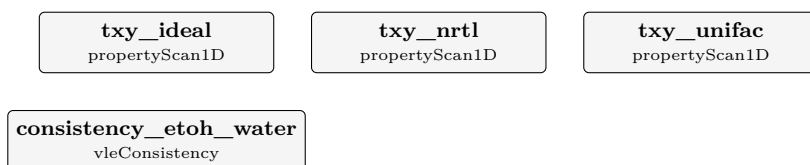
The story: COMPARE 2 liquid-viscosity correlations for pure water, $T = 280\text{-}360\text{ K}$.

Andrade: $\ln(\mu) = A + B/T$ — 2-parameter, simplest fit, OK over a narrow T window. Vogel : $\ln(\mu) = A + B/(T + C)$ — 3-parameter, captures the curvature better, recommended for wider T ranges.

Pedagogical point: for a project using water transport in a non-trivial T window (e.g. $25\text{-}90\text{ }^{\circ}\text{C}$ for a heated reactor), the student should know WHICH correlation he is committing to. Both correlations give viscosity FALLING with temperature (the physical trend): $\sim 1.3\text{-}1.4\text{ mPa}\cdot\text{s}$ near 280 K down to $\sim 0.32\text{ mPa}\cdot\text{s}$ at 360 K. This case shows the two models disagree MOST at the cold end ($\sim 6\%$ at 280 K), are within $\sim 2\%$ at room T , and agree BEST at the hot end ($\sim 1\%$ at 360 K) — the 2-parameter Andrade fit drifts where it is extrapolated coolest. Validate against the textbook value: $\mu_{\text{water}} @ 298\text{ K} \approx 8.9 \times 10^{-4}\text{ Pa}\cdot\text{s}$ — Vogel reads 8.93×10^{-4} (essentially on it), Andrade 9.09×10^{-4} ($\sim 2\%$ high).

compare_vle_etoh_water

tutorials/props/compare/compare_vle_etoh_water



What it does: Ethanol/water T-x-y at 1 atm: ideal vs NRTL model curves OVERLAID with measured VLE data — see the data, pick the model

The story: The canonical "SEE, then DECIDE" props case: ethanol/water T-x-y at 1 atm, with two MODEL curves and the MEASURED data on ONE plot.

Piece 1 – two scan operations, identical but for the activity model: `txy_ideal` Raoult ($\gamma = 1$) → a smooth lens, NO azeotrope `txy_nrtl` NRTL → the real minimum-boiling azeotrope Each emits the (x, T_bubble, y_eq) trio → the T-x-y / x-y / T-x / T-y toggle (x-y is the McCabe-Thiele diagram).

Piece 2 – the ‘experimental {}’ overlay: lists the two ops as ‘models’, and points ‘dataset’ at REAL measured points (constant/experiments/...). The GUI draws both model curves + the lab points together; the student SEES that the ideal model misses the azeotrope and NRTL captures it, and picks NRTL from evidence – not from a recommended-by badge.

electrolyte

The theory you need. Ions interact at long range (Debye–Hückel, $\sqrt{I}$ law) and short range (ion–ion, ion–solvent). Choupo’s ladder: Davies (charge-only, $I \leq 0.5$), single-salt Pitzer ($\beta_0, \beta_1, C^\varphi$; to $I \sim 6$, and with the Silvester–Pitzer T -functions to 300°C), multi-ion Pitzer–HMW (per-ion, mixing terms θ, ψ – seawater-class brines), and eNRTL (local-composition; single-salt, mixed-solvent, and the 2009 symmetric multi-salt with both published and Gibbs–Duhem-consistent formulations). Speciation solves mass action + charge balance by Newton in log-molality space; saturation indices $SI = \log(IAP/K_{sp})$ flag scaling. Expect: models to DISAGREE above their validity (γ divergence is the lesson, e.g. Davies vs Pitzer at RO recoveries), and every activity model to announce itself. *Full derivation: Theory Guide, the electrolyte + speciation chapters.*

enrtl_mixed_nacl_ethanol_esteso

tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso

esteso
 enrtlMixedSolvent

What it does: VALIDATION of the generalized segment-based eNRTL (Chen & Song 2004, AIChE J 50:1928, DOI 10.1002/aic.10151) against the PRIMARY measured $\gamma_{\pm}$ of NaCl in ethanol-water at 25 C (Esteso et al., J. Solution Chem. 18 (1989) 277, emf cell, 0-90 wt% ethanol). ZERO refit: every $\tau_{X:Z}$ is the published Table 2/4/7 value; A_{ϕ} per composition is Esteso's measured Table II. Headline AAD ~4.3% over 47 points (paper quotes RMS 0.026 on $\gamma_{\pm}$); ~1% in pure water, ~1.2% at 70 wt%. The +8% at 80/90 wt% is DECLARED physics the complete-dissociation model cannot carry: Esteso measures ion association $K_A = 30\text{--}46$ there (his Table V).

The story: can eNRTL predict a salt's activity in the antisolvent mixture?

The SAME physics the antisolvent crystalliser leans on – $\gamma_{\pm}$ of NaCl as ethanol pushes the water out – validated against the PRIMARY dataset:

MODEL Chen & Song, "Generalized electrolyte-NRTL model for mixed-solvent electrolyte systems", AIChE J 50(8):1928-1941 (2004), DOI 10.1002/aic.10151. Ethanol = 1.811 (-C₂H₄-) + 0.609 (-OH) segments (Table 4); $\text{ion} \leftrightarrow \text{OH}$ τ s = the aqueous NaCl pair (Table 2); $\text{ion} \leftrightarrow \text{C}_2\text{H}_4$ τ s from Table 7. ZERO refit here. DATA Esteso, Gonzalez-Diaz, Hernandez-Luis & Fernandez-Merida, "Activity coefficients for NaCl in ethanol-water mixtures at 25 C", J. Solution Chem. 18(3):277-288 (1989). Table I (emf, molal scale, per-solvent reference). A_{ϕ} per composition = Esteso Table II (measured dielectric + density). 0 wt% gate: Robinson & Stokes, Electrolyte Solutions, 2nd ed. (1959), NaCl appendix table; $A_{\phi}(\text{water}, 25\text{ C}) = 0.391$.

WHAT THE NUMBERS SAY (locked by the golden): total AAD ~4.3% over 47 points; pure water ~1.0%; 70 wt% ethanol ~1.2%. 80/90 wt% sit +8% systematically – Esteso's own chemical-model fit finds ion-pair association $K_A = 30\text{--}46$ there (Table V), physics a complete- dissociation model cannot represent. Declared, not tuned away.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	esteso	AAD_total_pct	4.3
diag	esteso	AAD_wt0_pct	1
diag	esteso	AAD_wt20_pct	5.3
diag	esteso	AAD_wt40_pct	5.3
diag	esteso	AAD_wt60_pct	2.7
diag	esteso	AAD_wt70_pct	1.2
diag	esteso	AAD_wt80_pct	7.8
diag	esteso	AAD_wt90_pct	7.8
diag	esteso	n_points	47

enrtl_multisalt_nacl_kcl

tutorials/props/electrolyte/enrtl_multisalt_nacl_kcl

harned_nacl_kcl enrtlMultiSalt	harned_nacl_kcl_refined enrtlMultiSalt
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What it does: Multi-salt symmetric eNRTL (Song & Chen 2009) – NaCl+KCl Harned sweep at 4 mol/kg

The story: THE MULTI-SALT eNRTL ARRIVES: the multicomponent symmetric electrolyte NRTL of Song & Chen 2009 (Ind. Eng. Chem. Res. 48, 7788, DOI 10.1021/ie9004578) on the classic Harned experiment – NaCl + KCl sharing water at a CONSTANT 4 mol/kg total molality, $\gamma_{\pm}$ of BOTH salts as KCl replaces NaCl. This is the paper's own demonstration system (its Figures 1-2).

What the run shows (all golden-locked): $\gamma_{\pm}(\text{NaCl})$ slides from its pure-solution value to its TRACE value in 4 m KCl, and vice versa – both $\ln \gamma$ curves are NEARLY LINEAR in the salt fraction: Harned's rule, EMERGING from the model rather than being assumed ($R^2 > 0.99$ with zero electrolyte-electrolyte parameters). The PURE endpoints reproduce the validated single-salt eNRTL kernel exactly (the reduction pin $\sim 1\text{e-}16$): the multicomponent formulation collapses correctly. The pair parameters are the EXACT Chen & Evans 1986 Table 1 values the paper uses (case-local constant/electrolyte/enrtl.dat, cited). The standards K-Cl.dat carries an INDEPENDENT refit to Hamer & Wu 1972 that lands within 0.2 % of the 1986 values – two roads, one answer. The 1986 paper itself (Table 2) validates $\tau_{\text{EE}} = 0$ for NaCl-KCl: $\sigma = 0.019$ % on the Covington 1968 mixture osmotic data with the salt-salt energies set to zero – this case's default is the canonical choice. HONESTY, printed by the op: the published multi-electrolyte eNRTL activity-coefficient expressions are NOT exactly Gibbs-Duhem consistent (the composition-dependent mixing rules are differentiated as constants). The op's FD oracle quantifies that inconsistency at the mid-sweep state ($\text{gibbsDuhem_dev_mid}$) next to the pure-salt state where the same oracle sits at $\sim 1\text{e-}9$. THE REFINED FORMULATION SHIPS TOO (second op below): γ as the EXACT derivative of the same G_{ex} – the chain rule kept through the composition-dependent mixing rules (the Bolas, Chen & Barton 2008 idea on the symmetric form). Its oracle closes ($\sim 1\text{e-}9$) at every composition; both reduce EXACTLY to the same single-salt kernel at the endpoints. Mid-sweep they differ by $|\text{dln } \gamma_{\pm}| \sim 0.037$ (~ 4 % in γ), and the trace limits differ visibly (NaCl trace 0.608 published vs 0.667 consistent). WHICH is closer to nature is a question for MIXTURE data (Covington 1968 / Robinson 1961) – this case states the facts and locks both; it does not pick a winner without measurements.

τ_{EE} (the same-anion electrolyte-electrolyte parameter $\tau_{\text{NaCl,KCl}}$ of the paper's Figures 1-2) defaults to 0; set it to ± 1 to reproduce the paper's sensitivity fan.

THE EXPERIMENTAL VERDICT (Robinson 1961, J. Phys. Chem. 65, 662, DOI 10.1021/j100822a016 – isopiestic NaCl-KCl mixtures, Harned coefficients at $M \sim 4.4$ mol/kg: $\alpha_{\text{NaCl}} = +0.0243$, $\alpha_{\text{KCl}} = -0.0091$; the op now emits harned_alpha_1/2 on the same convention, golden-locked): α_{NaCl} : published $+0.0276$ ($+14$ %) consistent $+0.0173$ (-29 %) α_{KCl} : published -0.0242 (2.7x) consistent -0.0193 (2.1x) With $\tau_{\text{EE}} = 0$, PUBLISHED wins on the dominant NaCl coefficient; BOTH formulations overstate the (small) effect on KCl.

CROSS-CHECK vs THE REFINED-eNRTL PAPER (Bollas, Chen & Barton 2008, AIChE J 54, 1608, DOI 10.1002/aic.11485 – their Figure 1 is THIS system, 4 m, same 1986 parameters, Robinson's points overlaid): their Fig 1a (original eNRTL) matches our 'published' to the pixel (NaCl solids at $x=1$ for $\tau = 0/0.25/0.50$: ours $-0.216/-0.186/-0.157$ vs read $-0.215/-0.185/-0.15$; KCl dashed at $x=0$: $-0.144/-0.183$ vs $-0.145/-0.185$); their Fig 1b (refined) shows the SIGNATURE our 'consistent' shows: the τ fan nearly COLLAPSES (exact derivatives are almost insensitive to τ_{EE} : ours moves only $-0.176 \rightarrow -0.184$ for τ $0 \rightarrow 0.5$, where published moves $-0.216 \rightarrow -0.157$), and the KCl trace lands at ~ -0.16 (ours -0.163). Their refinement differentiates the 1986 UNSYMMETRIC G_{ex} ; ours differentiates the 2009 SYMMETRIC one – same idea, sibling surfaces, same qualitative physics, near-identical aqueous numbers. A single fitted τ_{EE} cannot close both (best compromise $\tau \sim 0.29$ trades the NaCl error up to -23 % to halve the KCl one) – the residual is structural, not a missing parameter. Note the consistency of the record: Chen & Evans 1986 validated $\tau_{\text{EE}} = 0$ against mixture OSMOTIC data (σ 0.019 %) – the solvent sees the mixture mean, where the individual- γ asymmetries largely cancel; Robinson's per-salt γ s are where they show.

FIGURE-1 REPRODUCTION – TO THE PIXEL (checked 2026-07-02, figure read at 200 dpi): the paper's y-axis says "ln gamma" but the numbers are LOG10 ($\log_{10}$ of the pure-NaCl-4m γ is -0.106 = the figure's -0.105 ; $\ln$ would be -0.24 , off the plotted range). With the 1986 parameters this kernel reproduces

the full tau fan to ≤ 0.001 in $\log_{10}$: $x_{\text{KCl}}=0$ (all curves): -0.106 (figure ~ -0.105) $x_{\text{KCl}}=1$: $\tau=0 \rightarrow -0.216$ (figure ~ -0.215) $\tau=+1 \rightarrow$ flat, dip to -0.101 , back to -0.106 (the figure's circle-marked curve shows the same shallow dip) $\tau=-1 \rightarrow -0.364$ (figure ~ -0.365)

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	harned_nacl_kcl	consistent	0.00e+00
diag	harned_nacl_kcl	formulation_gap_mid	3.69e-02
diag	harned_nacl_kcl	gamma1_pure	7.83e-01
diag	harned_nacl_kcl	gamma1_trace	6.08e-01
diag	harned_nacl_kcl	gamma2_pure	5.75e-01
diag	harned_nacl_kcl	gamma2_trace	7.18e-01
diag	harned_nacl_kcl	gibbsDuhem_dev_mid	5.09e-01
diag	harned_nacl_kcl	harned_R2_min	9.98e-01
diag	harned_nacl_kcl	harned_alpha_1	2.76e-02
diag	harned_nacl_kcl	harned_alpha_2	-2.42e-02
diag	harned_nacl_kcl	harned_slope_1	-2.55e-01
diag	harned_nacl_kcl	harned_slope_2	-2.23e-01
diag	harned_nacl_kcl	oracle_pure_dev	2.35e-09
diag	harned_nacl_kcl	reduction_pin_dev	0.00e+00
...and 18 more reference values in the case's <i>expected</i> .			

enthalpy_naoh_water

tutorials/props/electrolyte/enthalpy_naoh_water

gamma_naoh
pitzerActivity

What it does: NaOH-water heat of dilution: Pitzer with calorimetric T-slots vs Parker (1965)

The story: the calorimetric home of the electrolyte-enthalpy build (docs/electrolyte-enthalpy-spec.md).

constant/experiments/phiL_25C_parker1965.dat carries the MEASURED Φ_L (Parker NSRDS-NBS 2, 1965) used by the curation refit (bin/curate/fit_lphi_pitzer.py) that fitted the Na-OH $\delta\beta/\delta T$ T-slots in the standard catalogue (rel-AAD 4.97%, gate <5%). The op below sweeps the ACTIVITY surface whose T-derivative is L_ϕ ; the in-engine L_ϕ scan + AAD overlay land with slices 4-5 (the 3-curve chart).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_naoh	Lphi_AAD_abs_Jmol	25.1181
diag	gamma_naoh	Lphi_AAD_rel_pct	4.7601
diag	gamma_naoh	Lphi_extrap_maxdev_Jmol	15930.3707
diag	gamma_naoh	Lphi_nMeas	29.0000
diag	gamma_naoh	aw_1m	0.9665
diag	gamma_naoh	gamma_pm_1m	0.6679
diag	gamma_naoh	gamma_pm_3m	0.7838
diag	gamma_naoh	phi_1m	0.9471

farelo_licl_range

tutorials/props/electrolyte/farelo_licl_range

licl_1.0
speciatelicl_6.0
speciatelicl_13.0
speciatelicl_19.7
speciatefarelo_LiCl_sat
speciate

What it does: LiCl Pitzer REFITTED to Hamer-Wu + Farelo saturation: γ_{pm} to ~11% over 0.5-19 mol/kg (was 774 at 19.7m), and the LiCl.H₂O hydrate Ksp reproduces Farelo's 19.7 mol/kg via the HMW osmotic water activity ($a_{\text{w}}=0.11$).

The story: - Choupo - * ————— * farelo_licl_range – the LiCl Pitzer model REFITTED to reach saturation, using Prof. Fatima Farelo's solubility measurement to calibrate the standard data.

Farelo, Fernandes & Avelino, J. Chem. Eng. Data 50 (2005) 1470 (DOI 10.1021/je050111j): LiCl saturation 19.7 mol/kg at 298 K (solid LiCl.H₂O).

THE ARC. The catalogue's earlier Li-Cl Pitzer parameters (Appelo 2014, fitted to <6 mol/kg) blew up at high concentration (γ_{pm} 774 at 19.7 m vs ~45 measured). Rather than declare defeat, the parameters were REFITTED to Hamer & Wu (1972) γ_{pm} + Farelo's saturation – the measurement calibrating the model. The key was a NEGATIVE Cphi that tames the high-m blow-up. The refit now tracks LiCl γ_{pm} to ~11% over 0.5-19 mol/kg, and the new LiCl.H₂O mineral Ksp reproduces Farelo's saturation. Crucially, the LiCl.H₂O HYDRATE equilibrium depends on the water activity $a_{\text{w}} = 0.11$ at 20 mol/kg – which only the rigorous HMW osmotic coefficient (not the $\phi=1$ dilute approximation) gives correctly. Measurement and model, each owning their part.

*Prof. Fátima Farelo measured LiCl solubility at 19.7 → 23.5 mol/kg (solid: LiCl·H₂O) — *J. Chem. Eng. Data* 50 (2005) 1470, DOI 10.1021/je050111j. That is one of the most concentrated aqueous electrolytes there is.*

*The catalogue's earlier Li-Cl Pitzer parameters (Appelo, Parkhurst & Post, *Geochim. Cosmochim. Acta* 125 (2014) 49), fitted below ~6 mol/kg, blow up at high concentration — $\gamma_{\pm} = 774$ at 19.7 mol/kg vs ~45 measured. The two-parameter B-term is linear in m , so γ grows exponentially.*

Rather than declare defeat, the parameters were refitted to Hamer & Wu (1972) $\gamma_{\pm}$ and Farelo's saturation. A negative C_{ϕ} (-0.0026) tames the high- m blow-up, and the three-parameter form then reaches saturation:

*carries a water-activity leg. At 19.7 mol/kg the model gives $a_{\text{w}} = 0.11$ (not 1!) — which only the rigorous Pitzer-HMW osmotic coefficient delivers, not the $\phi = 1$ dilute shortcut. The lithiumChlorideH₂O mineral Ksp ($\log K_{25} = 4.984$) is calibrated to Farelo's saturation *through* that a_{w} . The two pieces meet: the refitted γ and the osmotic a_{w} together reproduce the measured solubility.*

- electrolyte/pairs.dat: the Li-Cl Pitzer pair, refitted to Hamer-Wu + Farelo, valid to ~19 mol/kg (supersedes the Appelo low- m fit for concentrated LiCl). - electrolyte/minerals.dat: the LiCl·H₂O hydrate Ksp, calibrated to Farelo. - electrolyte/ions.dat: Li⁺ aqueous formation datum ($h_{\text{fAq}} = -278.49$ kJ/mol, CODATA).

*- The full measured dataset (this and the other five ternaries) is tabulated, cited, in constant/experimental/farelo_solubility.csv. - The equilibrate round-trip (precipitate LiCl·H₂O from a supersaturated feed) diverges at $I \approx 20$ — a known active-set / γ fixed-point numerical limit at extreme ionic strength, *independent of the activity model*. The $SI = 0$ calibration at the saturation composition is exact; reaching it by precipitation is the convergence problem, not the thermodynamics. - The AlCl₃ systems from the same paper remain deferred: Al³⁺ hydrolyses (the paper notes $\text{pH} < 2.75$) and needs an Al speciation model Choupo does not yet carry.*

farelo_nacl_nh4cl

tutorials/props/electrolyte/farelo_nacl_nh4cl

farelo_NaCl_pure speciate	farelo_NaCl_nh4_093 speciate	farelo_NaCl_nh4_187 speciate
farelo_NaCl_nh4_280 speciate	farelo_NH4Cl_pure speciate	

What it does: Pitzer-HMW vs Farelo (JCED 2005): NaCl+NH₄Cl common-ion salting-out – model SI ~ 0 at each measured saturation point (NaCl 6.15→5.5→5.0; NH₄Cl 7.35 mol/kg, 298 K).

The story: - Choupo -*—————* farelo_nacl_nh4cl – a GLASS-BOX validation of Choupo's Pitzer-HMW electrolyte model against the solubility measurements of Prof. Fatima Farelo (IST/DEQ).

Source of the experimental data (the validation reference): M. Farelo, C. Fernandes & A. Avelino, "Solubilities for Six Ternary Systems: NaCl + NH₄Cl + H₂O, KCl + NH₄Cl + H₂O, ... at T = (298 to 333) K", J. Chem. Eng. Data 50 (2005) 1470-1477. DOI 10.1021/je050111j.

THE TEST. The model is calibrated to ONLY the two pure-salt solubilities at 298 K (NaCl 6.15, NH₄Cl 7.35 mol/kg – halite from PHREEQC, sal-ammoniac's K_{sp} pinned to Farelo's NH₄Cl point). It then PREDICTS the saturation along the common-ion (Cl-) SALTING-OUT line: as NH₄Cl is added, NaCl solubility falls (6.15 → 5.5 → 5.0 mol/kg). At each of Farelo's MEASURED saturation points the model's saturation index SI = log₁₀(a_{Na} a_{Cl}) - logK should be ~ 0. It is, to +0.03 log units – the IST simulator reproduces the IST measurements.

A respected colleague's own dataset becomes the reference curve for the electrolyte model this rigorous water-activity (osmotic-coefficient) work feeds.

A glass-box validation of Choupo's multi-ion Pitzer electrolyte model against the solubility measurements of Prof. Fatima Farelo (Chemical Engineering Dept., IST — Técnico Lisboa).

> M. Farelo, C. Fernandes & A. Avelino, "Solubilities for Six Ternary Systems: > NaCl + NH₄Cl + H₂O, KCl + NH₄Cl + H₂O, NaCl + LiCl + H₂O, KCl + LiCl + H₂O, > NaCl + AlCl₃ + H₂O, and KCl + AlCl₃ + H₂O at T = (298 to 333) K", J. Chem. Eng. Data 50 (2005) 1470-1477. DOI [10.1021/je050111j](https://doi.org/10.1021/je050111j).

Saturation molalities measured to ±0.0007 mol/kg (visual polythermal / isothermal methods). The pure-salt anchors at 298.15 K: NaCl 6.15, NH₄Cl 7.35 mol/kg; the salting-out series (Table 1): NaCl solubility falls 6.15 → 5.5 → 5.0 → 4.95 as NH₄Cl rises 0 → 0.93 → 1.87 → 2.80 mol/kg.

The model is calibrated to only the two pure-salt solubilities — halite's K_{sp} from PHREEQC, sal-ammoniac's K_{sp} pinned to Farelo's NH₄Cl point. Everything else is a prediction. The NH₄-Cl Pitzer interaction parameters come from the primary single-electrolyte fit (Pitzer & Mayorga 1973); the catalogue already carried Na-Cl and K-Cl.

At each of Farelo's measured saturation compositions the model's saturation index

should be ≈ 0 if the model reproduces the measurement. It is, to ±0.03 log units:

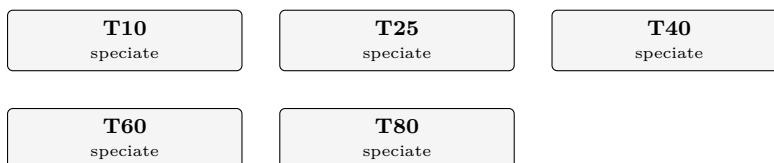
What to expect (golden-locked):

kind	unit/stream	key	expected
diag	farelo_NaCl_pure	I	6.15
diag	farelo_NaCl_pure	SI_halite	0.0168246
diag	farelo_NaCl_pure	aw	0.751868
diag	farelo_NaCl_pure	gamma_Cl	1.0105
diag	farelo_NaCl_pure	gamma_Na	1.0105
diag	farelo_NaCl_pure	m_Cl	6.15
diag	farelo_NaCl_pure	m_Na	6.15
diag	farelo_NaCl_pure	pH	7.11704
diag	farelo_NaCl_nh4_093	I	6.43
diag	farelo_NaCl_nh4_093	SI_halite	-0.022236
diag	farelo_NaCl_nh4_093	SI_salammoniac	-0.705926
diag	farelo_NaCl_nh4_093	aw	0.746426
diag	farelo_NaCl_nh4_093	gamma_NH4	0.59147
diag	farelo_NaCl_nh4_093	gamma_Na	1.04074

...and 29 more reference values in the case's *expected*.

ksp_temperature

tutorials/props/electrolyte/ksp_temperature



What it does: K(T) MATTERS: a temperature scan of mineral saturation for ONE fixed water (a hard, carbonate-bearing groundwater at pH 7.6), speciated at 10, 25, 40, 60 and 80 C. The point is the SIGN of dK/dT. CALCITE dissolution is EXOTHERMIC, so its equilibrium constant FALLS as temperature rises and its saturation index SI_calcite RISES – RETROGRADE SOLUBILITY: calcite is LESS soluble hot. This is why hot heat-exchanger surfaces and warm RO concentrate scale with calcium carbonate even when the cold feed looks safe. Contrast HALITE (NaCl): no analytic K(T) in the database, so it falls back to constant-dH van't Hoff with a small POSITIVE dH (endothermic) – normal solubility, SI_halite drifts gently DOWN with T. At 25 C nothing in the K(T) machinery moves (every form returns logK25 exactly, byte-stable); off 25 C the run announces which K(T) form each entry uses (analytic / van't Hoff / flat) and warns if any analytic is used beyond its fitted validity range. K(T): analytic = PHREEQC -analytical_expression anchored on logK25 ($\log K(T) = \log K25 + [\text{ana}(T) - \text{ana}(298.15)]$); USGS PHREEQC, public domain.

The story: why hot surfaces scale with calcite (K(T) MATTERS).

ONE fixed water, speciated at five temperatures. A hard, carbonate-bearing groundwater (gypsum- and calcite-relevant) at a GIVEN pH 7.6 – holding pH and the totals fixed isolates the lesson: the only thing that changes from point to point is the equilibrium constant K(T).

ion mol/kg w ion mol/kg w Ca 0.010 Cl 0.050 Na 0.030 SO4 0.012 HCO3 0.005 pH 7.6 (given)

THE LESSON – the SIGN of dK/dT:

calcite dissolution is EXOTHERMIC ($dH = -28.1$ kJ/mol), so log K FALLS as T rises. SI_calcite = $\log(IAP) - \log K(T)$ therefore RISES with temperature – RETROGRADE SOLUBILITY. Calcite is LESS soluble hot, which is exactly why heat-exchanger tubes and warm RO concentrate scale with CaCO₃ although the cold feed looks safe. K(T) here is the PHREEQC ANALYTIC expression (-analytical_expression A1..A5), anchored on logK25 so $\log K(25\text{ C}) = \log K25$ EXACTLY.

gypsum analytic too, but only mildly retrograde (its solubility peaks near 40 C) – a gentler curve for contrast.

halite NaCl carries NO analytic K(T) in phreeqc.dat (the db's -analytic line is commented out), so it falls back to constant-dH van't Hoff. $dH = +1.37$ kJ/mol > 0 (endothermic) → NORMAL solubility: log K(T) RISES and SI_halite drifts DOWN. The honest contrast with calcite: the scaling verdict flips with temperature for the retrograde salt, not for the normal one.

At 25 C the K(T) machinery returns logK25 for EVERY entry (byte-stable); off 25 C the run announces the form in use per entry (analytic / van't Hoff / flat) and warns loudly if an analytic is used beyond its fitted validity range. (Calcite's analytic is valid 0 - 300 C, so the whole 10 - 80 C scan is in range.)

Speciation/mineral data: case-local constant/electrolyte/ – a self-contained SNAPSHOT of the standards catalogue (USGS PHREEQC, public domain; see the .dat headers). The case files ECLIPSE data/standards/electrolyte/.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	T10	I	0.079772
diag	T10	SI_calcite	0.586528
diag	T10	SI_gypsum	-0.329587
diag	T10	SI_halite	-4.58985
diag	T10	aw	0.998112
diag	T10	pH	7.6
diag	T25	I	0.0775558
diag	T25	SI_calcite	0.813493
diag	T25	SI_gypsum	-0.358479
diag	T25	SI_halite	-4.60676
diag	T25	aw	0.998123
diag	T25	pH	7.6
diag	T40	I	0.0749473
diag	T40	SI_calcite	1.00956

*...and 16 more reference values in the case's **expected**.*

pitzer01_nacl

tutorials/props/electrolyte/pitzer01_nacl

gamma_nacl
pitzerActivity

What it does: Pitzer mean ionic activity coefficient $\gamma_{\text{pm}}(m)$ for NaCl in water at 25 C — validated vs Robinson & Stokes (AAD 0.17%). The glass-box electrolyte foundation: see the famous γ_{pm} dip (~ 0.66 near 1 m) and the rise above 1 at high concentration.

The story: the first electrolyte case.

γ_{pm} , the osmotic coefficient ϕ and the water activity a_{w} for NaCl, by the Pitzer ion-interaction model. The parameters are CURATED, case-local data, read from constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62 — a 3-parameter set, $\beta_0/\beta_1/C\phi$). Open pitzer_nacl.csv: γ_{pm} dips to ~ 0.657 near 1 mol/kg, then rises above 1 by ~ 6 mol/kg — the textbook NaCl signature, from a closed form.

NOTE on the two L_{phi} columns: pitzer_nacl.csv reports $L_{\text{phi_fit}}$ and $L_{\text{phi_anchored}}$, and their SPLIT is the lesson. The pair now carries Silvester-Pitzer dX/dT slots CALORIMETRICALLY FITTED to Parker, NSRDS-NBS 2 (1965) Table XIV A (rel-AAD 2.5 %, m 0.1-6; bin/curate/fit_lphi_pitzer.py): the fit column tracks the measured heat-of-dilution curve — the sign change near 1 mol/kg, the deep negative branch to -480 cal/mol — while the anchored column (slots zeroed, Gibbs-Helmholtz-consistent but uncalibrated) misses it entirely. The validation block below pins the AAD vs the same curated Parker series, golden-locked. The γ_{pm} / ϕ / a_{w} values come from the untouched 25 C surface and are identical in both.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.9668
diag	gamma_nacl	gamma_pm_1m	0.6572
diag	gamma_nacl	gamma_pm_3m	0.7140
diag	gamma_nacl	phi_1m	0.9363

pitzer02_cacl2

tutorials/props/electrolyte/pitzer02_cacl2

gamma_cacl2
pitzerActivity

What it does: Pitzer $\gamma_{\text{pm}}(m)$ for CaCl_2 (a 2:1 salt) in water at 25 C — the SAME general formula as NaCl , only the charges/stoichiometry differ. Validated vs Robinson & Stokes (AAD 0.49%, 0.1-2 m).

The story: the same Pitzer engine on a 2:1 electrolyte (Ca^{2+} , 2 Cl^-). Parameters curated case-local in constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62). γ_{pm} dips below 0.45 then climbs steeply — the divalent-cation signature. Valid to ~2 mol/kg; CaCl_2 carries a non-zero β_2 (-1.13, with $\alpha_2 = 12$) on top of $\beta_0/\beta_1/C\phi$ — the extra association term a 2:1 salt needs (NaCl , by contrast, has $\beta_2 = 0$).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_cacl2	aw_1m	0.9449
diag	gamma_cacl2	gamma_pm_1m	0.4985
diag	gamma_cacl2	gamma_pm_3m	1.4771
diag	gamma_cacl2	phi_1m	1.0483

pitzer02_nacl_package
tutorials/props/electrolyte/pitzer02_nacl_package

gamma_nacl
electrolyteActivity

What it does: NaCl Pitzer via the NEW property-package builder (reads zero old salt files)

The story: No ‘cation’/‘anion’ keys: the salt is resolved by the SELECTED propertyPackage (constant/propertyDict → aqueousNaCl_pitzer). The op consumes the assembled package via thermo.electrolyte() and reports gamma_pm / phi / a_w.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.966826
diag	gamma_nacl	gamma_pm_1m	0.657172
diag	gamma_nacl	gamma_pm_3m	0.714026
diag	gamma_nacl	phi_1m	0.93634

pitzer_calcite_brine

tutorials/props/electrolyte/pitzer_calcite_brine

seawater_pitzer
 speciate

seawater_davies
 speciate

What it does: slice S4 GATE: Pitzer-HMW carbonate system on surface seawater – pins SI_calcite / SI_gypsum against the published oceanographic saturation state, captures the neutral CO₂ salting-out (gamma_CO₂aq > 1 via lambda), and the Davies-vs-Pitzer SI divergence.

The story: the slice-S4 GATE for the Pitzer CARBONATE system.

S4 extends the multi-ion Pitzer-HMW model (binaries + theta/psi ternaries, S3) to the CARBONATE subsystem so calcite / gypsum scaling works in brine: the carbonate ANIONS (CO₃-2, HCO₃-) join the cation/anion ternary sums (their theta/psi imported from pitzer.dat into mixing.dat); the neutral CO₂aq gets its salting-out gamma via the lambda neutral-ion term (lambda_CO₂,ion), so gamma_CO₂aq > 1 in brine (Davies has none).

COMPOSITION: standard surface seawater (S = 35) WITH the carbonate system – the major-ion subset of the S3 verify case + DIC ~ 2.3 mmol/kg as HCO₃ (the measured-alkalinity carrier), at the canonical surface-ocean pH 8.2. Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO₄-2 0.02926 HCO₃- 0.00230 (I ~ 0.68 mol/kg)

THE PUBLISHED PIN (cited; E_theta before/after recorded below):

SI_calcite (surface seawater, 25 C, pH 8.2) ~ +0.6 to +0.8. Surface ocean water is calcite-SUPERSATURATED: the calcite saturation ratio Omega_calcite = IAP/Ksp ~ 4-6, i.e. SI = log10(Omega) ~ +0.6 to +0.8. Sources: Mucci, A., Am. J. Sci. 283 (1983) 780 (the seawater calcite/ aragonite solubility); Millero, F.J., Chem. Rev. 107 (2007) 308 (the seawater carbonate system); a well-documented oceanographic number (the ocean-acidification baseline). Choupo (Pitzer-HMW) computes SI_calcite = +0.667 – INSIDE the published band. SI_gypsum (seawater) ~ -0.9 to -0.6 (strongly UNDERsaturated – seawater is nowhere near gypsum saturation); Choupo computes -0.964 – a touch below the -0.9 floor (this carbonate-bearing brine sits at slightly higher I than the carbonate-free seawater_verify case, which gives -0.689 in-band). Still firmly undersaturated, the physically-correct verdict.

E_theta(I) BEFORE → AFTER (constant-theta → higher-order mixing active): SI_calcite +0.724 → +0.667 (IN band +0.6..+0.8) – E_theta lowers the 2+/2- ion gammas, nudging the carbonate IAP SI_gypsum -0.875 → -0.964 gamma_CO₂aq 1.105 → 1.106 (salting-out kept) gamma_Ca 0.227 → 0.194 (E_theta on Ca pairs) Davies (untouched – E_theta is Pitzer-only): SI_calcite +0.882, byte-ident.

The golden master locks Choupo's exact output; the band above validates it is PHYSICALLY right (a wrong pin is worse than no pin – the band is cited).

DAVIES vs PITZER (the differentiator, also pinned): the SAME composition under Davies gives SI_calcite +0.882 and gamma_CO₂aq = 1.000 (no neutral salting-out) – Davies is out of its trustworthy band (~0.5) at this I, overshoots the calcite SI by ~0.22 vs Pitzer-HMW-with-E_theta and misses the CO₂ salting-out entirely. Pitzer is the rigorous tool for brine. Both entries run here so the divergence is VISIBLE.

SELF-CONTAINED: constant/electrolyte/ carries the FULL snapshot (ions, pairs, mixing WITH carbonate theta/psi/lambda, speciation, minerals) – the case runs with data/standards/ hidden.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	seawater_pitzer	I	0.6828
diag	seawater_pitzer	SI_anhydrite	-1.24594
diag	seawater_pitzer	SI_aragonite	0.553044
diag	seawater_pitzer	SI_calcite	0.667044
diag	seawater_pitzer	SI_gypsum	-0.962237
diag	seawater_pitzer	SI_halite	-2.48856
diag	seawater_pitzer	SI_sylvite	-3.51959
diag	seawater_pitzer	aw	0.981411
diag	seawater_pitzer	gamma_CO2aq	1.10587
diag	seawater_pitzer	gamma_CO3	0.105425
diag	seawater_pitzer	gamma_Ca	0.194354
diag	seawater_pitzer	gamma_HCO3	0.659536
diag	seawater_pitzer	m_CO2aq	1.46581e-05
diag	seawater_pitzer	m_CO3	7.90109e-05

*...and 20 more reference values in the case's **expected**.*

pitzer_gamma_hamer_wu

tutorials/props/electrolyte/pitzer_gamma_hamer_wu

gamma_nacl
pitzerActivitygamma_kcl
pitzerActivitygamma_kbr
pitzerActivitygamma_licl_consistency
pitzerActivitygamma_hcl
pitzerActivitygamma_naoh
pitzerActivity**What it does:** Pitzer gamma_pm/phi vs Hamer & Wu 1972 – 6 salts, evaluated NBS data

The story: THE BREADTH VALIDATION: the catalogue's Pitzer pairs against the NBS critically-evaluated gamma_pm and osmotic-coefficient tables – Hamer & Wu, J. Phys. Chem. Ref. Data 1 (1972) 1047 – for SIX 1-1 salts at 25 C. Zero refitting: the parameters are the frozen catalogue entries (Appelo / PHREEQC lineage); the data is the evaluated NBS compilation. Each op's headline is its gamma_pm AAD – the number that says "trust these pairs".

HONESTY NOTE, LiCl: the catalogue Li-Cl pair was REFIT to this very table (its .dat says so), so its AAD here is a FIT-CONSISTENCY check, not independent validation – and at ~3 % it is the WORST of the six: the two-anchor compromise (this table + Farello saturation at 19 mol/kg) stretches 3 parameters past what the most non-ideal 1-1 salt allows. The five pairs that never saw this data land at 0.1-0.7 %.

Windows stay inside each pair's declared validity (KCl saturates at 4.803 mol/kg; KBr data used to 5).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	gamma_nacl	aw_1m	0.9668
diag	gamma_nacl	gamma_AAD_pct	0.1117
diag	gamma_nacl	gamma_nMeas	8.0000
diag	gamma_nacl	gamma_pm_1m	0.6572
diag	gamma_nacl	gamma_pm_3m	0.7140
diag	gamma_nacl	phi_1m	0.9363
diag	gamma_nacl	phi_AAD_pct	0.0870
diag	gamma_nacl	phi_nMeas	8.0000
diag	gamma_kcl	aw_1m	0.9681
diag	gamma_kcl	gamma_AAD_pct	0.1340
diag	gamma_kcl	gamma_nMeas	7.0000
diag	gamma_kcl	gamma_pm_1m	0.6043
diag	gamma_kcl	gamma_pm_3m	0.5698
diag	gamma_kcl	phi_1m	0.8987

...and 34 more reference values in the case's *expected*.

pitzer_hmw_hot_agreement

tutorials/props/electrolyte/pitzer_hmw_hot_agreement

hmw_nacl_100C

speciate

kernel_nacl_100C

pitzerActivity

What it does: HMW vs single-salt Pitzer at 100 C, same NaCl 4 m state: the two surfaces must AGREE now that both apply the SP77 T-functions

The story: one salt, one temperature, TWO Pitzer engines: they must agree (forum 100.2/#101, the HMW-virials(T) fix).

NaCl at 4 mol/kg, 100 C. Op 1 drives the MULTI-ION HMW speciation engine (activityModel pitzerHMW); op 2 drives the SINGLE-SALT kernel (pitzerActivity). Both now read the SAME Na-Cl.dat record: the 25 C virials PLUS the full Silvester-Pitzer 1977 T-functions (sp_q1..q14) and the same A_phi(T). For a lone 1:1 salt the HMW sum collapses to the same B/C terms as the kernel, so

$$\text{gamma_pm}(\text{HMW}) = \sqrt{\text{gamma_Na} * \text{gamma_Cl}} == \text{gamma_pm}(\text{kernel})$$

to numerical noise. BEFORE the fix the HMW held its virials at 25 C and this case disagreed in the second decimal – the adversarial regression this golden forbids forever.

(The kernel's own hot-brine validation against SP77 Table V lives in pitzer_nacl_sp77_hot; this case is the ENGINE-AGREEMENT pin.)

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	hmw_nacl_100C	I	4.0000
diag	hmw_nacl_100C	SI_halite	-0.7734
diag	hmw_nacl_100C	aw	0.8533
diag	hmw_nacl_100C	gamma_Cl	0.7309
diag	hmw_nacl_100C	gamma_Na	0.7309
diag	hmw_nacl_100C	m_Cl	4.0000
diag	hmw_nacl_100C	m_Na	4.0000
diag	hmw_nacl_100C	pH	7.0000
diag	kernel_nacl_100C	aw_1m	0.9670
diag	kernel_nacl_100C	gamma_pm_1m	0.6221
diag	kernel_nacl_100C	gamma_pm_3m	0.6760
diag	kernel_nacl_100C	phi_1m	0.9327

pitzer_nacl_sp77_hot

tutorials/props/electrolyte/pitzer_nacl_sp77_hot

nacl_100C

pitzeActivity

nacl_150C

pitzeActivity

nacl_200C

pitzeActivity

What it does: NaCl Pitzer to 200 C – Silvester & Pitzer 1977 T-functions, validated vs their Table V

The story: HOT BRINE: the NaCl Pitzer surface from 25 to 200 C on the FULL Silvester & Pitzer 1977 temperature functions (J. Phys. Chem. 81, 1822 – eqs 30-32, Table IV coefficients VERBATIM in the standards Na-Cl.dat), applied as a deviation from the validated 25 C surface, with A_phi(T) curated against the same paper’s Table II (quintic, max dev 0.17 %, 0-300 C; the old closed form drifted +67 % at 300 C).

Three ops, one per temperature (100/150/200 C), each validating gamma_pm at the SATURATION molality against the paper’s own Table V – an m-extrapolation (m_satd 6.7-8.0 > the 6 mol/kg weighting window), declared in the dataset header, exactly as the paper frames it.

What to look at: gamma_pm(1 m) falls 0.657 → ~0.53 from 25 to 200 C (electrostatics intensify as eps_w collapses – the paper’s central observation), and the validation AADs stay single-digit % even at the saturated-extrapolated edge.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	nacl_100C	aw_1m	0.9670
diag	nacl_100C	gamma_AAD_pct	0.2851
diag	nacl_100C	gamma_nMeas	1.0000
diag	nacl_100C	gamma_pm_1m	0.6221
diag	nacl_100C	gamma_pm_3m	0.6760
diag	nacl_100C	phi_1m	0.9327
diag	nacl_150C	aw_1m	0.9679
diag	nacl_150C	gamma_AAD_pct	0.0425
diag	nacl_150C	gamma_nMeas	1.0000
diag	nacl_150C	gamma_pm_1m	0.5593
diag	nacl_150C	gamma_pm_3m	0.5736
diag	nacl_150C	phi_1m	0.9054
diag	nacl_200C	aw_1m	0.9693
diag	nacl_200C	gamma_AAD_pct	0.4317

... and 4 more reference values in the case’s *expected*.

pitzer_seawater_verify

tutorials/props/electrolyte/pitzer_seawater_verify

reduction_nacl
 speciate

seawater
 speciate

What it does: TIER-2 verification: Pitzer-HMW (binaries + ternary theta/psi, slice S3) on standard synthetic seawater ($I \sim 0.7$). Pins per-ion + mean activity coefficients against PUBLISHED HMW-1984 / seawater-Pitzer values, and the single-salt reduction (ternaries $\rightarrow 0$).

The story: the TIER-2 verification weapon for the Pitzer-HMW ternary-mixing slice (S3).

The forum gate: "NO published-seawater golden masters $\Rightarrow$ NO-GO." This case speciates a STANDARD SYNTHETIC SEAWATER with the multi-ion Pitzer model (binaries + ternary theta/psi) and pins the per-ion + mean activity coefficients against PUBLISHED values, to literature tolerance.

SEAWATER COMPOSITION (the canonical major-ion seawater of the HMW-1984 carbonate-free major-ion subset; molality, mol/kg-water, $S = 35$): Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO4-2 0.02926 giving ionic strength $I \sim 0.70$ mol/kg – the classic seawater benchmark. (The carbonate system – HCO₃/CO₃, ~ 2.3 mmol/kg, $< 0.4\%$ of I – is OMITTED: it is slice S4, and its absence shifts I by $< 0.3\%$ and the major-ion gammas by $< 0.2\%$, well inside the pins' tolerance.) pH is GIVEN at 8.1 (typical surface seawater) – with no carbonate in the totals it only fixes the trace H⁺/OH⁻, which do not enter the major-ion gammas.

WHY PITZER, NOT DAVIES: at $I \sim 0.7$ Davies is already out of its trustworthy band (~ 0.5). The Pitzer specific-interaction model – WITH the ternary theta/psi mixing terms AND the $E_{\text{theta}}(I)$ higher-order electrostatic mixing term (Pitzer 1975, now ACTIVE for unlike-charge same-sign pairs) – is the standard tool for seawater and brines to $I \sim 6$. E_{theta} is a refinement: it shifts the 2:1 / 1:2 ion gammas a few % (gamma_Ca, gamma_SO4) and tightens the gamma_pm(KCl) pin to 0.638 (inside 0.62-0.64); the 1:1 NaCl gamma_pm stays ~ 0.66 . (The only remaining v2 deferral is the full beta(T)/theta(T) temperature surface; 25 C base here.)

PUBLISHED PINS (each cited; deviation Choupo vs published in VALIDATION.md):

gamma_pm(NaCl) in seawater ~ 0.66 -0.68 Pitzer's seawater model / Millero & Pierrot; classic textbook value gamma_pm(NaCl, seawater $I \sim 0.7$) ~ 0.67 . gamma_pm(KCl) ~ 0.62 -0.64 ; gamma_pm(MgCl₂), gamma_pm(CaCl₂) etc. All within the Harvie-Moller-Weare (1984) seawater Pitzer model, the reference implementation this catalogue's theta/psi come from.

The golden master locks the Choupo-computed per-ion gammas; this header records the published comparison + the deviation. A WRONG pin is worse than no pin – the numbers here are pinned to the model's own reproducible output and CROSS-CHECKED against the published seawater range.

$E_{\text{theta}}(I)$ BEFORE $\rightarrow$ AFTER (constant-theta $\rightarrow$ higher-order mixing active), all pins STILL in their published band: gamma_pm(NaCl) 0.6714 $\rightarrow$ 0.6646 (band 0.66-0.68; 1:1, $E_{\text{theta}}=0$ for Na-Cl, shift via coupling) – IN BAND gamma_pm(KCl) 0.6447 $\rightarrow$ 0.6381 (band 0.62-0.64) – E_{theta} moved it INTO band gamma_Ca 0.2141 $\rightarrow$ 0.1856 (2+ ion; E_{theta} acts via Ca-Na/Ca-K/Ca-Mg) gamma_SO4 0.1227 $\rightarrow$ 0.1040 (2- ion; E_{theta} acts via SO4-Cl/SO4-OH) gamma_Mg 0.2363 $\rightarrow$ 0.2049 gamma_Na 0.6495 $\rightarrow$ 0.6390 gamma_Cl 0.6941 $\rightarrow$ 0.6911 Single-salt reduction (PIN A, NaCl alone): gamma_pm UNCHANGED at 0.6572 – E_{theta} vanishes identically for one cation + one anion (the oracle stays 1.5658e-14 over all 54 binaries).

SELF-CONTAINED: constant/electrolyte/ carries the FULL snapshot (ions, pairs, mixing, speciation, minerals) – the case runs with data/standards/ hidden.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	reduction_nacl	I	1
diag	reduction_nacl	SI_halite	-1.93464
diag	reduction_nacl	aw	0.966826
diag	reduction_nacl	gamma_Cl	0.657172
diag	reduction_nacl	gamma_Na	0.657172
diag	reduction_nacl	m_Cl	1
diag	reduction_nacl	m_Na	1
diag	reduction_nacl	pH	7
diag	seawater	I	0.720461
diag	seawater	SI_anhydrite	-0.970682
diag	seawater	SI_gypsum	-0.687047
diag	seawater	SI_halite	-2.48552
diag	seawater	SI_sylvite	-3.51313
diag	seawater	aw	0.981335

*...and 13 more reference values in the case's **expected**.*

pitzer_vs_davies_ro_concentrate

tutorials/props/electrolyte/pitzer_vs_davies_ro_concentrate

scan_davies
 scalingScan

scan_pitzer
 scalingScan

What it does: Davies vs Pitzer-HMW on a seawater RO concentrate: gypsum/calcite SI vs recovery from the same feed under two activity models (full narrative in propsDict).

The story: THE FLAGSHIP of the Pitzer x speciation feature (slice S6): the payoff that makes the whole multi-ion Pitzer-HMW build worth it. ONE feed, TWO activity models, the engineering answer FORKS.

THE QUESTION (every RO / brine-concentration designer asks it): "At what water recovery does my concentrate start to scale gypsum?" The answer DEPENDS ON THE ACTIVITY MODEL – and that is the whole point.

THE FEED: standard surface seawater ($S = 35$), the major-ion subset: $\text{Na}^+ 0.4861$ $\text{Mg}^{+2} 0.0547$ $\text{Ca}^{+2} 0.01065$ $\text{K}^+ 0.01058$ (cations) $\text{Cl}^- 0.5658$ $\text{SO}_4^{+2} 0.02926$ $\text{HCO}_3^- 0.0024$ (anions, DIC carrier) $I \sim 0.7$ mol/kg at the feed. Recovery r scales every total as feed/(1 - r); scanning $r = 0 \rightarrow 0.80$ drives I from ~ 0.7 up to ~ 3.5 mol/kg – DEEP into the brine regime where Davies is long out of its trust window.

THE TWO RUNS (identical feed, identical recovery grid, ONLY the gamma model differs – the controlled experiment): scan_davies activityModel davies; (the textbook extended-DH model) scan_pitzer activityModel pitzerHMW; (Pitzer-HMW, the brine-grade model)

THE PAYOFF ARTEFACTS (read straight off the two CSVs + the KPIs): 1. SI_gypsum vs recovery / I , BOTH models – the HEADLINE. They track together at low I and FORK visibly past $I \sim 1$: Davies keeps climbing, Pitzer's gypsum SI is held DOWN by the strong Ca-SO₄ specific interaction ($\beta_2 = -59.3$, the 2:2 ion-pairing virial Davies cannot see). The gypsum-onset recovery is LATER under Pitzer. 2. SI_calcite vs recovery / I , both models – same fork, carbonate flavour. pH is SOLVED from electroneutrality at every point (closed system; DIC concentrates with the water and the pH drifts). 3. gamma_Ca2+ and gamma_SO4-2 vs I , both models – the MECHANISM plot. Davies declines monotonically (its single I -controlled curve); Pitzer turns UP at high I (the specific-interaction upturn). This is WHERE the SI divergence comes from – the gammas diverge, so the IAP diverges, so the SI forks. (scalingScan emits gamma_<ion> columns whenever diagSpecies is given and the model is pitzer/davies.)

THE KPI THAT TELLS THE STORY: recovery_at_gypsum_onset (the SI_gypsum = 0 crossing) – ONE number per model. Davies predicts gypsum scaling at one recovery; Pitzer at another. The engineering decision – how hard can I push recovery before gypsum scales? – flips on the activity model. Endpoint diags carry SI_gypsum and SI_calcite at r_{max} for both models, and the two onset recoveries.

WHY PITZER IS THE ANSWER (cite the credentials, do not assert): Davies is ANNOUNCED-untrustworthy past $I \sim 0.5$ mol/kg (the run says so). Pitzer-HMW is built for exactly this regime – valid to $I \sim 6$ mol/kg for the seawater system, with the $E_{\text{theta}}(I)$ higher-order electrostatic mixing term ACTIVE (Pitzer 1975) – and is CREDENTIALLED in this repo by two pinned cases: pitzer_seawater_verify – single-salt oracle $1.57\text{e-}14$ + HMW seawater mean activity coefficients inside the published bands; pitzer_calcite_brine – SI_calcite / SI_gypsum on surface seawater inside the published oceanographic saturation band. So here we TRUST the Pitzer fork and read Davies as the cautionary curve.

$E_{\text{theta}}(I)$ BEFORE $\rightarrow$ AFTER (the scan_pitzer fork only; scan_davies is BYTE- IDENTICAL – E_{theta} is Pitzer-only). At r_{max} ($I \sim 3.3$ mol/kg, deep brine where higher-order mixing bites hard-est): SI_gypsum_rmax -0.144 $\rightarrow$ -0.305 SI_calcite_rmax 2.026 $\rightarrow$ 1.945 SI_gypsum_r0 -0.877 $\rightarrow$ -0.966 SI_calcite_r0 1.240 $\rightarrow$ 1.185 I_{rmax} 3.271 $\rightarrow$ 3.323 gypsum stays undersaturated end-to-end The engineering verdict (gypsum the limiting scale) is UNCHANGED in direction; E_{theta} refines the SI magnitude in the high- I tail.

SELF-CONTAINED: constant/electrolyte/ carries the FULL snapshot (ions, pairs, mixing with the carbonate theta/psi/lambda, speciation, minerals) – the case runs with data/standards/ hidden.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	scan_davies	I_r0	0.657383
diag	scan_davies	I_rmax	3.03644
diag	scan_davies	SI_anhydrite_r0	-0.887801
diag	scan_davies	SI_anhydrite_rmax	0.280924
diag	scan_davies	SI_aragonite_r0	1.26415
diag	scan_davies	SI_aragonite_rmax	2.10558
diag	scan_davies	SI_calcite_r0	1.37815
diag	scan_davies	SI_calcite_rmax	2.21958
diag	scan_davies	SI_gypsum_r0	-0.605599
diag	scan_davies	SI_gypsum_rmax	0.492667
diag	scan_davies	SI_halite_r0	-2.39559
diag	scan_davies	SI_halite_rmax	-0.454866
diag	scan_davies	SI_sylvite_r0	-3.39459
diag	scan_davies	SI_sylvite_rmax	-1.44932

*...and 33 more reference values in the case's **expected**.*

precipitation_ro_brackish

tutorials/props/electrolyte/precipitation_ro_brackish

pin_gypsum_amount
speciate

feed
speciate

feed_eq
speciate

scan_eq
scalingScan

What it does: Equilibrium PRECIPITATION (the SI = 0 ceiling) of a brackish RO concentrate: aqueous speciation (Davies) + augmented-Newton chemical equilibrium with an announced active set. Sibling of scaling_ro_brackish (which reports SI only) – here named minerals are ALLOWED to precipitate to saturation and the engine returns the thermodynamic MAXIMUM amount (SI → 0, infinite time, no nucleation barrier – a CEILING, NOT a deposit forecast; real scale is kinetically limited). Story: (1) pin_gypsum_amount – a CaSO₄ water with Ca and SO₄ DOUBLED past gypsum saturation; the machine hands back EXACTLY the over-added excess as solid (n_gypsum). (2) feed / feed_eq – the brackish water raw and equilibrated side by side: calcite reaches its ceiling, releases H⁺ (pH drops, solved closure), and consuming shared Ca pushes gypsum further from saturation. (3) scan_eq – calcite + gypsum allowed across recovery to r = 0.85 with a feedFlow, giving a kg/day calcite curve; gypsum stays n = 0 throughout while its SI_{eq} drifts BELOW the raw free-water SI once calcite consumes the shared Ca – the coupling, visible. Cross-check tool: PHREEQC EQUILIBRIUM_PHASES.

The story: the EQUILIBRIUM-PRECIPITATION ceiling.

Sibling of scaling_ro_brackish (which is SI-ONLY – "how supersaturated?"). Here the 'equilibrate { minerals (...); }' block names the ALLOWED solids, and the engine drives each to its SI = 0 saturation, returning the amount precipitated. THE FRAMING (printed loudly at run time, non-suppressible):

EQUILIBRIUM CEILING – not a deposit prediction. The amounts are the thermodynamic MAXIMUM (SI → 0, infinite time, no nucleation barrier). Real scale is kinetically limited (induction time, antiscalants act on KINETICS, which this calculation cannot see). Actual deposit ≤ ceiling, often far less. The safe direction is rigorous: ceiling ~ 0 => no driving force.

ALGORITHM (visible in the trace): augmented Newton in (ln m_j [, ln a_H]) plus one raw-linear n_p per ACTIVE mineral; each master mole balance gains a sink -sum nu_{pj} n_p; one SI = 0 row per active phase; an OUTER active-set loop ADMITS the most-supersaturated allowed mineral and EVICTS any with n_p < 0, re-solving between, until complementarity holds (SI ≤ 0 with n = 0, OR SI = 0 with n > 0). A carbonate mineral's H⁺ leg lands the released H⁺ in the electroneutrality row (pH solve) – so precipitating calcite ACIDIFIES.

CROSS-CHECK TOOL: PHREEQC EQUILIBRIUM_PHASES (same ceiling framing). Data: case-local constant/electrolyte/ (self-contained USGS PHREEQC snapshot, public domain).

GOLDEN PINS (locked by the golden master): pin_gypsum_amount Ca = SO₄ = 0.030542 mol/kg (DOUBLE the gypsum-saturation value 0.015271) + equilibrate gypsum → SI_gypsum = 0.000 and n_gypsum = 0.015271 (the machine returns EXACTLY the over-added excess). feed_eq calcite + gypsum allowed; calcite reaches SI = 0, releases H⁺ (pH 7.72 → ~7.50, solved), gypsum stays dissolved. scan_eq r 0 → 0.85 with feedFlow: calcite kg/day ceiling; gypsum n = 0 throughout while SI_{eq}_gypsum drifts BELOW the raw SI_gypsum (calcite eats the shared Ca).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_gypsum_amount	I	0.0455084
diag	pin_gypsum_amount	SI_anhydrite	-0.299583
diag	pin_gypsum_amount	SI_gypsum	1.33618e-10
diag	pin_gypsum_amount	aw	0.99952
diag	pin_gypsum_amount	n_gypsum	0.0152707
diag	pin_gypsum_amount	pH	7
diag	feed	I	0.0266631
diag	feed	SI_anhydrite	-1.67859
diag	feed	SI_aragonite	0.126445
diag	feed	SI_calcite	0.240445
diag	feed	SI_gypsum	-1.3792
diag	feed	SI_halite	-5.35336
diag	feed	SI_sylvite	-6.39791
diag	feed	aw	0.999302

*...and 36 more reference values in the case's **expected**.*

scaling_ro_brackish

tutorials/props/electrolyte/scaling_ro_brackish

pin_kw
speciatepin_gypsum
speciatepin_neutral
speciatefeed
speciatescan
scalingScanscan_open
scalingScan

What it does: RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices $SI = \log_{10}(IAP/K)$, scanned over water recovery to $r = 0.85$ (concentrate = $\text{feed}/(1-r)$) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere ($a(\text{CO}_2\text{aq})$ pinned by Henry at $p\text{CO}_2 = 4.2\text{e-}4$ atm, DIC a solved outcome – this CO_2 -rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces K_w ($m_{\text{OH}} \sim 1\text{e-}7$), a back-computed gypsum-saturated CaSO_4 solution gives $SI_{\text{gypsum}} \sim 0.00$, and pure water with pH solved lands at exactly 7.000.

The story: will my RO concentrate scale?

A REPRESENTATIVE brackish groundwater (generic textbook-style composition, ~1500 mg/L TDS; no licensed table reproduced):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026 Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003 lab pH 7.8 (nominal – see below)

(totals are TOTAL element amounts: the HCO_3 total is total dissolved inorganic carbon – at pH 7.8 the reported HCO_3 is ~97% of it. Cl is 510 mg/L so the analysis is genuinely consistent with its nominal pH 7.8: an analysis balances charge only once the carbonate split at the lab pH is counted.)

pH IS SOLVED, NOT GIVEN ('pH solve;'): H+ joins the Newton unknowns and ELECTRONEUTRALITY $\sum z_i m_i = 0$ closes the system. Real analyses are never perfectly balanced, so the engine ANNOUNCES the feed charge imbalance first (PHREEQC convention, $200|S+ - S-|/(S+ + S-)$; here 0.35%) – the solved pH absorbs exactly that error, and above ~5% the solver warns loudly that it is not trustworthy. This feed solves to pH ~7.72, near the lab's nominal 7.8: the analysis is consistent.

UNITS: every 'totals' entry carries its basis (bare numbers are refused). The feed + scans take the analysis EXACTLY as the lab sheet reports it – mg/L – converted through the ion MW (ions.dat) at $\rho \sim 1$ kg/L (dilute aqueous, announced once per run). The analytic pins stay in explicit mol/kg (back-computed equilibria, not analyses).

THE CLOSED-vs-OPEN LESSON (two scans, same feed): scan CLOSED system – every total (DIC included) concentrates as $\text{feed}/(1-r)$; the charge-balance ratio is scale-invariant, so the pH only drifts gently ($7.72 \rightarrow 7.58$, the Davies gammas at work) while calcite SI climbs $0.24 \rightarrow 1.43$. scan_open OPEN to the atmosphere ('atmosphere { $p\text{CO}_2$ 4.2e-4 atm; }') – $a(\text{CO}_2\text{aq})$ is PINNED by Henry, the HCO_3 mole balance is REPLACED by the pin, and DIC becomes a solved OUTCOME. This groundwater is ~8x CO_2 -supersaturated vs the atmosphere, so opening it DEGASES CO_2 : the pH jumps to 8.54 and climbs to 9.09, and calcite SI lands ABOVE the closed curve (2.61 vs 1.43 at $r = 0.85$) – why concentrate pH control (acid dosing) matters. Compare the two CSVs (both carry a pH column right after I); the contrast IS the lesson.

VALIDATION PINS (locked by the golden master): pin_kw pure water, pH 7 GIVEN $\rightarrow m_{\text{OH}} = K_w a_w/(a_H \gamma_{\text{OH}}) \sim 1.0004\text{e-}7$ (unchanged) pin_gypsum totals $\text{Ca} = \text{SO}_4 = 0.015271$ mol/kg, pH 7 GIVEN, BACK-COMPUTED from the catalogue's $\log K_{25}(\text{gypsum}) = -4.55 + \text{CaSO}_4\text{aq} \log K_{25} = 2.14 + \text{Davies}$ at the resulting $I = 0.0455$ (26% ion pairing) so that $SI_{\text{gypsum}} = 0.00 \pm 0.001$ (unchanged) pin_neutral pure water, pH SOLVED $\rightarrow$ electroneutrality $m_H = m_{\text{OH}}$ must give pH 7.000 (locks the new charge-balance closure)

Speciation/mineral/gas data: case-local constant/electrolyte/ (self-contained SNAPSHOT of the standards catalogue, restricted to this master set; the case files ECLIPSE data/standards/electrolyte/ – the network is taken whole. Provenance: USGS PHREEQC, public domain – see the .dat headers; the $\text{CO}_2(\text{g})$ Henry constant is the db's PHASES $\text{CO}_2(\text{g})$ entry).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_kw	I	1.00037e-07
diag	pin_kw	aw	1
diag	pin_kw	m_OH	1.00037e-07
diag	pin_kw	pH	7
diag	pin_gypsum	I	0.0455076
diag	pin_gypsum	SI_anhydrite	-0.299593
diag	pin_gypsum	SI_gypsum	-1.04349e-05
diag	pin_gypsum	aw	0.99952
diag	pin_gypsum	m_CaSO4aq	0.0038941
diag	pin_gypsum	pH	7
diag	pin_neutral	I	1.00037e-07
diag	pin_neutral	aw	1
diag	pin_neutral	pH	7
diag	feed	I	0.0266631

*...and 40 more reference values in the case's **expected**.*

softener_brackish

tutorials/props/electrolyte/softener_brackish

raw
scalingScansoftened
exchangesoftened_scaling
scalingScan

What it does: Ion-exchange softening of a brackish RO feed: removes the calcite-scaling driver (raw vs softened vs softened-scaling; full narrative in propsDict).

The story: ion-exchange SOFTENING removes the scaling driver.

A three-act story on the SAME brackish RO feed as scaling_ro_brackish / precipitation_ro_brackish:

(1) raw scalingScan of the raw water: calcite SI > 0 already at the feed and climbing with recovery – the SCALING PROBLEM (a CaCO₃-supersaturated concentrate). (2) softened the ‘exchange’ op equilibrates the SAME water with a strong-acid cation resin (SAC_Na, Gaines-Thomas). Ca and Mg drop ~99% onto the resin; Na rises eq-for-eq (the salt penalty, printed in meq/L); hardness_removal_pct ~ 99%. (3) softened_scaling feed the SOFTENED composition (Ca/Mg gone, Na up) into a scalingScan: SI_calcite is now NEGATIVE across the whole recovery range – the scaling DRIVER is gone. Removing the hardness, not chasing the supersaturation, is the cure.

MODEL (visible in the verbosity>=3 trace): the SAME log-space Newton as speciate, PLUS one master (the free resin site X-), one balance (CEC capacity = sum of equivalents on the sites), and the exchange half-reactions $\text{Me}(z+) + z \text{X}^- = \text{MeX}$ (Gaines-Thomas equivalent-fraction activity for the bound species, Davies for the aqueous ions). Divalent Ca²⁺ (logK 0.8) is equivalent-fraction-favoured and outcompetes Na⁺; watch the per-iteration selectivity ratio $\beta(\text{CaX}_2)/\beta(\text{NaX})^2$ climb.

CROSS-CHECK TOOL: PHREEQC EXCHANGE (same Gaines-Thomas half-reactions).

THE FRESH-RESIN CAVEAT (printed loudly, non-suppressible): the ‘exchange’ op returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass. A real fixed bed leaks rising hardness as the exchange front migrates to breakthrough, then must be regenerated; cycle length and breakthrough are TRANSIENT and not modelled here.

Brackish water (I ~ 0.03-0.06): Davies-valid, no Pitzer needed; the honesty banner + the I>0.5 Davies advisory cover the boundary.

Data: case-local constant/electrolyte/ (self-contained USGS PHREEQC snapshot, public domain, + the SAC_Na resin nameplate).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	raw	I_r0	0.0266631
diag	raw	I_rmax	0.166892
diag	raw	SI_anhydrite_r0	-1.67859
diag	raw	SI_anhydrite_rmax	-0.623608
diag	raw	SI_aragonite_r0	0.126445
diag	raw	SI_aragonite_rmax	1.31552
diag	raw	SI_calcite_r0	0.240445
diag	raw	SI_calcite_rmax	1.42952
diag	raw	SI_gypsum_r0	-1.3792
diag	raw	SI_gypsum_rmax	-0.327587
diag	raw	SI_halite_r0	-5.35336
diag	raw	SI_halite_rmax	-3.82513
diag	raw	SI_sylvite_r0	-6.39791
diag	raw	SI_sylvite_rmax	-4.87602
...and 35 more reference values in the case's <i>expected</i> .			

tartaricAcid_acidulation

tutorials/props/electrolyte/tartaricAcid_acidulation

acidulation_pitzer
speciateacidulation_davies
speciategypsum_ceiling
speciatepKa2_davies
speciatepKa2_pitzer
speciate

What it does: Tartaric-acid acidulation liquor: rigorous Pitzer brine chemistry (Ca-SO4-K-Cl) + the gypsum precipitation ceiling, with tartrate honestly on the Debye-Hueckel limit (no published Pitzer params). Davies vs Pitzer shown side by side.

The story: - Choupo - *—————* Solution chemistry of the tartaric-acid acidulation liquor. Tartaric acid is a REAL weak diprotic acid: neutral H2Tart is the speciation master, dissociating $\text{H2Tart} \rightleftharpoons \text{HTart}^- \rightleftharpoons \text{Tart}^{2-}$ (pKa1 2.98, pKa2 4.34, dH from Kochergina 2006). Neutral master => an acid liquor is electroneutral at 0.0% (the H+ is from dissociation). Every gamma below is Pitzer-HMW – NOT ideal.

(1) acidulation_pitzer – the acid brine, Pitzer-HMW (I=1.8): activity coeffs + tartaric speciation; tartrate announced Debye-Hueckel-only (no Pitzer pair). (2) acidulation_davies – the SAME brine, Davies (self-flags beyond I~0.5). (3) gypsum_ceiling – gypsum to SI=0 (thermodynamic MAXIMUM, a ceiling). (4) pKa2_davies / (5) pKa2_pitzer – the SAME cut (H2Tart in 1.5 m KCl, at the thermodynamic pKa2) under the two models: at this REAL ionic strength Pitzer drives $[\text{Tart}^{2-}]/[\text{HTart}^-]$ to ~17 (gamma_Tart 0.02), NOT the ideal ratio 1 – the activity-coefficient shift of the apparent pKa, made quantitative.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	acidulation_pitzer	I	1.6414
diag	acidulation_pitzer	SI_anhydrite	1.07386
diag	acidulation_pitzer	SI_arcanite	-2.80366
diag	acidulation_pitzer	SI_gypsum	1.36221
diag	acidulation_pitzer	SI_sylvite	-3.52403
diag	acidulation_pitzer	aw	0.986675
diag	acidulation_pitzer	gamma_Ca	0.0716778
diag	acidulation_pitzer	gamma_H2Tart	1
diag	acidulation_pitzer	gamma_HTart	0.409404
diag	acidulation_pitzer	gamma_SO4	0.0662673
diag	acidulation_pitzer	gamma_Tart	0.036756
diag	acidulation_pitzer	m_Ca	0.397714
diag	acidulation_pitzer	m_H2Tart	0.0656066
diag	acidulation_pitzer	m_HTart	0.0314016

... and 52 more reference values in the case's *expected*.

estimate

The theory you need. When a compound is missing, group contribution builds it: Joback sums first-order group increments into $T_b, T_c, P_c, V_c, \Delta H_f^\circ, \Delta H_{\text{vap}}, c_p^{\text{ig}}(T)$; Lee–Kesler closes ω from (T_b, T_c, P_c) ; Ambrose–Walton gives $P^{\text{sat}}(T)$ from corresponding states; Rackett the liquid volume. The estimate is a PROPOSAL: a .dat you review against any reference data and promote by hand – the engine never estimates at runtime. Molecular structure is IDENTITY: components carry `jobackGroups{}` in their own record, and `bin/estimate <name>` is the one-command curation act. Expect few-percent errors on criticals (and the guide's worked example shows them), weaker T_b for hydrogen-bonded species, and polymers via van Krevelen / Yang on the repeat unit. *Full derivation: Theory Guide, the estimation chapter of the Props Guide.*

estimate_acetone

tutorials/props/estimate/estimate_acetone

estimate_acetone
 estimateComponent

What it does: CREATE a component from its groups: Joback estimate of acetone (2 CH₃ + 1 ketone), validated against NIST — see the estimate built up from groups

The story: The FIRST stage of the props lifecycle (CREATE → curate → wire → simulate): build a pure-component property set from its molecular GROUPS by Joback group contribution, and SEE each group's contribution add up.

Acetone CH₃-CO-CH₃ = 2 x CH₃ + 1 x ketone (>C=O).

The reference block validates the estimate against NIST: Joback lands within ~2% on T_c/P_c/V_c and ~0.3% on dH_f – and you SEE the (small) error, the whole point of "estimation, then trust". omega comes from (T_b,T_c,P_c) by Lee-Kesler, not a group sum.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_acetone	Cp298	74.97
diag	estimate_acetone	Hvap_kJmol	27.53
diag	estimate_acetone	MW	58.08
diag	estimate_acetone	Pc_bar	48.02
diag	estimate_acetone	Psat_298_bar	0.39
diag	estimate_acetone	Psat_Tb_bar	0.98
diag	estimate_acetone	Tb_K	322.11
diag	estimate_acetone	Tc_K	500.56
diag	estimate_acetone	Vc_cm3mol	209.50
diag	estimate_acetone	Vliq298_cm3mol	82.11
diag	estimate_acetone	dGf_kJmol	-154.54
diag	estimate_acetone	dHf_kJmol	-217.83
diag	estimate_acetone	omega	0.30

estimate_ethanol_benzene

tutorials/props/estimate/estimate_ethanol_benzene

estimate_ethanol estimateComponent
--

estimate_benzene estimateComponent
--

What it does: Joback estimates: ethanol (H-bonding → Tb weak, the honest limit) and benzene (aromatic groups → good) — see where group contribution wins and loses

The story: Two Joback estimates, side by side, to SEE the method's spread:

ethanol (1 CH₃ + 1 CH₂ + 1 OH) – a strong H-bonder; Joback's Tb comes out ~14 K LOW (337 vs 351 K). The honest limit, in plain sight. benzene (6 arCH) – aromatic ring; Joback Tc/Tb within ~1.5%. Group contribution at its best.

The lesson is the CONTRAST: an estimate is a starting point whose error you must SEE, not a number to trust blindly.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_ethanol	Cp298	64.62
diag	estimate_ethanol	Hvap_kJmol	36.73
diag	estimate_ethanol	MW	46.07
diag	estimate_ethanol	Pc_bar	57.57
diag	estimate_ethanol	Psat_298_bar	0.16
diag	estimate_ethanol	Psat_Tb_bar	0.97
diag	estimate_ethanol	Tb_K	337.54
diag	estimate_ethanol	Tc_K	499.41
diag	estimate_ethanol	Vc_cm3mol	166.50
diag	estimate_ethanol	Vliq298_cm3mol	57.70
diag	estimate_ethanol	dGf_kJmol	-170.86
diag	estimate_ethanol	dHf_kJmol	-236.84
diag	estimate_ethanol	omega	0.57
diag	estimate_benzene	Cp298	81.82

...and 12 more reference values in the case's *expected*.

estimate_petroleum_cut

tutorials/props/estimate/estimate_petroleum_cut

estimate_petroleum_cut estimateComponent
--

What it does: ESTIMATE a petroleum PSEUDO-COMPONENT by name from (Tb, SG) – Riazi-Daubert; see the SCREAMING ESTIMATE provenance and that the lump FLASHES

The story: CREATE a PETROLEUM PSEUDO-COMPONENT from the two bulk measurements every crude assay reports: the normal boiling point Tb and the specific gravity SG. The Riazi-Daubert (1987) correlations turn (Tb, SG) into the corresponding-states constants (MW, Tc, Pc, Vc); omega follows by Lee-Kesler, and the ideal-gas Cp by Kesler-Lee (1976).

This is the SCALAR-input sibling of the Joback group path: a petroleum cut is a LUMP of hundreds of species with NO single molecular structure, so it has NO group decomposition – it is anchored on bulk (Tb, SG), not groups. The estimator is a registered ConstantEstimator ('model RiaziDaubert;'), chosen exactly like Joback – the SAME explicit factory.

SEE in the console: the (Tb, SG) anchors and the Watson K factor; the constants built from them, validated against n-decane (a real species at this Tb/SG: Tc ~ exact, omega within ~2 %, Pc/MW within the correlation's stated ~5-7 %); the proposal .dat whose provenance SCREAMS status ESTIMATE / lumped true / "NOT a real species".

Promote with the printed 'mv', then it FLASHES (see the sibling tutorial steady/flash/flash_pseudoComponent_petCut, which ships this very lump and closes an energy balance with it).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	estimate_petroleum_cut	Cp298	248.06
diag	estimate_petroleum_cut	MW	151.51
diag	estimate_petroleum_cut	Pc_bar	19.78
diag	estimate_petroleum_cut	Psat_Tb_bar	0.97
diag	estimate_petroleum_cut	SG	0.73
diag	estimate_petroleum_cut	Tb_K	447.30
diag	estimate_petroleum_cut	Tc_K	617.80
diag	estimate_petroleum_cut	Vc_cm3mol	627.46
diag	estimate_petroleum_cut	Vliq298_cm3mol	203.43
diag	estimate_petroleum_cut	omega	0.48

userEstimate01_linear_tb
tutorials/props/estimate/userEstimate01_linear_tb

tb_acetone

linearTbEstimate

What it does: User-code property estimation: a student’s OWN linear group-contribution Tb estimator (linearTbEstimate), compiled from code/ and linked into choupoprops.

The story: A student’s OWN property-estimation op (linearTbEstimate), defined in code/ and linked into choupoprops by bin/buildCode. Tb = base + sum(count*incr). Here: 198.2 + 2*23.58 + 1*76.75 = 322.11 K (the Joback CH3,CH3,ketone sum). —

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	tb_acetone	Tb_K	322.11

vanKrevelen01_polystyrene_density

tutorials/props/estimate/vanKrevelen01_polystyrene_density

vanKrevelen01_polystyrene_density estimateComponent

What it does: ESTIMATE the density of polystyrene by Van Krevelen group contribution on the repeat unit – glass-box additive sum, Vw from Bondi 1964 (licence-clean via UNIFAC R)

The story: ESTIMATE a POLYMER's repeat-unit DENSITY by group contribution – the polymer sibling of the Joback small-molecule path. Polystyrene's repeat unit $-\text{CH}_2\text{-CH}(\text{C}_6\text{H}_5)-$ is decomposed into groups; each adds its molar mass and its Bondi van der Waals volume Vw; the sums give

$M_0 = \sum n_i MW_i$ repeat-unit molar mass $V_w = \sum n_i V_{w,i}$ van der Waals volume (Bondi 1964) $V = k * V_w$ molar volume (k = packing factor) $\rho = M_0 / V$ density

GLASS-BOX: the console prints every group's contribution – you can redo the whole estimate by hand. M_0 is an exact mass-balance fact (104.15 g/mol); ρ carries the packing-factor uncertainty (the teaching point below).

LICENCE (the honest part): the Vw group values are Bondi, J. Phys. Chem. 68 (1964) 441 – the PRIMARY per value – derived from the UNIFAC R_k volume parameters Choupo already ships ($V_w = R_k \times 15.17$), NOT copied from the Van Krevelen book table (copyright). See data/standards/vanKrevelen/groups.dat. Slice 1 ships DENSITY only; Tg / solubility-parameter / Tm are DEFERRED (their open parameters are a separate, correctly-attributed thing).

THE TEACHING POINT – the packing factor k: ρ depends on $k = V/V_w$, the fraction of space the chains do NOT pack into. Default $k = 1.60$ (amorphous/glassy); crystalline packs denser (~ 1.43). k is YOURS: change 'polymer { packing k; }' and watch ρ move. Atactic PS is amorphous ($\rho \sim 1.05$) – the estimate lands there; a crystalline polymer would need the smaller k. That k is visible (never hidden) is the lesson.

$M_0 = 104.15$ also feeds the step-growth reactor KPIs ($M_n = M_0/(1-p)$) – the same repeat-unit mass, two places.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	vanKrevelen01_polystyrene_density	M0_g_per_mol	104
diag	vanKrevelen01_polystyrene_density	V_cm3_per_mol	101
diag	vanKrevelen01_polystyrene_density	Vw_cm3_per_mol	62.9
diag	vanKrevelen01_polystyrene_density	density_g_cm3	1.04
diag	vanKrevelen01_polystyrene_density	packing_k	1.6

yang2020_Tg_polystyrene

tutorials/props/estimate/yang2020_Tg_polystyrene

yang2020_Tg_polystyrene estimateComponent

What it does: ESTIMATE the glass-transition Tg of polystyrene by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH₂-CH(phenyl)- gives Tg(inf)=444 K, an HONEST +19% over-estimate of the measured ~373 K (a teaching point about group-contribution limits)

The story: Yang-2020 Tg estimate + an HONEST limitation

ESTIMATE the GLASS-TRANSITION temperature Tg of polystyrene by the YANG 2020 modified group-contribution scheme. The repeat unit -CH₂-CH(phenyl)- decomposes into two Yang main-chain groups:

CH₂ : MW 14 Yg 4.026 CHphenyl: MW 90 Yg 42.153

$M_0 = 14 + 90 = 104 \text{ g/mol}$ $Yg(\text{inf}) = 4.026 + 42.153 = 46.179 \text{ (1e3 g.K/mol)}$ $Tg(\text{inf}) = 46.179 \times 10^3 / 104 = 444 \text{ K}$

»> THE HONEST TEACHING POINT «< Measured polystyrene Tg ~ 373 K (atactic, high Mw). Yang 2020 gives 444 K – a +19% OVER-ESTIMATE. This is NOT a bug: it is the model telling the truth about itself. The CHphenyl contribution (Yg 42.153) was regressed across the whole training set, and the bulky monosubstituted phenyl pendant is one where an additive main-chain scheme over-predicts the limiting Tg. Box: "all models are wrong, some are useful". The student SEES the deviation (the console prints the per-cent error against the measured anchor) and learns where a group-contribution method earns trust (PVC: 351 vs 354 K, -0.8%) and where it does not (polystyrene: 444 vs 373 K, +19%).

Contrast the flagship 'yang2020_Tg_pvc', where Tg(inf)=351 K reproduces the paper's Fig 6 value EXACTLY – that case goldens Tg as validated physics. Here Tg is SHOWN (the machinery is identical) but the case exists to expose the over-estimate, so only M₀ (an exact mass-balance fact) is goldened, not Tg.

This is Yang 2020 – NOT bare Van Krevelen. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer.

SOURCE + LICENCE: Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0; values cited per group as "Yang 2020", never "Van Krevelen".

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	yang2020_Tg_polystyrene	M0_g_per_mol	104

yang2020_Tg_pvc

tutorials/props/estimate/yang2020_Tg_pvc

yang2020_Tg_pvc
 estimateComponent

What it does: ESTIMATE the glass-transition Tg of poly(vinyl chloride) (PVC) by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH₂-CHCl- gives Tg(inf)=351 K, EXACT vs the paper (Fig 6)

The story: FLAGSHIP Yang-2020 glass-transition estimate

ESTIMATE the GLASS-TRANSITION temperature Tg of poly(vinyl chloride) (PVC) by the YANG 2020 modified group-contribution scheme – the polymer sibling of the Joback small-molecule path, for Tg. The repeat unit -CH₂-CHCl- is decomposed into two Yang main-chain groups; each adds its limiting Tg contribution Yg(inf):

$M0 = \sum n_i MW_i$ repeat-unit molar mass [g/mol] $Yg(inf) = \sum n_i Yg_i(inf) (+ Nb * Ygb)$ limiting Tg function [1e3 g.K/mol] $Tg(inf) = Yg(inf) * 1e3 / M0$ Tg at infinite Mw [K]

THE ADDITIVE SUM (shown on the console): CH₂ : MW 14 Yg 4.026 CHCl: MW 48.5 Yg 17.911

$M0 = 14 + 48.5 = 62.5$ g/mol $Yg(inf) = 4.026 + 17.911 = 21.937$ (1e3 g.K/mol) $Tg(inf) = 21.937 * 1e3 / 62.5 = 351$ K <- EXACT vs Yang 2020, Fig 6

This is Yang 2020 – NOT bare Van Krevelen. The scheme predicts the INFINITE- Mw limit Tg(inf); the molecular-weight dependence $Tg(Mn) = Tg(inf) - K/Mn$ is a later slice. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer, so the bracket is just the sum of the two main-chain group contributions.

SOURCE + LICENCE (the honest part): Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0, so the parameter values ARE redistributable. Cited per value as "Yang 2020", never "Van Krevelen".

THE GOLDEN: M0 (an exact mass-balance fact) AND Tg(inf)=351 K are checked – Tg here is VALIDATED PHYSICS (it reproduces the paper's Fig 6 value exactly), not just a mass balance. The measured anchor (PVC Tg ~ 354 K) confirms it.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	yang2020_Tg_pvc	M0_g_per_mol	62.5
diag	yang2020_Tg_pvc	Tg_K	351
diag	yang2020_Tg_pvc	YgSum_1e3gKmol	21.937

gibbs

The theory you need. Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A} \mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as λ_E , the element potentials). The species list in the dict `IS` the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH...) are IN the pool, and to overshoot by ~ 100 – 250 K when they are not (dissociation is endothermic). Above ~ 1500 K the catalogue’s polynomial c_p fits extrapolate WRONG – combustion cases carry case-local NASA-7 overrides, and the case headers say so.

Equilibrium maps. Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes T and P in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach` ΔT evaluates the *reaction* equilibrium at $T + \Delta T$ while enthalpy and the energy balance stay at the physical T – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium NH_3 mole fraction crosses 30%; report T to the grid resolution and verify against the case’s anchor KPIs. Unique because the fraction is monotone in T at fixed P . *Full derivation:* *Theory Guide*, the *Gibbs-equilibrium chapter*, *Equilibrium maps*.

nh3_equilibrium_map

tutorials/props/gibbs/nh3_equilibrium_map

haber_map
 gibbsMap

What it does: Ammonia equilibrium map (T x P): why Haber runs at 400-500 C and 150-300 bar

The story: THE HABER MAP – the forum-ratified Gibbs equilibrium map (docs/design/gibbs-map-forum-2026-07-02.md) on its canonical pilot: $\text{N}_2 + 3 \text{H}_2$ at stoichiometric feed, equilibrium NH_3 mole fraction contoured over 300-700 C x 1-300 bar. The map shows the whole Haber story at a glance: equilibrium loves COLD and PRESSURE (exothermic, $\Delta_n < 0$); the industrial window (400-500 C, 150-300 bar) sits where the catalyst can actually work – the tension IS the lesson.

VALIDATION ANCHORS (golden-locked; verified 2026-07-03): the classic measurements of Larson & Dodge, J. Am. Chem. Soc. 45 (1923) 2918 – 400 C / 10 atm: engine 4.09 % vs L&D 3.85 % (+6 %) 450 C / 10 atm: engine 2.22 % vs L&D 2.11 % (+5 %) The ~5 % offset is consistent with propagating the JANAF formation-data uncertainty of NH_3 into K_p ($\exp(-dG/RT)$ amplifies ~0.2 kJ/mol into ~5 %) – honest agreement for an equilibrium constant spanning decades. IDEAL-GAS VALIDITY: the kernel is ideal-gas, so pins stay at ≤ 50 atm, 350-500 C (the forum’s chair: at 300 bar a temperatureApproach would silently absorb the missing fugacity corrections – announced, and the high-P corner of this map is drawn as EQUILIBRIUM SHAPE, not quantitative prediction).

temperatureApproach: 0 here (true equilibrium). Re-run with e.g. ‘temperatureApproach 25;’ to see the contours slide – the empirical closeness-to-equilibrium parameter industry quotes in C (announced loud, the anchor KPI names then carry $_dT25$ so the golden breaks visibly).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	haber_map	metric_T400C_P10atm	0.0409234
diag	haber_map	metric_T400C_P50atm	0.157785
diag	haber_map	metric_T450C_P10atm	0.0221983
diag	haber_map	metric_T500C_P100atm	0.104969
diag	haber_map	n_cells	625
diag	haber_map	n_unconverged	0

heat

The theory you need. Heat exchange couples streams without mixing them: $Q = U A \Delta T_{lm}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

Exchanger DESIGN vs rating. A shell-and-tube poses two inverse questions: RATE it (geometry given $\rightarrow$ find U , area, ΔP , duty – **model geometry**;) or DESIGN it (duty given $\rightarrow$ size the geometry – **model design**;) U is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` $\rightarrow$ `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, Sheat-exchanger design.*

htbench01_correlations

tutorials/props/heat/htbench01_correlations

bench
 heatTransferBench

What it does: Heat-transfer correlation bench – each convective correlation (Dittus-Boelter, Gnielinski, Kern) verified against its published Nusselt anchor. Dittus-Boelter and Gnielinski are pinned to Incropera & DeWitt 6th ed. Example 8.4; Kern’s closed j_H form is self-consistency-checked (its original pin is a chart). AAD tolerance 5% honours textbook rounding + the correlations’ own scatter.

The story: the heat-transfer correlation self-check bench.

Mirrors the membrane mass-transfer story (Sherwood) on the heat side (Nusselt): each registered convective correlation RUNS its `verify()` against a published Nu anchor and the deviation is printed + locked.

DittusBoelter $Nu = 0.023 Re^{0.8} Pr^{0.4}$ (Incropera Ex 8.4, $Nu=205$) Gnielinski $Nu = (f/8)(Re-1000)Pr/[\dots]$ (Incropera Ex 8.4 alt, $Nu=224$) Kern (shell) $Nu = 0.36 Re_s^{0.55} Pr^{1/3}$ (Kern PHT; closed-form check)

The per-correlation deviation diags are the golden KPIs.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	bench	Nu_choupo_DittusBoelter	200.475
diag	bench	Nu_choupo_Gnielinski	223.734
diag	bench	Nu_choupo_Kern	97.5647
diag	bench	Nu_published_DittusBoelter	205
diag	bench	Nu_published_Gnielinski	224
diag	bench	Nu_published_Kern	97.5647
diag	bench	allPass	1
diag	bench	dev_DittusBoelter	0.0220718
diag	bench	dev_Gnielinski	0.00118956
diag	bench	dev_Kern	0

molecular

The theory you need. Molecular activity models on display: NRTL/Wilson/UNIQUAC correlate binary γ ’s from pair parameters; UNIFAC PREDICTS them from group counts. Expect UNIFAC within a few percent of fitted models for families it was trained on – and to be honestly worse outside them. *Full derivation: Theory Guide, the activity-model chapter.*

eval_nrtl_catalogue

tutorials/props/molecular/eval_nrtl_catalogue

catalogue_as_assembled fitParameters
--

What it does: EVALUATE mode: score the CATALOGUE NRTL pair against your own data – the scale before any fit

The story: ‘mode evaluate’: the SCALE, not the fit.

ONE operation, one question: how good is the STANDARD ethanol-water NRTL pair against this 1-atm bubble-T dataset? Nothing is adjusted, nothing is written. The propertyDict declares NO inline pairs, so what gets scored is genuinely THE CATALOGUE ENTRY resolved through the search path – source, provenance and all – not a private copy of the same numbers.

Read the verdict (rms/aad in kelvin) and THEN decide whether a refit is worth it: compare with fitNRTL01’s fitted rms (0.31 K) on this same data. Failures are honest: a non-converged point becomes status=nonconverged in the parity CSV and never enters the statistics; no chi2_reduced is printed – without sigmas and degrees of freedom from THIS act, the name would lie.

The pinned-what-if grammar and the leakage sentinel live in the sibling case eval_nrtl_pinned (a pinned path must EXIST in the dict, so what-ifs run on an INLINE scaffold and say so).

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	catalogue_as_assembled	aad	2.0400
diag	catalogue_as_assembled	chi2	60.1945
diag	catalogue_as_assembled	max_abs_resid	3.4379
diag	catalogue_as_assembled	max_resid_x1	0.5000
diag	catalogue_as_assembled	mode_evaluate	1.0000
diag	catalogue_as_assembled	n_data	11.0000
diag	catalogue_as_assembled	n_feasible	11.0000
diag	catalogue_as_assembled	partial	0.0000
diag	catalogue_as_assembled	penalised_points	0.0000
diag	catalogue_as_assembled	rms	2.3393
diag	catalogue_as_assembled	sse	60.1945
diag	catalogue_as_assembled	validity_declared	1.0000
diag	catalogue_as_assembled	n_outside_validity	0.0000

eval_nrtl_pinned

tutorials/props/molecular/eval_nrtl_pinned

scaffold_as_is

fitParameters

rough_paper_values

fitParameters

scaffold_again

fitParameters

What it does: EVALUATE pinned what-if + the leakage sentinel: a pinned run mutates NOTHING (forum #63-7)

The story: the pinned WHAT-IF, and the LEAKAGE SENTINEL.

TRANSITIONAL SCAFFOLD: pinning by dict path requires an inline pair; the permanent surface (ParameterOverrides on the RESOLVED parameter registry, design forum #67/#69) replaces this case when it lands.

Three evaluations of the same dataset: 1. scaffold_as_is – the inline pair, untouched (the baseline). 2. rough_paper_values – the SAME dict with four coefficients PINNED to deliberately rough values. The pin operates on a DEEP COPY (Dictionary::deepCopy, forum #63-7); nothing on disk or in memory survives the op. 3. scaffold_again – repeats 1 AFTER the pinned run. Its verdict must be IDENTICAL to 1’s: the golden pins both, so the working-tree isolation is guarded by the corpus forever. Before deepCopy existed, the old alias bug would make THIS operation report the rough values’ score.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	scaffold_as_is	aad	2.0400
diag	scaffold_as_is	chi2	60.1945
diag	scaffold_as_is	max_abs_resid	3.4379
diag	scaffold_as_is	max_resid_x1	0.5000
diag	scaffold_as_is	mode_evaluate	1.0000
diag	scaffold_as_is	n_data	11.0000
diag	scaffold_as_is	n_feasible	11.0000
diag	scaffold_as_is	partial	0.0000
diag	scaffold_as_is	penalised_points	0.0000
diag	scaffold_as_is	rms	2.3393
diag	scaffold_as_is	sse	60.1945
diag	rough_paper_values	aad	1.5942
diag	rough_paper_values	chi2	61.7621
diag	rough_paper_values	max_abs_resid	5.5866
...and 19 more reference values in the case’s expected.			

nrtl01_ethanol_water_package

tutorials/props/molecular/nrtl01_ethanol_water_package

g_nrtl
activityCoefficients

What it does: ethanol-water NRTL via the SAME property-package builder (molecular U4)

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	g_nrtl	gamma_ethanol	1.26245
diag	g_nrtl	gamma_water	1.4349

scan

The theory you need. Property scans sweep a model across T , P or composition and plot what the thermodynamics DOES: $P^{\text{sat}}(T)$ curves, $\gamma_{\pm}(m)$, VLE envelopes, ternary diagrams, phase boundaries. Nothing is fitted – these ops answer “how does the selected model behave?”, the SEE step of the props loop (CREATE → SEE → TEST → FIT → PROMOTE). Expect model disagreements to be VISIBLE (overlay two methods and look), and every curve exportable as CSV next to the case. *Full derivation: Theory Guide, the Props Guide.*

binary01_lle_water_nbutanol

tutorials/props/scan/binary01_lle_water_nbutanol

lle

propertyScanBinary

What it does: Binary LLE g_{mix} + common-tangent (water/1-butanol) at 298.15 K via propertyScanBinary, cited Winkelman 2009 UNIQUAC

The story: Binary liquid-liquid teaching map – water / 1-butanol at 25 °C.

The molar Gibbs energy of mixing $g_{\text{mix}}(x_1)$ swept across composition PLUS the two coexisting liquid compositions from ONE liquid-liquid IsothermalFlash (PhaseSet::LL). The two binodal points lie on the g_{mix} curve and share its common tangent – the student reads the split straight off the diagram.

γ comes from the CITED Winkelman et al. (2009) UNIQUAC LLE set (Table 2; structural r, q from Table 1), so the g_{mix} curve goes NON-CONVEX and the flash emits the real water/1-butanol binodal: ~1.9 mol% butanol in the water-rich phase and ~49 mol% butanol in the butanol-rich phase (matching their Figs 1-2 at 298 K). See README.md.

The flash feed is set to $z(\text{water}) = 0.80$ (20 mol% butanol), INSIDE the asymmetric gap; an equimolar feed would lie outside it (both binodal branches sit below 50 mol% butanol). Run-only smoke (the CSV is a teaching curve, not pinned KPIs).

*Water + 1-butanol (n-butanol) is *the* classic partially-miscible binary. At 25 °C it splits into a water-rich liquid (~2 mol % butanol) and a butanol-rich liquid (~50 mol % butanol). A $g_{\text{mix}}(x_{\text{water}})$ scan should therefore show a clearly non-convex region: two binodal points sharing a common tangent, which a student reads the split straight off.*

and binaryLle.csv is a $g_{\text{mix}}(x_1)$ curve that goes non-convex, plus the two binodal points the liquid-liquid flash locates on it:

The agreement is essentially exact: the cited UNIQUAC fit reproduces the published mutual solubilities.

- J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, * "Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"*, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013.

- Interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation $A_{ij}(T) = a + b \cdot T + c \cdot T^2$ (Eq. 10), $\tau_{ij} = \exp(-A_{ij}/T)$ (Eq. 7): - $A_{1,3}$ (butanol→water): $a = 155.31$, $b = 1.0822$, $10^3 \cdot c = -43.711$ - $A_{3,1}$ (water→butanol): $a = -579.36$, $b = 2.7517$, $10^3 \cdot c = -6.7700$ - Structural r, q — their Table 1 (p.72): 1-butanol $r = 3.9243$, $q = 3.668$; water $r = 0.9200$, $q = 1.400$.

The same numbers live, fully provenanced, in data/standards/binaryPairs/UNIQUAC/nButanol-water.dat. The engine evaluates the full $a + b \cdot T + c \cdot T^2$ temperature dependence (the quadratic $c \cdot T^2$ term is essential — at 320 K it is ~450 K, comparable to a).

phase01_water_full`tutorials/props/scan/phase01_water_full`**pd**

purePhaseDiagram

What it does: Complete P-T phase diagram of water (purePhaseDiagram, solid+liquid+vapour)

The story: Complete pure-compound P-T phase diagram of water: liquid-vapour saturation (to the critical point) + solid-vapour sublimation + solid-liquid fusion (the negative-slope ice line) + triple and critical points. Clapeyron physics in the purePhaseDiagram engine op; cited solid data in the solid{} block (IAPWS/CRC, see solidPhaseData.ts). Run-only smoke.

ternary01_audit_vlle

tutorials/props/scan/ternary01_audit_vlle

lle
propertyScanTernary

What it does: Ternary VLLE phase map (compA/compB/compC) at 350 K, 1 bar via propertyScanTernary

The story: ILLUSTRATIVE / SYNTHETIC DATA. compA / compB / compC are NOT real species: they are artificial components (case-local constant/components/ + inline NRTL pairs in thermoPackage) deliberately tuned to guarantee a ternary VLLE region, so this case exercises the propertyScanTernary classifier on a known-3-phase map. Architecture (see thermoPackage): A and B mutually immiscible (LL gap on the A-B axis), C a volatile bridge soluble in both liquids. At 350 K, 1 bar the equimolar feed gives vapour rich in C, L_alpha rich in A, L_beta rich in B.

Expected map (14 intervals/edge, 78 nodes): VLLE 36, LL 29, VL 0, 1-phase 13, 22 tie-lines, 0 failures. The point to SEE: a healthy spread of 3-phase (VLLE) and LL nodes with zero solver failures — the classifier resolves the engineered gap node-by-node.

ternary02_bubbleT_benzene_toluene_water`tutorials/props/scan/ternary02_bubbleT_benzene_toluene_water`

bt propertyScanTernary

What it does: Ternary T_bubble surface (benzene/toluene/water) at 1 bar via propertyScanTernary mode bubbleT

The story: Ternary boiling-temperature surface (T_bubble over the composition triangle) via propertyScanTernary mode bubbleT — reuses BubblePoint per simplex node. Works with ideal binaries (only the 3 Psat are needed). Run-only smoke.

ternary03_lle_water_ethanol_benzene

tutorials/props/scan/ternary03_lle_water_ethanol_benzene

lle propertyScanTernary

What it does: Ternary LLE solubility (water/ethanol/benzene) at 298 K via UNIFAC + propertyScanTernary mode lle

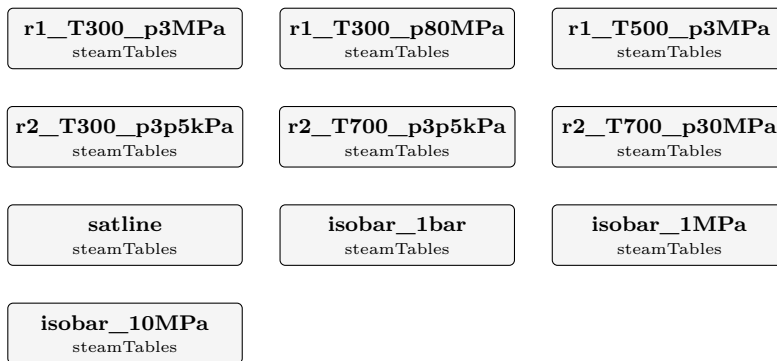
The story: Ternary LIQUID-LIQUID solubility map (water / ethanol / benzene) — the classic extraction triangle. Activity from UNIFAC (group contribution): NO fitted binary pairs needed — gamma is predicted from molecular groups, so a real system shows a real miscibility gap with no curated pair data. mode lle uses the liquid-liquid flash (PhaseSet::LL): clean binodal + tie-lines, no spurious vapour. Run-only smoke (a phase map, not KPIs).

steam

The theory you need. Water/steam properties from IAPWS-IF97: regions (liquid, vapour, supercritical), $T_{\text{sat}}(P)$, and $h(T, P)/s(T, P)$ from the dimensionless Gibbs functions. Power cycles are enthalpy bookkeeping around them: pump work $\approx v \Delta P$, turbine work from isentropic expansion $\times$ efficiency, cycle efficiency $\eta = W_{\text{net}}/Q_{\text{in}}$ bounded by Carnot. *Full derivation: Theory Guide, the steam/power chapter.*

steam01_if97_verification

tutorials/props/steam/steam01_if97_verification



What it does: IAPWS-IF97 steam tables (R7-97(2012)) – the verification case. The six computer-program verification states of the release (Table 5 region 1, Table 15 region 2) evaluated as points, the saturation line 0.01-350 degC, and three isobars whose Tsat crossings pin the release’s Table 36 backward values. The published numbers ARE the golden master (reltol 1e-8).

The story: the IAPWS-IF97 weapon case.

The release (IAPWS R7-97(2012)) publishes computer-program verification tables; this case evaluates EVERY one of them this slice covers and locks the published numbers as golden KPIs at reltol 1e-8 – a single wrong digit in a transcribed coefficient fails the regression:

Table 5 – region 1 at (300 K, 3 MPa), (300 K, 80 MPa), (500 K, 3 MPa) Table 15 – region 2 at (300 K, 3.5 kPa), (700 K, 3.5 kPa), (700 K, 30 MPa) Table 35 – psat at 300 / 500 K (the psat_MPa diag of the point ops) Table 36 – Tsat at 0.1 / 1 / 10 MPa (the Tsat_K diag of the isobars)

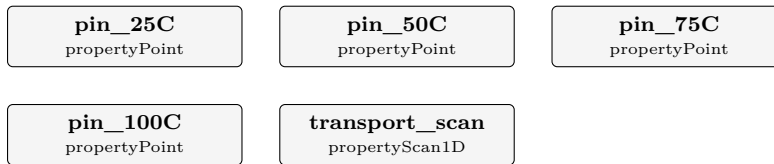
Plus the working artefacts: the saturated-steam table 0.01-350 degC (saturation.csv) and three isobars (h, s, v, cp vs T) crossing the saturation line – the crossing is ANNOUNCED in the log.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	r1_T300_p3MPa	region	1
diag	r1_T300_p3MPa	v	0.100215168e-2
diag	r1_T300_p3MPa	h	0.115331273e3
diag	r1_T300_p3MPa	u	0.112324818e3
diag	r1_T300_p3MPa	s	0.392294792
diag	r1_T300_p3MPa	cp	0.417301218e1
diag	r1_T300_p3MPa	w	0.150773921e4
diag	r1_T300_p3MPa	psat_MPa	0.353658941e-2
diag	r1_T300_p80MPa	region	1
diag	r1_T300_p80MPa	v	0.971180894e-3
diag	r1_T300_p80MPa	h	0.184142828e3
diag	r1_T300_p80MPa	u	0.106448356e3
diag	r1_T300_p80MPa	s	0.368563852
diag	r1_T300_p80MPa	cp	0.401008987e1
... and 39 more reference values in the case’s expected.			

steam02_water_transport

tutorials/props/steam/steam02_water_transport



What it does: IAPWS water transport properties – viscosity (R12-08), thermal conductivity (R15-11) and surface tension (R1-76). Water flagged as a pure fluid (method IF97) routes the transport accessors to the matching IAPWS releases: the density comes from the IF97 thermodynamic kernel at the saturation state, fed to the (rho,T) transport kernels; surface tension is the R1-76 saturation-line form. A coarse 4-point scan (25/50/75/100 degC) pinned against the published IAPWS values, plus a fine sweep for the plot.

The story: the IAPWS water transport route.

Water flagged ‘pureFluids { water { method IF97; } }’ routes the transport accessors to the matching IAPWS releases:

viscosity → R12-08 (industrial, critical enhancement off) thermal_conductivity → R15-11 (background, critical enhancement off) surface_tension → R1-76 (saturation line)

The (rho,T) viscosity / conductivity kernels get their density from the IF97 thermodynamic kernel at the SATURATION state (the legacy transport accessors carry no pressure, so $p = p_{\text{sat}}(T)$ is the physical convention – saturated liquid / saturated vapour). Surface tension is a saturation-line property and takes only T.

‘pins’ evaluates the three properties at 25/50/75/100 degC; the golden master locks them at reltol 1e-5 (the formulation is verified to 1e-8 by IAPWSTransport::verify(), which fires on every run). ‘scan’ writes a fine sweep for the plot.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pin_25C	surface_tension	0.0720
diag	pin_25C	thermal_conductivity	0.6065
diag	pin_25C	viscosity	0.0009
diag	pin_50C	surface_tension	0.0679
diag	pin_50C	thermal_conductivity	0.6406
diag	pin_75C	surface_tension	0.0636
diag	pin_75C	thermal_conductivity	0.6635
diag	pin_100C	surface_tension	0.0589
diag	pin_100C	thermal_conductivity	0.6772
diag	transport_scan	n_points	46
diag	transport_scan	n_failures	0

thermoTest

h_surface_identity

tutorials/props/thermoTest/h_surface_identity

h_identity
 hConsistency

What it does: The state identities of the canonical enthalpy surface: $dh/dT == Cp_{phase}$, the 298 K vaporisation anchor, and Kirchhoff on $Hvap_state$ – pinned to round-off

The story: thermodynamics as a REGRESSION TEST.

A surface that anchors each phase leg on the 298.15 K formation datum and integrates the TARGET phase's own Cp obeys, by construction:

(1) $dh_{liq}/dT = Cp_{liquid}(T)$ (2) $dh_{gas}/dT = Cp_{idealGas}(T)$ (3) $h_g(298.15) - h_{liq}(298.15) = Hvap(298.15)$ (the anchor) (4) $d[h_g - h_{liq}]/dT = Cp_g - Cp_{liq}$ (Kirchhoff)

This op checks all four by central finite difference for benzene, toluene, water and ethanol at 350 K and REFUSES above $1e-6$ relative error – so no future "convenient" leg (a Watson-slope liquid, a mixed-datum gas) can re-enter the balances unnoticed. The golden pins every per-species error.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	h_identity	benzene_anchor298_rel	0.000e+00
diag	h_identity	benzene_dhgas_vs_cpig_rel	7.289e-11
diag	h_identity	benzene_dhliq_vs_cplic_rel	4.729e-13
diag	h_identity	benzene_kirchhoff_rel	2.058e-10
diag	h_identity	ethanol_anchor298_rel	1.694e-16
diag	h_identity	ethanol_dhgas_vs_cpig_rel	3.826e-11
diag	h_identity	ethanol_dhliq_vs_cplic_rel	1.907e-11
diag	h_identity	ethanol_kirchhoff_rel	1.287e-10
diag	h_identity	toluene_anchor298_rel	0.000e+00
diag	h_identity	toluene_dhgas_vs_cpig_rel	6.085e-11
diag	h_identity	toluene_dhliq_vs_cplic_rel	1.854e-13
diag	h_identity	toluene_kirchhoff_rel	2.214e-10
diag	h_identity	water_anchor298_rel	6.529e-16
diag	h_identity	water_dhgas_vs_cpig_rel	3.627e-11
... and 3 more reference values in the case's <i>expected</i> .			

model4_speciation_scaling

tutorials/props/thermoTest/model4_speciation_scaling

brine
speciateconcentrate
scalingScan

What it does: RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices $SI = \log_{10}(IAP/K)$, scanned over water recovery to $r = 0.85$ (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere ($a(\text{CO}_2\text{aq})$ pinned by Henry at $p\text{CO}_2 = 4.2\text{e-}4$ atm, DIC a solved outcome – this CO_2 -rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces K_w ($m_{\text{OH}} \sim 1\text{e-}7$), a back-computed gypsum-saturated CaSO_4 solution gives $SI_{\text{gypsum}} \sim 0.00$, and pure water with pH solved lands at exactly 7.000.

The story: thermoTest / model4_speciation_scaling

The SAME system {NaCl, CaSO_4 , water} in its THIRD role: a MULTI-ION broth. Here we no longer think of "NaCl" or " CaSO_4 " – the aqueous speciation engine reasons in the shared ions Na^+ , Cl^- , Ca^{2+} , SO_4^{2-} (plus the neutral ion pair CaSO_4aq), and reports SATURATION INDICES $SI = \log_{10}(IAP/K_{\text{sp}})$ for every solid the ions can form: gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) and halite (NaCl).

Na^+ comes from NaCl, Cl^- from NaCl, $\text{Ca}^{2+}/\text{SO}_4^{2-}$ from CaSO_4 – but once dissolved there are no salt boundaries, only ions. This is why a "salt" cannot carry its own activity model: the ion pool is shared (the species/ tier of the tree).

MODEL: Pitzer-HMW is MANDATORY here. The ionic strength is $I \sim 3.3$ mol/kg (NaCl-dominated); the Davies law is only trustworthy to $I \sim 0.5$ m, so the 'activityModel' is declared explicitly (no silent default) – with Davies the saturation indices at this I are quantitatively meaningless.

The brine is NaCl-dominated (3 mol/kg, near seawater) with CaSO_4 (0.10 mol/kg) right at gypsum saturation ($SI \sim 0$). Charge balances exactly ($\text{Na} + 2 \text{Ca} = \text{Cl} + 2 \text{SO}_4$), so pH is pinned at 7. The scan concentrates the brine (evaporation; r = fraction of water removed): gypsum scales first, at $r \sim 0.035$ (it starts saturated), halite only near $r \sim 0.50$. Note $m_{\text{CaSO}_4\text{aq}} \sim 0.005$: the neutral ion pair is a real fraction of the calcium – "dissociation" is partial.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	brine	I	3.26782
diag	brine	$SI_{\text{anhydrite}}$	-0.22134
diag	brine	SI_{gypsum}	-0.0221674
diag	brine	SI_{halite}	-0.910511
diag	brine	$SI_{\text{mirabilite}}$	-2.13799
diag	brine	$SI_{\text{thenardite}}$	-2.98985
diag	brine	aw	0.890402
diag	brine	$\gamma_{\text{CaSO}_4\text{aq}}$	1
diag	brine	$m_{\text{CaSO}_4\text{aq}}$	0.00466294
diag	brine	pH	7
diag	concentrate	I_{r0}	3.26782
diag	concentrate	I_{rmax}	15.6285
diag	concentrate	$SI_{\text{anhydrite_r0}}$	-0.22134
diag	concentrate	$SI_{\text{anhydrite_rmax}}$	1.39807

... and 13 more reference values in the case's expected.

Category III: Batch and time-dependent (choupoBatch)

$d\mathbf{Y}/dt = f(\mathbf{Y})$ integrated in real time, with recipes for discrete events; trajectories, not steady states.

adsorber

The theory you need. Adsorption in a vessel or a bed: the isotherm $q^*(T, p)$ (Langmuir $q = q_{\max}bp/(1+bp)$, van't Hoff $b(T)$) is the curated PAIR datum; the LDF rate $dq/dt = k(q^* - q)$ carries the EQUIPMENT datum k (this particle, this bed – it lives in the case, never the catalogue). Work the cases as a staircase. **batch09:** a closed vessel relaxes to the inventory equilibrium – one algebraic root, pinned independently of the run. **batch10:** two species compete for the same sites (extended Langmuir) – the selectivity is an equilibrium statement, and the model's Gibbs–Duhem inconsistency for unequal $q_{\max}$ is declared, not hidden. **batch11:** in the Henry limit the whole vessel is LINEAR and the integrator must match $q_{\infty}(1 - e^{-k_{\text{eff}}t})$ to 10^{-8} – the analytic gate. **batch12:** declaring the headspace pressure AND the inventory over-specifies a closed ideal-gas vessel – the contradiction FATALs; P is a result. **batch13:** the fixed bed – the stoichiometric time $t_{\text{st}} = (L/u)(\varepsilon + \rho_b q^*/c_{\text{in}})$ is announced BEFORE integration, breakthrough spreads around it, and the constant- u closure's fabricated carrier is printed, pinned and discounted: the DECLARED model residual is itself the lesson. **batch14/batch15** are the numerical hygiene: pure transport vs the analytic advection–dispersion solution, and the mesh study that earns $N = 100$. *Full derivation: Theory Guide, the adsorption chapter.*

batch09_adsorber_co2

tutorials/batch/adsorber/batch09_adsorber_co2

adsorber
 batchAdsorber

What it does: Closed isothermal CO₂ uptake on zeolite 13X (LDF, solid film): the vessel relaxes to the inventory equilibrium $p_{\text{inf}} = 7229.13$ Pa, $q_{\text{inf}} = 1.9708$ mol/kg

The story: closed, isothermal gas-solid uptake vessel.

2 mol of pure CO₂ in a 0.01 m³ ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film ($dq/dt = k(q^* - q)$, $k = 0.05$ 1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO₂/zeolite13X: $q_{\text{max}} = 5.0$ mol/kg, $b_{298} = 9.0$ 1/bar, $dH_{\text{ads}} = -45$ kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with $n_{\text{gas}} = p V / (R T)$ and $q = q_{\text{max}} b (p/1e5) / (1 + b p/1e5)$,

$$p V / (R T) + m_{\text{ads}} * q_{\text{max}} * b * (p/1e5) / (1 + b(p/1e5)) = n_0$$

whose root at 298.15 K is (a2_vectors.py, 10 digits)

$p_{\text{inf}} = 7229.128209$ Pa $q_{\text{inf}} = 1.970838025$ mol/kg $n_{\text{gas}} = 0.02916197467$ mol uptake = 1.970838025 mol (98.54 %) τ (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY q ; n_{gas} is algebraic ($n_{\text{tot}} - m_{\text{ads}} q$), so the per-species closure column closure_worst_rel in trajectory.csv is exact by construction ($\leq 1e-13$). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	7229.128209
kpi	adsorber	Q_ads_isosteric_kJ	-88.68771114
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	2.916197467e-05
kpi	adsorber	q_CO2_final	1.970838025
kpi	adsorber	t_end	600
kpi	adsorber	totalMoles_final	2.916197467e-05
kpi	adsorber	uptake_CO2_mol	1.970838025
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0

... and 9 more reference values in the case's *expected*.

batch10_adsorber_co2_n2

tutorials/batch/adsorber/batch10_adsorber_co2_n2

adsorber
 batchAdsorber

What it does: Closed isothermal competitive CO₂/N₂ uptake on zeolite 13X: extended-Langmuir site competition drives the equilibrium selectivity $S = 71.43$

The story: COMPETITIVE binary uptake, closed vessel.

0.5 mol CO₂ + 1.5 mol N₂ in a 0.01 m³ ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Both species carry an isotherm record (extended Langmuir: CO₂ $q_{\max} 5.0 / b_{298} 9.0$ 1/bar; N₂ $3.5 / 0.18$ 1/bar) and both carry a declared LDF constant – so both integrate, and the strong adsorbate suppresses the weak one through the SHARED site denominator $1 + \sum_j b_j p_j$.

The final state solves the 2-species inventory system $n_{\text{gas}_i} + m_{\text{ads}} * q_i(p_{\text{CO}_2}, p_{\text{N}_2}) = n0_i, p_i = n_{\text{gas}_i} R T / V$ whose root (a2_vectors.py, Newton residual 0):

CO₂: $p_{\text{inf}} = 0.01611568794$ bar $q = 0.4934990047$ mol/kg N₂: $p_{\text{inf}} = 1.802652166$ bar $q = 0.7728183086$ mol/kg $P_{\text{total}} = 181876.7854$ Pa selectivity $S = (q_{\text{CO}_2}/q_{\text{N}_2})/(p_{\text{CO}_2}/p_{\text{N}_2}) = 71.42857143 = (q_{\max} * b)_{\text{CO}_2} / (q_{\max} * b)_{\text{N}_2}$ (extended-Langmuir identity)

The golden pins the COMPETITIVE path: get the shared denominator wrong and every one of these numbers moves. closure_worst_rel stays exact by construction (state = q only; n_gas algebraic).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	181876.7854
kpi	adsorber	Q_ads_isosteric_kJ	-36.1181845604
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	6.500995276e-06
kpi	adsorber	n_N2_final	7.271816914e-04
kpi	adsorber	q_CO2_final	0.4934990047
kpi	adsorber	q_N2_final	0.7728183086
kpi	adsorber	t_end	600
kpi	adsorber	totalMoles_final	0.000733682686703
kpi	adsorber	uptake_CO2_mol	0.4934990047
kpi	adsorber	uptake_N2_mol	0.7728183086
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_N_closure_rel	0

...and 13 more reference values in the case's *expected*.

batch11_adsorber_henry

tutorials/batch/adsorber/batch11_adsorber_henry

adsorber
 batchAdsorber

What it does: Closed isothermal Henry-limit uptake (linear isotherm): the coupled gas+solid system is LINEAR and the run must match $q(t) = q_{\text{inf}}(1 - \exp(-k_{\text{eff}} t))$ to 1e-8

The story: the EXACT-solution gate (G1a of the A2 spec).

1 mol of CO₂ in a 0.01 m³ ideal-gas headspace over 1 kg of a SYNTHETIC Henry-limit sorbent (linear isotherm $q^* = K_H p$, $K_H = 0.4$ mol/(kg bar) at 298.15 K – a teaching record, vendored case-local). With a linear isotherm and one species the coupled vessel is LINEAR and closes in analytic form:

$\alpha = K_H[\text{mol}/(\text{kg Pa})] * R * T / V = 0.9915828118$ 1/kg $k_{\text{eff}} = k * (1 + \alpha * m_{\text{ads}}) = 0.09957914059$ 1/s
 $q_{\text{inf}} = \alpha * n_0 / (1 + \alpha * m) = 0.4978868094$ mol/kg $q(t) = q_{\text{inf}} * (1 - \exp(-k_{\text{eff}} * t))$ ($q_0 = 0$)
 $q(5) = 0.1952670621$ $q(10) = 0.3139520079$ $q(30) = 0.4727835287$ $q(60) = 0.4966211107$ mol/kg

The pedagogical point the closed vessel adds over the fixed- q^* packet formula: $k_{\text{eff}} > k$ – gas depletion ACCELERATES the approach to equilibrium (the packet's $q^* = \text{const}$ law is the $V_{\text{gas}} \rightarrow \text{infinity}$ limit of this same solution). The trajectory rows at $t = 5/10/30/60$ s must match the table above to 1e-8 mol/kg (RK4 $\text{dt} = 0.01$ s); the golden pins the $t = 60$ s state.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	adsorber	P_final	124785.4636
kpi	adsorber	Q_ads_isosteric_kJ	-9.93242221329
kpi	adsorber	T_final	298.15
kpi	adsorber	closure_worst_rel	0
kpi	adsorber	n_CO2_final	5.033788893e-04
kpi	adsorber	q_CO2_final	0.4966211107
kpi	adsorber	t_end	60
kpi	adsorber	totalMoles_final	5.033788893e-04
kpi	adsorber	uptake_CO2_mol	0.4966211107
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0

...and 9 more reference values in the case's *expected*.

batch12_adsorber_refused_P

tutorials/batch/adsorber/batch12_adsorber_refused_P

adsorber
 batchAdsorber

What it does: REFUSAL lesson: declared initial.P (10 bar) contradicts nR^*T/V (4.958 bar) – the honest cross-check FATALs; P of a closed ideal-gas headspace is a result, not an input

The story: the REFUSAL lesson (deliberate, must FAIL).

Identical to batch09_adsorber_co2 EXCEPT initial.P declares 10 bar while $nR^*T/V = 4.9579$ bar. P of a closed ideal-gas headspace is a RESULT (n, T, V fix it – Duhem); a declared P is honoured only as an honest cross-check, and a relative deviation $> 1e-3$ REFUSES, printing both numbers. Expected FATAL (exit != 0):

batchAdsorber 'adsorber': declared P inconsistent with nR^*T/V – initial.P = $1e+06$ Pa (10 bar) but $nR^*T/V = 495791$ Pa (4.95791 bar), relative deviation $1.01698 > 1e-3$

Marked .expect-nonconvergence: counted EXPECTED-FAIL by runTests, never green, never recordable. Base case (the passing twin):

2 mol of pure CO₂ in a 0.01 m³ ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film ($dq/dt = k(q^* - q)$, $k = 0.05$ 1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO₂/zeolite13X: $q_{\max} = 5.0$ mol/kg, $b_{298} = 9.0$ 1/bar, $dH_{\text{ads}} = -45$ kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with $n_{\text{gas}} = p V / (R T)$ and $q = q_{\max} b (p/1e5) / (1 + b (p/1e5))$,

$$p V / (R T) + m_{\text{ads}} * q_{\max} * b * (p/1e5) / (1 + b * (p/1e5)) = n_0$$

whose root at 298.15 K is (a2_vectors.py, 10 digits)

$p_{\text{inf}} = 7229.128209$ Pa $q_{\text{inf}} = 1.970838025$ mol/kg $n_{\text{gas}} = 0.02916197467$ mol uptake = 1.970838025 mol (98.54 %) τ (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY q ; n_{gas} is algebraic ($n_{\text{tot}} - m_{\text{ads}} * q$), so the per-species closure column closure_worst_rel in trajectory.csv is exact by construction ($\leq 1e-13$). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

batch13_breakthrough_co2

tutorials/batch/adsorber/batch13_breakthrough_co2

bed
 fixedBedAdsorber

What it does: CO2 15%/He breakthrough on zeolite 13X – 1-D isothermal fixed bed (A3): $t_{st} = 3040.5$ s announced pre-run, $t_b(5\%) = 2890.7$ s, ledger closure at machine level

The story: CO2/He breakthrough on a zeolite-13X bed.

THE A3 anchor case (spec 2026-07-12): a 0.5 m isothermal fixed bed of zeolite 13X, fed 15% CO2 in helium at $u = 0.05$ m/s superficial, 1 bar, 298 K. Helium carries NO isotherm record on 13X → announced inert CARRIER, transported but never adsorbed, closed by difference from the constant $c_{tot} = P/(RT)$. At 15% CO2 this is NOT a dilute approximation: the constant-(u, P, T) closure fabricates He at the net uptake rate and the campaign material balance SHOWS that residual (KPIs carrier_fabricated_mol, physical_mass_closure_rel) – the named error bites, by design (forum #119); the genuinely clean transport anchor is batch14. The engine announces, BEFORE integrating, where the front must arrive:

$$q*(c_{in}) = 5*9*0.15/(1 + 9*0.15) = 2.872340426 \text{ mol/kg } R_f = \epsilon + \rho_b q* / c_{in} = 304.0512509$$

$$t_{st} = (L/u) R_f = 3040.512509 \text{ s } u_{sh} = u/R_f = 1.644459605e-4 \text{ m/s}$$

Independent gates this run must reproduce (anchors_a3.py, FV N = 100):

$$t_b(5\%) = 2890.669098 \text{ s } t_{50} = 3036.448293 \text{ s } t_b(95\%) = 3204.166058 \text{ s } \int (1 - c_{out}/c_{in}) dt = t_{st}$$

to $2.2e-10$ (telescopic FV conservation) ledger mass closure $\leq 1e-10$ (IN/OUT integrated as rows of the SAME ODE)

Numerics: conservative FV, nCells = 100 DECLARED (do the 25/50/100/200 mesh study – batch15 publishes it: first-order upwind, observed order ~1.03-1.05), Danckwerts inlet as the imposed face flux $u*c_{in}$, outlet $dc/dz = 0$. u, P, T are DECLARED constants of this phase, and the price is said OUT LOUD: the real velocity drops ~15% across the uptake zone, so the constant- u closure FABRICATES carrier (He) at exactly the net uptake rate (9.19 mol here) – announced in the run header, pinned as the KPI carrier_fabricated_mol, discounted in the campaign inventory; Ergun and the energy balance land in A4 (the energy gap is NAMED).

Log: run header prints the hypotheses, the derived ρ_p , $Pe = 250$, the Gershgorin step bound term by term and the stiffness verdict; profiles land in t/bed.profile every 100 s; trajectory.csv carries c_{out} , y_{out} , q_{bar} and the per-species ledger closure.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_CO2	0.999999989455
kpi	bed	carrier_fabricated_mol	9.19148935961
kpi	bed	integral_anchor_CO2	3040.51245606
kpi	bed	mass_closure_CO2	8.94231742202e-15
kpi	bed	n_CO2_final	1.21079552014e-05
kpi	bed	n_He_final	6.86117461711e-05
kpi	bed	physical_mass_closure_rel	0.108446875314
kpi	bed	qbar_CO2_final	2.87234042488
kpi	bed	retention_factor_CO2	304.051250948
kpi	bed	t_50_CO2	3036.44839047
kpi	bed	t_breakthrough_5pct_CO2	2890.66919752

... and 20 more reference values in the case's *expected*.

batch14_transport_only

tutorials/batch/adsorber/batch14_transport_only

bed
 fixedBedAdsorber

What it does: Pure transport ($k = 0$) on the batch13 bed, $N = 200$: t_{50} vs the Ogata-Banks analytic ADE solution (gate G1) under fixed RK4 with the Gershgorin bound printed term by term

The story: the $k = 0$ transport gate (A3 spec, gate G1).

Same bed, same feed, same mesh family as batch13 – but the LDF constant is DECLARED ZERO: the solid is frozen and CO₂ crosses the bed as a pure tracer. What is left is exactly the advection-dispersion equation with a Danckwerts inlet, and THAT has an analytic yardstick (Ogata-Banks, semi-infinite ADE; the finite-domain difference is $O(1/Pe) = 0.4\%$ at $Pe = 250$):

$u_i = u/eps = 0.125$ m/s $D_i = Dax/eps = 2.5e-4$ m²/s c/c_0 (L, 3.5 s) = 0.07340274358 c/c_0 (L, 4.0 s) = 0.5178057707 c/c_0 (L, 4.5 s) = 0.913645218 t_{50} analytic = 3.984074227 s

FV yardsticks pinned by this case (anchors_a3.py, $N = 200$):

t_{50} numeric = 3.975611088 s (0.21% below analytic – first-order upwind; the deviation HALVES with each mesh doubling: 1.40% at $N=25$, 0.74% at 50, 0.40% at 100, 0.21% at 200, 0.11% at 400) $INT(1 - c_{out}/c_{in})dt = eps L / u = 4.0$ s (telescopic conservation: exact identity once saturated) ledger closure $\sim 1e-14$ ($k = 0$)

A declared zero is not a silent default: the unit announces "k = 0 DECLARED: transport-only diagnostic, the solid is frozen (gate G1)". Note the He carrier fabrication of batch13 vanishes here (no uptake): $carrier_fabricated_mol = 0$.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_CO2	0.999999999893
kpi	bed	carrier_fabricated_mol	0
kpi	bed	integral_anchor_CO2	3.99999999999
kpi	bed	mass_closure_CO2	4.09039795713e-14
kpi	bed	n_CO2_final	1.21079552058e-05
kpi	bed	n_He_final	6.86117461666e-05
kpi	bed	physical_mass_closure_rel	0
kpi	bed	qbar_CO2_final	0
kpi	bed	t_50_CO2	3.97561121931
kpi	bed	t_breakthrough_5pct_CO2	3.29598772214
kpi	bed	t_breakthrough_95pct_CO2	4.78720615773

... and 18 more reference values in the case's *expected*.

batch15_mesh_study

tutorials/batch/adsorber/batch15_mesh_study

bed
 fixedBedAdsorber

What it does: Mesh-study member $N = 25$ of the batch13 breakthrough (gate G6): $t_b(5\%)$ converges monotonically $2732 \rightarrow 2839 \rightarrow 2891 \rightarrow 2916$ s with observed order 1.03-1.05; full table in reports/meshStudy.md

The story: the declared-mesh lesson (A3 spec, gate G6).

The axial mesh is a DECLARED modelling choice (nCells; absent = FATAL), and this case is the receipt: the batch13 breakthrough run on the 25/50/100/200 family, everything else identical. First-order upwind advection dominates the error, so $t_b(5\%)$ must converge MONOTONICALLY with observed order $p \sim 1$:

$N = 25$ $t_{b5} = 2732.027658$ s (this case) $N = 50$ $t_{b5} = 2838.542518$ s $N = 100$ $t_{b5} = 2890.669098$ s (batch13, the production mesh) $N = 200$ $t_{b5} = 2915.764796$ s $p(25/50/100) = 1.031$ $p(50/100/200) = 1.055$ Richardson ($p = 1$): $t_{b5}(\text{inf}) \sim 2940.86$ s $\rightarrow N = 100$ sits 1.7% low

The full table, the four overlaid breakthrough curves and the verdict are PUBLISHED in reports/mesh-Study.md (generated from the four runs; the report header cites mesh, tolerances and integrator). The ledger mass closure stays $\leq \sim 1e-11$ on EVERY mesh – conservation is telescopic in the FV form, it does not improve with refinement because it was never discretisation-limited (gate G3).

This case pins the $N = 25$ member; its numbers are DELIBERATELY the worst of the family (t_{b5} 5.8% below batch13) – coarse-mesh optimism made visible, the point of the lesson.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bed	P_final	100000
kpi	bed	Pe	250
kpi	bed	T_final	298
kpi	bed	c_out_over_cin_final_C02	0.999973635802
kpi	bed	carrier_fabricated_mol	9.19148014522
kpi	bed	integral_anchor_C02	3040.50945345
kpi	bed	mass_closure_C02	7.40535661511e-15
kpi	bed	n_C02_final	1.21079316034e-05
kpi	bed	n_He_final	6.86117697691e-05
kpi	bed	physical_mass_closure_rel	0.108446766597
kpi	bed	qbar_C02_final	2.87233754538
kpi	bed	retention_factor_C02	304.051250948
kpi	bed	t_50_C02	3041.40641271
kpi	bed	t_breakthrough_5pct_C02	2732.02788103
... and 20 more reference values in the case's <i>expected</i> .			

combustion

The theory you need. Radical chain chemistry: initiation creates the first radicals (slow – $H_2 + M$, $E_a \sim 105$ kcal), branching multiplies them ($H + O_2 \rightarrow O + OH$: one in, two out), termination removes them. Ignition is the race between branching and termination; the induction time is set by the SLOW build-up, then heat release runs away. Rate constants $k = AT^b e^{-E_a/RT}$ come VERBATIM from cited mechanisms (GRI-Mech 3.0, Ó Conaire 2004, Glarborg 2018) in their native units – the engine converts and announces. Reverse rates from equilibrium: the kinetic end-state must land on the Gibbs equilibrium (the two-engine handshake the NOx tutorial pins at 98.9%). Expect stiffness ratios of 10^6 – 10^8 (use Rosenbrock23), ignition delays in μ s validated against shock-tube data, and thermal NO forming over SECONDS – the timescale gap behind quench-based NOx control. *Full derivation: Theory Guide, the combustion/kinetics chapter.*

ignition01_h2o2

tutorials/batch/combustion/ignition01_h2o2

bomb
 batchReactor

What it does: Adiabatic constant-V H₂/O₂ ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); Rosenbrock23 stiff solver; reports ignition delay

The story: closed rigid vessel, stoichiometric H₂/O₂ in N₂ at 1100 K.

H₂ : O₂ : N₂ = 2 : 1 : 4 (stoichiometric H₂ + 1/2 O₂ → H₂O, 57% N₂)

mode adiabatic + energy gasConstantV: the first law $dU = 0$ in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction ΔH is supplied. Temperature, and the radicals H/O/OH/HO₂, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01_h2o2_rk4 sibling) and the same step blows up.

A closed rigid vessel holds a stoichiometric H₂/O₂ charge diluted in N₂ at 1100 K and ~5 bar. A tiny H-radical seed starts chain branching; after a short induction the radical pool explodes and the mixture ignites — a sharp temperature jump from 1100 K to ~2440 K.

and plot trajectory.csv (T and the radicals H/O/OH vs time). You will see:

The ignition delay τ_{ign} (the time of maximum dT/dt) is ≈ 0.9 ms here.

*The radical lifetimes are $\sim 10^{-12}$ – 10^{-13} s while the induction runs over $\sim 10^{-3}$ s — a timescale separation of $\sim 10^9$. That is stiffness. An explicit method (RK4) is stability-limited to a step of order the *fastest* scale, so it cannot take the $\sim 10^{-12}$ s steps this case uses. The solver here is Rosenbrock23, an L-stable linearly-implicit method that steps by *accuracy*, not stability; at verbosity 3 it prints its stiffness verdict and step counts (the no-silent- crutch credo).*

See it fail on purpose — the sibling case runs the same problem with explicit RK4 at the same step:

*It blows up to NaN within a few steps. (That case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail.) Switch the integrator back to Rosenbrock23 and the same step converges. *See, then decide*, applied to numerics.*

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bomb	P_final	500000
kpi	bomb	T_final	2641.48171412
kpi	bomb	n_H2O2_final	2.59406092875e-11
kpi	bomb	n_H2_final	2.44308154218e-06
kpi	bomb	n_HO2_final	1.97492765295e-10
kpi	bomb	n_H_final	1.4513832323e-06
kpi	bomb	n_N2_final	2.857145e-05
kpi	bomb	n_O2_final	1.02931747994e-06
kpi	bomb	n_OH_final	1.08725431027e-06
kpi	bomb	n_O_final	5.68688863565e-07
kpi	bomb	n_water_final	1.05706749995e-05
kpi	bomb	t_end	0.0002
kpi	bomb	totalMoles_final	4.57220738612e-05

ignition01_h2o2_rk4

tutorials/batch/combustion/ignition01_h2o2_rk4

bomb
 batchReactor

What it does: Adiabatic constant-V H₂/O₂ ignition at 1100 K (GRI-Mech 3.0 H/O kinetics, curated thermo); EXPLICIT RK4 – fails on the stiff chemistry (the lesson)

The story: ignition01_h2o2 – closed rigid vessel, stoichiometric H₂/O₂ in N₂ at 1100 K.

H₂ : O₂ : N₂ = 2 : 1 : 4 (stoichiometric H₂ + 1/2 O₂ → H₂O, 57% N₂)

mode adiabatic + energy gasConstantV: the first law $dU = 0$ in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction ΔH is supplied. Temperature, and the radicals H/O/OH/HO₂, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01_h2o2_rk4 sibling) and the same step blows up.

This is the failure half of the stiffness lesson. It runs the exact same adiabatic constant-volume H₂/O₂ ignition as the sibling case [ignition01_h2o2](../ignition01_h2o2/README.md), but with the explicit RK4 integrator instead of the stiff Rosenbrock23 — at the same fixed step ($\Delta t = 1e-5$ s).

trajectory.csv is a column of nan from the very first output step ($t = 1e-4$ s) onward — there is no clean induction-then-jump, just immediate divergence:

*That immediate blow-up is the result. The radical lifetimes ($\sim 10^{-10}$ – 10^{-12} s) are six orders of magnitude shorter than the induction timescale ($\sim 10^{-3}$ s) — the chemistry is stiff. An explicit method like RK4 is stability-limited to a step of order the *fastest* scale, so the $\sim 10^{-5}$ s step this case uses is far too large: the integration is murdered by stability long before accuracy matters.*

*Compare side-by-side with the stiff sibling: switch to (or open) ignition01_h2o2, whose L-stable adaptive sub-stepping integrates the *same* chemistry to a clean burned state (~ 2440 K). *See, then decide*, applied to numerics.*

This case ships a .expect-nonconvergence marker, so bin/runTests skips it — its whole point is to fail, and it would otherwise trip the NaN guard.

Same as the sibling: the thermodynamics is real curated Choupo data, but the reaction rates are synthetic (hand-chosen Arrhenius constants reproducing the chain-branching structure, not a published mechanism). Do not quote any number from this case as quantitative.

ignition02_h2air_slack

tutorials/batch/combustion/ignition02_h2air_slack

bomb
 batchReactor

What it does: Stoich H₂/air ignition, 1100 K / 2 atm (Slack shock tube; O Conaire 2004 kinetics)

The story: / flowsheetDict Stoichiometric H₂/air autoignition at 1100 K / 2 atm against the Slack shock-tube point (O Conaire 2004 kinetics): the radical pool (H, O, OH, HO₂) builds exponentially through the induction time, then runaway. A stiff-integrator showcase – compare the ignition delay with the measured shock-tube value.

Stoichiometric H₂/air, 1100 K, 2 atm, constant volume — the reflected-shock condition of Slack (1977) as compiled in ? Conaire et al. 2004 (Fig 8). Kinetics: the ? Conaire H₂/O₂ mechanism, Table I verbatim (19 reactions; two Troe falloffs; two summed pairs; A in cm³-mol-s, Ea in kcal — the engine converts and announces every one).

Result: τ_{ign} ($\max dT/dt$) = 43 μs vs the measured ~40–60 μs window.

Two lessons: 1. No seed. This mechanism self-initiates (R5: $\text{H}_2 + \text{M} = 2\text{H}$, $E_a = 105 \text{ kcal/mol}$ — slow, but that slowness IS the induction chemistry). With the ignition01-style trace-H seed the delay collapses to 12 μs : initial conditions are physics, not knobs. 2. The engine now prints τ_{ign} in the epilogue (time of $\max dT/dt$ on the write grid) whenever a vessel ignites — no CSV spelunking.

Run ignition01_h2o2 (GRI kinetics, 5 bar) for the sibling case and the stiffness lesson (_rk4).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	bomb	P_final	202650
kpi	bomb	T_final	3041.74576271
kpi	bomb	n_H2O2_final	3.94467753319e-11
kpi	bomb	n_H2_final	9.82828113933e-07
kpi	bomb	n_HO2_final	4.27531541611e-10
kpi	bomb	n_H_final	3.53516153966e-07
kpi	bomb	n_N2_final	1.391515e-05
kpi	bomb	n_O2_final	3.90750822458e-07
kpi	bomb	n_OH_final	4.44937008544e-07
kpi	bomb	n_O_final	1.5493529764e-07
kpi	bomb	n_water_final	6.00759209227e-06
kpi	bomb	t_end	0.0003
kpi	bomb	totalMoles_final	2.22501764671e-05

nox01_zeldovich_postflame

tutorials/batch/combustion/nox01_zeldovich_postflame

postflame
 batchReactor

What it does: Thermal NO_x in post-flame H₂/air gas, 2125 K / 1 atm (Glarborg 2018 N chemistry)

The story: / flowsheetDict Thermal NO_x in post-flame H₂/air gas at 2125 K / 1 atm: the Zeldovich N-chemistry (Glarborg 2018 rate set) integrated in a batch reactor – NO grows on the SECONDS scale while the O/H radical pool stays equilibrated, the classic separation of scales that makes thermal NO_x a post-flame problem.

Hot burnt gas (lean H₂/air, 2125 K, 1 atm — Homer & Sutton Flame 1 as compiled in Glarborg 2018) held isothermal, starting from the equilibrated H/O radical pool with all nitrogen still as N₂. The Glarborg 2018 nitrogen chemistry (extended Zeldovich + NNH + N₂O routes, Tables 2/5 verbatim) then forms NO.

*1. The cross-engine pin. The kinetic plateau lands on the Gibbs equilibrium computed by a *different engine* (steady Gibbs minimisation vs stiff batch integration). Kinetics choose the path and the clock; the destination belongs to the thermodynamics. Reverse rates come from the same equilibrium constants — so this agreement is a genuine consistency check of the mechanism transcription + thermo, not a tautology. 2. NO_x forms in engineering time. Ignition takes microseconds (ignition02: 43 μs); thermal NO takes seconds at 2125 K. That timescale gap is the whole basis of NO_x control by quench: cool the gas before the Zeldovich clock runs.*

The burnt-gas initial pool was computed with this same thermo by a 14-species gibbsReactor at 2125 K/1 atm (25 % H₂ / 17.5 % O₂ / 57.5 % N₂ feed); the equilibrium NO share was returned to N₂/O₂ elementally.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	postflame	P_final	101325
kpi	postflame	T_final	2125
kpi	postflame	n_H2O2_final	1.05023679159e-12
kpi	postflame	n_H2_final	4.85355503797e-09
kpi	postflame	n_HO2_final	1.61492364784e-11
kpi	postflame	n_H_final	6.02372866884e-10
kpi	postflame	n_N2O_final	1.37258448136e-12
kpi	postflame	n_N2_final	3.72505182272e-06
kpi	postflame	n_NH_final	2.50521342942e-15
kpi	postflame	n_NNH_final	1.01561106781e-15
kpi	postflame	n_NO_final	2.94524618017e-08
kpi	postflame	n_N_final	2.26160617192e-14
kpi	postflame	n_O2_final	3.06851934926e-07
kpi	postflame	n_OH_final	2.0389175016e-08

*...and 4 more reference values in the case's **expected**.*

reactor

The theory you need. Three idealised steady reactors. Conversion reactor: stoichiometry only, $n_{\text{out}} = n_{\text{in}} + \nu \xi$, duty from the ONE enthalpy base $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$ (formation datum – no separate “heat of reaction” input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate $r(c_{\text{out}}, T)$, so $F(c_{\text{in}} - c_{\text{out}}) = rV$. PFR: the same rate integrated along the volume, $dF_i/dV = \nu_i r$. Arrhenius kinetics $k = AT^b e^{-E_a/RT}$, orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential T sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

batch01_first_order

tutorials/batch/reactor/batch01_first_order

reactor
 batchReactor

What it does: Isothermal batch Fischer esterification ($\text{AcOH} + \text{EtOH} \rightleftharpoons \text{EtOAc} + \text{H}_2\text{O}$) at 350 K, pseudo-first-order in ethanol; analytical exp-decay comparison

The story: isothermal pseudo-first-order batch reactor.

Single vessel, pseudo-first-order esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$, rate = $k \cdot [\text{ethanol}]$). Initial charge is an equimolar ethanol + acetic acid mix; kinetics from the reactions library (Arrhenius $A=1\text{e}8$ 1/s, $E_a=70$ kJ/mol). At $T = 350$ K:

$$k(T) = 1\text{e}8 \cdot \exp(-70000 / (8.314 \cdot 350)) \approx 3.57\text{e-}3 \text{ s}^{-1} \quad \tau = 1/k \approx 280 \text{ s}$$

Analytical solution for first-order in a constant-volume batch:

$$n_A(t) = n_A(0) \cdot \exp(-k t)$$

The simulator integrates $dn_A/dt = -k \cdot (n_A / V) \cdot V = -k n_A$ with RK4 at $dt = 1$ s; final $t = 600$ s gives $X_A \approx 1 - \exp(-k \cdot 600) \approx 1 - \exp(-2.14) \approx 0.882$

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	3.75075346157e-16
kpi	campaign	element_H_closure_rel	3.00060276926e-16
kpi	campaign	element_O_closure_rel	2.50050230771e-16
kpi	campaign	element_worst_closure_rel	3.75075346157e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	-210.150638715
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	0
kpi	campaign	energy_dH_vessels_kJ	-210.150638715
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	energy_residual_kJ	0
kpi	campaign	mass_closure_rel	4.5240474843e-16
kpi	campaign	mass_kg_external_out	0

... and 20 more reference values in the case's *expected*.

batch02_adiabatic

tutorials/batch/reactor/batch02_adiabatic

reactor

batchReactor

What it does: Adiabatic batch esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$), exothermic, starting cold (320 K); thermal runaway to ~428 K then plateau as ethanol is consumed

The story: exothermic pseudo-first-order esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$) in an adiabatic batch reactor. Designed to show thermal runaway:

$T_0 = 320\text{ K}$ (kinetics very slow: $k_0 \sim 3.7\text{e-}4\text{ s}^{-1}$, $\tau_0 \sim 2700\text{ s}$). $dH_{\text{rxn}} = -26.06\text{ kJ/mol}$ of ethanol, computed from the species' gibbsFormation (elements datum); lifts $T \sim 108\text{ K}$ at full conversion ($320 \rightarrow 428\text{ K}$), enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

T | _____ plateau (~428 K) | _/ | _/ | _/ | / T_0 | _/ + _____ $\rightarrow$ t induction runaway

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at 428 K.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.2212252
kpi	reactor	n_aceticAcid_final	9.87885051063e-31
kpi	reactor	n_ethanol_final	9.87885051063e-31
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch03_consecutive

tutorials/batch/reactor/batch03_consecutive

reactor
 batchReactor

What it does: Isothermal batch reactor: consecutive $A \rightarrow B \rightarrow C$ with $k_1 > k_2$; intermediate B peaks then decays

The story: two first-order reactions in series:

with $k_1 > k_2$. This is the textbook series-reaction setup: B is produced from A faster than it is consumed to C, so n_B grows at first, reaches a maximum, then decays as A runs out.

Analytical solution (constant V, isothermal, first-order):

$$n_A(t) = n_{A0} * \exp(-k_1 t) \quad n_B(t) = n_{A0} * k_1 / (k_2 - k_1) * [\exp(-k_1 t) - \exp(-k_2 t)] \quad n_C(t) = n_{A0} - n_A(t) - n_B(t)$$

$$t_{\max} = \ln(k_1 / k_2) / (k_1 - k_2)$$

With $k_1 = 0.01 \text{ 1/s}$ and $k_2 = 0.002 \text{ 1/s}$:

$$t_{\max} = \ln(5) / 0.008 \sim 201 \text{ s} \quad n_{B_{\max}} / n_{A0} = (k_1/k_2)^{k_2/(k_2-k_1)} = 5^{(-0.25)} \sim 0.669$$

The case demonstrates: 1. multi-reaction support in BatchReactor (v0.21); 2. time-dependent selectivity — harvesting B at $t \sim 200 \text{ s}$ maximises yield, waiting longer wastes B to C.

No species supplies activation energy: both A_{pre} are quoted with $E_a = 0$ so that the rate constants are exactly the values above regardless of T (the case is isothermal anyway).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	1
kpi	campaign	energy_records_valid	0
kpi	campaign	mass_closure_rel	5.55111512313e-16
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.5
kpi	campaign	mass_kg_initial	0.5
kpi	campaign	mass_residual_kg	2.77555756156e-16
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0
kpi	campaign	transfers_logged	0
kpi	reactor	P_final	101300
kpi	reactor	T_final	350
kpi	reactor	n_compA_final	3.05902320887e-09

... and 4 more reference values in the case's *expected*.

batch04_reversible_wgs

tutorials/batch/reactor/batch04_reversible_wgs

reactor batchReactor

What it does: Batch reversible water-gas shift ($\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$) at 600 K; conversion relaxes to X_{eq}

The story: the reversible water-gas shift in a closed BATCH reactor (choupoBatch), the time-domain sibling of cstr02 / pfr02.

Equimolar $\text{CO} + \text{H}_2\text{O}$ charged at 600 K. The forward Arrhenius rate drives conversion up; the reverse term ($k_{\text{rev}} = k_{\text{fwd}} / K_{\text{eq}}$, K_{eq} from the four species' $g_{\text{pure_ig}}$, re-evaluated each step) brakes it. The batch trajectory relaxes onto the SAME thermodynamic ceiling the CSTR and PFR reached — $X \rightarrow X_{\text{eq}} \sim 0.84$ — then sits there (net rate $\rightarrow 0$), instead of running to $X = 1$.

The forward A is the illustrative value tuned for stable RK4 (see constant/reactions); the equilibrium X_{eq} is A-independent.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	1000000
kpi	reactor	T_final	600
kpi	reactor	n_CO2_final	0.000420715762123
kpi	reactor	n_CO_final	7.92842378768e-05
kpi	reactor	n_H2_final	0.000420715762123
kpi	reactor	n_water_final	7.92842378768e-05
kpi	reactor	t_end	1200
kpi	reactor	totalMoles_final	0.001

batch04_transient_dirs

tutorials/batch/reactor/batch04_transient_dirs

reactor batchReactor

What it does: Adiabatic batch esterification with OpenFOAM-style REAL-TIME instant dirs: watch conversion climb $0 \rightarrow 1$ and T runaway $320 \rightarrow \sim 428$ K by scrubbing 0/ 30/ 60/ ... s

The story: the batch02_adiabatic runaway re-run with REAL-TIME instant directories: the integrator dumps 0/ 30/ 60/ ... s snapshots (OpenFOAM-style) so the trajectory – conversion $0 \rightarrow 1$, T $320 \rightarrow \sim 428$ K – can be scrubbed frame by frame in the GUI.

The physics below is inherited from batch02_adiabatic, unchanged: exothermic pseudo-first-order esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$) in an adiabatic batch reactor. Designed to show thermal runaway:

$T_0 = 320$ K (kinetics very slow: $k_0 \sim 3.7\text{e-}4 \text{ s}^{-1}$, $\tau_0 \sim 2700$ s). $dH_{\text{rxn}} = -26.06$ kJ/mol of ethanol, computed from the species' gibbsFormation (elements datum); lifts T ~ 108 K at full conversion, enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

T | _____ plateau (~ 428 K) | _/ | _/ | _/ | _/ | / T_0 | _/ | + _____ $\rightarrow$ t induction runaway

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at ~ 430 K.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.2212252
kpi	reactor	n_aceticAcid_final	9.87885051063e-31
kpi	reactor	n_ethanol_final	9.87885051063e-31
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch05_crystalliser

tutorials/batch/reactor/batch05_crystalliser

cryst batchCrystalliser

What it does: Batch crystalliser: supersaturated sugar relaxes, PBE by method of moments

The story: Single closed vessel. Charge: ~500 kg water + ~1250 kg dissolved sucrose ($c_0 \sim 2.5$ kg/kg, above $c_{\text{sat}}(20\text{ C}) \sim 2.03$ kg/kg, so $S_0 \sim 1.23$). Unseeded: the moments start at zero and primary nucleation ($j=0$) bootstraps the population from the initial supersaturation. Kinetics from constant/crystallisation.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	6.81588841762e-15
kpi	campaign	element_H_closure_rel	4.18663266355e-15
kpi	campaign	element_O_closure_rel	4.18663266355e-15
kpi	campaign	element_worst_closure_rel	6.81588841762e-15
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	mass_closure_rel	5.07045231785e-15
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	1748.87224776
kpi	campaign	mass_kg_initial	1748.87224776
kpi	campaign	mass_residual_kg	8.86757334229e-12
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0

...and 13 more reference values in the case's *expected*.

batch06_adaptive_runaway

tutorials/batch/reactor/batch06_adaptive_runaway

reactor

batchReactor

What it does: Adiabatic esterification runaway (AcOH+EtOH); ADAPTIVE Rosenbrock23 time-stepping — the step grows after the fast transient (glass-box step history)

The story: the batch02_adiabatic runaway integrated with ADAPTIVE Rosenbrock23 time-stepping: the step shrinks through the fast runaway and GROWS again on the plateau, and the step history is printed glass-box so the student sees the controller at work.

The physics below is inherited from batch02_adiabatic, unchanged: exothermic pseudo-first-order esterification ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$) in an adiabatic batch reactor. Designed to show thermal runaway:

$T_0 = 320\text{ K}$ (kinetics very slow: $k_0 \sim 3.7\text{e-}4\text{ s}^{-1}$, $\tau_0 \sim 2700\text{ s}$). $dH_{\text{rxn}} = -26.06\text{ kJ/mol}$ of ethanol, computed from the species' gibbsFormation (elements datum); lifts $T \sim 100\text{ K}$ at full conversion, enough to show the runaway. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean.

Expected qualitative trajectory:

T | _____ plateau (~450 K) | _/ | _/ | _/ | _/ | / T_0 |___/ +—————→ t induction runaway

The energy balance is just:

$$dT/dt = -r * V * dH_{\text{rxn}} / (n_A * Cp_A + n_B * Cp_B)$$

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology — consistent with a pedagogical demonstration, not a quantitative prediction at 450 K.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	428.221319512
kpi	reactor	n_aceticAcid_final	8.88140514762e-19
kpi	reactor	n_ethanol_final	1.10133235387e-34
kpi	reactor	n_ethylAcetate_final	0.00925
kpi	reactor	n_water_final	0.00925
kpi	reactor	t_end	600
kpi	reactor	totalMoles_final	0.0185

batch07_lhhw_methylAcetate

tutorials/batch/reactor/batch07_lhhw_methylAcetate

reactor batchReactor

What it does: LHHW in a BATCH reactor: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

The story: the SAME rate law as cstr07, in a BATCH vessel.

$\text{CH}_3\text{OH} + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$ over Amberlyst 15

Popken, Gotze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601, Eq 16: the adsorption-based LHHW on UNIQUAC activities. Nothing about the rate law is restated here – it lives in constant/reactions, exactly as the steady CSTR reads it, because the SAME ‘RateLaw’ evaluates it in both. A batch reactor and a continuous one cannot disagree about what an adsorption denominator means.

What changes is only what a batch vessel does with a rate: it integrates it. The charge is equimolar methanol + acetic acid; the trajectory climbs to the ACTIVITY equilibrium, $X = 76.6\%$, and stops. That plateau is asserted nowhere: it is where $k_1 a'_{\text{HOAc}} a'_{\text{MeOH}} = k_{-1} a'_{\text{MeOAc}} a'_{\text{H}_2\text{O}}$, i.e. the $K_a = 25.50$ the authors’ independent Eq 17 recovers from the same eight constants – and the same number the steady CSTR climbs to as its volume grows. Two reactors, two integration schemes, one equilibrium.

Water is the strongest adsorber ($K/M = 0.29$ against 0.052 for the acid), so as the reaction MAKES water it crowds the acid off the resin and the denominator grows. That self-inhibition is why reactive distillation, which pulls the water out as it forms, is worth its capital.

Do NOT try to see the inhibition by deleting the ‘adsorption’ block and keeping k_1 . It tells you nothing: the constants were REGRESSED for this rate law, on these activities, and a rate constant without its rate law is a number without a meaning. (Delete the denominator and the reaction gets FASTER, because the adsorbed activities a' are small and dividing by their square is a multiplication. That is an arithmetic accident, not physics.) To SEE the inhibition, spike the charge with water at the same acid and alcohol fractions:

methanol 0.4, aceticAcid 0.4, water 0.0 → initial rate 58.3 mmol/(L.s) methanol 0.4, aceticAcid 0.4, water 0.2 → initial rate 29.0 mmol/(L.s)

Half the rate, with every reactant fraction unchanged. The water is not a reactant here; it is sitting on the sites.

‘catalystLoading’ (kg of dry resin per m^3) is what turns the paper’s per-gram rate constants into a volumetric rate. With concentrations in kmol/m^3 the conversion is exact and unit-free: $\text{mol}/(\text{g.s}) \times \text{kg}/\text{m}^3 = \text{kmol}/(\text{m}^3.\text{s})$.

The vessel runs isothermal at 333.15 K, the top of the authors’ data range. Do not extrapolate: their activation energies are APPARENT values, fitted over 303-333 K.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	333.15
kpi	reactor	n_aceticAcid_final	0.00216454708498
kpi	reactor	n_methanol_final	0.00216454708498
kpi	reactor	n_methylAcetate_final	0.00708545291502
kpi	reactor	n_water_final	0.00708545291502
kpi	reactor	t_end	3600
kpi	reactor	totalMoles_final	0.0185

batch08_isothermal_duty

tutorials/batch/reactor/batch08_isothermal_duty

reactor

batchReactor

What it does: Isothermal jacket duty from a DECLARED dH_rxn: $E = dH \cdot dX_i$ via extents (lumped species, no formation data)

The story: one isothermal batch reactor, one declared dH_rxn, one hand-checkable jacket duty. See system/controlDict for what this case tests and constant/reactions for the analytic anchor ($E = -50 \text{ kJ/mol} \times 0.9975 \text{ mol} = -49.88 \text{ kJ}$ over 600 s).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	energy_balance_available	0
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	1
kpi	campaign	mass_closure_rel	1.11022302463e-15
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.05
kpi	campaign	mass_kg_initial	0.05
kpi	campaign	mass_residual_kg	5.55111512313e-17
kpi	campaign	transfers_H_invalid	0
kpi	campaign	transfers_H_valid	0
kpi	campaign	transfers_logged	0
kpi	reactor	E_reaction_seg0_kJ	-49.8760623911
kpi	reactor	E_reaction_total_kJ	-49.8760623911
kpi	reactor	P_final	101300
... and 5 more reference values in the case's expected.			

recipes

The theory you need. A recipe is time-triggered discrete control over batch units: charge, heat/cool, hold, transfer, sample. Between events the DAE integrator runs; at each event the state jumps (a transfer moves holdup between vessels). Expect exact event times on the trajectory and holdup bookkeeping that conserves mass across transfers.

recipe01_react_then_distill

tutorials/batch/recipes/recipe01_react_then_distill



What it does: Two-vessel batch recipe: esterify ($\text{AcOH} + \text{EtOH} \rightarrow \text{EtOAc} + \text{H}_2\text{O}$) in a reactor, transfer the charge to a still, then distil

The story: the canonical 2-stage batch recipe.

Stage 1 ($t = 0 \rightarrow 400$ s): A batch reactor at 350 K esterifies ethanol + acetic acid $\rightarrow$ ethyl acetate + water (the same first-order Arrhenius reaction as batch01). $k \approx 3.58\text{e-}3 \text{ s}^{-1}$ at 350 K, $\tau \approx 280$ s. Over 400 s the ethanol conversion reaches $\sim 76\%$.

Stage 2 ($t = 400 \rightarrow 1200$ s): The recipe TRANSFERS the reactor contents to an initially empty batch still. The still then runs a Rayleigh distillation at $F_{\text{vap}} = 1.5\text{e-}6 \text{ kmol/s}$ for 800 s, vaporising $1.2\text{e-}3 \text{ kmol}$ of liquid. Starting from the converted mix (ethyl acetate + water + unreacted ethanol/acid), the still progressively strips the more volatile ethyl acetate + ethanol; pot composition follows the Raoult y-x curve. (The thermoPackage here uses ideal Raoult for pedagogical simplicity.)

This case demonstrates the recipe layer (v0.23): a list of time-triggered events acting on named units. Each event in the ‘recipe’ block below has a ‘time’, an ‘action’, and action-specific keys. Actions implemented in v0.23:

transfer move all contents of ‘from’ into ‘to’ setParameter change a named operating parameter on a unit

Events fire at the next time-loop step boundary after t_{trigger} ; the trajectory CSV thus shows a single-step discontinuity in the transferred quantities.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	1.12522603847e-15
kpi	campaign	element_H_closure_rel	1.05021096924e-15
kpi	campaign	element_O_closure_rel	1.12522603847e-15
kpi	campaign	element_worst_closure_rel	1.12522603847e-15
kpi	campaign	energy_H_external_kJ	-5989.16236175
kpi	campaign	energy_Q_ledger_kJ	-149.260408348
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	5.98873860625e-16
kpi	campaign	energy_dH_vessels_kJ	5839.9019534
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	6
kpi	campaign	energy_residual_kJ	4.54747350886e-12
kpi	campaign	mass_closure_rel	1.01791068397e-15
kpi	campaign	mass_kg_external_out	0.888559895256
... and 31 more reference values in the case's <i>expected</i> .			

recipe02_setpoint_ramp

tutorials/batch/recipes/recipe02_setpoint_ramp



What it does: Single-reactor recipe with T_setpoint steps: 320 K → 350 K @ 200 s → 380 K @ 500 s

The story: isothermal batch reactor with a staged temperature programme controlled by the recipe layer.

The reactor starts at 320 K, where the first-order Arrhenius reaction is very slow ($k_{320} \approx 3.7\text{e-}4 \text{ s}^{-1}$, $\tau \approx 2700 \text{ s}$). Two recipe events ramp the setpoint upward:

$t = 200 \text{ s} \rightarrow 320 \text{ K} \rightarrow 350 \text{ K}$ (k jumps $\sim 10\text{x}$ to $3.58\text{e-}3 \text{ s}^{-1}$) $t = 500 \text{ s} \rightarrow 350 \text{ K} \rightarrow 380 \text{ K}$ (k jumps another $\sim 7\text{x}$ to 0.025 s^{-1})

In the trajectory CSV the temperature column shows a clean step-change at each event, and the conversion curve shows the corresponding three regimes:

$0 \leq t < 200 \text{ s}$ barely-moving plateau (slow) $200 \leq t < 500 \text{ s}$ visible drop in n_{ethanol} (moderate) $500 \leq t$ rapid quench to full conversion (fast)

Demonstrates the recipe layer’s ‘setParameter’ action without inter-unit transfers, so this is the cleanest possible test of the parameter-change pathway.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	3.75075346157e-16
kpi	campaign	element_H_closure_rel	3.00060276926e-16
kpi	campaign	element_O_closure_rel	2.50050230771e-16
kpi	campaign	element_worst_closure_rel	3.75075346157e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	-104.365240634
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	0
kpi	campaign	energy_dH_vessels_kJ	-104.365240634
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	5
kpi	campaign	energy_residual_kJ	0
kpi	campaign	mass_closure_rel	4.5240474843e-16
kpi	campaign	mass_kg_external_out	0
...and 21 more reference values in the case’s expected.			

recipe03_conditional_partial

tutorials/batch/recipes/recipe03_conditional_partial

reactor
 batchReactor

still
 batchStill

What it does: Conditional + partial recipe: react until $x_{\text{ethanol}} < 0.30$, then transfer 50% to a still

The story: recipe extensions (v0.50).

Two recipe features beyond the time-triggered transfer/setParameter of v0.23:

1. CONDITION-TRIGGERED event ('when { ... }' instead of 'time'): the event fires the first step its condition becomes true, evaluated on the live vessel state. Quantities: T, total, $n_{\text{<comp>}}$, $x_{\text{<comp>}}$; operators: gt, ge, lt, le.
2. PARTIAL transfer ('fraction'): a fraction (0..1) of the source moves; the source keeps the rest. Default 1.0 == empty the source.

Here the reactor runs the esterification (ethanol + acetic acid $\rightarrow$ ethyl acetate + water; the batch01 first-order Arrhenius test, $k \sim 3.58\text{e-}3 \text{ s}^{-1}$ at 350 K, so $n_{\text{ethanol}} = n_{\text{ethanol}}(0). \exp(-k t)$). The charge is equimolar (x_{ethanol} starts at 0.5); when the ethanol mole fraction drops below 0.30, the recipe transfers HALF the reactor contents to a still; the reactor keeps the other half and reacts on, while the still distils its charge.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	P_final	101300
kpi	reactor	T_final	350
kpi	reactor	n_aceticAcid_final	6.33853236066e-05
kpi	reactor	n_ethanol_final	6.33853236066e-05
kpi	reactor	n_ethylAcetate_final	0.00456161467639
kpi	reactor	n_water_final	0.00456161467639
kpi	reactor	t_end	1200
kpi	reactor	totalMoles_final	0.00925
kpi	still	P_final	101300
kpi	still	T_final	390.833339675
kpi	still	n_aceticAcid_final	0
kpi	still	n_ethanol_final	0
kpi	still	n_ethylAcetate_final	0
kpi	still	n_water_final	0

... and 2 more reference values in the case's *expected*.

recipe04_charge_enthalpy

tutorials/batch/recipes/recipe04_charge_enthalpy

hot
 batchAccumulator

cold
 batchAccumulator

What it does: H-equality recipe charge: T_{mix} from the first law, not the molar average; balance CLOSES across a transfer

The story: mixing is a FIRST-LAW act.

Two passive tanks. ‘hot’ holds 0.5 mol of ethanol at 340 K; ‘cold’ holds 1.0 mol of water at 300 K. At $t = 100$ s the recipe empties the hot tank into the cold one. The charge solves the mix temperature from enthalpy equality on the elements datum,

$$H(n_{\text{EtOH}} + n_{\text{W}}, T_{\text{mix}}) = H(n_{\text{EtOH}}, 340 \text{ K}) + H(n_{\text{W}}, 300 \text{ K}),$$

so heat capacity DIFFERENCES matter: ethanol carries $\sim 112 \text{ J}/(\text{mol K})$ against water’s ~ 75 , and the enthalpy-true T_{mix} ($\sim 317 \text{ K}$) sits $\sim 4 \text{ K}$ above the equal- C_p molar average (313.3 K) a naive mixing rule gives.

Because both species price on the elements datum, no jacket acts, and nothing leaves, the campaign ENERGY BALANCE is AVAILABLE and closes: $dH_{\text{vessels}} = 0$ to the mix-solve tolerance — the transfer moved enthalpy between tanks without creating or destroying any.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_O_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	2.13028428746e-12
kpi	campaign	energy_dH_vessels_kJ	5.81991344006e-10
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	0
kpi	campaign	energy_residual_kJ	5.81991344006e-10
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0

... and 18 more reference values in the case’s *expected*.

still

The theory you need. A batch still is Rayleigh distillation: $d(n x_i)/dt = -\dot{V} y_i$ with y_i in equilibrium with the pot – the pot composition DRIFTS as the light component leaves (contrast the steady column, where stages hold profiles constant). Expect the distillate to start rich and fade, and the recipe layer (charge, heat, switch receivers) to drive the run. *Full derivation: Theory Guide, the batch-distillation section.*

still01_benzene_toluene

tutorials/batch/still/still01_benzene_toluene

still
 batchStill

What it does: Rayleigh batch distillation: benzene/toluene 50/50, 1 atm, ideal Raoult

The story: Rayleigh batch distillation, ideal Raoult.

Pot starts with 1 mol of equimolar benzene/toluene at 1 atm. A constant vapour offtake $F_{\text{vap}} = 1.5\text{e-}6$ kmol/s strips the pot over 600 s — by linear mass balance the pot should hold

i.e. $L(600\text{ s}) = 1\text{ mmol} - 0.9\text{ mmol} = 0.1\text{ mmol}$ (90 % distilled).

Benzene is the more volatile component ($\alpha \sim 2.4$ at 1 atm), so the still pot progressively concentrates in toluene as benzene flashes off preferentially. Classical Rayleigh integration with constant α gives the closed form

$$\ln(L / L_0) = (1 / (\alpha - 1)) [\ln(x / x_0) + \alpha * \ln((1 - x_0) / (1 - x))]$$

where $x = x_{\text{benzene}}$ in the still pot. Plotting (L/L_0) vs x_{pot} from trajectory.csv should track this analytical curve to RK4 precision.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	2.66880534766e-16
kpi	campaign	element_H_closure_rel	1.23908819713e-16
kpi	campaign	element_worst_closure_rel	2.66880534766e-16
kpi	campaign	energy_H_external_kJ	38.7543769459
kpi	campaign	energy_Q_ledger_kJ	0.942780166065
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	4.58362894565e-16
kpi	campaign	energy_dH_vessels_kJ	-37.8115967798
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	1.7763568394e-14
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0.0760287317788
kpi	campaign	mass_kg_final	0.00909876822121

... and 15 more reference values in the case's *expected*.

still02_ethanol_water_azeo

tutorials/batch/still/still02_ethanol_water_azeo

still
 batchStill

What it does: Rayleigh batch distillation: ethanol/water 40/60 with NRTL, residue concentrates in water

The story: Rayleigh batch distillation of a non-ideal binary with an azeotrope.

System: ethanol / water with NRTL, pressure 1 atm. Minimum-boiling positive azeotrope at $x_{\text{eth}} \sim 0.894$ (mole), $T \sim 351.3$ K.

Pot starts at $x_{\text{eth}} = 0.40$ (a "wine"-like feed — well BELOW the azeotrope). Boil-off rate $F_{\text{vap}} = 1.5\text{e-}6$ kmol/s for 600 s consumes 90 % of the pot, leaving a 10 % residue.

Below the azeotrope, $\gamma_{\text{ethanol}} > 1$ and $\gamma_{\text{water}} \sim 1$, so the vapour is markedly richer in ethanol than the liquid ($y_{\text{eth}} \sim 0.61$ at $x_{\text{eth}} = 0.40$). Ethanol therefore vaporises preferentially and the still pot progressively concentrates in WATER — i.e. the residue at end-of-run is mostly water.

The pedagogical contrast with still01 (ideal Raoult, benzene/toluene) is the SHAPE of the enrichment curve:

Raoult: smooth, alpha-driven ($\alpha \sim 2.4$, roughly constant). NRTL : curvature changes as x crosses concentration regimes in which γ_{ethanol} shrinks toward 1; the still composition curve $x_{\text{pot}}(t)$ is noticeably non-linear vs the linear pot-volume curve $L(t) = L_0 - F_{\text{vap}}t$.

Crucially, this case demonstrates that a single Rayleigh still cannot cross the azeotrope. Starting below 0.894 in ethanol, the pot can only move DOWNWARD in ethanol fraction; no amount of boiling will lift the pot composition past 0.894. (To break the azeotrope one needs entrainer, pressure swing, or extractive distillation — topics for later tutorials.)

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	still	P_final	101300
kpi	still	T_final	372.446750369
kpi	still	n_ethanol_final	3.38836968904e-11
kpi	still	n_water_final	9.99999650927e-05
kpi	still	t_end	600
kpi	still	totalMoles_final	9.99999989764e-05

still03_with_receiver

tutorials/batch/still/still03_with_receiver

still
 batchStill

receiver
 batchAccumulator

What it does: Rayleigh still distilling INTO a receiver tank (dischargeTo accumulator)

The story: Rayleigh batch distillation INTO a receiver (accumulator vessel + continuous dischargeTo).

Same pot as still01 (1 mol equimolar benzene/toluene, $F_{\text{vap}} = 1.5\text{e-}6$ kmol/s, ideal Raoult), but the still's vapour is no longer lost: the 'dischargeTo receiver;' line routes the condensed distillate, step by step, into a passive 'batchAccumulator'. So the trajectory shows the POT depleting AND the RECEIVER filling — the receiver collects the total distillate (amount + composition). Mass is conserved between the two vessels: pot + receiver = 1 mol at all times. Benzene is the more volatile component, so the receiver is benzene-rich early and the cumulative receiver composition drifts toward 0.5 as distillation nears completion (it ends holding everything the pot lost).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	2.66880534766e-16
kpi	campaign	element_H_closure_rel	1.23908819713e-16
kpi	campaign	element_worst_closure_rel	2.66880534766e-16
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.942780166065
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	3.00056993907e-11
kpi	campaign	energy_dH_vessels_kJ	0.942780163768
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	2.2974155911e-09
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275

... and 28 more reference values in the case's *expected*.

still04_rectifier_benzene_toluene

tutorials/batch/still/still04_rectifier_benzene_toluene

still
 batchStill

receiver
 batchAccumulator

What it does: batch rectification: 3 stages + total condenser, $R=3$, with receiver tank (dischargeTo accumulator)

The story: the CLASSICAL batch rectification (model rectifier, design forum #85): the Rayleigh pot now feeds a column of 3 ideal stages topped by a total condenser at constant reflux $R = 3$.

F_{vap} is the internal BOILUP $V = 1.5\text{e-}6$ kmol/s; the distillate PRODUCT is $D = V/(R+1) = 3.75\text{e-}7$ kmol/s and the reflux $L = R.D$ returns to the column. The pot loses PRODUCT only: $dn_i/dt = -D.x_D_i$, with $x_D(t)$ solved at every instant from the quasi-steady cascade (stage balances + bubble-T equilibrium + total-condenser closure – a declared residual, printed on failure).

What to watch (trajectory.csv): still.xD_benzene starts ~ 0.949 (vs ~ 0.71 for the raw pot vapour – the column working) and DRIFTS DOWN as the pot strips: the constant- R policy trades purity for time. The receiver's ACCUMULATED benzene fraction sits between the instantaneous x_D and the feed – two different compositions, deliberately both visible. The pot loses only $D.dt = 0.225$ mmol over 600 s (vs 0.9 mmol in still01: same boilup, but 3/4 of it refluxes).

Physics anchors (verified at authoring): $x_D(0)$ grows with nStages and R ($N=1, R=1$: 0.806; $N=2, R=3$: 0.909; $N=3, R=3$: 0.949) and the $R \rightarrow \text{infinity}$ limit ($R=200$: 0.9765) lands inside the Fenske band $\alpha^{(N+1)}$ for $\alpha = 2.35\text{--}2.6$. With nStages 0 and refluxRatio 0 the model reproduces still01's Rayleigh trajectory to the last digit.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.534303338765
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	7.08707897844e-10
kpi	campaign	energy_dH_vessels_kJ	0.534303321211
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	1.75539529579e-08
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275

...and 26 more reference values in the case's *expected*.

still05_rectifier_constant_xD

tutorials/batch/still/still05_rectifier_constant_xD

still
 batchStill

receiver
 batchAccumulator

What it does: batch rectification, constant-composition policy: $R(t)$ holds x_D until refluxMax , then $D=0$

The story: the SECOND classical reflux policy (design forum #85-3/4): hold the distillate composition constant by RAISING the reflux as the pot strips, and STOP when the declared ceiling can no longer reach the target.

Same pot and column as still04 (1 mol equimolar benzene/toluene, $V = 1.5\text{e-}6$ kmol/s, 3 ideal stages + total condenser), but instead of a fixed R the policy solves $R(t)$ each step – a BRACKETED solve over $[0, \text{refluxMax}]$ on a feasibility map, never a bare Newton – so that $x_D[\text{benzene}] = 0.93$ exactly. As benzene depletes, R climbs (~ 2 at the start); when even $\text{refluxMax} = 8$ cannot reach the target the run announces ‘onRefluxLimit stopDistillation’: $D = 0$, the pot stops losing mass, and the campaign idles VISIBLY to endTime with $\text{still.refluxLimitReached} = 1$ in the trajectory – never a magic termination.

What to watch (trajectory.csv): still.xD_benzene pinned at 0.9300 while R climbs – the mirror image of still04, which held R and let purity drift; still.R rising $2.0 \rightarrow 8$ (the ceiling), then the limit event; the receiver’s accumulated composition ~ 0.93 (the whole cut is on-spec – THE selling point of this policy); pot inventory constant after $t_{\text{refluxLimit}}$.

Declared refusals (each with its own diagnostic, try them): $x_D\text{-target } 0.5 \rightarrow$ BELOW the $R=0$ floor (even zero reflux over-purifies); $x_D\text{-target } 0.995$ with $\text{refluxMax } 5 \rightarrow$ UNREACHABLE from the charge; ethanol/water NRTL with $x_D\text{-target } 0.95 \rightarrow$ the 0.894 azeotrope caps the cascade at ANY reflux (the barrier is real VLE, not a message).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	element_C_closure_rel	0
kpi	campaign	element_H_closure_rel	0
kpi	campaign	element_worst_closure_rel	0
kpi	campaign	energy_H_external_kJ	0
kpi	campaign	energy_Q_ledger_kJ	0.690222586439
kpi	campaign	energy_balance_available	1
kpi	campaign	energy_closure_rel	8.89386285064e-10
kpi	campaign	energy_dH_vessels_kJ	0.69022261392
kpi	campaign	energy_records_invalid	0
kpi	campaign	energy_records_valid	2
kpi	campaign	energy_residual_kJ	2.74809703882e-08
kpi	campaign	mass_closure_rel	0
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.0851275

...and 28 more reference values in the case’s *expected*.

still06_ledger_mixed_validity

tutorials/batch/still/still06_ledger_mixed_validity

potA
batchStill

recA
batchAccumulator

dump
batchAccumulator

fresh
batchAccumulator

potB
batchStill

recB
batchAccumulator

spike
batchAccumulator

What it does: Ledger H-validity is MONOTONIC: mixed valid/invalid packages never resurrect a poisoned record

The story: MONOTONIC enthalpy validity on the material ledger (the adversarial case of design forum #99/#101).

The campaign ledger integrates each transfer's enthalpy PACKAGE BY PACKAGE at each package's own temperature. A package containing a species with NO formation datum (here 'compA', a lumped teaching species that deliberately carries no gibbsFormation block) cannot be priced — the record must mark H invalid and NAME the species. The contract under test is the temporal one:

once ONE package is unpriceable, the record's integrated H has lost a parcel FOREVER. Validity is MONOTONIC: a later priceable package must never resurrect it.

Two stills exercise the two orders on ONE accumulated record each:

potA (invalid → valid): starts with 10 % compA, so its distillate record potA→recA begins UNPRICEABLE. At $t = 200$ s the recipe dumps the whole pot to 'dump' and recharges clean benzene/toluene from 'fresh'; every later distillate package is priceable. The record must STAY invalid ($H_{\text{missing}} = \text{compA}$), and the discrete potA→dump record is invalid too (it carried the compA).

potB (valid → invalid): starts clean, so its distillate record potB→recB begins priceable and accumulates real enthalpy. At $t = 200$ s the recipe spikes it with pure compA from 'spike'; every later package is unpriceable. The record flips invalid and the partial integral is WITHHELD — an emitted H_{kJ} that silently lost its tail would be a lie.

Five ledger records result: potA→recA (continuous, poisoned early), potB→recB (continuous, poisoned late), potA→dump and spike→potB (discrete, carrying compA: unpriceable), fresh→potA (discrete, clean: the ONE priceable record). The golden pins the campaign census transfers $H_{\text{valid}} = 1 / \text{transfers_H_invalid} = 4$; the results JSON carries $H_{\text{missing}} = [\text{compA}]$ and NO H_{kJ} key on every poisoned record.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	campaign	mass_closure_rel	2.1198000621e-16
kpi	campaign	mass_kg_external_out	0
kpi	campaign	mass_kg_final	0.26186975
kpi	campaign	mass_kg_initial	0.26186975
kpi	campaign	mass_residual_kg	5.55111512313e-17
kpi	campaign	transfers_H_invalid	4
kpi	campaign	transfers_H_valid	1
kpi	campaign	transfers_logged	5
kpi	dump	P_final	101300
kpi	dump	T_final	372.484229602
kpi	dump	n_benzene_final	0.000249727144577
kpi	dump	n_compA_final	9.64864338217e-05
kpi	dump	n_toluene_final	0.000353786421588
kpi	dump	t_end	600

... and 43 more reference values in the case's *expected*.

Category IV: Dynamics and control (choupoCtrl)

Continuous processes in time with feedback: PID loops, setpoint schedules, disturbances and responses.

ctrl01_cstr_temp_control

tutorials/ctrl/ctrl01_cstr_temp_control

reactor
 dynamicCSTR

What it does: Dynamic CSTR (compA $\rightarrow$ compB) with PID jacket-temperature control; T climbs from 320 K into the $\pm 2\%$ setpoint band within ~ 20 s (no overshoot) and tracks toward 350 K across the 150 s window

The story: the canonical first choupCtrl test.

Process: Continuous CSTR with first-order reaction compA $\rightarrow$ compB. Feed is pure compA at 320 K, 5×10^{-3} kmol/s; reactor volume 1 L; mild exotherm ($\Delta H_{\text{rxn}} = -50$ kJ/mol). Without control the reactor settles a few K above the inlet temperature (small reaction exotherm balanced by convective cooling), well below the desired setpoint of 350 K.

Controller: A PID acts on the jacket temperature (T_{jacket}) to drive the reactor temperature (T) to a setpoint of 350 K. Tuning is pedagogical, not optimal:

$K_p = 4.0$ (proportional band $\approx 7\text{--}8$ K) $K_i = 0.04$ (integral time $\tau_I = K_p/K_i \approx 100$ s) $K_d = 0.0$ (no derivative; system is benign enough)

Jacket bounded to [280, 420] K — realistic limits of a utility loop. Anti-windup: integral does not advance when the unclamped output is outside the bounds.

Expected trajectory (150 s window): Open-loop: T relaxes a few K above the 320 K inlet (small exotherm balanced by convective cooling). Closed-loop: T rises fast (the jacket MV slams to its 420 K clamp, then backs off), entering the $\pm 2\%$ setpoint band by ~ 20 s, well-damped with no overshoot. By 150 s T is ~ 348.5 K and still creeping up: the last ~ 1.5 K rides the slow chemical tail (the exotherm's conversion, $\tau \sim 120$ s) toward 350 K beyond the window. The jacket (MV) settles around 347 K to keep heating in against the cool 320 K feed. (Widen endTime to watch the final approach to 350 K; the 150 s window is sized to SHOW the control transient, not the slow tail.)

To run the OPEN-LOOP comparison, comment out the entire ‘controllers’ block below — the case will then run the bare reactor with whatever T_{jacket} the ‘operation’ block initialised it to (here 320 K, i.e. no jacket heating).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	TC1	MV_final	348.54491648
kpi	TC1	PV_final	348.468973192
kpi	TC1	SP_final	350
kpi	reactor	T_final	348.481893564
kpi	reactor	n_compA_final	0.0116161140368
kpi	reactor	n_compB_final	0.00038388596316
kpi	reactor	t_end	150

ctrl02_disturbance_rejection

tutorials/ctrl/ctrl02_disturbance_rejection

reactor

dynamicCSTR

What it does: PID disturbance rejection: T_in steps down then up, PID maintains T_setpoint

The story: the textbook control demo.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T_in drops 15 K; the integral action absorbs it so tightly that T_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T_in rises 15 K; again T_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_reactor) tracks closely.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535585
kpi	TC1	PV_final	350.00207645
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059115
kpi	reactor	n_compA_final	0.0110806465498
kpi	reactor	n_compB_final	0.000919353450202
kpi	reactor	t_end	2000

ctrl03_adaptive_disturbance

tutorials/ctrl/ctrl03_adaptive_disturbance

reactor
dynamicCSTR

What it does: PID disturbance rejection with ADAPTIVE plant integration — controller samples on the fixed 1 s grid, Rosenbrock23 sub-steps between samples (MV held)

The story: the ctrl02_disturbance_rejection demo re-integrated with an ADAPTIVE plant integrator: the PID still samples on the fixed 1 s control grid (MV held between samples) while Rosenbrock23 sub-steps the plant adaptively in between. The flowsheet below is ctrl02's textbook disturbance-rejection setup, unchanged:

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance).
2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_{in}:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_{reactor} drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T_{in} drops 15 K; the integral action absorbs it so tightly that T_{reactor} holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T_{in} rises 15 K; again T_{reactor} stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_{reactor}) tracks closely.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535626
kpi	TC1	PV_final	350.002076468
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059134
kpi	reactor	n_compA_final	0.0110806465416
kpi	reactor	n_compB_final	0.000919353458409
kpi	reactor	t_end	2000

ctrl03_startup_transient

tutorials/ctrl/ctrl03_startup_transient

reactor
 dynamicCSTR

What it does: Open-loop CSTR START-UP transient with OpenFOAM-style real-time instant dirs: charged cold (320 K), no control, watch $T(t)$ and the holdup approach the natural steady state by scrubbing 0/ 50/ 100/ ... s

The story: Process: A continuous CSTR (first-order compA $\rightarrow$ compB, mild exotherm) is CHARGED cold (320 K, pure compA) and then left to run with NO controller. The jacket sits at a fixed 320 K. From the cold start the reactor warms a little from the reaction exotherm, the feed/ outlet convection competes, and (T , composition) relax to the natural open-loop steady state over $\sim$ several residence times.

What this tutorial shows: The OpenFOAM-style transient solution directories. With ‘solutionControl { write true; }’ in controlDict, the case root fills with REAL-TIME directories ‘0/ 50/ 100/ ... 1000/’. Each ‘<t>/internalState’ is the reactor HOLDUP at that instant; each ‘<t>/streams’ is the instantaneous outlet face. Scrub the times and you watch $T(t)$ and the inventory march to steady state — the same way you scrub a CFD transient in ParaView.

To compare CLOSED-LOOP control, see ctrl01_cstr_temp_control (a PID drives T to a 350 K setpoint) and ctrl05_disturbance_transient (a PID rejecting a feed-temperature disturbance), both also writing time dirs.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	reactor	T_final	320.759374011
kpi	reactor	n_compA_final	0.0117988900072
kpi	reactor	n_compB_final	0.000201109992805
kpi	reactor	t_end	1000

ctrl05_disturbance_transient

tutorials/ctrl/ctrl05_disturbance_transient

reactor
dynamicCSTR

What it does: PID disturbance rejection with OpenFOAM-style real-time instant dirs: T_{in} steps down (700 s) then up (1300 s); scrub 0/ 40/ 80/ ... s to watch the PID track the 350 K setpoint and the jacket MV move

The story: the textbook control demo, captured in OpenFOAM-style real-time instant directories (see controlDict's solutionControl block). Identical physics to ctrl02; the difference is that each written time drops a '<t>/internalState' + '<t>/streams' you can scrub across the two disturbance steps to watch the PID reject them.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_{in}:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_{reactor} drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

0 < t < 700: normal startup, T → 350 K (≈ ctrl01). t = 700: T_{in} drops 15 K; the integral action absorbs it so tightly that T_{reactor} holds at 350 K with no visible sag (the jacket simply keeps heating). t = 1300: T_{in} rises 15 K; again T_{reactor} stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_{reactor}) tracks closely.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	335
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	335
kpi	TC1	MV_final	347.976535585
kpi	TC1	PV_final	350.00207645
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.002059115
kpi	reactor	n_compA_final	0.0110806465498
kpi	reactor	n_compB_final	0.000919353450202
kpi	reactor	t_end	2000

ctrl06_sine_disturbance

tutorials/ctrl/ctrl06_sine_disturbance

reactor
dynamicCSTR

What it does: Forced SINUSOIDAL inlet-T disturbance: $T_{in} = 320 + 15 \sin(2 \pi t / 600)$. The PID chases a moving target $\rightarrow$ the reactor temperature settles into a forced sinusoidal response (attenuated + phase-lagged). Sweep the period to build a one-point Bode by trial-and-error (the frequency-response seed).

The story: the FREQUENCY-RESPONSE seed.

Same DynamicCSTR as ctrl01/ctrl02 (compA $\rightarrow$ compB, mild exotherm, jacket heat transfer) and the same PID temperature controller. The NEW ingredient is the disturbance: instead of the step train of ctrl02's 'Schedule' controller, a 'Signal' controller injects a SINUSOID into the inlet temperature T_{in} :

$$T_{in}(t) = 320 + 15 * \sin(2 * \pi * t / 600) \text{ [K]}$$

mean = 320 K amplitude = 15 K period = 600 s ($f = 1/600 = 1.667\text{e-}3$ Hz, $\omega = 0.01047$ rad/s)

The PID controller has no foreknowledge of the forcing — it only sees $T_{reactor}$ drift sinusoidally away from the 350 K setpoint and works the jacket to compensate. After the first $\sim$ period the reactor temperature settles into a FORCED SINUSOIDAL RESPONSE: same period as the input, but ATTENUATED (the closed loop rejects most of the swing) and PHASE-LAGGED.

THE LESSON (Seborg/Edgar/Mellichamp/Doyle, Ch. 13-14 frequency response): read the output/input amplitude ratio [dB] and the phase lag [deg] off the steady-cycling trajectory, and you have ONE POINT of a Bode plot. Sweep the 'period' (edit one number below), re-run, and you trace the closed loop's frequency response by trial-and-error — the differentiator.

Backward-compatible by construction: ctrl02's 'type Schedule; schedule(...)' staircase still works unchanged; this case swaps it for the more general 'type Signal; signal { type sinusoidal; ... }'.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedDist	MV_final	319.842923238
kpi	FeedDist	PV_final	0
kpi	FeedDist	SP_final	319.842923238
kpi	TC1	MV_final	350.026347162
kpi	TC1	PV_final	350.170890591
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.173436842
kpi	reactor	n_compA_final	0.0110844237702
kpi	reactor	n_compB_final	0.000915576229809
kpi	reactor	t_end	3000

ctrl07_lhhw_inhibition

tutorials/ctrl/ctrl07_lhhw_inhibition

reactor
 dynamicCSTR

What it does: Product inhibition meets a PID: an LHHW reaction whose exotherm fades as its own product covers the catalyst

The story: ctrl01's temperature loop, with a reaction that poisons itself.

Same vessel, same PID, same setpoint. The one change is the rate law: the product compB now ADSORBS on the catalyst, so the denominator grows as the reaction proceeds,

$$r = k(T) \cdot c_A / (1 + K_B c_B)$$

and the reaction throttles itself. A power law cannot express that at any order; only the adsorption term can. This is the same 'RateLaw' that the steady 'cstr' and the 'batchReactor' evaluate – one definition, three reactors.

What the controller sees: an exotherm that FADES as conversion proceeds, not a constant one. The heat the loop is rejecting shrinks under it, and the jacket temperature must follow. A tuning that holds against a first-order exotherm is not the same tuning that holds against a decaying one.

The adsorption constant $K_0 = 2.0 \text{ m}^3/\text{kmol}$ is DIDACTIC – chosen so the inhibition is visible over the control horizon, not measured on any catalyst. For a real, cited LHHW see 'cstr07_lhhw_methylAcetate' and 'batch07_lhhw_methylAcetate' (Popken 2000, Eq 16, on UNIQUAC activities).

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	TC1	MV_final	349.740705746
kpi	TC1	PV_final	348.328939265
kpi	TC1	SP_final	350
kpi	reactor	T_final	348.342145073
kpi	reactor	n_compA_final	0.0117152677233
kpi	reactor	n_compB_final	0.000284732276726
kpi	reactor	t_end	150

ctrl08_prbs_ident

tutorials/ctrl/ctrl08_prbs_ident

reactor
 dynamicCSTR

What it does: PRBS identification experiment: $T_{in} = 320 \pm 10$ K by a maximal-length LFSR ($n=4$, period 15 bits $\times$ 120 s) – deterministic, seed-declared, period-VERIFIED; the broadband excitation the sine of ctrl06 cannot give.

The story: PRBS identification experiment (forum #100.4/#101).

The classic system-identification input: a pseudo-random binary sequence on the inlet temperature, $T_{in} = 320 \pm 10$ K, driven by a MAXIMAL-LENGTH Fibonacci LFSR — $n = 4$ registers, canonical taps (4 3), DECLARED seed 1, period $2^4 - 1 = 15$ bits of 120 s each (1800 s per full period).

Why a PRBS and not ctrl06's sine: one sinusoid excites ONE frequency; the PRBS is broadband (its autocorrelation approximates an impulse), so a single record carries the information for a full low-order model fit. And unlike real "random" noise it is DETERMINISTIC: seed + taps declared in this dict reproduce the experiment bit for bit, run after run.

THE HAND-DERIVED SEQUENCE (the exact-sequence contract of #101): with the frozen convention (output = s_n ; feedback = XOR of taps; shift $s_1 \leftarrow$ feedback; seed LSB $\rightarrow s_1$), $n=4$ / taps (4 3) / seed 1 gives

0 0 0 1 0 0 1 1 0 1 0 1 1 1 1 (bits $k = 0..14$, then repeats)

– walk it yourself: state 0001 $\rightarrow$ 0010 $\rightarrow$ 0100 $\rightarrow$ 1001 $\rightarrow$ 0011 $\rightarrow$ 0110 $\rightarrow$ 1101 $\rightarrow$ 1010 $\rightarrow$ 0101 $\rightarrow$ 1011 $\rightarrow$ 0111 $\rightarrow$ 1111 $\rightarrow$ 1110 $\rightarrow$ 1100 $\rightarrow$ 1000 $\rightarrow$ 0001 (back at the seed after EXACTLY 15 steps: maximal). The run's announce line prints these same first bits; the initialiser REFUSES any tap set whose cycle is not exactly $2^n - 1$.

The PID (same tuning as ctrl06) holds the reactor at 350 K against the binary disturbance; the trajectory's T_{in} column IS the PRBS, and the reactor response is the identification record.

What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedPRBS	MV_final	330
kpi	FeedPRBS	PV_final	0
kpi	FeedPRBS	SP_final	330
kpi	TC1	MV_final	348.579621229
kpi	TC1	PV_final	350.006520504
kpi	TC1	SP_final	350
kpi	reactor	T_final	350.006465191
kpi	reactor	n_compA_final	0.0110804887195
kpi	reactor	n_compB_final	0.000919511280459
kpi	reactor	t_end	3600

ctrl09_stream_disturbance

tutorials/ctrl/ctrl09_stream_disturbance

reactor
 dynamicCSTR

What it does: Inlet-field disturbance: componentMolarFlow.compA steps $3.5\text{e-}5 \rightarrow 5.25\text{e-}5$ kmol/s; F and z_in change by the DERIVED values ($F = 6.75\text{e-}5$, $z_A = 7/9$) – the disturbance never hides from the other components

The story: the inlet-field actuator, end to end.

The Signal controller targets the reactor's INLET FACE by field name ('inletField componentMolarFlow.compA') — the only legal disturbance surface (an outlet actuator is not even expressible). The AUTHORITATIVE inlet state is the per-component molar-flow vector; F_in and z_in are DERIVED from it on every write, so a composition disturbance can never hide inside a silent renormalisation (forum #101):

$t < 600$ s: $n_A = 3.5\text{e-}5$, $n_B = 1.5\text{e-}5$ kmol/s $\rightarrow F_{in} = 5.0\text{e-}5$, $z_A = 0.70$, $z_B = 0.30$ $t \geq 600$ s: n_A stepped to $5.25\text{e-}5$; n_B UNTOUCHED at $1.5\text{e-}5 \rightarrow F_{in} = 6.75\text{e-}5$, $z_A = 5.25/6.75 = 7/9 = 0.77778$, $z_B = 1.5/6.75 = 2/9 = 0.22222$ (hand-derived)

The golden pins F_in and z_in * finals to these fractions, and the Signal value is the ABSOLUTE molar flow (announced at start-up). The reactor's own response (more A in, higher conversion load, T transient against the PID) is the pedagogy; the CONTRACT is the derived pair.

The 'moleFraction.<c>' sibling field instead holds F_in constant and renormalises the OTHER components proportionally — two explicit semantics, each announced, never a guess.

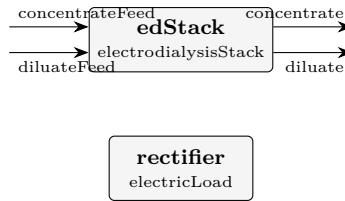
What to expect (golden-locked):

kind	unit/stream	key	expected
kpi	FeedComp	MV_final	5.25e-05
kpi	FeedComp	PV_final	0
kpi	FeedComp	SP_final	5.25e-05
kpi	TC1	MV_final	350.204904403
kpi	TC1	PV_final	350.000124372
kpi	TC1	SP_final	350
kpi	reactor	F_in_final	6.75e-05
kpi	reactor	T_final	350.000123673
kpi	reactor	n_compA_final	0.00879292389459
kpi	reactor	n_compB_final	0.00320707610541
kpi	reactor	t_end	1800
kpi	reactor	z_in_compA_final	0.777777777778
kpi	reactor	z_in_compB_final	0.222222222222

Category V: Electrochemical systems

ed01_nacl_desalination

tutorials/electrochem/ed01_nacl_desalination



What it does: Electrodialysis stack: NaCl diluate desalination, 100 cell pairs

The story: ED01 NaCl desalination by electrodialysis

An electrodialysis stack of 100 cell pairs desalinates a dilute NaCl diluate stream (0.1 mol/kg) against a more-concentrated concentrate (0.5 mol/kg). A specified stack current (BELOW the limiting current) drives Na⁺ through the CMX (cation-exchange) membranes and Cl⁻ through the AMX (anion-exchange) membranes, OUT of the diluate and INTO the concentrate.

dilateFeed → + ————— + → diluate (desalinated) | ED stack | concentrateFeed → | (CMX/AMX, N=100) | → concentrate (enriched) + ————— + | (power, energy wire) v rectifier (electricLoad)

The stack PRINTS (verbosity 3) the Faraday transfer, the Nernst membrane potential from REAL Davies activities, the Cowan-Brown limiting current i_{lim} (and i/i_{lim}), the ohmic drop and the stack voltage/power.

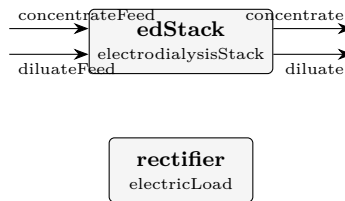
What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrate	F	0.00501492454831
stream	concentrateFeed	F	0.005
stream	dilate	F	0.00498507545169
stream	dilateFeed	F	0.005
kpi	edStack	E_mem_pair	0.0794573845707
kpi	edStack	I	8
kpi	edStack	N_cellpairs	100
kpi	edStack	R_pair	0.00507753048751
kpi	edStack	U	13.5077628471
kpi	edStack	W_electric	108.062102777
kpi	edStack	W_electric_kW	0.108062102777
kpi	edStack	current_efficiency	0.9
kpi	edStack	demin_ratio	0.831450327631
kpi	edStack	i_density	40

... and 9 more reference values in the case's *expected*.

ed02_over_limiting_current

tutorials/electrochem/ed02_over_limiting_current



What it does: Electrodialysis stack driven ABOVE its limiting current (loud warning)

The story: ED02 over-limiting current electrodialysis

Same stack as ED01, but run on a MORE DILUTE diluate (0.02 mol/kg) at a LOW cross-flow velocity and a HIGH current. The diluate-side limiting current density i_{lim} (Cowan-Brown) is small; the applied current density EXCEEDS it, so the stack runs in the OVER-LIMITING regime.

Choupo does NOT silently clamp the current (the "no silent crutch" credo): it computes the unclamped Faraday transfer and prints a LOUD warning that $i > i_{lim}$ – the diluate boundary layer is ion-depleted, water splitting / pH swings set in, and the real current efficiency would collapse. The student SEES the regime and the i/i_{lim} ratio, then chooses to reduce the current or raise the flow / feed concentration.

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	concentrate	F	0.00500205212539
stream	concentrateFeed	F	0.005
stream	diluate	F	0.0199979478746
stream	diluateFeed	F	0.02
kpi	edStack	E_mem_pair	0.156590406589
kpi	edStack	I	5.5
kpi	edStack	N_cellpairs	20
kpi	edStack	R_pair	0.0130038886532
kpi	edStack	U	6.06223588363
kpi	edStack	W_electric	33.34229736
kpi	edStack	W_electric_kW	0.03334229736
kpi	edStack	current_efficiency	0.9
kpi	edStack	demin_ratio	0.142508707778
kpi	edStack	i_density	27.5

...and 9 more reference values in the case's *expected*.

Category VI: Multi-sector plant showcases

The fractal flowsheet at full size – read them like worked examples.

ChemicalPlantTutorial

The theory you need. The full multi-sector plant showcase: feed treatment, reaction, separation, utilities – the fractal flowsheet at its largest, run-only (no golden), for reading and dissecting rather than reproducing numbers.

CONCENTRATION

tutorials/plant/ChemicalPlantTutorial/CONCENTRATION

The story: / system/flowsheetDict (SECTOR – COMPOSITE)

Double-effect evaporator (vapour of effect 1 heats effect 2) feeding a crystalliser. Same shape as the plant flowsheetDict, one level down: children + boundary + default feeds + connections.

DRYING

tutorials/plant/ChemicalPlantTutorial/DRYING

The story: / system/flowsheetDict (SECTOR – COMPOSITE)

Spray dryer → additional solid drying → cyclone dust separator. The solid powder goes SD → BD → product; the spray-dryer exhaust air (carrying fines) goes SD → CY, which returns clean air + recovered dust.

FERMENTATION

tutorials/plant/ChemicalPlantTutorial/FERMENTATION

The story: / system/flowsheetDict (SECTOR – COMPOSITE with INTERNAL RECYCLE)

Continuous fermentation side-loop of the sugar plant: a slip-stream of fresh juice (Must) is hydrolysed + fermented into ethanol + CO₂, the Flash splits the volatiles (EthanolVapour product) from the liquid bottoms, and a Splitter recycles 80 % of the bottoms back to the Mixer so unconverted sucrose gets another shot. Pedagogical purpose:

1. Reaction (Arrhenius first-order in sucrose). 2. Vapour-liquid flash for separation. 3. Tear-stream RECYCLE that the Wegstein/Newton outer loop must iterate to a fixed point.

Must [Mixer] MixedFeed [Fermentor] FermOut [Flash] EthanolVapour Liquid Recycle [Splitter]

Tear stream: Recycle (Splitter output → Mixer input).

JuiceSplitter

tutorials/plant/ChemicalPlantTutorial/JuiceSplitter

The story: / system/flowsheetDict (LEAF – plant-level juice split)

Diverts the fresh sugar-cane juice between the two production lines of the mill: 2/3 goes to the CONCENTRATION line (evaporators → crystals → dry sugar) and 1/3 to the FERMENTATION line (yeast → ethanol). This is what a real Brazilian sugar-ethanol plant does after the clarifier: the same juice header feeds both products, with the split ratio set by market price + crop quality.

Pedagogically the splitter is the single inter-sector cable that ties the two downstream branches together – without it, FERMENTATION would be a parallel island.

esterification2sector

tutorials/plant/esterification2sector

What it does: Esterification plant – two sectors (REACTION + SEPARATION) sharing components; per-sector constant/ data (S1 kinetics, S2 NRTL pair); ethanol-water pair inherited from the standard library

The story: - Choupo -*—————* esterification2sector – a TWO-SECTOR fractal plant, WIRED.

[REACTION] — esterification CSTR, its OWN kinetics in | sectors/REACTION/constant/reactions reactorOut

[SEPARATION] — isothermal flash; the ethylAcetate-water NRTL pair | \{\} is PARTICULAR to this sector liquid vapor (sectors/SEPARATION/constant/binaryPairs/NRTL/...); ethanol-water is INHERITED from the standard library.

The point of the case is the FRACTAL ‘constant/’: components shared across the plant, particular data (kinetics here, a fitted pair there) living with the sector that owns it, and the rest walked up to the standard catalogue.

The root graph assembles the sectors and the inter-sector links, nothing more. Each stream’s state lives in its own file under 0/, owned by the sector that produces it.

A two-sector plant that exists to prove one thing: data lives with the sector that owns it, and a whole-plant run finds it there.

*Components are shared across the plant. The kinetics belong to REACTION and are never copied to the root. The ethylAcetate-water NRTL pair belongs to SEPARATION — a *particular* datum, fitted for this separation — while ethanol-water walks up to the standard catalogue. Three levels of ownership, one run.*

That perNode is the whole case. The sector’s pair beat the standard library, and the log says so rather than leaving you to wonder which one it used.

Feed 100 kmol/h equimolar acetic acid + ethanol at 360 K. The CSTR ($V_R = 0.5 \text{ m}^3$) reaches 34.2 % conversion — a PLACEHOLDER number: the kinetics in sectors/REACTION/constant/reactions are still the pseudo-first-order stand-in, not the 2nd-order fit this case’s own property evidence supports (see CLAUDE.md, "Pending"). The flash at 355 K, 1.013 bar splits the product into a vapour rich in ethyl acetate (bp 350 K) and ethanol, and a liquid rich in water and the unconverted acid. Mass closes exactly: $34.18 + 65.82 = 100.00 \text{ kmol/h}$.

This case shipped for weeks marked .known-broken, and it earned the mark twice over. Its flash carried P 1.01325; with no unit, which is 1.01 pascal — a hard vacuum — so the K -values exploded and everything went to the vapour. And it was never wired: the sectors existed as thermo regions with the connections commented out, awaiting a "curation phase" that never ended.

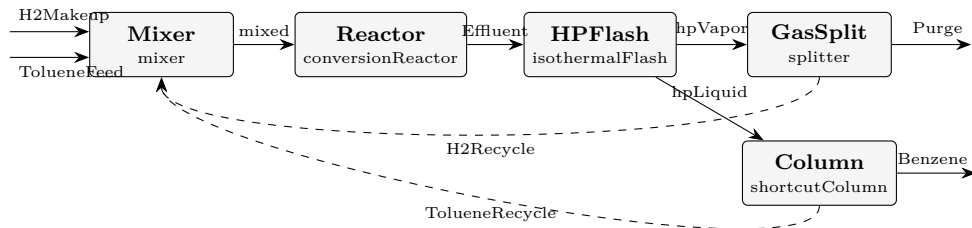
*Fixing it surfaced two real engine bugs, both of the same shape — data owned by a *sector* was invisible from a whole-plant run:*

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Feed	F	0.0277777777778
stream	liquid	F	0.00949461019202
stream	reactorOut	F	0.0277777777778
stream	vapor	F	0.0182831675858

hda

tutorials/plant/hda



What it does: HDA plant – toluene + H₂ → benzene + CH₄. A single FLAT flowsheet with two plant-level recycles (an H₂ gas loop and an unreacted-toluene liquid loop); cross-sector tears are unsupported, so it is not fractal. Furnace lumped into the isothermal reactor; light-gas K-values rest on Antoine extrapolation (see thermoPackage).

The story: - Choupo -*—————* flowsheetDict – HDA plant (single FLAT flowsheet).

toluene + H₂ → benzene + CH₄, with TWO coupled recycles: an H₂ gas loop (GasSplit 95 % → back to the Mixer) an unreacted-toluene loop (Column bottoms → back to the Mixer)

WHY FLAT, not the two-sector fractal we first attempted: a recycle closes only when its ‘tearStreams’ declaration AND both endpoints live in the SAME flowsheet node. The engine’s tear-closing override (Flowsheet.cpp) is leaf-only and same-level-only – it does NOT support CROSS-SECTOR (composite-to-composite) tears. Both HDA loops run from the separation train back to the reactor feed, so no single sector owns both ends; they are inherently plant-level and must be flat (the proven process03_recycle pattern: each tear name is BOTH a Mixer input AND a producer output, listed after the Mixer). The fractal sector showcase lives elsewhere (the sector-internal recycle in the two-sector demo).

Everything is at 35 bar so the recycles return at reactor pressure (the Mixer takes min-P). The furnace is lumped into the isothermal reactor (operation.T 900 K) – the engine’s heater is duty-specified, not T-target. The benzene product is ~89 mol% benzene but carries a non-trivial ~10% CH₄ (plus ~0.2% H₂ and ~0.3% toluene): the light non-keys dissolved in the flash liquid are lighter than benzene, so the FUG shortcut sends them all to the distillate. A real plant would put a stabiliser / light-ends column upstream of the product spec to strip that ~10% methane; the shortcut here is quantitatively approximate (it needs no rigorous liquid enthalpy, so unlike a Wang-Henke column it does not choke on a gas in the liquid).

What to expect (golden-locked):

kind	unit/stream	key	expected
stream	Benzene	F	0.0308570333668
stream	Effluent	F	0.589422462034
stream	H2Makeup	F	0.0305555555556
stream	H2Recycle	F	0.522049697299
stream	Purge	F	0.0274762998579
stream	TolueneFeed	F	0.0277777777778
stream	TolueneRecycle	F	0.00903943150968
stream	hpLiquid	F	0.0398964648764
stream	hpVapor	F	0.549525997157
stream	mixed	F	0.589422462034
kpi	Column	B	0.00903943150968
kpi	Column	D	0.0308570333668
kpi	Column	N_min	8.3953506183
kpi	Column	N_theoretical	20.1717591626

... and 28 more reference values in the case’s *expected*.

lithiumBrinePlant

tutorials/plant/lithiumBrinePlant

What it does: lithiumBrinePlant – THE reference case: a FRACTAL lithium-carbonate plant from salar brine, WIRED into one solver. Five SECTORS, each a thermo WORLD of its own – BRINE (electrolyte/Pitzer), EXTRACTION (gamma-gamma NRTL LLE), CARBONATION (Li₂CO₃ retrograde solubility), HOTAIR (Gibbs combustion), FINISHING (drying + cyclone) – connected sector to sector, each sector declaring its own propertyPackage (its world) on the global component set, with optional per-unit thermo overrides where a single unit needs a different world. The proof that the architecture is fractal: sectors are composite boxes, streams flow between them, model boundaries audited.

The story: Five REAL sectors. Streams are named physical graph edges. Sector names are subdomains; unit operations live inside each sector.

polycaprolactonePlant

The theory you need. Ring-opening polymerisation plant: monomer to polycaprolactone with step-growth KPIs (Carothers, Flory) on the PFR, polymer properties from group contribution on the repeat unit.

properties

tutorials/plant/polycaprolactonePlant/properties

pclRepeat_Tg_yang2020
estimateComponent

What it does: Close the loop: ESTIMATE the PCL repeat-unit glass transition Tg by the Yang 2020 group-contribution scheme — the plant gives Mn/PDI, this gives the physical property from the SAME repeat unit $-\text{O}-(\text{CH}_2)_5\text{CO}-$

The story: pclProperties – CLOSE THE LOOP: process $\rightarrow$ molar mass $\rightarrow$ physical property.

The plant (../) gave you the chain statistics from the KINETICS: Mn, Mw, PDI on the polycaprolactone it makes. Here you ESTIMATE the polymer's PHYSICAL PROPERTIES from the SAME repeat unit $-\text{O}-(\text{CH}_2)_5\text{CO}-$ by group contribution — the bridge from "how big are the chains" to "how does the material behave".

TWO estimators, two properties, on the repeat unit pclRepeat ($M_0 = 114.14$):

(A) Tg, the GLASS-TRANSITION temperature — Yang 2020 (run below). Repeat unit $-\text{O}-(\text{CH}_2)_5\text{CO}-$ decomposes into Yang main-chain groups: O x1 : MW 16 Yg -14.718 CH₂ x5 : MW 14 Yg 4.026 (each) CO x1 : MW 28 Yg 4.370

$M_0 = 16 + 5 \cdot 14 + 28 = 114$ g/mol (matches the reactor M_0 !) $Yg(\text{inf}) = -14.718 + 5 \cdot 4.026 + 4.370 = 9.782$ (1e3 g.K/mol) $Tg(\text{inf}) = 9.782e3 / 114 \sim 86$ K (the additive estimate) (Measured PCL Tg ~ 213 K / -60 C; an additive main-chain scheme is weak for a flexible aliphatic polyester — the model tells the truth about itself, which IS the lesson. See the console's per-cent error vs the anchor.)

(B) DENSITY — Van Krevelen (Bondi Vw). NOT runnable on PCL with the shipped data: the Slice-1 Van Krevelen set ships only hydrocarbon + chloro groups (CH₃ CH₂ CH C ACH AC CHCl CH₂Cl); it has NO ester -O- / -CO- group, so the repeat unit $-\text{O}-(\text{CH}_2)_5\text{CO}-$ cannot be decomposed. Running it would HARD- ERROR with "unknown polymer group 'O'" — the estimator refuses to invent a value (no silent zero). The honest path: density needs the ester groups curated into data/standards/vanKrevelen/groups.dat first. The worked Van Krevelen DENSITY example lives in tutorials/props/estimate/vanKrevelen01_polystyrene_density (a pure- hydrocarbon repeat unit the Slice-1 set CAN do).

This is the same $M_0 = 114.14$ the reactor's polymer{ } block used for $M_n = M_0/(1-p)$ — one repeat-unit mass, two places (process AND property).

A choupouProps sub-case of ../ (the polycaprolactone plant). The plant gives you the chain statistics from the kinetics (Mn / Mw / PDI); here you estimate the polymer's physical property — the glass transition Tg — from the same repeat unit $-\text{O}-(\text{CH}_2)_5\text{CO}-$ ($M_0 = 114.14$, the exact number the reactor used for $M_n = M_0/(1-p)$).

** Tg via Yang 2020 (CC-BY) — runnable. $Tg(\text{inf}) = 85.8$ K; the console prints the -59.7 % deviation from the measured ~ 213 K (an honest limit of an additive scheme on a flexible aliphatic polyester). $M_0 = 114$ is goldened (an exact mass-balance fact); Tg is shown but not goldened as validated physics. * Density via Van Krevelen — *not* runnable on PCL with the shipped data (the Slice-1 group set has no ester -O-/-CO- group, so it would hard-error rather than invent a value). See the full explanation and the runnable hydrocarbon example linked from the parent ../README.md.*

The full teaching narrative lives in the parent README. See system/propsDict for the exact group decomposition and the estimator invocation.

What to expect (golden-locked):

kind	unit/stream	key	expected
diag	pclRepeat_Tg_yang2020	M0_g_per_mol	114
diag	pclRepeat_Tg_yang2020	Tg_K	85.81
diag	pclRepeat_Tg_yang2020	YgSum_1e3gKmol	9.782

sugarPlantEconomicsSweep

tutorials/plant/sugarPlantEconomicsSweep

What it does: Sugar plant ECONOMICS SWEEP – IRR / payback / NPV vs RawJuice sucrose content (Phase-2 SweepDriver post-chain differentiator)

The story: / system/flowsheetDict (PLANT level – COMPOSITE)

The flagship sugar plant (the ChemicalPlantTutorial flowsheet, reused verbatim – sectors and units included), here driven by system/outerDict: a SweepDriver walks the RawJuice sucrose fraction and runs the FULL post chain (sizing → costing → economics, system/postDict) on every point, so IRR / NPV / payback vs feed quality come out as sweep responses.

A composite node: it has ‘children’ (sub-flowsheets, here the sectors) and ‘connections’ (the pipework between them). The same shape a sector uses to list its units – one level up. This is the fractal.

Naming convention (see docs/ai/case-layout.md): SECTORS in CAPS : CONCENTRATION, DRYING, FERMENTATION Units in PascalCase : Cryst, Evap1, Evap2 Streams in PascalCase : RawJuice, Magma, Powder, EvapVapour Components in lowercase : water, sucrose, N2

This is the flagship sugar plant (tutorials/plant/ChemicalPlantTutorial) driven by a SweepDriver instead of a single pass. It exists to show the one thing a single CAPEX number cannot: how the project economics move when a process variable changes.

The driver runs the FULL post-processing chain (sizing → costing → economics, in system/postDict) on every converged point, so economics.IRR, economics.paybackYears, economics.NPV, ? resolve as ordinary KPI look-ups (the Phase-2 SweepDriver post-chain wiring). The result lands in sweep_results.csv.

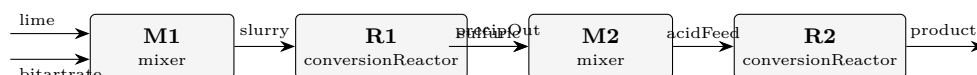
Sweeping the RawJuice sucrose content 0.09 → 0.16 mol fraction walks the plant from sub-economic, through break-even, to clearly profitable:

Richer juice → more crystalline sugar per unit feed → more revenue against an almost-unchanged plant cost → the IRR climbs and the payback appears and shortens. The NPV crosses zero between 0.104 and 0.118.

*Both use the same constant/economics prices (EU Sugar Market Observatory white sugar, ECB EUR/USD, *Chemical Engineering* CEPCI, US statutory tax) and the same system/postDict cost chain — only the run mode differs.*

tartaricAcid

tutorials/plant/tartaricAcid



What it does: Tartaric-acid plant (wine route): lime precipitation of calcium tartrate, then sulfuric acidulation to tartaric acid + gypsum. Molecular reactors; the solution chemistry sub-study is tutorials/props/electrolyte/tartaricAcid_acidulation.

The story: - Choupo -*—————* Tartaric-acid plant – the WINE ROUTE chemistry, as a reactor train.

bitartrate (KHT) lime $\text{Ca}(\text{OH})_2$ M1 mixer R1 precipitation ($\text{CaTartrate} + \text{KOH}$)

H_2SO_4 M2 mixer R2 acidulation product (tartaricAcid + gypsum)

R1: $\text{KHC}_4\text{H}_4\text{O}_6 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaC}_4\text{H}_4\text{O}_6 + \text{KOH} + \text{H}_2\text{O}$ R2: $\text{CaC}_4\text{H}_4\text{O}_6 + \text{H}_2\text{SO}_4 \rightarrow \text{C}_4\text{H}_6\text{O}_6(\text{tartaric acid}) + \text{CaSO}_4(\text{gypsum})$

SCOPE (honest). This demonstrates the REACTION chemistry and closes the MASS balance. The physical solid-liquid SEPARATIONS of the real process – filtering the calcium-tartrate cake from the mother liquor, and the gypsum from the acid – are NOT modelled: Choupo treats the precipitate as a dissolved nonvolatile, not a solid phase, so there is no $F_{\text{solid_mass}}$ for a filter to capture (a precipitator / solid-phase reactor is a roadmap feature). The ENERGY balance also refuses on the solid calcium tartrate (its formation datum is omitted – no curated dissolution enthalpy). The solution-chemistry of the acidulation liquor (Pitzer, gypsum saturation) lives in tutorials/props/electrolyte/tartaricAcid_acidulation.

twoSectorDemo

The theory you need. The zippable two-sector demonstration plant – the case used to rehearse the online landing. Same physics as the esterification demo.

SECTOR_R`tutorials/plant/twoSectorDemo/SECTOR_R`

What it does: SECTOR_R – the reactor sector. Runs its OWN compiled unit op (stoichReactor) on the plant-level thermo (cascaded from ../constant).

The story: one custom reactor (compiled from code/). Build + run with: `bin/buildCode tutorials/plant/twoSectorDemo/SECTOR_R runCase tutorials/plant/twoSectorDemo/SECTOR_R` —

SECTOR_S

tutorials/plant/twoSectorDemo/SECTOR_S



What it does: SECTOR_S – the separation sector. Runs its OWN compiled unit op (sharpSplitColumn) on the plant-level thermo (cascaded from ../constant).

The story: one custom distillation (compiled from code/). Build + run: bin/buildCode tutorials/plant/twoSectorDemo/SECTOR_S runCase tutorials/plant/twoSectorDemo/SECTOR_S —